

New Developments of the Sequential Probability Ratio Test Control Chart



by

TEOH JING WEI

Submitted for the degree of

Doctor of Philosophy

HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

April 2024

The copyright in this thesis is owned by the author. Any quotation from the thesis or use of any of the information contained in it must acknowledge this thesis as the source of the quotation or information.

ABSTRACT

This thesis aims to enrich the literature on the SPRT control chart following two major contributions made in the late 1990s and early 2010s. The SPRT chart has been chosen as the central premise of this thesis due to its strong detection performance as well as high sampling efficiency. Two major research gaps have been identified. The first gap is the lack of documentation on the SPRT chart with estimated process parameters and its statistical design. The second gap is the lack of a well-motivated SPRT chart for joint monitoring of the mean and dispersion of a process. To fill the literature gaps, we formulate the theoretical framework for the SPRT chart with estimated process parameters, as well as develop a new SPRT chart for joint monitoring of the mean and dispersion. Optimisation designs based on various industrial objectives have been developed in this thesis. Real industrial examples involving a variety of destructive and non-destructive tests are also presented in this thesis to illustrate the implementation of the proposed SPRT charts. The thesis should serve as a reference to researchers working in the field of statistical quality control, as well as practitioners seeking to improve the performance of their processes.

ACKNOWLEDGEMENTS

After being turned down on a scholarship opportunity, I was attracted to a different research project advertised under a professor who works on statistical process control. Motivated by the ambition to pursue an academic career, I decided to pull myself together and give it a try. The project supervisor was quickly convinced that I would be competent for the role, thus she (and the School of Mathematical and Computer Sciences) hired me as a graduate research assistant and granted me a scholarship for my Ph.D. study. The pivotal moment has then led to the creation of this thesis.

Everything would not have been achieved if it was not with my primary supervisor, Dr. Teoh Wei Lin. I would like to express my heartfelt gratitude towards her. Her unwavering commitment and guidance have shaped me as a better researcher and learner throughout my Ph.D. journey. She has always emphasised the importance of doing quality research, and explained how research advancements can contribute to increased productivity and a more sustainable industrialisation. One of my proudest achievements was to be involved (together with my supervisor) in a research and development agreement between Heriot-Watt University Malaysia and a renowned multinational technology company, which proves the novelty of our work alongside the impact that our research has created. I would also like to thank my secondary supervisor, Dr. Laila El-Ghandour from the Heriot-Watt University, Edinburgh, and my external supervisor, Dr. Chong Zhi Lin from Universiti Tunku Abdul Rahman, for their guidance throughout my Ph.D. journey.

I would like to express my appreciation to the Ministry of Higher Education Malaysia for funding this research project under the Fundamental Research Grant Scheme (FRGS), no. FRGS/1/2021/STG06/HWUM/02/1. The grant has allowed me to disseminate my findings in multiple international conferences, which helps to build my research profile as well as the soft skills necessary to be a good researcher.

I would also like to thank Heriot-Watt University Malaysia for waiving the tuition fee of my Ph.D. study and providing appropriate resources to facilitate the completion of this research project.

I would like to thank all the academic staff from the School of Mathematical and Computer Sciences for their assistance with regard to my postgraduate affairs. I am also thankful to the staff from the administrative departments, such as the student services, human resources, and finance departments, for approving my application to attending academic conferences as well as giving advice on related matters. I also appreciate their effort in organising events (such as seminars and talks) that bring like-minded Ph.D. students together. During the write-up of this thesis, I have communicated with some of my peers and colleagues. I am truly grateful for their contributions and ideas to help improve the style and content of the thesis. A special shout out to Kwok Wai Meng, whom I have discussed a particular part of my thesis (Markov chain Monte Carlo) with. *A fun fact: certain sections of this thesis were a complete revamp due to the lack of clarity in the original writing.* My appreciation to Professor Sarat Chandra Dass for guiding me in the creation of a thesis that I will forever be proud of.

Finally, I would like to thank my parents for believing in me, being supportive of all my decisions, and continuously encouraging me to pursue my aspirations. It goes beyond words to describe the looks on their faces when I told them that I would be graduating with a Ph.D. soon. *It seemed to them like I have just started it!* However, the end of my Ph.D. journey does not mark the end of my research journey. I will continue to do research and (hopefully) make it to the top of the list in my field. Besides, I would also like to thank all my friends who have motivated me to keep going during my Ph.D. endeavour. Their encouragements have certainly brought me to where I am now.

Inclusion of Published Works Form

Please note you are only required to complete this form if your thesis contains published works. If this is the case, please include this form within your thesis before submission.

Declaration

This thesis contains one or more multi-author published works. I hereby declare that the contributions of each author to these publications is as follows:

Citation details	Teoh, J. W., Teoh, W. L., El-Ghandour, L., Chong, Z. L., Teh, S. Y., & You, H. W. (2022). New Insights on the SPRT Control Chart when the Process Parameters are Unknown. In <i>2022 IEEE 5th International Symposium in Robotics and Manufacturing Automation (ROMA)</i> (pp. 1-6). IEEE. https://doi.org/10.1109/ROMA55875.2022.9915700
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Supervision, Writing – review & editing
Author 4	Supervision, Writing – review & editing
Author 5	Writing – review & editing
Author 6	Writing – review & editing

Citation details	Teoh, J. W., Teoh, W. L., Khoo, M. B. C., Castagliola, P., & Moy, W. H. (2022). On designing an optimal SPRT control chart with estimated process parameters under guaranteed in-control performance. <i>Computers & Industrial Engineering</i> , 174(12), Article 108806. https://doi.org/10.1016/j.cie.2022.108806
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Validation, Writing – review & editing
Author 4	Validation, Writing – review & editing
Author 5	Resources, Validation, Writing – review & editing

Citation details	Teoh, J. W., Teoh, W. L., El-Ghandour, L., Chong, Z. L., & Teh, S. Y. (2023). A Study on the Performances of the SPRT Control Chart When Estimating Process Parameters. In <i>International Conference on Computational Science and Technology</i> (pp. 449-463). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-19-8406-8_36
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing

Author 3	Supervision, Writing – review & editing
Author 4	Supervision, Writing – review & editing
Author 5	Writing – review & editing

Citation details	Teoh, J. W., Teoh, W. L., El-Ghandour, L., Chong, Z. L., & Teh, S. Y. (2023). A Thorough Comparison of the Variable-Sample-Size Weighted-Loss CUSUM and ABS-SPRT Control Charts. In <i>2023 IEEE 13th Symposium on Computer Applications & Industrial Electronics (ISCAIE)</i> (pp. 268-273). IEEE. https://doi.org/10.1109/ISCAIE57739.2023.10165300
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Supervision, Writing – review & editing
Author 4	Supervision, Writing – review & editing
Author 5	Writing – review & editing

Citation details	Teoh, J. W., Teoh, W. L., Chong, Z. L., Chew, X. Y., & Teh, S. Y. (2024). Redesigning the omnibus SPRT control chart for simultaneous monitoring of the mean and dispersion of Weibull processes. <i>International Journal of Integrated Engineering</i> , 16(5), 201-213. https://doi.org/10.30880/ijie.2024.16.05.016
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Supervision, Validation, Writing – review & editing
Author 4	Validation, Writing – review & editing
Author 5	Validation, Writing – review & editing

Citation details	Teoh, J. W., Teoh, W. L., Chong, Z. L., Lee, M. H., & Khaw, K. W. (2024). A robust design for the omnibus SPRT control chart under skewed data distributions. <i>Sains Malaysiana</i> , 53(6), 1441-1461. https://doi.org/10.17576/jsm-2024-5306-17
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Supervision, Validation, Writing – review & editing
Author 4	Validation, Writing – review & editing

Author 5	Validation, Writing – review & editing
----------	--

Citation details	Teoh, J. W., Teoh, W. L., Chong, Z. L., Lee, M. H., & Teh, S. Y. (2024). Re-Evaluating the SPRT Chart with Estimated Process Parameters When the Underlying Distributions Are Gamma, Lognormal, and Weibull. <i>International Journal of Electrical and Electronic Engineering and Telecommunications</i> , 13(1), 80-89. https://doi.org/10.18178/ijeetc.13.1.80-89
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Supervision, Validation, Writing – review & editing
Author 4	Validation, Writing – review & editing
Author 5	Validation, Writing – review & editing

Citation details	Teoh, J. W., Teoh, W. L., El-Ghandour, L., Chong, Z. L., & Teh, S. Y. (2024). An Analysis of the Performance of the SPRT Chart with Estimated Parameters Under the Weibull Distribution. In <i>International Conference on Robotics, Vision, Signal Processing and Power Applications</i> (pp. 175-182). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-99-9005-4_22
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Supervision, Writing – review & editing
Author 4	Supervision, Writing – review & editing
Author 5	Writing – review & editing

Citation details	Teoh, J. W., Teoh, W. L., El-Ghandour, L., Chong, Z. L., & Teh, S. Y. (2024). Redesigning the Joint SPRT Mean and Dispersion Control Chart Based on Mismatched In-control Performances. In <i>2024 20th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)</i> (pp. 12-17). IEEE. https://10.1109/CSPA60979.2024.10525643
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Supervision, Writing – review & editing
Author 4	Supervision, Writing – review & editing
Author 5	Writing – review & editing

Citation details	Teoh, J. W., Teoh, W. L., Khoo, M. B. C., Celano, G., & Chong, Z. L. (2024). Optimal designs of the omnibus SPRT control chart for joint monitoring of process mean and dispersion. <i>International Journal of Production Research</i> , 62(12), 4202-4225. https://doi.org/10.1080/00207543.2023.2254855
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Validation, Writing – review & editing
Author 4	Validation, Writing – review & editing
Author 5	Supervision, Writing – review & editing

Citation details	Teoh, J. W., Teoh, W. L., Khoo, M. B. C., Tran, K. P., & Lee, M. H. (2024). An integrated optimal-GICP design for the SPRT control chart with estimated process parameters based on the average number of observations to signal. <i>Quality Technology & Quantitative Management</i> , In Press. https://doi.org/10.1080/16843703.2024.2312010
Author 1	Conceptualisation, Methodology, Software, Formal analysis, Writing – original draft
Author 2	Conceptualisation, Formal analysis, Funding acquisition, Supervision, Writing – review & editing
Author 3	Validation, Writing – review & editing
Author 4	Validation, Writing – review & editing
Author 5	Validation, Writing – review & editing

TABLE OF CONTENTS

Abstract	i
Acknowledgements	ii
Table of Contents	iv
List of Tables	viii
List of Figures	xviii
List of Notations	xxi
1 Introduction	1
1.1 Statistical Quality Control (SQC)	1
1.2 Problem Statements	4
1.3 Objectives of the Thesis	8
1.4 Structure of the Thesis	8
2 Literature Review	11
2.1 Introduction	11
2.2 Some Mainstream Control Charts	11
2.2.1 The Shewhart Control Chart	12
2.2.2 The Exponentially Weighted Moving Average Control Chart	15
2.2.3 The Cumulative Sum Control Chart	17
2.3 Developments of the Sequential Probability Ratio Test (SPRT) Control Chart	18
2.3.1 The SPRT for Hypothesis Testing	19
2.3.2 The SPRT Chart for an Arbitrary Parameter	24
2.3.3 The SPRT Chart for Process Location	26
2.3.4 The SPRT Chart for Process Dispersion	28
2.4 Performance Measures of a Control Chart	29
2.4.1 The Run Length and the Time to Signal	30

2.4.2	The Sample Number	33
2.4.3	The Number of Observations to Signal	34
2.4.4	Some Measures of the Overall Performance	36
2.5	Control Charts with Estimated Process Parameters	38
2.5.1	Sample Size Determination	40
2.5.2	Selection of Estimators	43
2.5.3	Unconditional In-control Metric Matching Design	45
2.5.4	Guaranteed In-control Performance Design	47
2.6	Control Charts for Joint Monitoring of the Process Mean and Dispersion	52
2.6.1	Multi-chart Control Schemes	53
2.6.2	Single-chart Control Schemes	55
2.7	Conclusions	58
3	The SPRT Chart with Estimated Process Parameters	60
3.1	Introduction	60
3.2	The SPRT Chart with Known Process Parameters	60
3.3	The Proposed SPRT Chart with Estimated Process Parameters	71
3.4	An Evaluation of the Performances of the Traditional SPRT Chart with Estimated Process Parameters	81
3.4.1	Optimal Design Based on the AEQL	82
3.4.2	Optimal Design Based on the EANOS	91
3.5	An Evaluation of the Performances of the SPRT Chart with Estimated Process Parameters Designed under the Unconditional Metric Matching Approach	95
3.5.1	Optimal Design Based on the AAEQL	96
3.5.2	Optimal Design Based on the EAANOS	100
3.6	Optimal Designs of the SPRT Chart with Estimated Process Parameters under the Guaranteed In-Control Performance Approach	103

3.6.1	Optimal Design Based on the AATS	105
3.6.2	Optimal Design Based on the AAQL	118
3.6.3	Optimal Design Based on the ANOS	128
3.6.4	Optimal Design Based on the EANOS	135
3.7	A Comparison of the Performances between the SPRT Chart and the CUSUM Chart with Estimated Process Parameters	142
3.8	Real Industrial Examples	148
3.8.1	Epitaxial Process of Silicon Wafers	148
3.8.2	Wire Bonding Process	153
3.9	Conclusions	156
4	The SPRT Chart for Joint Monitoring of Process Mean and Dispersion	158
4.1	Introduction	158
4.2	The Proposed Omnibus SPRT (OSPRT) Chart	159
4.2.1	Theoretical Framework and Working Mechanism of the OSPRT Chart	159
4.2.2	Run Length Properties of the OSPRT Chart	164
4.3	Optimal Designs of the OSPRT Chart	170
4.3.1	Optimal Design Based on the ATS	171
4.3.2	Optimal Design Based on the AQL	182
4.3.3	Optimal Design Based on the ANOS	188
4.3.4	Optimal Design Based on the EANOS	193
4.4	A Brief Review on Some Joint Monitoring Control Schemes	198
4.4.1	The Shewhart $\bar{X}$ -S Chart	199
4.4.2	The Weighted Loss CUSUM (WLC) and Variable-sample-size WLC Charts	200
4.4.3	The Absolute-value SPRT Chart	202

4.5 A Comparison of the Performances between the OSPRT Chart and the Four Competing Charts	203
4.5.1 Comparative Analyses Based on the ATS-optimal and AEQL-optimal Designs	204
4.5.2 Comparative Analyses Based on the ANOS-optimal and EANOS-optimal Designs	215
4.6 Real Industrial Examples	225
4.6.1 Wire Pull Test	225
4.6.2 Ball Bond Shear Test	230
4.7 Conclusions	233
5 The SPRT Charts under Non-normal Conditions	235
5.1 Introduction	235
5.2 A Review on Some Non-normal Distributions	236
5.2.1 The Gamma Distribution	236
5.2.2 The Lognormal Distribution	237
5.2.3 The Weibull Distribution	238
5.3 The Performances of the SPRT Chart with Estimated Process Parameters under Non-normal Conditions	239
5.4 The Performances of the OSPRT Chart under Non-normal Conditions	249
6 Conclusions	257
6.1 Introduction	257
6.2 Summary of Research Findings and Contributions	257
6.3 Future Research Directions	264
References	267
Appendix A	286

LIST OF TABLES

Table 3.1	The exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$ and the unconditional metrics (AATS, ASDTS, SDATS) of the SPRT chart designed under the traditional approach based on the minimisation of the AEQL, when $\tau_{\text{ATS}} = 370.40$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$.	89
Table 3.2	The exceedance probability $\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$ and the unconditional metrics (AANOS, SDANOS) of the SPRT chart designed under the traditional approach based on the minimisation of the EANOS, when $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = \bar{n} = 5$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$.	94
Table 3.3	The optimal charting parameters ($\text{AASN}_0, \gamma, d, g, h$) and the resulting AAEL value, the exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$, and the (AATS, ASDTS, SDATS) of the SPRT chart designed by minimising the AAEL, when $\tau_{\text{ATS}} = 370.40$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$.	98
Table 3.4	The optimal charting parameters (γ, g, h) and the resulting EAANOS value, the exceedance probability $\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$, and the (AANOS, SDANOS) of the SPRT chart designed by minimising the EAANOS, when $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$.	102
Table 3.5	A two-way table showing the industrial objective each optimisation criterion aims to achieve, as well as the assumption each criterion makes about the shift sizes.	104

Table 3.6	The adjusted control limits (g, h) of the SPRT chart designed under the EPC constraint, together with the AATS, ASDTS, and SDATS values, when $p_e = 0.05$, $\varepsilon = 0$, $\tau_{\text{ATS}} = 370.4$, $\text{AASN}_0 = 2.5$, $\gamma = 0.3$, $\text{IR} = 5$, $d = 0.5$, and $m \in \{50, 100, 150, 200, 250\}$.	110
Table 3.7	Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$ and the corresponding AATS_1 , ASDTS_1 , and SDATS_1 values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.05$, $\varepsilon = 0$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.	112
Table 3.8	Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$ and the corresponding AATS_1 , ASDTS_1 , and SDATS_1 values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.05$, $\varepsilon = 0.2$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.	113
Table 3.9	Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$ and the corresponding AATS_1 , ASDTS_1 , and SDATS_1 values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.10$, $\varepsilon = 0$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.	114
Table 3.10	Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$ and the corresponding AATS_1 , ASDTS_1 , and SDATS_1 values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.10$, $\varepsilon = 0.2$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.	115

Table 3.11	Optimal charting parameters ($AASN_0$, γ , d , g , h), the minimum AAEQL, and the resulting AATS, ASDTS, and SDATS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.05$, $\varepsilon = \{0.0, 0.2\}$, $\tau_{ATS} = 370.4$, $IR = 5$, $\bar{n}_{max} = 15$, $d_{min} = 0.2$, $\delta_{min} = 0.1$, $\delta_{max} = 2.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.	119
Table 3.12	Optimal charting parameters ($AASN_0$, γ , d , g , h), the minimum AAEQL, and the resulting AATS, ASDTS, and SDATS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.10$, $\varepsilon = \{0.0, 0.2\}$, $\tau_{ATS} = 370.4$, $IR = 5$, $\bar{n}_{max} = 15$, $d_{min} = 0.2$, $\delta_{min} = 0.1$, $\delta_{max} = 2.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.	120
Table 3.13	Optimal charting parameters ($AASN_0$, γ , d , g , h), the minimum AAEQL, and the resulting AATS, ASDTS, and SDATS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.05$, $\varepsilon = \{0.0, 0.2\}$, $\tau_{ATS} = 370.4$, $IR = 5$, $\bar{n}_{max} = 15$, $d_{min} = 0.2$, $\delta_{min} = 0.1$, $\delta_{max} = 3.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.	121
Table 3.14	Optimal charting parameters ($AASN_0$, γ , d , g , h), the minimum AAEQL, and the resulting AATS, ASDTS, and SDATS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.10$, $\varepsilon = \{0.0, 0.2\}$, $\tau_{ATS} = 370.4$, $IR = 5$, $\bar{n}_{max} = 15$, $d_{min} = 0.2$, $\delta_{min} = 0.1$, $\delta_{max} = 3.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.	122
Table 3.15	Optimal charting parameters (γ , g , h) and the corresponding $AANOS_1$ and $SDANOS_1$ values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.05$, $\varepsilon = 0$, $\tau_{ANOS} = 1500$, $AASN_0 = \bar{n} = 5$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{opt} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.	130

Table 3.16	Optimal charting parameters (γ, g, h) and the corresponding AANOS ₁ and SDANOS ₁ values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.05$, $\varepsilon = 0.2$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.	131
Table 3.17	Optimal charting parameters (γ, g, h) and the corresponding AANOS ₁ and SDANOS ₁ values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.10$, $\varepsilon = 0$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.	132
Table 3.18	Optimal charting parameters (γ, g, h) and the corresponding AANOS ₁ and SDANOS ₁ values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.10$, $\varepsilon = 0.2$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.	133
Table 3.19	Optimal charting parameters (γ, g, h) , the minimum EAANOS, and the resulting AANOS and SDANOS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.05$, $\varepsilon \in \{0.0, 0.2\}$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.	136
Table 3.20	Optimal charting parameters (γ, g, h) , the minimum EAANOS, and the resulting AANOS and SDANOS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.10$, $\varepsilon \in \{0.0, 0.2\}$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.	137

Table 3.21	Optimal charting parameters (γ, g, h) , the minimum EAANOS, and the resulting AANOS and SDANOS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.05$, $\varepsilon \in \{0.0, 0.2\}$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 3.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.	138
Table 3.22	Optimal charting parameters (γ, g, h) , the minimum EAANOS, and the resulting AANOS and SDANOS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.10$, $\varepsilon \in \{0.0, 0.2\}$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 3.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.	139
Table 3.23	Phase-II summary statistics for monitoring resistivity measurements using the AAELQ-optimal SPRT chart with GICP-adjusted control limits.	151
Table 3.24	Phase-II summary statistics for monitoring loop height measurements using the EAANOS-optimal SPRT chart with GICP-adjusted control limits	155
Table 4.1	Optimal charting parameters $(\text{ASN}_0, k, \gamma, d, g, h)$, and the corresponding $(\text{ATS}_1, \text{SDTS}_1)$ values of the OSPRT chart, when $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 3$, $\bar{n}_{\max} = 15$, $d_{\min} = 0.2$, $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta_{\text{opt}} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.	178
Table 4.2	Optimal charting parameters $(\text{ASN}_0, k, \gamma, d, g, h)$, and the corresponding $(\text{ATS}_1, \text{SDTS}_1)$ values of the OSPRT chart, when $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\max} = 15$, $d_{\min} = 0.2$, $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta_{\text{opt}} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.	179
Table 4.3	Optimal charting parameters $(\text{ASN}_0, k, \gamma, d, g, h)$, and the corresponding optimum AEQL value of the OSPRT chart, when $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 3$, $\bar{n}_{\max} = 15$, $d_{\min} = 0.2$, $\delta_{\max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$, and $\eta_{\max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$.	184

Table 4.4	Optimal charting parameters (ASN_0 , k , γ , d , g , h), and the corresponding optimum AEQL value of the OSPRT chart, when $\tau_{ATS} = 370.4$, $IR = 5$, $\bar{n}_{max} = 15$, $d_{min} = 0.2$, $\delta_{max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$, and $\eta_{max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$.	185
Table 4.5	(ATS, SDTS) values of the AEQL-optimal OSPRT chart designed using $\tau_{ATS} = 370.4$, $IR = 5$, $\bar{n}_{max} = 15$, $d_{min} = 0.2$, $\delta_{max} = 2.00$, and $\eta_{max} = 2.00$, evaluated at $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$.	186
Table 4.6	(ATS, SDTS) values of the AEQL-optimal OSPRT chart designed using $\tau_{ATS} = 370.4$, $IR = 5$, $\bar{n}_{max} = 15$, $d_{min} = 0.2$, $\delta_{max} = 3.00$, and $\eta_{max} = 3.00$, evaluated at $\delta \in \{0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.4, 1.7, 2.0, 2.5, 3.0\}$.	187
Table 4.7	Optimal charting parameters (k , γ , g , h), and the corresponding (ARL_1 , $ANOS_1$) values of the OSPRT chart, when $\tau_{ANOS} = 1500$, $ASN_0 = 3$, $\delta_{opt} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta_{opt} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.	191
Table 4.8	Optimal charting parameters (k , γ , g , h), and the corresponding (ARL_1 , $ANOS_1$) values of the OSPRT chart, when $\tau_{ANOS} = 1500$, $ASN_0 = 5$, $\delta_{opt} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta_{opt} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.	192
Table 4.9	Optimal charting parameters (k , γ , g , h), and the corresponding optimum EANOS value of the OSPRT chart, when $\tau_{ANOS} = 1500$, $ASN_0 = 3$, $\delta_{max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$, and $\eta_{max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$.	194
Table 4.10	Optimal charting parameters (k , γ , g , h), and the corresponding optimum EANOS value of the OSPRT chart, when $\tau_{ANOS} = 1500$, $ASN_0 = 5$, $\delta_{max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$, and $\eta_{max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$.	195

Table 4.11	(ARL, ANOS) values of the EANOS-optimal OSPRT chart designed using $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 5$, $\delta_{\text{max}} = 2.00$, and $\eta_{\text{max}} = 2.00$, evaluated at $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$.	196
Table 4.12	(ARL, ANOS) values of the EANOS-optimal OSPRT chart designed using $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 5$, $\delta_{\text{max}} = 3.00$, and $\eta_{\text{max}} = 3.00$, evaluated at $\delta \in \{0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.4, 1.7, 2.0, 2.5, 3.0\}$.	197
Table 4.13	Optimal charting parameters of the ATS-optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts, optimised for $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$, when $\tau_{\text{ATS}} = 370.4$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, and $\text{IR} = 5$.	206
Table 4.14	(ATS ₁ , SDTS ₁) values of the ATS-optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts, optimised for $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$, when $\tau_{\text{ATS}} = 370.4$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, and $\text{IR} = 5$.	207
Table 4.15	Optimal charting parameters of the Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS SPRT, and OSPRT charts designed for $\tau_{\text{ATS}} = 370.4$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\text{IR} = 5$, and $(\delta_{\text{max}}, \eta_{\text{max}}) \in \{(2.0, 2.0), (3.0, 3.0)\}$.	209
Table 4.16	(ATS ₁ , SDTS ₁) values of the optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\text{max}}, \eta_{\text{max}}) = (2.0, 2.0)$, for shift sizes $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta = 1$.	210
Table 4.17	(ATS ₁ , SDTS ₁) values of the optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\text{max}}, \eta_{\text{max}}) = (3.0, 3.0)$, for shift sizes $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta = 1$.	210
Table 4.18	(ATS ₁ , SDTS ₁) values of the optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\text{max}}, \eta_{\text{max}}) = (2.0, 2.0)$, for shift sizes $\delta = 0$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.	211

Table 4.19	(ATS ₁ , SDTS ₁) values of the optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta = 0$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$.	212
Table 4.20	(ATS ₁ , SDTS ₁) values of the optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.	213
Table 4.21	(ATS ₁ , SDTS ₁) values of the optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$.	214
Table 4.22	Optimal charting parameters of the ATS-optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts, optimised for $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$, when $\tau_{\text{ANOS}} = 1500$ and $\text{ASN}_0 = 5$.	216
Table 4.23	(ARL ₁ , ANOS ₁) values of the ANOS-optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts, optimised for $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$, when $\tau_{\text{ANOS}} = 1500$ and $\text{ASN}_0 = 5$.	217
Table 4.24	Optimal charting parameters of the Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS SPRT, and OSPRT charts designed for $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 5$, and $(\delta_{\max}, \eta_{\max}) \in \{(2.0, 2.0), (3.0, 3.0)\}$.	219
Table 4.25	(ARL ₁ , ANOS ₁) values of the optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta = 1$.	220
Table 4.26	(ARL ₁ , ANOS ₁) values of the optimal Shewhart $\bar{X}$ -S, WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta = 1$.	220

Table 4.27	(ARL_1 , $ANOS_1$) values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta = 0$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.	221
Table 4.28	(ARL_1 , $ANOS_1$) values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta = 0$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$.	221
Table 4.29	(ARL_1 , $ANOS_1$) values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.	223
Table 4.30	(ARL_1 , $ANOS_1$) values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$.	224
Table 4.31	Phase-II summary statistics for monitoring pull force readings using the AEQL-optimal OSPRT chart.	228
Table 4.32	Phase-II summary statistics for monitoring ball shear force readings using the EANOS-optimal OSPRT chart.	232
Table 5.1	Optimal charting parameters ($AASN_0$, γ , d , g , h) of the AAEL-optimal SPRT chart with GICP-adjusted limits for $m \in \{200, 400, 600, 1000, +\infty\}$ when the underlying data follow the normal distribution.	240
Table 5.2	The exceedance probability $\Pr(CATS_0 \geq \tau_{ATS})$, $AATS_0$, and $ASDTS_0$ values of the AAEL-optimal SPRT chart with GICP-adjusted limits, for $\rho \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the gamma, lognormal, and Weibull distributions.	241

Table 5.3	The AATS ₁ and ASDTS ₁ values of the AAEQL-optimal SPRT chart with GICP-adjusted limits evaluated at $\delta \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, when $\rho \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and the underlying data follow the gamma distribution.	245
Table 5.4	The AATS ₁ and ASDTS ₁ values of the AAEQL-optimal SPRT chart with GICP-adjusted limits evaluated at $\delta \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, when $\rho \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and the underlying data follow the lognormal distribution.	246
Table 5.5	The AATS ₁ and ASDTS ₁ values of the AAEQL-optimal SPRT chart with GICP-adjusted limits evaluated at $\delta \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, when $\rho \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and the underlying data follow the Weibull distribution.	247
Table 5.6	The (ATS, SDTS) values of the AEQL-optimal OSPRT chart evaluated for $\rho = 0.0$, $\delta \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the normal, gamma, lognormal, and Weibull distributions.	251
Table 5.7	The (ATS, SDTS) values of the AEQL-optimal OSPRT chart evaluated for $\rho \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, $\delta \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the gamma distribution.	252
Table 5.8	The (ATS, SDTS) values of the AEQL-optimal OSPRT chart evaluated for $\rho \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, $\delta \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the lognormal distribution.	253
Table 5.9	The (ATS, SDTS) values of the AEQL-optimal OSPRT chart evaluated for $\rho \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, $\delta \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the Weibull distribution.	254

LIST OF FIGURES

Figure 2.1	The two-sided Shewhart $\bar{X}$ control chart for monitoring the mean of the photoresist thickness of 125-mm silicon wafers.	12
Figure 2.2	The two-sided EWMA control chart for monitoring simulated dataset.	16
Figure 2.3	The two one-sided CUSUM control chart for monitoring simulated dataset.	18
Figure 2.4	The upper one-sided SPRT control chart.	26
Figure 2.5	The distributions of the sample mean when the sample sizes are 100, 400, and 1600 (left to right).	40
Figure 2.6	The distribution of ARL_0 values for various parameter estimates.	49
Figure 2.7	The distribution of ARL_0 values for various parameter estimates after the GICP design is applied.	49
Figure 3.1	The visualisation of a single run of a control chart, with indications of the out-of-control event and signalling time.	68
Figure 3.2	Trace plot of a sequence of h values generated from the modified stochastic approximation algorithm with 31,000 simulation runs.	109
Figure 3.3	Boxplots of the $CATS_0$ distributions for the SPRT chart designed under the traditional approach, AATS-matching approach, and GICP approach with $p_e \in \{0.05, 0.10\}$ and $\varepsilon \in \{0.0, 0.2\}$.	125
Figure 3.4	Boxplots of the $CATS_1$ distributions for the SPRT chart designed under the traditional approach, AATS-matching approach, and GICP approach with $p_e \in \{0.05, 0.10\}$ and $\varepsilon \in \{0.0, 0.2\}$ when (a) $\delta = 0.2$, (b) $\delta = 0.4$, (c) $\delta = 0.6$, and (d) $\delta = 0.8$.	127

Figure 3.5	Boxplots of the $CANOS_0$ distributions for the SPRT chart designed under the traditional approach, AANOS-matching approach, and GICP approach with $p_e \in \{0.05, 0.10\}$ and $\varepsilon \in \{0.0, 0.2\}$.	140
Figure 3.6	Boxplots of the $CANOS_1$ distributions for the SPRT chart designed under the traditional approach, AANOS-matching approach, and GICP approach with $p_e \in \{0.05, 0.10\}$ and $\varepsilon \in \{0.0, 0.2\}$, when (a) $\delta = 0.2$, (b) $\delta = 0.4$, (c) $\delta = 0.6$, and (d) $\delta = 0.8$.	141
Figure 3.7	Boxplots of the $CATS_0$ distributions for the AAQL-optimal CUSUM and SPRT charts designed under the GICP approach.	143
Figure 3.8	Boxplots of the $CATS_1$ distributions for the AAQL-optimal CUSUM and SPRT charts designed under the GICP approach when (a) $\delta = 0.2$, (b) $\delta = 0.4$, (c) $\delta = 0.6$, and (d) $\delta = 0.8$.	144
Figure 3.9	Boxplots of the $CANOS_0$ distributions for the EAANOS-optimal CUSUM and SPRT charts designed under the GICP approach.	146
Figure 3.10	Boxplots of the $CANOS_1$ distributions for the EAANOS-optimal CUSUM and SPRT charts designed under the GICP approach when (a) $\delta = 0.2$, (b) $\delta = 0.4$, (c) $\delta = 0.6$, and (d) $\delta = 0.8$.	147
Figure 3.11	The Phase-I AMR chart for $m = 200$ individual resistivity measurements.	149
Figure 3.12	The Phase-II upper- and lower-sided SPRT charts for monitoring resistivity measurements of epitaxial wafers.	152
Figure 3.13	The Phase-I AMR chart for $m = 200$ individual loop height measurements.	153

- Figure 3.14 The Phase-II upper- and lower-sided SPRT charts for monitoring loop height measurements of the wire bonding process. 156
- Figure 4.1 The optimum ATS_1 values achieved by the OSPRT chart for $\delta_{\text{opt}} \in \{0.8, 1.0, 1.5, 2.0\}$ against the standard deviation shift size $1.2 \leq \eta_{\text{opt}} \leq 2.0$, when $IR = 3$ (left) and $IR = 5$ (right). 180
- Figure 4.2 A visualisation of the VSS WLC control chart and its sampling mechanism. 202
- Figure 4.3 The Phase-I Shewhart (a) $\bar{X}$ chart and (b) S chart for monitoring 50 Phase-I samples, each consisting of five pull force measurements. 226
- Figure 4.4 The Phase-II OSPRT chart for monitoring pull force measurements of the wire bonding process. 229
- Figure 4.5 The Phase-I Shewhart (a) $\bar{X}$ chart and (b) S chart for monitoring 50 Phase-I samples, each consisting of five shear force measurements. 231
- Figure 4.6 The Phase-II OSPRT chart for monitoring ball shear force measurements of the wire bonding process. 233

LIST OF NOTATIONS

The notations and abbreviations used in this thesis are listed below.

$\mathbf{I}_2$	2×2 identity matrix
ABS-CUSUM	Absolute value cumulative sum
ABS-SPRT	Absolute value sequential probability ratio test
d_2	Adjustment factor for the sample range
c_4	Adjustment factor for the sample standard deviation
AEQL	Average extra quadratic loss
AMR	Average moving range
ANOS	Average number of observations to signal
AAEQL	Average of the average extra quadratic loss
AANOS	Average of the conditional average number of observations to signal
AARL	Average of the conditional average run length
AASN	Average of the conditional average sample number
AATS	Average of the conditional average time to signal
ARL	Average run length
ASN	Average sample number
$\bar{R}$	Average sample range
$\bar{S}$	Average standard deviation
ASDTS	Average standard deviation of the time to signal
ATS	Average time to signal
$F_{\chi_1^2}(\cdot ncp)$	Cdf of the chi-squared distribution with one degree of freedom and non-centrality parameter ncp
$\Phi(\cdot)$	Cdf of the standard normal distribution
CL	Centre line
CANOS	Conditional average number of observations to signal
CARL	Conditional average run length
CASN	Conditional average sample number
CATS	Conditional average time to signal

CSDRL	Conditional standard deviation of the run length
CSDTS	Conditional standard deviation of the time to signal
$L_{\bar{X}}$	Control limit multiplier of the $\bar{X}$ chart
L_{EWMA}	Control limit multiplier of the EWMA chart
L_S	Control limit multiplier of the S chart
H_{WLC}	Control limit of the WLC and VSS WLC charts
$U_{i,j}^{\text{ABS}}$	Control statistic of the ABS-SPRT chart
$U_{i,j}^-, \hat{U}_{i,j}^-$	Control statistic of the lower one-sided SPRT chart with known or estimated process parameters
$U_{i,j}, \hat{U}_{i,j}$	Control statistic of the SPRT chart with known or estimated process parameters
$U_{i,j}^+, \hat{U}_{i,j}^+$	Control statistic of the upper one-sided SPRT chart with known or estimated process parameters
$U_{i,j}^{+-}, U_{i,j}^{--}$	Control statistics for the OSPRT chart for detecting an increasing mean shift and a decreasing mean shift, respectively, alongside a lower-sided variance shift
$U_{i,j}^{++}, U_{i,j}^{+-}$	Control statistics for the OSPRT chart for detecting an increasing mean shift and a decreasing mean shift, respectively, alongside an upper-sided variance shift
c	Count of defective items
u	Count of defective items per inspection unit
$c_R, \tilde{c}_R$	Critical values of the FSS likelihood ratio test
cdf	Cumulative distribution function
CUSUM	Cumulative sum
p_e	Error probability in EPC
$\hat{\mu}_0, \hat{\sigma}_0$	Estimated in-control mean and standard deviation of a quality characteristic
EPC	Expected probability criterion
ESS	Expected sample size
EAANOS	Expected value of the AANOS
EANOS	Expected value of the ANOS
EARL	Expected value of the ARL
EATS	Expected value of the ATS

EWMA	Exponentially weighted moving average
FSS	Fixed sample size
P	Full transition probability matrix of the absorbing Markov chain for the SPRT / OSPRT chart
$\Gamma(\cdot)$	Gamma function
X_γ	Gamma random variable
GLR	Generalised likelihood ratio
g	gram
gf	gram-force
GICP	Guaranteed in-control performance
u_0	Head start value in the fast initial response strategy
$\widehat{OC}_0, \widehat{OC}_1$	In-control and out-of-control $\widehat{OC}$
AANOS ₀ , AANOS ₁	In-control and out-of-control AANOS
AASN ₀ , AASN ₁	In-control and out-of-control AASN
AATS ₀ , AATS ₁	In-control and out-of-control AATS
ANOS ₀ , ANOS ₁	In-control and out-of-control ANOS
ARL ₀ , ARL ₁	In-control and out-of-control ARL
ASDTS ₀ , ASDTS ₁	In-control and out-of-control ASDTS
ASN ₀ , ASN ₁	In-control and out-of-control ASN
ATS ₀ , ATS ₁	In-control and out-of-control ATS
CANOS ₀ , CANOS ₁	In-control and out-of-control CANOS
CARL ₀ , CARL ₁	In-control and out-of-control CARL
CATS ₀ , CATS ₁	In-control and out-of-control CATS
CSDTS ₀ , CSDTS ₁	In-control and out-of-control CSDTS
μ_0, μ_1	In-control and out-of-control mean of a quality characteristic
OC ₀ , OC ₁	In-control and out-of-control OC
RL ₀ , RL ₁	In-control and out-of-control run lengths
SDANOS ₀ , SDANOS ₁	In-control and out-of-control SDANOS

$\text{SDATS}_0, \text{SDATS}_1$	In-control and out-of-control SDATS
$\text{SDRL}_0, \text{SDRL}_1$	In-control and out-of-control SDRL
$\text{SDTS}_0, \text{SDTS}_1$	In-control and out-of-control SDTS
σ_0, σ_1	In-control and out-of-control standard deviation of a quality characteristic
TS_0, TS_1	In-control and out-of-control times to signal
Y_ζ	Individual Phase-I observations
IR	Inspection rate
$\Phi^{-1}(\cdot)$	Inverse cdf of the standard normal distribution
I_i	Kullback–Leibler information numbers
$\mu_{\text{LN}}, \sigma_{\text{LN}}$	Location and scale parameters of the lognormal distribution
L_j	Log-likelihood ratio of j independent observations
X_{LN}	Lognormal random variable
$k_{\min}, k_{\max}$	Lower and upper bounds for the reference parameter k
$\gamma_{\min}, \gamma_{\max}$	Lower and upper bounds for the reference parameter γ
A, B	Lower and upper constants for Wald’s SPRT
g, h	Lower and upper control limits of the SPRT chart
LCL	Lower control limit
g_{ABS}	Lower control limit of the ABS-SPRT chart
Q	Matrix containing vector of probabilities from the transient states to the acceptance and the rejection states for the SPRT / OSPRT chart with known process parameters
0	Matrix with all zero entries
$\bar{n}_{\max}$	Maximum allowable ASN_0
μ_0^*	Mean of a non-normal distribution
μ	Mean of a quality characteristic
δ_{opt}	Mean shift size for which a quick detection is required
MSE	Mean square error
O_a	Midpoint of transient state T_a
$d_{\min}$	Minimum allowable sampling interval
$\delta_{\min}, \delta_{\max}$	Minimum and maximum allowable mean shifts

$\eta_{\min}, \eta_{\max}$	Minimum and maximum allowable standard deviation shifts
MR	Moving range
NP	Neyman-Pearson
$\mathcal{N}(\mu, \sigma^2)$	Normal distribution with mean μ and standard deviation σ
P_0, P_1	Null and alternative distributions
H_0, H_1	Null and alternative hypothesis
np	Number of defective units
NOS	Number of observations to signal
ξ	Number of quadrature nodes in methods of integral equations/ number of transient states in the Markov chain approach
ohm-cm	ohm-centimetre
OSPRT	Omnibus sequential probability ratio test
$f_0(u)$	One-step transition density of the control statistic from the initial value 0 to an arbitrary value u
P_0	One-step transition probability from the initial state to the acceptance state for the SPRT / OSPRT chart with known process parameters
$\hat{P}_0$	One-step transition probability from the initial state to the acceptance state for the SPRT chart with estimated process parameters
c_a	One-step transition probability from the initial state to transient state T_a for the SPRT / OSPRT chart with known process parameters
$\hat{c}_a$	One-step transition probability from the initial state to transient state T_a for the SPRT chart with estimated process parameters
$q_{a,(\xi+1)}$	One-step transition probability from transient state T_a to the acceptance state for the SPRT / OSPRT chart with known process parameters
$\hat{q}_{a,(\xi+1)}$	One-step transition probability from transient state T_a to the acceptance state for the SPRT chart with estimated process parameters
$q_{a,(\xi+2)}$	One-step transition probability from transient state T_a to the rejection state for the SPRT / OSPRT chart with known process parameters

$r_{a,b}$	One-step transition probability from transient state T_a to transient state T_b for the SPRT / OSPRT chart with known process parameters
$\hat{r}_{a,b}$	One-step transition probability from transient state T_a to transient state T_b for the SPRT chart with estimated process parameters
mils	One-thousandth of an inch
OC	Operating characteristic
$\widehat{\text{OC}}$	Operating characteristic in the case where process parameters are estimated
Θ	Parameter space for θ
$f_{\chi^2_{df}}(\cdot)$	Pdf of the chi-squared distribution with df degrees of freedom
$f_{\delta}(\delta)$	Pdf of the mean shift size δ
$f_V(v)$	Pdf of the random variable V
$f_W(w)$	Pdf of the random variable W
ρ	Pearson's moment coefficient of skewness
m	Phase-I sample size
W	Pivotal quantity $(\hat{\mu}_0 - \mu_0)/(\sigma_0/\sqrt{m})$
V	Pivotal quantity $\hat{\sigma}_0/\sigma_0$
V'	Pivotal quantity $(m - 1)\hat{\sigma}_0^2/\sigma_0^2$
S_{pooled}	Pooled sample standard deviation
pdf	Probability density function
$f(X \mid \theta)$	Probability density function of the random variable X given the parameter θ
p	Proportion defect
R	Range
τ_{ANOS}	Recommended value of ANOS_0 in the design of the SPRT / OSPRT chart
τ_{ATS}	Recommended value of ATS_0 in the design of the SPRT / OSPRT chart
γ_{ABS}	Reference parameter of the ABS-SPRT chart
γ_C	Reference parameter of the CUSUM chart
k	Reference parameter of the OSPRT chart

γ	Reference parameter of the SPRT / OSPRT chart
γ_{WLC}	Reference parameter of the WLC and VSS WLC charts
n_1, n_2	Relax and alert sample sizes of the VSS WLC chart
RHS	Right hand side
RL	Run length
$\overline{MR}$	Sample average of the moving range
$\bar{X}$	Sample mean
$\bar{Y}$	Sample mean of the Phase-I observations
N_i	Sample number of the i -th SPRT
n	Sample size in fixed-sample-size control charts
S	Sample standard deviation
S^2	Sample variance
d	Sampling interval
SPRT	Sequential probability ratio test
α', β'	Shape and scale parameters of the gamma distribution
k', λ'	Shape and scale parameters of the Weibull distribution
δ	Size of mean shift
η	Size of standard deviation shift
σ_0^*	Standard deviation of a non-normal distribution
σ	Standard deviation of a quality characteristic
SDANOS	Standard deviation of the conditional average number of observations to signal
SDARL	Standard deviation of the conditional average run length
SDATS	Standard deviation of the conditional average time to signal
SDRL	Standard deviation of the run length
SDTS	Standard deviation of the time to signal
η_{opt}	Standard deviation shift sizes for which a quick detection is required
$\mathcal{U}$	Standard uniform random variable used to model the time of shift
$Z_{i,j}$	Standardised measurement

T_a	States of an absorbing Markov chain for the SPRT / OSPRT chart
SQC	Statistical quality control
θ_0, θ_1	Target value and shifted value of θ
$X_{(i)}$	The i -th order statistic of the sample $\mathbf{X}$
$Y_{i,j}$	The j -th observation in the i -th subgroup of Phase-I samples
$X_{i,j}$	The j -th observation of the i -th sample
θ	The parameter of an arbitrary distribution
$\bar{n}$	The potential optimum value of the ASN_0
ω_a	The quadrature weights in methods of integral equations
u_a	The values of the control statistic in methods of integral equations
d_0	Time period allocated immediately after the initialisation of the process
TS	Time to signal
ε	Tolerance term in EPC
$\mathbf{R}$	Transition probability matrix between the transient states for the SPRT / OSPRT chart with known process parameters
$\hat{\mathbf{R}}$	Transition probability matrix between the transient states for the SPRT chart with estimated process parameters
CC	Two-sided CUSUM mean and variance chart
EE	Two-sided EWMA mean and variance chart
α, β	Type-I and Type-II error probabilities of a hypothesis test
UCL	Upper control limit
h_{ABS}	Upper control limit of the ABS-SPRT chart
H_{CUSUM}	Upper control limit of the CUSUM chart
$\text{UCL}_{\bar{X}}$	Upper control limit of the upper-sided $\bar{X}$ chart
UCL_S	Upper control limit of the upper-sided S chart
VSS	Variable sample size
VSI	Variable sampling interval
$\mathbf{C}$	Vector containing probabilities from the initial state to the transient states for the SPRT / OSPRT chart with known process parameters

$\hat{\mathbf{c}}$	Vector containing probabilities from the initial state to the transient states for the SPRT chart with estimated process parameters
$\mathbf{q}_1$	Vector containing probabilities from the transient states to the acceptance state for the SPRT / OSPRT chart with known process parameters
$\hat{\mathbf{q}}_1$	Vector containing probabilities from the transient states to the acceptance state for the SPRT chart with estimated process parameters
$\mathbf{q}_2$	Vector containing probabilities from the transient states to the rejection state for the SPRT / OSPRT chart with known process parameters
$\mathbf{1}$	Vector with entries filled with ones
W_{WLC}	Warning limit of the VSS WLC chart
X_W	Weibull random variable
WL	Weighted loss function
WLC	Weighted-loss CUSUM
λ_{WL}	Weighting factor of the WLC and VSS WLC charts
Δ	Width of each transient state
$\mathbf{I}_\xi$	$\xi \times \xi$ identity matrix

1 Introduction

1.1 Statistical Quality Control (SQC)

The term ‘quality’ originates from the Latin word ‘*qualitas*’, which represents the inherent characteristics of an entity that define its nature and value. While the term has old linguistic origins, the concept of statistical quality control (SQC) only started gaining prominence in the 1920s, with major contributions from figures like Walter A. Shewhart. Working as a physicist at the Bell Telephone Laboratories, Walter A. Shewhart created the very first control chart, known as the Shewhart $\bar{X}$ chart, with simple rules that underlie the fundamental principles of today’s SQC practices. The control chart, also known as the process-behaviour chart, is a graphical tool used to monitor processes over time. Shewhart (1931) discovered that all processes are subjected to a certain degree of random variation, known as *common cause* variation. As common cause variations are inherent to a process, Shewhart (1931) claimed that they should be left to “chance”. However, there should be some criteria to distinguish the unusual variations in a process from those inherent ones. These unusual variations are known as *special cause* variations, and they may arise due to a number of reasons, such as changes in raw materials, human error, machine failure, etc. (Montgomery, 2019). With meticulously chosen control limits, these special causes can be swiftly detected by a control chart, and subsequently eliminated to bring the process back to normal.

Before we visit the main objectives of this thesis, let us paint a picture depicting the current landscape in academic and industrial research. Based on our observation, (almost) all research developments can be fundamentally separated into two main types. The first type is known as vertical development, where the main goal of the development is to devise a(n) model/ invention with an improved performance compared to some existing benchmark. The second type is known as

expansive development, where a practical challenge is often identified prior to the introduction of some new strategies that will help increase the robustness of a model. To elaborate, vertical developments aim to enhance the performance of the state-of-the art technique as far as possible. For instance, the introduction of ChatGPT 4.0 as an improved version of ChatGPT 3.0 – promising a better, quicker, and more accurate chat response – is an example of a vertical development. On the other hand, expansive developments focus on addressing external or practical issues that relate to the performance and functionality of a model. For instance, the introduction of a new feature for interpreting image-based queries in ChatGPT 4.0 can be thought of as an expansive development, since it addresses the practical challenge with translating image-based tasks into text faced by users.

It should be noted that vertical developments are restrictive in nature. For example, we know that the shortest time a chat bot takes to respond to queries can only be zero (and not lower), the accuracy of the best predictive model can only be 100% (and not higher), etc. On the other hand, expansive developments have unlimited potential as practical issues will surface infinitely often across the timeline.

There have been concentrated efforts devoted to a range of vertical and expansive developments within the SQC regimen over the past century. For vertical developments of control charts, researchers either derive a new statistic to be used with a control chart, or introduce additional features that help boost the detection performance of a control chart. One of the most prominent examples is the introduction of the variable-sample-size and variable-sampling-interval strategies. As control schemes usually operate on a fixed sample size and a fixed inspection interval (for the ease of administration), these strategies encourage a dynamic on-line monitoring procedure based on past measurements from a process. In particular, a small sample size and/or a low sampling frequency is employed when the preceding measurement lies “close” to its target value, whereas a large sample

size and/or a high sampling frequency is used when an observation lands “far” from the target. Many researchers have reported that the variable-sample-size and variable-sampling-interval control charts indeed outperform standard control charts based on fixed sample sizes and fixed sampling intervals, in terms of their detection sensitivity and speed (see, e.g., Costa, 1994; Reynolds *et al.*, 1988). Aside from enhancement features, there are numerous research works based on the development of memory statistics. By memory statistics, we refer to statistics that accumulate data values over time, hence each statistic gains more “memory” as time progresses. Some well-known examples include the cumulative sum (CUSUM) statistic (Page, 1954) and the exponentially weighted moving average (EWMA) statistic (Roberts, 1959). Researchers have shown that control charts based on the CUSUM procedure and EWMA procedure are substantially more responsive than Shewhart control charts when process deviations are small or moderate (see, e.g., Hawkins & Olwell, 1998; Chen *et al.*, 2001; Reynolds & Stoumbos, 2005; Saleh *et al.*, 2015a). The interested reader should refer to Montgomery (2019) for a comprehensive review of these procedures.

Expansive developments differ from vertical developments in a way that they are motivated by the observation of a real industrial challenge, rather than solely based on the aim to improve the performance of a control chart. For example, engineers noticed that not all industrial processes possess the “nice” qualities, e.g., independence, normality, etc. When the distribution of the data deviates from normality, it is likely that the performances of a control chart would be adversely affected (see Chan *et al.*, 1988). Another widely identified issue is the violation of the assumption that process parameters (i.e., mean and standard deviation) are known. Quesenberry (1993) inferred that, in many applications, the parameters of a process should be estimated from historical data rather than being given as a definite value. Hence, practitioners face the risk of obtaining parameter estimates that deviate considerably from their true values, which can lead to unexpected

performances during SQC applications. Besides, there are also various interpretations to what defines the “best-performing” control chart. Some researchers argued that the best control chart is the one that achieves the shortest detection time for a particular process shift size, while some researchers argued that the best control chart should be one that performs well on average over a range of permissible shift sizes (see Sparks, 2000; Reynolds and Stoumbos, 2004a). Many scholars have since developed statistical designs for control charts based on various performance metrics, as well as designs catered for different industrial objectives, each holding to their unique interpretation to what defines a good control chart.

In this section, we have briefly discussed the history of SQC, as well as the importance of both vertical and expansive developments in advancing the state-of-the-art control charting techniques. The subsequent sections will reveal how the research gaps in both vertical and expansive developments of the control chart chosen in this thesis are identified.

1.2 Problem Statements

In this thesis, we focus on new developments of the sequential probability ratio test (SPRT) control chart. The SPRT chart is a sampling scheme based on the most powerful sequential test, and it was first introduced by Abraham Wald in 1945. It was chosen in our research for three main reasons:

- (i) The SPRT chart has a solid track record of being a consistently better performer than many mainstream control charts in the literature, e.g., the CUSUM and EWMA charts stated in Section 1.1. Stoumbos and Reynolds (1997a) showed that the SPRT chart has a higher sensitivity than the Shewhart $\bar{X}$ and CUSUM charts for detecting shifts in the process mean. In terms of variance shifts, Godase and Mahadik (2019) showed that the SPRT

chart outperforms a range of variance-based control charts for almost all magnitudes of shifts. Many such comparisons have been documented in Deore *et al.* (2023), Haridy *et al.* (2013), Ou *et al.* (2012a, 2012b, 2012c, 2015a), and we direct the interested reader to these references.

- (ii) The SPRT is found to be useful in many industrial and non-industrial applications. For example, Rasay and Alinezhad (2022) proposed an adaptable SPRT for monitoring the strength of deep groove ball bearings. Solak and Öner (2020) integrated the SPRT with nanoscale networks to detect unusual patterns in fluids, and it was shown to be effective in medical surveillance and disease control.
- (iii) Despite the usefulness of the SPRT, there have been limited major developments related to the SPRT chart in the past decade. The first major development of the SPRT chart occurred in the late 90s, with contributions from Stoumbos and Reynolds (1996, 1997a, 1997b, 2001). After close to a decade, a new group of researchers, i.e., Ou *et al.* (2010, 2012a, 2012b, 2012c), have devoted concentrated efforts in the development of optimisation designs for the SPRT control chart. After almost a decade of ceased activity, the SPRT chart regains attention from another group of scholars, i.e., Godase and Mahadik (2019, 2022), Mahadik *et al.* (2021), Mahadik and Godase (2023a, 2023b).

Motivated by the three reasons stated above, we have decided to make the SPRT control chart the central premise of our research, considering that there is plenty of room for development and improvement in this area. In the subsequent paragraphs, we will explain the literature gaps and the motivations of our study, citing key publications that contribute to the basis of our research work.

Traditionally, in the design of control charts, it is assumed that the parameters of a process (mean and standard deviation) are known and fixed. This

is relevant to, for example, a chocolate beverage production company, where the product specifications are usually stated upfront and made known to the consumers. However, in many other quality applications, process parameters are unknown and need to be estimated from a separate set of in-control data, known as Phase-I data. Quesenberry (1993) inferred that, when the process parameters are estimated, there exists practitioner-to-practitioner variability that can distort the performances of control charts. Capizzi and Masarotto (2020) further indicated that, organisations tend to establish multiple production lines that switch between different products in response to rapid changes in the market trend and demand, specifically those in the fashion industry, manufacturing industry, etc. As the Phase-I data available for these new products are limited, the design of control charts with estimated process parameters becomes more challenging than ever. Hence, this thesis delves into the investigation of control charts with estimated process parameters and robust designs for those control charts. After an extensive literature survey, it is found that none of the existing works have contributed to the development of the SPRT chart with estimated process parameters. Hence, this thesis aims to develop the full theoretical framework of the SPRT chart with estimated process parameters and its statistical design.

In standard SQC applications, joint monitoring of the mean and variance of a process is crucial to ensure that production outputs are stable and reliable (Gan *et al.*, 2004). To demonstrate the importance of joint control schemes, Gan *et al.* (2004) cited an example in the circuitry industry, describing how an improperly adjusted stencil may lead to a joint shift in the mean and variability of the solder paste thickness of printed circuit boards. Based on our knowledge, the SPRT mean and variance charts had been fully developed by Stoumbos and Reynolds (1997a) and Godase and Mahadik (2019), respectively. However, there is very little documentation on the SPRT control chart designed for monitoring both the mean and variance simultaneously, with the exception of one single-chart joint monitoring

scheme developed by Ou *et al.* (2011a), known as the absolute-value SPRT (ABS-SPRT) chart. While the formulation of the ABS-SPRT chart seems reasonable, there are some concerns regarding its optimality properties, since the ABS-SPRT chart was simply developed by amending the original SPRT mean chart (instead of deriving from first principles). Hence, we attempt to develop a new single-chart SPRT scheme with stronger optimality properties, which we name as the omnibus SPRT chart, for joint monitoring of the mean and variance. The performances of the proposed omnibus SPRT chart are then compared to the ABS-SPRT chart as well as other benchmarks in the literature.

The vast majority of the developments of control charts assume that quality measurements follow the normal distribution. Qiu (2018) inferred that, in practice, the majority of the quality variables can be adequately modelled by the normal distribution due to the central limit theorem. Naderkhani (2022) further stated that the normality assumption is reasonable after pre-processing methods have been applied to numerical data. While the normality assumption seems a convenient assumption to make, Qiu (2018) noted that, in fields such as the economic and environmental industries, variables often follow a distribution that differs from the normal distribution in terms of its shape and skewness. Bai and Choi (1995) discovered that control charts designed under the normal distribution are susceptible to high false alarm rates when the underlying distribution is positively skewed. As the SPRT charts developed in this thesis are based on the normal distribution, we are interested in analysing their performances under cases where the distributions of data are skewed. Some preliminary results through simulation are provided.

1.3 Objectives of the Thesis

This thesis aims to enrich the literature of the SPRT control chart by including a combination of vertical and expansive developments. The objectives of this thesis are listed below:

- (i) To develop the full theoretical framework of the SPRT chart with estimated process parameters, and to propose four new optimal designs based on the state-of-the-art method. The proposed designs are subsequently compared to some existing designs in the literature.
- (ii) To develop a new omnibus SPRT chart for joint monitoring of the mean and dispersion, and to propose four new optimal designs for the proposed chart. The proposed chart is then compared to four benchmarks in the existing literature.
- (iii) To investigate the performances of the SPRT chart with estimated process parameters and the omnibus SPRT chart under skewed distributions.

1.4 Structure of the Thesis

Chapter 1 provides comprehensive insights into the current research trends and developments in SQC, as well as outlining the core motivation of the research work proposed in this thesis. In Chapter 2, a thorough literature review on the development of the SPRT chart and other related topics is presented. The chapter begins with a review of some basic control charts in the literature. It is then followed by a description of the SPRT as a sequential hypothesis test, as well as the development of the SPRT control chart in chronological order. The chapter also introduces some key metrics used for examining the performances of these control charts. In the subsequent sections, we provide a comprehensive review on the

evolution of design methodologies for control charts with estimated process parameters, outlining their benefits and pitfalls at various stages of development. Finally, the chapter discusses some control schemes for joint monitoring of the process location and dispersion, separating them into single-chart schemes and multi-chart schemes.

Chapter 3 begins with the full construction of the SPRT chart with known process parameters. It is then followed by the full theoretical derivation of the SPRT chart assuming that process parameters are estimated. To compare the effectiveness of various design methodologies, the performances of the SPRT chart with estimated process parameters are evaluated under three designs, i.e., two alternative designs and our adopted state-of-the-art design. Four optimisation algorithms are developed for the SPRT chart with estimated process parameters based on four different performance metrics, respectively, each aiming to achieve a different industrial objective (e.g., detection speed, inspection efficiency). To enhance the value of the thesis, a comparison between the performances of the proposed SPRT chart and those of the CUSUM chart is presented.

Chapter 4 introduces a new single-chart SPRT scheme, i.e., the omnibus SPRT (OSPRT) chart, for joint monitoring of the mean and dispersion. We present the theoretical derivation of the statistic of the OSPRT chart, as well as its working mechanism and run length properties. The chapter also outlines the full optimisation algorithms for the OSPRT chart based on four performance metrics, respectively. The effectiveness of the OSPRT chart is then highlighted via a comparison with four other joint monitoring control schemes in the literature.

In Chapter 5, the effects of skewness on the performances of the SPRT chart with estimated process parameters as well as the OSPRT chart are investigated. The distributions considered in this thesis are the gamma, lognormal, and Weibull distributions, tuned for a multitude of different shapes and skewness. Finally, in

Chapter 6, the research findings and contributions made throughout the thesis are summarised. Suggestions and recommendations for future research are listed to motivate continuous efforts in the advancement of SPRT-related charting techniques.

The computer programmes developed for this research project are written in C++, and are to be compiled in RStudio via the **sourceCpp()** function from the **Rcpp** package. All the codes can be found on the following Github repository: <https://github.com/teohjingwei/New-Developments-of-the-Sequential-Probability-Ratio-Test-Control-Chart>. The document “Code Documentation.pdf” provides information on how the functions can be implemented to obtain the desired results. The main purpose of the computer programmes includes computing the run length properties of the SPRT chart with estimated process parameters and the OSPRT chart, as well as computing the optimal charting parameters for various choices of objective functions. The codes also provide facility for parallel computing, which is aimed at speeding up the Monte Carlo simulation as well as the computation of the optimal charting parameters.

2 Literature Review

2.1 Introduction

This chapter presents a thorough literature study on the historical origins of the SPRT chart, as well as other topics related to the new developments of the SPRT chart proposed in this thesis. The chapter begins with a brief review of three mainstream control charts in the literature, i.e., the Shewhart control chart, exponentially weighted moving average (EWMA) control chart, and cumulative sum (CUSUM) control chart, provided in Section 2.2. In Section 2.3, we discuss the development of the SPRT chart in chronological sequence, revealing how the SPRT is beneficial in the context of hypothesis testing. The SPRT chart for monitoring an arbitrary parameter, the mean, and the dispersion of a process are subsequently discussed in the section. In Section 2.4, we provide the definitions of the performance metrics used throughout the thesis. In Section 2.5, some literature related to control charts with estimated process parameters is studied. The evolution of the design methodologies for control charts with estimated process parameters at various stages are also critically analysed and reviewed. In Section 2.6, we provide a literature survey on multi- and single-chart joint monitoring control schemes, highlighting their pros and cons in various different circumstances.

2.2 Some Mainstream Control Charts

This section briefly introduces the Shewhart control chart (the first control chart ever proposed in history), as well as two mainstream memory-type control charts in the literature, i.e., the EWMA and CUSUM charts. By “memory-type” control charts, we refer to control charts that rely on memory statistics (see Section 1.1) to make control decisions. In each subsection, we detail the control statistic

and working mechanism of each control chart, as well as including a schematic diagram that depicts how each control chart works.

2.2.1 The Shewhart Control Chart

The Shewhart control chart is constructed based upon the idea that “a point landing far from its target value indicates evidence of an out-of-control event” (Montgomery, 2019). A typical Shewhart control chart contains a centre line (CL), an upper control limit (UCL), and a lower control limit (LCL), as shown in Figure 2.1. The data used in Figure 2.1 consist of photoresist thickness measurements (measured in 10^{-7} of a metre) taken from 125-mm silicon wafers, and they can be found in Montgomery (2019). In fact, the points plotted on the Shewhart control chart can be based on any suitable statistic (e.g., the sample mean $\bar{X}$ and sample standard deviation S) of a quality characteristic. It is clear that the choice of the statistic would depend on the intended aspect of monitoring: If the practitioner intends to monitor the mean of a quality variable, then the Shewhart $\bar{X}$

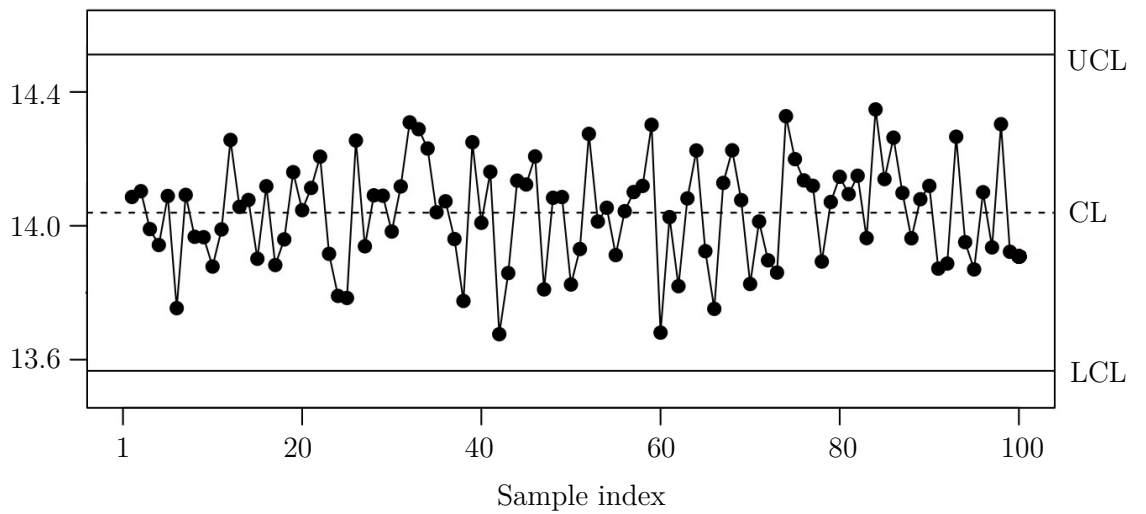


Figure 2.1 The two-sided Shewhart $\bar{X}$ control chart for monitoring the mean of the photoresist thickness of 125-mm silicon wafers.

chart would be used; if the practitioner intends to monitor the variability of a quality variable, then the Shewhart S chart would be used.

Let us define the Shewhart $\bar{X}$ chart. Suppose that, at the i -th sampling point ($i = 1, 2, \dots$), a sample of n observations ($X_{i1}, X_{i2}, \dots, X_{in}$) is taken from the normal distribution with mean μ_0 and standard deviation σ_0 , i.e., $\mathcal{N}(\mu_0, \sigma_0^2)$. The statistic $\bar{X}_i$ of the Shewhart $\bar{X}$ chart is calculated as

$$\bar{X}_i = \frac{1}{n} \sum_{j=1}^n X_{ij}, \quad (2.1)$$

for $i = 1, 2, \dots$. The UCL, CL, and LCL of the Shewhart $\bar{X}$ chart, denoted as $\text{UCL}_{\bar{X}}$, $\text{CL}_{\bar{X}}$, and $\text{LCL}_{\bar{X}}$, respectively, are expressed as

$$\text{UCL}_{\bar{X}} = \mu_0 + L_{\bar{X}} \frac{\sigma_0}{\sqrt{n}}, \quad (2.2)$$

$$\text{CL}_{\bar{X}} = \mu_0, \quad (2.3)$$

and

$$\text{LCL}_{\bar{X}} = \mu_0 - L_{\bar{X}} \frac{\sigma_0}{\sqrt{n}}, \quad (2.4)$$

where $L_{\bar{X}} > 0$ is a control limit multiplier of the Shewhart $\bar{X}$ chart. At any sampling point i , the process is said to be in-control if $\bar{X}_i$ falls between $\text{UCL}_{\bar{X}}$ and $\text{LCL}_{\bar{X}}$. If the statistic $\bar{X}_i$ goes above $\text{UCL}_{\bar{X}}$ (or goes below $\text{LCL}_{\bar{X}}$), then the process is flagged as out-of-control with a probable upward mean shift (or downward mean shift). A signal is produced, and the relevant engineer carries out investigations on the root causes of the process shift and eliminate them.

On the other hand, if the quality characteristic to be monitored is the process variability, then the Shewhart S chart is used. For the Shewhart S chart, the standard deviation of the i -th sample, S_i , is calculated as

$$S_i = \sqrt{\frac{1}{n-1} \sum_{j=1}^n (X_{i,j} - \bar{X}_i)^2}. \quad (2.5)$$

The UCL, CL, and LCL of the Shewhart S chart, denoted as UCL_S , CL_S , and LCL_S , respectively, are expressed as

$$\text{UCL}_S = c_4 \sigma_0 + L_S \sigma_0 \sqrt{1 - c_4^2}, \quad (2.6)$$

$$\text{CL}_S = c_4 \sigma_0, \quad (2.7)$$

and

$$\text{LCL}_S = \max\{0, c_4 \sigma_0 - L_S \sigma_0 \sqrt{1 - c_4^2}\}, \quad (2.8)$$

where

$$c_4 = \sqrt{\frac{2}{n-1}} \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})} \quad (2.9)$$

is a bias adjustment factor for the standard deviation, and $\Gamma(\cdot)$ denotes the gamma function. At any sampling point i , the process is said to be in-control if S_i falls between UCL_S and LCL_S . If the statistic S_i goes above UCL_S (or goes below LCL_S), then the process is flagged as out-of-control with a probable upward variance shift (or downward variance shift). Note that there have been several arguments concerning the necessity of including a LCL for the variance chart, since a decrease in the variance often reveals an improvement in the process stability, and would require minimal (or no) attention. Some of these arguments can be found in Wu *et al.* (2002, 2010), and Ou *et al.* (2015b).

Note that, apart from the $\bar{X}$ and S statistics, the Shewhart control chart can also be used to monitor other types of statistics. A comprehensive review on Shewhart control charts for monitoring other statistics, e.g., the range (R), the proportion defect (p), number of defective items (np), number of nonconformities

in an inspection unit (c), and average number of nonconformities in n inspection units (u), can be found in Montgomery (2019).

2.2.2 The Exponentially Weighted Moving Average Control Chart

Roberts (1959) introduced the exponentially weighted moving average (EWMA) control chart as an alternative to the Shewhart control chart, since the Shewhart control chart is known to be relatively insensitive to small and moderate shifts in a quality characteristic. Suppose that a sequence of *individual* (i.e., $n = 1$) measurements $X_1, X_2, \dots$ are taken from $\mathcal{N}(\mu_0, \sigma_0^2)$ at the sampling points $i = 1, 2, \dots$, respectively. The EWMA statistic is defined recursively as

$$\text{EWMA}_i = \lambda X_i + (1 - \lambda)\text{EWMA}_{i-1}, \quad (2.10)$$

where $0 < \lambda \leq 1$ is a smoothing constant, and the initial value of the statistic is $\text{EWMA}_0 = \mu_0$.

The UCL, CL, and LCL of the EWMA chart are computed as

$$\text{UCL} = \mu_0 + L_{\text{EWMA}} \sigma_0 \sqrt{\frac{\lambda}{2-\lambda}}, \quad (2.11)$$

$$\text{CL} = \mu_0, \quad (2.12)$$

and

$$\text{LCL} = \mu_0 - L_{\text{EWMA}} \sigma_0 \sqrt{\frac{\lambda}{2-\lambda}}, \quad (2.13)$$

respectively, where $L_{\text{EWMA}} > 0$ is a control limit multiplier for the EWMA chart.

Figure 2.2 shows an illustration of the EWMA chart for monitoring individual measurements simulated from $\mathcal{N}(10, 1)$ for the first 20 sampling points, and from $\mathcal{N}(11, 1)$ for the last 10 sampling points. The full dataset is available in

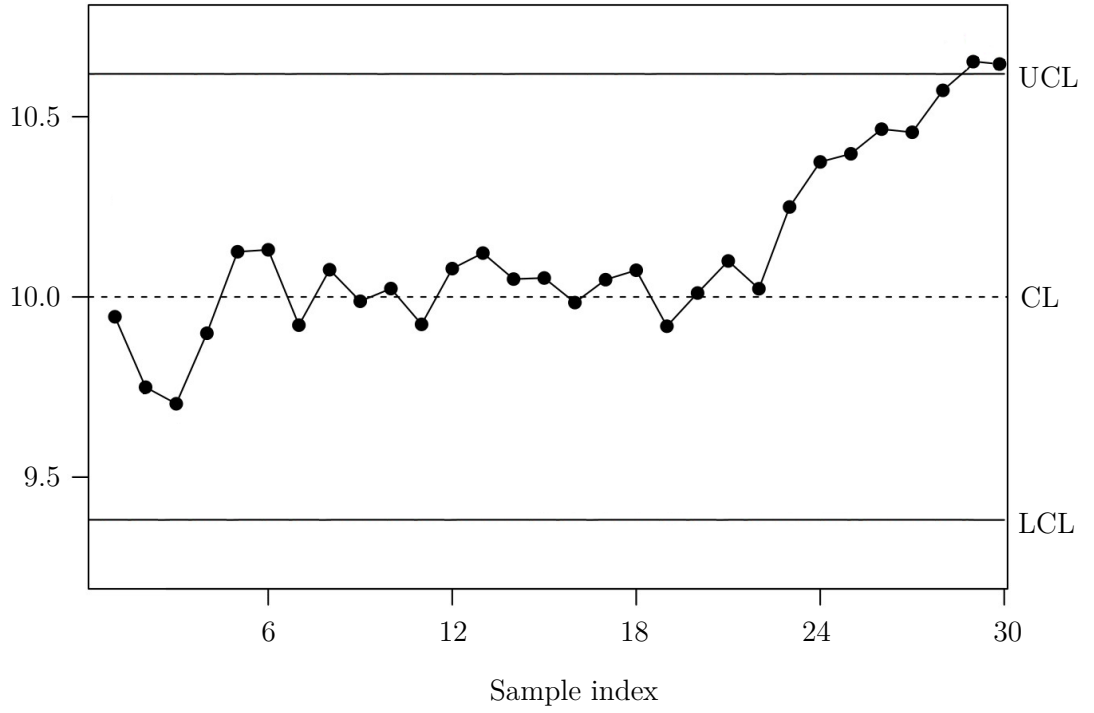


Figure 2.2 The two-sided EWMA control chart for monitoring simulated dataset.

Montgomery (2019). At any sampling point i , the process is said to be in-control if $EWMA_i$ falls between UCL and LCL. If the statistic $EWMA_i$ goes above UCL (or goes below LCL), then the process is flagged as out-of-control with a probable upward mean shift (or downward mean shift). For instance, from Figure 2.2, the process is declared as out-of-control at sampling point $i = 29$, and investigations are conducted immediately to identify the cause of the shift.

In the case where subgrouping is possible (i.e., $n > 1$), practitioners may replace the individual measurement X_i in Equation (2.10) by the sample average of n measurements $\bar{X}_i$, and replace σ_0 in Equations (2.11) and (2.13) by $\sigma_0/\sqrt{n}$. More details can be found in Saccucci *et al.* (1992), Jones *et al.* (2001), Reynolds and Stoumbos (2005), and Teoh *et al.* (2019).

2.2.3 The Cumulative Sum Control Chart

The cumulative sum (CUSUM) control chart proposed based on Page's procedure (Page, 1954) is also an excellent alternative to the Shewhart control chart. Assume that individual measurements $X_1, X_2, \dots$ are taken from $\mathcal{N}(\mu_0, \sigma_0^2)$ at the sampling points $i = 1, 2, \dots$, respectively. We must evaluate two statistics for the CUSUM chart, i.e., the upper and lower one-sided statistics. The upper and lower one-sided CUSUM statistics are defined as

$$\text{CUSUM}_i^+ = \max\left\{0, \text{CUSUM}_{i-1}^+ + \frac{X_i - \mu_0}{\sigma_0} - \gamma_c\right\} \quad (2.14)$$

and

$$\text{CUSUM}_i^- = \min\left\{0, \text{CUSUM}_{i-1}^- + \frac{X_i - \mu_0}{\sigma_0} + \gamma_c\right\}, \quad (2.15)$$

respectively, where $\gamma_c > 0$ is called the reference value of the CUSUM chart (Hawkins & Olwell, 1998), and the initial values of the statistics are set as $\text{CUSUM}_i^+ = \text{CUSUM}_i^- = 0$.

Note that there are no exact formulae to compute the UCL and LCL of the upper and lower one-sided CUSUM chart, respectively. We shall denote these control limits as H_{CUSUM} and $-H_{\text{CUSUM}}$, respectively, since both the upper- and lower-sided CUSUM charts are symmetric. The reader should refer to Hawkins and Olwell (1998) for guidelines on determining the value of H_{CUSUM} via numerical methods.

Figure 2.3 shows an illustration of the CUSUM chart for monitoring the same dataset used in Section 2.2.2. At any sampling point i , the process is said to be in-control if $\text{CUSUM}_i^+ \leq H_{\text{CUSUM}}$ and $\text{CUSUM}_i^- \geq -H_{\text{CUSUM}}$. If the statistic CUSUM_i^+ goes above H_{CUSUM} (or CUSUM_i^- goes below $-H_{\text{CUSUM}}$), then the process is flagged as out-of-control with a probable upward mean shift (or downward mean shift). For instance, from Figure 2.3, the process is declared as out-of-control at sampling point $i = 29$ with the indication of an upward mean shift.

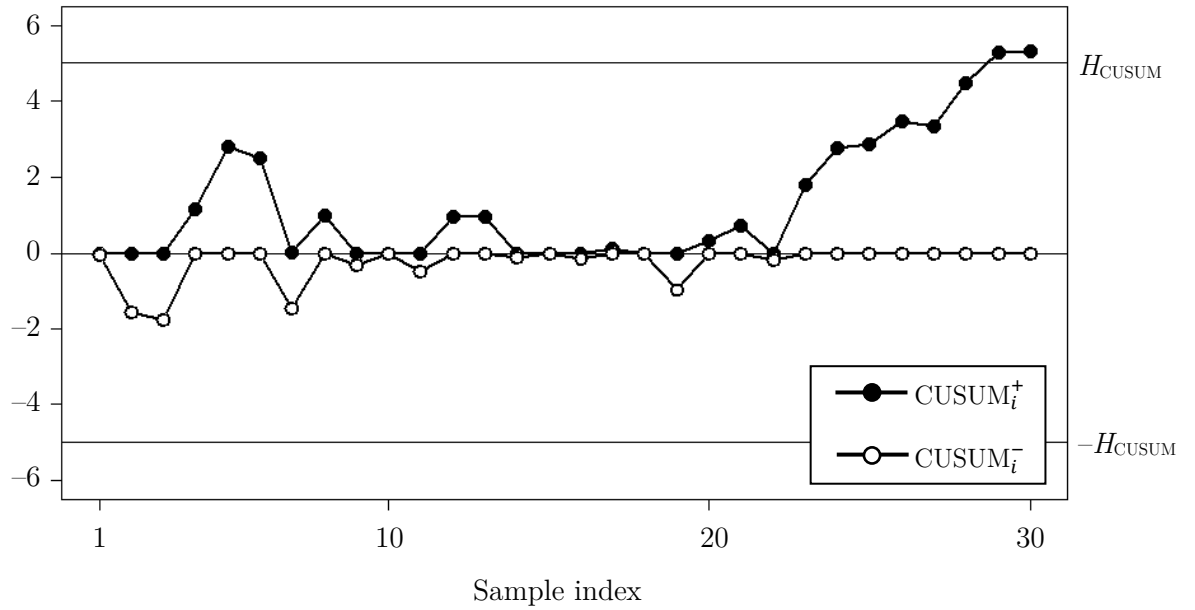


Figure 2.3 The two one-sided CUSUM control chart for monitoring simulated dataset.

Similar to the EWMA chart, in the case where subgrouping is possible (i.e., $n > 1$), practitioners may replace the standardised measurement $(X_i - \mu_0)/\sigma_0$ in Equations (2.14) and (2.15) by $(\bar{X}_i - \mu_0)/(\sigma_0/\sqrt{n})$. Some discussions can be found in Annadi *et al.* (1995), Reynolds and Stoumbos (2004a, 2004b), and Zhang and Wu (2007).

2.3 Developments of the Sequential Probability Ratio Test (SPRT)

Control Chart

The SPRT was originally developed for the purpose of statistical hypothesis testing (Wald, 1945). Unlike traditional tests based on fixed sample sizes, the SPRT allows for the continuous monitoring of incoming data, making it well-suited for applications where decisions need to be made promptly. Besides, the SPRT offers the advantage of flexibility in sample size determination, as it allows for termination of the test as soon as sufficient evidence is accumulated to make a conclusive

decision, thereby minimising unnecessary data collection. This feature is particularly valuable in scenarios where sampling efficiency and swift decision-making are crucial, such as in quality control processes or medical diagnostics.

2.3.1 The SPRT for Hypothesis Testing

Inspired by the Neyman-Pearson theory, Wald (1945) formulated the idea of a sequential test based on likelihood ratios, where the number of observations sought is treated as a random variable instead of a constant. Unlike standard tests based on fixed sample sizes, the sequential test reassesses its decision each time after one observation is sought, and the process of sampling terminates after a decision is reached.

Before we define the SPRT, let us outline the framework of a simple hypothesis test. Let the quality variable of interest X be distributed according to the probability distribution P . It is convenient to represent a candidate P within the collection of parametric distributions $\{P_\theta\}_{\theta \in \Theta}$ associated with the parameter $\theta \in \mathbb{R}$. In the context of SQC, one can think of θ as an important quality parameter of a process (e.g., the mean, standard deviation, proportion defect, etc.), while θ_0 and θ_1 can be thought of as the target and off-target values of the quality parameter. We consider the following definition of a simple hypothesis.

Definition 2.3.1. A simple hypothesis H is an assumption made on the distribution P_θ which corresponds to a specific value of $\theta \in \mathbb{R}$.

Let us restrict our hypothesis test to two distinct distributions, i.e., P_0 and P_1 . Our aim is to determine whether we can reject the default (i.e., null) distribution in favour of another (i.e., alternative) distribution, i.e., reject “ $H_0 : P = P_0$ ” in favour of “ $H_1 : P = P_1$ ”. In the parametric version of a hypothesis test, we consider

the simple hypotheses “ $H_0 : \theta = \theta_0$ ” versus “ $H_1 : \theta = \theta_1$ ”, where θ_0 and θ_1 are fixed points in the parameter space Θ .

Let us now define the SPRT for testing the hypotheses $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$. Suppose that a sequence of independent observations $X_1, X_2, \dots$ is sampled from an identical distribution with probability function (mass function in the discrete case and density function in the continuous case) $f(X | \theta)$ with respect to a σ -finite measure (e.g., Lebesgue measure). For an integer $j \geq 1$, we define the log-likelihood ratio for the sample $(X_1, \dots, X_j)$ as

$$L_j = \ln \left[\frac{f(X_1|\theta_1) \cdots f(X_j|\theta_1)}{f(X_1|\theta_0) \cdots f(X_j|\theta_0)} \right] = \sum_{j'=1}^j \ln \left[\frac{f(X_{j'}|\theta_1)}{f(X_{j'}|\theta_0)} \right]. \quad (2.16)$$

Let A and B be two constants such that $A < B$. For natural numbers j , the SPRT

- (i) terminates and rejects H_0 if $L_j \geq B$,
- (ii) terminates and accepts (fails to reject) H_0 if $L_j \leq A$,
- (iii) continues sampling if $A < L_j < B$.

To assist the reader in better understanding the sequential nature of the SPRT, we describe a complete run-through of the test. Suppose that the first observation X_1 is taken from a process. We calculate the value of L_1 from Equation (2.16) and compare it against A and B . If $L_1 \geq B$, the test is terminated with the decision to reject the null hypothesis H_0 ; whereas if $L_1 \leq A$, the test is terminated with the decision to accept the null hypothesis H_0 . In the case where $A < L_j < B$, we should sample another measurement X_2 and calculate the value of L_2 . The procedure is repeated until a decision is made.

Wald (1947) proved that the SPRT terminates with probability one, given that $f(X | \theta_0) \neq f(X | \theta_1)$ almost everywhere. Hence, in most practical applications, practitioners should be assured that the sampling procedure will eventually stop at

some point. Often, practitioners are interested in understanding the number of observations required (i.e., the sample number) for the SPRT to terminate. We provide the following definition.

Definition 2.3.2. The sample number, or the stopping time of the SPRT, is defined as

$$N = \inf\{n \geq 1 : L_n \in (-\infty, A] \cup [B, +\infty)\}. \quad (2.17)$$

Note that the sample number N is a random variable of the data.

Clearly, each SPRT can be characterised by a 2-tuple $(N, \mathcal{T})$, where $\mathcal{T}$ denotes a binary variable that takes on the value 1 if $L_N \geq B$, and takes on the value 0 if $L_N \leq A$. For example, the SPRT characterised by $(5, 1)$ rejects H_0 after taking five observations, whereas the SPRT characterised by $(7, 0)$ accepts H_0 after taking seven observations.

To determine the appropriate limits for the SPRT, we need to first specify the error probabilities of the test. We provide the following definitions for the Type-I and Type-II errors of the SPRT.

Definition 2.3.3. Denote by α and β the Type-I and Type-II error probabilities of the SPRT, respectively. The error probabilities are defined as

$$\alpha = \Pr(\mathcal{T} = 1 | \theta = \theta_0) \quad (2.18)$$

and

$$\beta = \Pr(\mathcal{T} = 0 | \theta = \theta_1). \quad (2.19)$$

Given the values of α and β , Wald (1945) derived the appropriate bounds for the constants A and B as

$$A \geq \ln\left(\frac{\beta}{1-\alpha}\right) \quad (2.20)$$

and

$$B \leq \ln\left(\frac{1-\beta}{\alpha}\right), \quad (2.21)$$

respectively. Wald (1945) ascertained that, by setting $A = \ln[\beta/(1 - \alpha)]$ and $B = \ln[(1 - \beta)/\alpha]$, the true Type-I and Type-II errors of the SPRT should not exceed α and β by a large degree. He explained that, since the desired values of α and β are set to be very small in practice (< 0.05), the true Type-I and Type-II errors can be reasonably assumed to be lower than α and β . Hence, the bounds in Equations (2.20) and (2.21) can be used as the approximate limits of the SPRT.

Wald (1947) defined the probability that the SPRT accepts H_0 for a given θ as the operating characteristic (OC) of the SPRT. By ignoring the excess in Equations (2.20) and (2.21), i.e., by setting $A = \ln[\beta/(1 - \alpha)]$ and $B = \ln[(1 - \beta)/\alpha]$, we have

$$\text{OC}(\theta_0) \approx \frac{e^B - 1}{e^B - e^A} \quad (2.22)$$

and

$$\text{OC}(\theta_1) \approx \frac{e^{-B} - 1}{e^{-B} - e^{-A}}. \quad (2.23)$$

Wald (1947) also derived the expected sample size (ESS) of the SPRT, which is essentially the expected value of the sample number N given in Definition 2.3.2. To derive approximate expressions for the ESS, we first define the Kullback–Leibler information numbers $I_i = E[L_1 \mid \theta_i]$, for $i = 0, 1$. The ESS of the SPRT under either hypotheses can be approximated via the equations:

$$\text{ESS}(\theta_0) \approx \frac{\text{OC}(\theta_0)A + [1 - \text{OC}(\theta_0)]B}{I_0} \quad (2.24)$$

and

$$\text{ESS}(\theta_1) \approx \frac{\text{OC}(\theta_1)A + [1 - \text{OC}(\theta_1)]B}{I_1}, \quad (2.25)$$

respectively, where $A = \ln[\beta/(1 - \alpha)]$ and $B = \ln[(1 - \beta)/\alpha]$ are chosen as the limits of the SPRT. According to Wald (1945), the best sequential test should be the one that minimises the expected number of observations taken up to the point where sampling is terminated. It was proven that, among all sequential tests of a given strength, the SPRT attains the smallest $\text{ESS}(\theta_0)$ and $\text{ESS}(\theta_1)$, in cases where H_0 is true and H_1 is true, respectively (Wald, 1945, 1947; Wald & Wolfowitz, 1948).

Equipped with the statistical framework given earlier, Wald (1947) subsequently formulated the SPRT for testing a variety of statistical variables. Some examples include testing the mean of a normal distribution with unknown variance, testing the variance of a normal distribution, etc. In the case where $A < 0 < B$, it is possible to obtain asymptotic approximations for the OC and the ESS of the SPRT, given that the absolute values of A and B are large. Siegmund (1975) derived asymptotic expressions for the error probabilities of the SPRT by considering a renewal-theoretic approach alongside Wald's formulae. Recognising that the sequence of log-likelihood ratios can be modelled by a Brownian motion, Siegmund (1979) developed approximations for the OC and ESS based on a corrected version of the Brownian motion. The interested reader is directed to Siegmund (1975, 1979) for additional details.

As computing technologies improved over decades, the constants (i.e., A and B), OC, and ESS of the SPRT could be computed with a very high degree of accuracy via standard numerical techniques. Kemp (1958) suggested that the expressions for the OC and ESS can be calculated by first constructing integral equations for the two quantities. The equations can then be solved via the Gaussian quadrature method or some other numerical methods (Goel & Wu, 1971). Despite the efficiency of machine computations, the approximations serve as an important cornerstone to the development of theoretical calculations of the SPRT and its properties. For instance, Godase and Mahadik (2019) stated that the lower and

upper bounds of Equations (2.20) and (2.21), respectively, can be used as starting values of A and B to reduce the number of iterations run by a numerical algorithm.

2.3.2 The SPRT Chart for an Arbitrary Parameter

Stoumbos and Reynolds (1996) were the first to incorporate the SPRT into a quality control chart for monitoring the parameter θ of an arbitrary distribution. To assess the stability of a process over time, they proposed applying a sequence of independent SPRTs at regular time intervals d , where $d > 0$ is a real number. The mathematical formulation of the SPRT chart is as follows. Suppose that we are interested in testing the hypotheses $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1 > \theta_0$. Suppose further that, at the sampling points $i = 1, 2, \dots$, a sequence of samples $(X_{1,1}, X_{1,2}, \dots, X_{1,N_1}), (X_{2,1}, X_{2,2}, \dots, X_{2,N_2}), \dots$ is obtained from the process. Here, $N_1, N_2, \dots$ are the sample numbers of the sequence of SPRTs (see Definition 2.3.2). The construction of the SPRT chart relies on the following two assumptions.

Assumption 1. The within-sample and between-sample observations are independent. It is easy to see that Equation (2.16) is a consequence of this assumption, since the log-likelihood ratio of the joint distribution of the observations simplifies to the sum of individual likelihood ratios corresponding to each observation.

Assumption 2. The sequence of observations within each sample are obtained in negligible time. In other words, we assume that the time between the observations $X_{i,j-1}$ and $X_{i,j}$ is substantially smaller than d , for $i = 1, 2, \dots$ and $j = 1, 2, \dots$.

The one-sided control statistic of the SPRT chart can be calculated via the following recursive formula

$$U_{i,j} = U_{i,j-1} + \ln \left[\frac{f(X_{i,j}|\theta_1)}{f(X_{i,j}|\theta_0)} \right], \quad (2.26)$$

for $i = 1, 2, \dots$ and $j = 1, 2, \dots, N_i$, where the value of the initial control statistic is set as $U_{i0} = 0$.

The decision criterion of the SPRT chart is established through two control limits g and h . The lower control limit g is also known as the acceptance limit, and the region below g is known as the acceptance region. On the contrary, the upper control limit h is known as the rejection limit, and the region above h is known as the rejection region. At any sampling point i ($i = 1, 2, \dots$), for $j = 1, 2, \dots, N_i$,

- if $U_{i,j}$ falls in the region $(-\infty, g]$, sampling is terminated and the process is accepted as in-control,
- if $U_{i,j}$ falls in the region $[h, +\infty)$, sampling is terminated and the process is declared as out-of-control,
- if $U_{i,j}$ falls in the region (g, h) , the status of the process is inconclusive, and we should proceed with sampling another measurement to calculate $U_{i,j+1}$.

The visualisation of the upper one-sided SPRT chart is given in Figure 2.4. In Figure 2.4, the process is concluded as in-control in the first three sampling points (i.e., $i = 1, 2, 3$). At the fourth sampling point (i.e., $i = 4$), the process is declared as out-of-control, since the statistic $U_{i,j}$ exceeds h . The sample numbers of the four SPRTs are $N_1 = 4$, $N_2 = 7$, $N_3 = 3$, and $N_4 = 12$, respectively.

To safeguard against issues that might surface during the initialisation of a process, Stoumbos and Reynolds (1996) suggested to set the initial control statistic as $U_{1,0} = u_0 \in (0, h)$ and $U_{i,0} = 0$ for $i = 2, 3, \dots$. They argued that setting a head start $u_0 > 0$, also known as the fast initial response strategy, is effective in flagging potential start-up problems of a process. Furthermore, it is also advised to allocate a short time period d_0 ($\leq d$) immediately after the initialisation of the process to allow for start-up intermission, such as machinery warm up (no sampling takes place during this period). In practice, it is reasonable to assume that the process

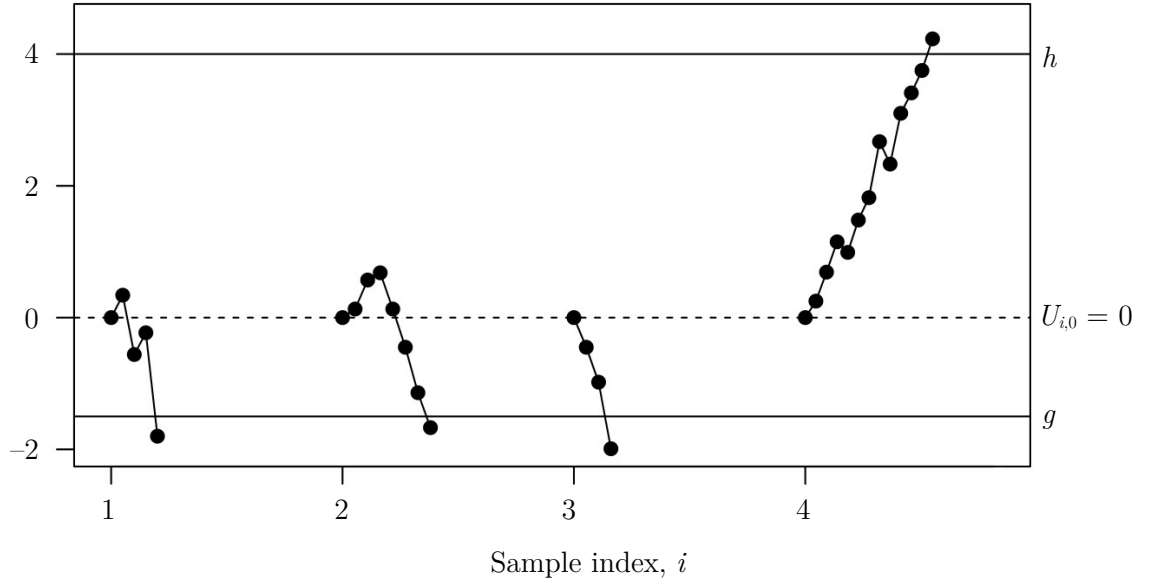


Figure 2.4 The upper one-sided SPRT control chart.

has been running for a long period of time, hence the aforementioned issues can be avoided (Stoumbos & Reynolds, 1997a). Throughout the thesis, it is implicitly assumed that $U_{i,0} = 0$ for $i = 1, 2, \dots$ and $d_0 = 0$.

2.3.3 The SPRT Chart for Process Location

Stoumbos and Reynolds (1997a) developed the SPRT chart for monitoring the mean of a quality characteristic $X \sim \mathcal{N}(\mu, \sigma_0^2)$. We assume that the mean μ of the distribution is unknown, whereas the variance σ_0^2 is known and constant. The hypotheses are now specified as $H_0 : \mu = \mu_0$ versus $H_1 : \mu = \mu_1$, where μ_0 represents the target mean value, and μ_1 represents the shifted mean value. While there are various ways to quantify deviations from the target mean μ_0 , it is conventional to define $\mu_1 = \mu_0 + \delta\sigma_0$, where δ is the size of the mean shift expressed in units of σ_0 . Consider the case where an upper-sided mean shift persists, i.e., $\delta > 0$. By substituting the probability density function (pdf) of the normal distribution into

Equation (2.26), Stoumbos and Reynolds (1997a) showed that the upper one-sided control statistic of the SPRT chart can be simplified as

$$U_{i,j}^+ = U_{i,j-1}^+ + \left(\frac{X_{i,j} - \mu_0}{\sigma_0} - \gamma \right), \quad (2.27)$$

where $U_{i,0}^+ = 0$, and $\gamma > 0$ is a reference parameter of the SPRT chart, whose expression is given as $\gamma = (\mu_1 - \mu_0)/(2\sigma_0) = \delta/2$. In the case where a lower-sided mean shift is expected, i.e., $\delta < 0$, the lower one-sided control statistic of the SPRT chart is given as

$$U_{i,j}^- = U_{i,j-1}^- + \left(\frac{X_{i,j} - \mu_0}{\sigma_0} + \gamma \right), \quad (2.28)$$

where $U_{i,0}^- = 0$, and $\gamma = -(\mu_1 - \mu_0)/(2\sigma_0) = -\delta/2 > 0$. Note that in both Equations (2.27) and (2.28), the reference parameter takes the value $\gamma = |\delta|/2$. It was proven that the SPRT chart achieves its optimum detection speed for the shift size δ when γ is set to be equal to $|\delta|/2$ (Moustakides, 1986; Geyer *et al.*, 1996; Stoumbos & Reynolds, 1997a).

Remark 2.3.1. Under the same in-control conditions, the upper- and lower-sided SPRT charts are symmetric.

Note that Remark 2.3.1 implies that the orientation and decision criteria for both the upper- and lower-sided SPRT charts are exactly the opposite. For example, the process is said to have sustained an(a) upper(lower)-sided mean shift if the control statistic $U_{i,j}^+ > h$ ($U_{i,j}^- < -h$); whereas the process is said to be in-control if the control statistic $U_{i,j}^+ < g$ ($U_{i,j}^- > -g$).

There have been numerous documentations on the superiority of the SPRT chart over other control charts, such as the Shewhart $\bar{X}$, EWMA, and CUSUM control charts, in terms of the detection performance. These documentations can be found in Stoumbos and Reynolds (1997a), Ou *et al.* (2012c), Mahadik and Godase (2023a, 2023b). There are also multiple extensions and enhancement designs

developed for the SPRT mean chart in the past few decades. The interested reader should find the works by Stoumbos and Reynolds (2001), Li *et al.* (2009), Ou *et al.* (2010, 2011b, 2012b, 2015a) useful.

2.3.4 The SPRT Chart for Process Dispersion

The SPRT chart for monitoring process variability was first proposed by Chou *et al.* (2006). Let us now suppose that the standard deviation σ of the distribution of X is unknown, whereas the mean μ is known and equal to μ_0 . The hypotheses are set up as $H_0 : \sigma = \sigma_0$ versus $H_1 : \sigma = \sigma_1$, where σ_0 and σ_1 represent the target and shifted values of σ , respectively. It is customary to define $\sigma_1 = \eta\sigma_0$, where η is the standard deviation shift size expressed as a ratio of the shifted standard deviation to the target level. Note that Chou *et al.* (2006) only considered the case $\eta > 1$. Substituting the pdf of the normal distribution into Equation (2.26), Chou *et al.* (2006) derived the control statistic of the upper one-sided SPRT chart for monitoring process dispersion as

$$U_{i,j} = U_{i,j-1} + \left[\left(\frac{X_{i,j} - \mu_0}{\sigma_0} \right)^2 - \gamma \right], \quad (2.29)$$

where $U_{i,0} = 0$, and $\gamma = 2\sigma_1^2 \ln(\sigma_1/\sigma_0) / (\sigma_1^2 - \sigma_0^2) = 2\eta^2 \ln(\eta) / (\eta^2 - 1)$ is the reference parameter of the SPRT chart. It is easy to show that, when the standard deviation is expected to increase, we have $\gamma > 1$.

Godase and Mahadik (2019) noted that Chou *et al.* (2006) had not derived the exact statistical properties of the SPRT dispersion chart, nor considered its lower-sided counterpart, i.e., when $\eta < 1$. To fill the research gap, Godase and Mahadik (2019) provided the full derivation for the statistical properties of the SPRT dispersion chart. They inferred that the lower one-sided control statistic for the SPRT dispersion chart is identical to that in Equation (2.29), and the control

limits of the lower one-sided chart have an opposite orientation to those of the upper one-sided chart. It is also easy to show that, when the standard deviation is expected to decrease, we have $0 < \gamma < 1$. Hence, the reference parameter of the SPRT dispersion chart is always positive.

Remark 2.3.2. Under the same in-control conditions, the upper- and lower-sided SPRT dispersion charts are not symmetric.

Remark 2.3.2 should be obvious, since a percentage increase in σ cannot be compensated by the same percentage decrease in σ . For example, in the case of the mean shift, $\delta = 0.5$ can be simply countered by an equivalent shift in the negative direction $\delta = -0.5$. However, if one increases σ by 20% (i.e., $\eta = 1.2$), it is not possible to bring the process standard deviation back to its original value by shrinking the shifted value by the same percentage (i.e., $\eta = 0.8$). In current literature, many researchers have only necessitated the detection of an upper-sided shift in the variance, since a lower-sided shift in the variance usually reveals an improvement in the process stability and would require minimal (or no) attention. The arguments can be found in Wu *et al.* (2002, 2010), and Ou *et al.* (2015b).

2.4 Performance Measures of a Control Chart

Just like any algorithm or prototype, the effectiveness of a control chart needs to be assessed in order to provide a comprehensive knowledge of its advantages and limitations. In the following subsections, we define some metrics used to measure the performances of a control chart, i.e., the run length, time to signal, sample number, and the number of observations to signal. As the aforementioned metrics are localised performance measures (i.e., they are evaluated for a particular shift size), we also discuss some metrics for evaluating the overall performance of a control chart, given a range of shift sizes.

2.4.1 The Run Length and the Time to Signal

One of the most crucial properties of a control chart is its run length. The run length is defined as the number of samples required until a signal is emitted by a control chart. Consider a sequence of samples $(X_{1,1}, X_{1,2}, \dots, X_{1,N_1}), (X_{2,1}, X_{2,2}, \dots, X_{2,N_2}), \dots$ taken at the sampling points $i = 1, 2, \dots$, respectively, which we denote as $\mathbf{X}_1, \mathbf{X}_2, \dots$. Consider also a statistic ζ for the sample. We define a non-empty set $\mathfrak{R}$, such that $\zeta(\mathbf{X}_i) \in \mathfrak{R}$ leads to the rejection of H_0 , and subsequently the production of a signal by the control chart. We give the following definition.

Definition 2.4.1. The run length (RL) is defined as

$$\text{RL} = \inf\{n \geq 1 : \zeta(\mathbf{X}_n) \in \mathfrak{R}\}. \quad (2.30)$$

Note that the sufficiency of ζ is desirable but not necessary. For instance, we know that the usual $\bar{X}$ and S statistics are sufficient statistics for the process mean and standard deviation, respectively. Therefore, the run length properties of the Shewhart $\bar{X}$ and S charts can be derived by citing the sufficiency of these statistics. On the other hand, Ou *et al.* (2012a) proposed the Shewhart $|\bar{X}|$ chart for joint monitoring of process mean and standard deviation. Clearly, the $|\bar{X}|$ statistic is not a jointly sufficient statistic for the mean and standard deviation, but there have been no issues with defining the run length properties of the Shewhart $|\bar{X}|$ chart.

Since RL is a random variable, it is often useful to understand its average value, i.e., the average run length. The average run length (ARL) is defined as

$$\text{ARL} = E[\text{RL}], \quad (2.31)$$

and it measures the average number of samples taken until a signal is produced. In other words, the ARL tells us how many samples we should expect to take on average, up to (and including) the point where an alarm is given. When the process is in-control, the in-control ARL (ARL_0) tells us how frequently we should expect

to obtain false alarms (or false signals). When the process is out-of-control, the out-of-control ARL (ARL_1) tells us how quickly we should expect to detect the process shift. Thus, we have the following remark.

Remark 2.4.1. An effective control chart has a large ARL_0 and a small ARL_1 .

Note that Remark 2.4.1 is consistent with the philosophy “If it ain’t broken, don’t fix it”. In particular, a large ARL_0 is desired to avoid excessive false alarms (which can diminish an inspector’s confidence), whereas a small ARL_1 is desired to promote a spontaneous shift detection. In the context of hypothesis testing, a large ARL_0 is equivalent to a small Type-I error, whereas a small ARL_1 is equivalent to a small Type-II error.

Another useful property is the standard deviation of the run length (SDRL). It measures the variability of the run length values obtained by practitioners throughout their course of monitoring. The formula for the SDRL is given as

$$SDRL = \sqrt{\text{Var}(RL)}. \quad (2.32)$$

It should be noted that, under both the in-control and out-of-control cases, we desire a small in-control SDRL ($SDRL_0$) value and a small out-of-control SDRL ($SDRL_1$) value. This implies that the performance of a control chart is stable over time.

In many applications of a control chart, the sampling interval is involved. Hence, it may be useful to calculate the time to signal instead of the run length, which has the following definition.

Definition 2.4.2. Given a fixed sampling interval $d > 0$, the time to signal (TS) is defined as

$$TS = d \times RL. \quad (2.33)$$

The TS measures the number of time units required until a signal is produced by a control chart. The average time to signal (ATS) and standard deviation of the time to signal (SDTS) are defined as

$$\text{ATS} = E[\text{TS}] \quad (2.34)$$

and

$$\text{SDTS} = \sqrt{\text{Var}(\text{TS})}, \quad (2.35)$$

respectively. Note that Remark 2.4.1 also applies to the ATS. That is, we desire a large in-control ATS (ATS_0) and a small out-of-control ATS (ATS_1). Similarly, both a small in-control SDTS (SDTS_0) and a small out-of-control SDTS (SDTS_1) are desirable.

In current literature, there are two common ways of evaluating the out-of-control performance (e.g., the ARL_1 , ATS_1) of a control chart, i.e., the zero-state mode and the steady-state mode. We provide the definition of a zero-state performance.

Definition 2.4.3. The *zero-state* performance refers to the performance of a control chart measured under the assumption that the out-of-control event occurs immediately after the control chart is initiated.

Reynolds *et al.* (1990) stated that, in most applications, the control chart should have operated in the in-control state for a long period of time before an out-of-control event emerges. Hence, the zero-state performance tends to overestimate the true performance of a control chart in practical settings, since the charting statistic may have attained a value greater than the initial statistic by the time the shift occurs. To provide a more realistic assessment of the performance of a control chart, they introduced the steady-state performance, which has the following definition.

Definition 2.4.4. The *steady-state* performance refers to the performance of a control chart measured under the assumption that the charting statistic has reached its stationary distribution when the out-of-control event occurs.

Remark 2.4.2. The steady-state ARL_1 (ATS_1) is less than or equal to the zero-state ARL_1 (ATS_1).

One way to think of Remark 2.4.2 is to imagine that the control statistic has gained some “momentum” under the steady state. Hence, when an out-of-control event emerges, it is likely that the control statistic will reach the rejection region quicker under the steady state compared to that under the zero state.

2.4.2 The Sample Number

Suppose that a control chart takes a sequence of samples of sizes $N_1, N_2, \dots$ from a process at regular time intervals. Here, $\{N_i\}_{i \geq 1}$ is the sequence of sample numbers of the control chart. For fixed-sample-size (FSS) control charts with sample size n , such as the Shewhart $\bar{X}$ chart stated in Section 2.2.1, it is clear that

$$N_1 = N_2 = \dots = n.$$

In the case where the sample numbers N_i can vary, it is often useful to calculate the average sample number (ASN), which tells us how many observations we should expect to take in each sample. The ASN is defined as

$$ASN = E[N_i]. \tag{2.36}$$

Note that the ASN is primarily used to evaluate the sampling efficiency of a variable-sample-size (VSS) control chart (Annadi *et al.*, 1995). In the case where the ASN of a VSS control chart is smaller than the sample size n of a FSS control chart, we say that the VSS control chart is more efficient (in terms of sampling)

than the FSS control chart. Conversely, if the ASN of a VSS control chart is larger than the sample size n of a FSS control chart, we say that the VSS control chart is less efficient than the FSS control chart.

For a FSS control chart with subgroup n , it is easy to see that $ASN = n$. For a VSS control chart, the calculation of the ASN may require a bit more context. For example, consider the VSS $\bar{X}$ chart that samples n_1 observations when the $\bar{X}$ values are “close” to the centre line, and samples n_2 observations when the $\bar{X}$ values lie “far” from the centre line. Assuming that the decision rules are well-defined, we should be able to derive the probabilities $\Pr(N = n_1)$ and $\Pr(N = n_2)$. Thus, the ASN can be calculated as

$$ASN = n_1 \times \Pr(N = n_1) + n_2 \times \Pr(N = n_2).$$

Suppose now that the sample number can take any non-negative integer values, i.e., $\text{Im}(N) = \{1, 2, \dots\}$. In this case, the derivation of the ASN becomes slightly more tricky. For memory-type control charts, e.g., the SPRT chart, the distribution of N is usually intractable, and we are not able to calculate the probabilities $\Pr(N = i)$ directly, where $i = 1, 2, \dots$. This implies that the expected value of N needs to be calculated via some other methods (e.g., by means of the Markov chain). We will provide the full derivation of the ASN of the SPRT chart later in Chapter 3.

2.4.3 The Number of Observations to Signal

Stoumbos (1993) was the first to propose the number of observations to signal metric. As the metric combines both the information on the run length and the sampling efficiency, it is especially effective for measuring the performance of a VSS control chart. Suppose that a control chart emits a signal after RL number of

samples, where the i -th sample contains N_i number of observations. We provide the following definition.

Definition 2.4.5. The number of observations to signal (NOS) is defined as

$$\text{NOS} = \sum_{i=1}^{\text{RL}} N_i. \quad (2.37)$$

In other words, the NOS counts the total number of individual observations taken until a signal is produced by a control chart. Note that a large NOS can be a consequence of either a large run length, or a sequence of large sample numbers, or both.

It is of interest to evaluate the average value of the NOS. The average number of observations to signal (ANOS) is calculated as

$$\text{ANOS} = \text{E}[\text{NOS}]. \quad (2.38)$$

Lemma 2.4.1. Given that RL and $\{N_i\}_{i \geq 1}$ are mutually independent, the ANOS is expressed as

$$\text{ANOS} = \text{E}[\text{RL}] \times \text{E}[N_i] = \text{ARL} \times \text{ASN}. \quad (2.39)$$

Proof. The statement follows from the Wald's identity, which states that the expected value of the sum of a random number of random quantities is equal to the product of the expected value of the number and the expected value of the quantity, given that both the number and quantity have finite means. Since the $\text{ARL} < \infty$ and $\text{ASN} < \infty$, the Wald's identity applies and the result follows. $\square$

Remark 2.4.3. Lemma 2.4.1 can also be applied when RL and $\{N_i\}_{i \geq 1}$ are not mutually independent.

Discussions on Remark 2.4.3 can be found in Annadi *et al.* (1995), where the control procedure is said to resemble a renewal reward process.

When the process is in-control, the in-control ANOS (ANOS_0) tells us how many observations we should expect to take until a false alarm is produced. When the process is out-of-control, the out-of-control ANOS (ANOS_1) tells us how many observations we should expect to check until the shift is detected. Generally, we desire a small ANOS_1 . However, there is no specific requirement for how large or small should the ANOS_0 be, since the value of the ANOS_0 depends on the values of the ARL_0 and in-control ASN (ASN_0). For example, one might decide to use the combination of a large ARL_0 and a small ASN_0 , which results in a moderate value for the ANOS_0 .

2.4.4 Some Measures of the Overall Performance

While understanding the localised performance of a control chart (see Sections 2.4.1 to 2.4.3) helps with better decision making, it is important to note that a change in the process distribution is almost always unpredictable in practice. Hawkins and Olwell (1998) stated that, if practitioners have a specialised understanding of a manufacturing process (and hence know the most probable shift size), then it is possible to tune the charting parameters in a way so that the control chart reacts swiftly to the particular shift size of interest.

However, process shifts often occur sparsely in practice, and it is quite difficult to gauge the actual magnitude of the shift to which a process is likely to move (Wu *et al.*, 2010; Calzada & Scariano, 2013; Chong *et al.*, 2022). Calzada and Scariano (2013) stated that the assumption of a deterministic shift size is unrealistic in practice, since process shifts usually occur in a stochastic manner and leave no traceable trend. They explained that, when the control chart is designed to detect a particular shift size, it may lose its detection ability towards other sizes of process shifts for which it is not optimised. Hence, some performance metrics for measuring the overall performance of a control chart should be considered.

Without loss of generality, suppose that the shift size of an arbitrary parameter is denoted by δ , and $[\delta_{\min}, \delta_{\max}]$ represents the interval that contains all the possible values of δ . Here, $\delta_{\min}$ and $\delta_{\max}$ represent the minimum and maximum possible shift sizes. The expected value of the ARL (EARL) is calculated as

$$\text{EARL} = \int_{\delta_{\min}}^{\delta_{\max}} \text{ARL} f_{\delta}(\delta) d\delta, \quad (2.40)$$

where $f_{\delta}(\delta)$ is the pdf of the shift size δ . Owing to the lack of information about δ , researchers have suggested modelling δ using a flat prior, such as the uniform distribution. More details can be found in Sparks (2000), Reynolds and Stoumbos (2004a), Wu *et al.* (2010), and Celano and Castagliola (2016).

Aside from the EARL, some other metrics, such as the expected value of the ATS (EATS) (Khaw *et al.*, 2017; Saha *et al.*, 2019) and the expected value of the ANOS (EANOS) (Saha *et al.*, 2018; Malela-Majika *et al.*, 2021), have also been widely used. It is clear that the EATS and EANOS metrics can be computed using the formula in Equation (2.40), except that the ARL should be replaced by the ATS and the ANOS, respectively.

Recognising that quality loss has a proportional relationship to the square deviation of an observation from its target mean, Reynolds and Stoumbos (2004a) advocated the use of the quadratic loss (Taguchi, 1980) for evaluating the performance of control charts. The quadratic loss (QL) is defined as

$$\text{QL} = \text{E}\left\{\int_0^T \text{E}\left[\left(\frac{x-\mu_0}{\sigma_0}\right)^2\right] dt\right\} = \text{E}\left[\left(\frac{x-\mu_0}{\sigma_0}\right)^2\right] \text{E}[T], \quad (2.41)$$

where the integral domain $t \in [0, T]$ represents the duration from the start of the shift to the time T when a signal is produced. Note that T can be either the run length RL or the time to signal TS.

The average extra quadratic loss (AEQL), assuming that only the mean of X has shifted (by δ units), can be calculated as (Wu *et al.*, 2009)

$$\text{AEQL} = \int_{\delta_{\min}}^{\delta_{\max}} \delta^2 E[T] f_{\delta}(\delta) d\delta, \quad (2.42)$$

where $E[T] = \text{ARL}$ if $T = \text{RL}$, and $E[T] = \text{ATS}$ if $T = \text{TS}$. Here, both $\text{ARL} = \text{ARL}(\delta)$ and $\text{ATS} = \text{ATS}(\delta)$ are functions of δ based on their definitions in Equations (2.30) and (2.33), respectively. Note that the AEQL assigns the weight δ^2 to each ARL or ATS along the shift range $[\delta_{\min}, \delta_{\max}]$, justified on the grounds that larger weights should be given to the performances at a larger shift size. The AEQL metric has been widely used in the design of various control charts, with proven effectiveness in safeguarding against both small and large shift sizes, see for example, Ou *et al.* (2012b, 2012c), Khoo *et al.* (2013), Adeoti and Malela-Majika (2020). The metric will be used in the optimal designs of the SPRT control charts later in Chapters 3 and 4.

2.5 Control Charts with Estimated Process Parameters

In this section, we discuss how the design methodologies for control charts with estimated process parameters have evolved since the problem was first identified by Hillier (1969). As stated in Section 1.2, the values of the location and dispersion parameters of a process are not usually available to practitioners. Thus, it is customary for practitioners to compute the estimates of these parameters using a small set of in-control samples taken from the process. Due to the randomness in these estimates, it is likely that the performances of control charts would differ across practitioners, even though they are using the same control chart to monitor the same process. Some common issues, such as uncertainties due to small reference samples, inefficient estimators, and excessive false alarms produced by control

charts, have been extensively discussed and addressed by many scholars over the past few decades.

The implementation of SQC can be divided into two main phases, i.e., Phase-I chart calibration and Phase-II on-line monitoring. In Phase I, the process is preliminarily assessed for its stability from a modest number of historical observations via some appropriate Phase-I control charts. Any unknown parameters of the process (such as the mean and variance) are estimated from those observations, provided that they are declared as in-control under Phase-I analysis. Chakraborti *et al.* (2008) ascertained that an accurate Phase-I analysis is crucial to guarantee the success of Phase-II control charts, since Phase-II control charts ultimately rely on the parameters estimated in Phase I to produce control decisions. More details about the appropriate choices of Phase-I control charts and their design guidelines can be found in Chakraborti *et al.* (2008) and Atalay *et al.* (2020).

In the following subsections, we present the four stages of evolution of control charting methods with estimated process parameters. The four stages (discussed in chronological order) are:

- Stage 1 : Sample size determination,
- Stage 2 : Selection of estimators,
- Stage 3 : Unconditional in-control metric matching design,
- Stage 4 : Guaranteed in-control performance design.

It is important to note that the timelines spanned by the stages above can overlap due to the varying progresses of different scholars working on the same problem. Some scholars may have worked on hybrid developments, e.g., they may have considered a robust estimator for the standard deviation (Stage 2) alongside the guaranteed in-control performance design (Stage 4) for their choice of control chart. The following literature review aims to discuss the four stages in a sequential and independent manner.

2.5.1 Sample Size Determination

Increasing the sample size is perhaps the most natural resolution to challenges associated with Phase-I parameter estimation. We know, by the law of large numbers, that an estimated parameter of a distribution tends towards its true value as sample size increases to infinity. However, in most industrial practices, pursuing an indefinitely large Phase-I sample size is not realistic, and practitioners often need to start the Phase-II monitoring procedure as long as a sufficient amount of data is gathered for parameter estimation. Figure 2.5 shows the distributions of the sample mean $\bar{X}$ when 100, 400, and 1600 observations (left to right) were used to estimate the process mean, respectively. From Figure 2.5, it is clear that the variability of the estimate diminishes as the sample size increases.

Other than the variability in the estimators, researchers are often more interested in understanding the variability in the performances of control charts when different sample sizes were used, known as *practitioner-to-practitioner variability*. Quesenberry (1993) was among the first few to notice that the number of Phase-I samples has a direct influence on the performances of the Shewhart $\bar{X}$ and X control charts with estimated process parameters. They found that, with a small number of Phase-I samples, e.g., 20 or 30 samples of subgroup five, the

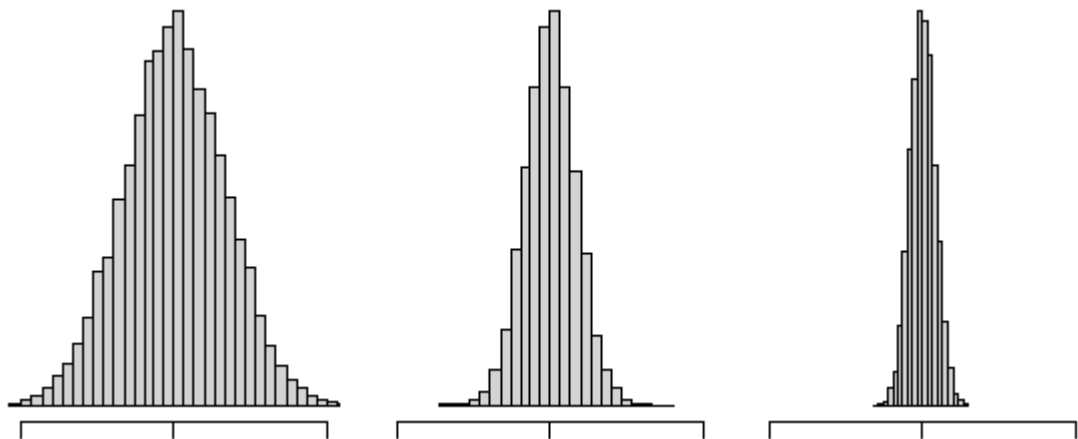


Figure 2.5 The distributions of the sample mean when the sample sizes are 100, 400, and 1600 (left to right).

variability in the run length distribution can be strikingly large. To achieve a performance (e.g., ARL, SDRL) that is “close” to that based on truly known process parameters, Quesenberry (1993) suggested to take at least 100 Phase-I samples of subgroup five to establish the $\bar{X}$ chart, and at least 300 individual Phase-I measurements to establish the X chart. Chen (1997) suggested that, for the $\bar{X}$ chart with estimated process parameters to perform as well as that in the known-parameter case, at least 100 Phase-I samples should be taken when the subgroup size is five, and at least 50 Phase-I samples should be taken when the subgroup size is ten. Sample size recommendations for various other Shewhart and memory-type control charts (e.g., the EWMA and CUSUM charts) can be found in Chen (1998), Braun (1999), Maravelakis *et al.* (2002), Jensen *et al.* (2006), Testik *et al.* (2006), and Testik (2007).

Furthermore, researchers discovered that control charts tend to produce too many false alarms when a small number of Phase-I samples is used. Note that excessive false alarms in a charting program can be endangering to a business as it leads to a lowered confidence of an inspecting personnel, which in turn leads to conclusions about the charting program being ineffective and useless. Quesenberry (1993) noticed that taking fewer than 100 Phase-I samples of subgroup five could lead to more than 10% increase in the early false alarm rates of the $\bar{X}$ chart. Similar findings have been found by Jones *et al.* (2001), who reported a substantial increase in the false alarm probabilities of the EWMA chart when a small sample size is used. Jones *et al.* (2001) discovered that the minimum sample size required to achieve a small increase ($< +10\%$) in the false alarm probability also depends on the smoothing parameter λ of the EWMA chart (see Section 2.2.2). In particular, when a smaller λ is used, a larger sample size would be required to keep the false alarm probability at the same level. To reduce both the practitioner-to-practitioner variability as well as the false alarm probabilities, Saleh *et al.* (2015a) provided sample-size recommendations for the EWMA chart based on various ARL_0 values

and smoothing parameters. They revealed that the required sample sizes can reach up to 5000 observations in total. Other sample-size recommendations based on different control charts can be found in the works of Yang *et al.* (2002), Jones *et al.* (2004), Ozsan *et al.* (2010), Saleh *et al.* (2013), Zhang *et al.* (2014a), and Saleh *et al.* (2015a, 2015b).

While the preceding works attempt to make recommendations by experimenting with different sample sizes, other scholars have developed methods to calculate the exact minimum sample size required to achieve a desired criterion. These efforts often require the researcher to derive the exact run length property (e.g., the ARL) of a control chart as a function of the Phase-I sample size, and subsequently solve (the root of) the function via some analytical methods or numerical methods. For example, Epprecht *et al.* (2015) was able to derive the exact number of Phase-I samples required for the Shewhart S chart, so that the true false alarm rate does not exceed a specified threshold with a high probability, say 95%. Some other related works can be found in Capizzi and Masarotto (2010), Jeske (2016), and Kumar and Chakraborti (2016).

Note that the recommended sample sizes suggested in most literature are too large to be applied in practice. In reality, practitioners are often reluctant to collect a large number of observations (say, 500) before proceeding to Phase-II on-line monitoring. Some recommendations based on stricter criteria may even require sample sizes up to 5000, which is impractical in most applications. Many researchers have since been looking out for other alternative methods, e.g., parameter updating based on self-starting control charts. The interested reader is directed to the works of Keefe *et al.* (2015) and Capizzi and Masarotto (2020) for some discussions.

2.5.2 Selection of Estimators

Apart from the sample size, researchers noticed that the choice of estimators also have a direct influence on the performance of control charts. Ideally, a good estimator should possess two properties: (1) the estimator is unbiased, i.e., the expected value of the estimator is equal to the value of the parameter, (2) the estimator has minimum variance, i.e., the estimator attains the smallest variance among all other estimators (Montgomery, 2019). It is also customary to evaluate the mean square error (MSE) of an estimator, which is defined as

$$\text{MSE} = \text{Bias}^2 + \text{Variance}. \quad (2.43)$$

The estimator with the lowest MSE among all other estimators considered is called the most *efficient* estimator.

In current literature, the discussions related to estimators are mostly for the standard deviation rather than for the mean, since the process mean can usually be adequately estimated by the sample mean. There are several ways to estimate the standard deviation σ of a normal distribution, as documented by Montgomery (2019) and many other authors. One approach is to estimate σ by using the sample range R of the data. Suppose that we have a sample of n measurements ($X_1, X_2, \dots, X_n$). The sample range R is calculated as

$$R = X_{(n)} - X_{(1)}, \quad (2.44)$$

where $X_{(i)}$ represents the i -th order statistic of the sample.

It is worth noting that the ratio R/σ is a pivotal quantity (known as the relative range), and its distribution is well-defined in many standard SQC textbooks. Thus, it is possible to estimate the standard deviation of the process using $\hat{\sigma} = R/d_2$, where d_2 is a bias adjustment factor for σ , and its value (as a function of the sample size n) is tabulated in standard SQC textbooks. In the case

where data are grouped, i.e., there are m number of samples each of size n ($X_{1,1}, X_{1,2}, \dots, X_{1,n}, \dots, (X_{m,1}, X_{m,2}, \dots, X_{m,n})$), the average sample range statistic $\bar{R} = \sum_{i=1}^m R_i / m$ can be used, where R_i denotes the range of the i -th sample.

Another (more common) approach is to estimate σ by using the sample standard deviation S . An unbiased estimator for σ is $\hat{\sigma} = S/c_4$, where the value of c_4 is defined based on Equation (2.9). In the case where data are grouped, the average standard deviation $\bar{S} = \sum_{i=1}^m S_i / m$ or the pooled standard deviation $S_{\text{pooled}} = \{\sum_{i=1}^m S_i^2 / m\}^{1/2}$ can be used to estimate σ . Note that in our research, we use only the sample standard deviation S in the construction of the SPRT chart with estimated process parameters. Therefore, in this thesis, we will avoid discussing other range-based or standard-deviation-based estimators in detail, and we direct the reader to the works of Del Castillo (1996), Mahmoud *et al.* (2010), and Saleh *et al.* (2015a, 2015b) for thorough discussions of those estimators.

There have been quite a number of discussions on the efficiencies of different standard deviation estimators. For example, Del Castillo (1996) and Chen (1997) discovered that the pooled standard deviation estimator S_{pooled} has a lower mean square error compared to the unbiased average standard deviation estimator $\bar{S}/c_4$, even though S_{pooled} is a biased estimator of σ . Mahmoud *et al.* (2010) found out that, as sample size increases, all estimators based on the sample range R become less efficient than standard-deviation-based estimators (i.e., $S, \bar{S}, S_{\text{pooled}}$). They advised against the use of all range-based estimators, and they ascertained that standard-deviation-based estimators should be adopted in all SQC applications considering that computing technologies have improved substantially. However, Mahmoud *et al.* (2010) also noted that, as the sample size increases (> 25 individual observations or > 20 samples of subgroup five), there is hardly any difference found between the efficiencies of any two standard-deviation-based estimators. Therefore, it seems that practitioners should devote more effort to other aspects of the design of control

charts with estimated process parameters, rather than solely focusing on searching the “best” estimator for a parameter.

Note that the aforementioned discussions were made based on the assumption that the Phase-I data follow the normal distribution. In cases where these “ideal” conditions are not met, the sample mean and the sample standard deviation estimators may fail to accurately capture the true values of the process parameters, which in turn leads to unsatisfactory performances in the subsequent Phase-II on-line monitoring. To safeguard against potential outliers and disturbances in Phase-I data, Schoonhoven *et al.* (2011a, 2011b) investigated a range of robust estimators for the location and dispersion parameters of a process. In terms of location estimators, Schoonhoven *et al.* (2011a) proposed a trimmed mean coupled with a screening procedure to avoid contamination to the location estimate. In terms of dispersion estimators, Schoonhoven *et al.* (2011b) concluded that the median absolute deviation estimator coupled with a screening procedure is the most robust method among all other methods studied. The interested reader may refer to the works of Schoonhoven *et al.* (2011a, 2011b) for a thorough analysis of various location and dispersion estimators.

2.5.3 Unconditional In-control Metric Matching Design

After much effort on determining the suitable sample size and appropriate estimators for the parameters, researchers realised that it would probably be wise to focus on the control chart itself instead of constantly dwelling on the Phase-I estimates. In this stage, researchers attempt to modify (or adjust) the control limits of a Phase-II control chart so that the chart would perform as desired, regardless of the sample size and estimators used. The idea of the design is simple. Suppose that our choice of control chart is designed to have an ARL_0 of 370.4 (this is equivalent to a false alarm rate of 0.27%), assuming that process parameters are

known. When process parameters are estimated, our control chart would almost always perform differently from $ARL_0 = 370.4$, say, our control chart produces $ARL_0 = 500$. Another practitioner, who has acquired a different set of parameter estimates, may have their control chart producing $ARL_0 = 235$. In this case, it makes sense to ask the following question: “Is the expected performance (i.e., ARL_0) across practitioners the same as 370.4?” Certainly, we cannot expect the performance of each practitioner to be exactly 370.4, since that could only be achieved if an infinite number of Phase-I samples is available. However, we can set the expected performance to be equal to 370.4, so that practitioners are aware that, in the long run, their ARL_0 performances would have an average value of about 370.4. This perspective is known as the *unconditional perspective*, where the word “unconditional” is used to indicate “average” over an infinite number of possible control charts/ parameter estimates.

Let us denote by Θ the vector of process parameters of interest, and $\hat{\Theta}$ the estimators of Θ . Without loss of generality, we consider the performance metric ARL_0 , which is expressed as a function of the estimated process parameters, i.e., as a function of $\hat{\Theta}$. Then, the unconditional value of the ARL_0 , or the average value of the ARL_0 ($AARL_0$) across all parameter estimates, is defined as

$$AARL_0 = \int_{\Omega} ARL_0 f_{\hat{\Theta}}(\Theta) d\Theta, \quad (2.45)$$

where Ω is the set that contains all possible realisations of $\hat{\Theta}$, and $f_{\hat{\Theta}}(\cdot)$ is the (joint) density function of $\hat{\Theta}$. The unconditional in-control metric matching design involves solving a simple equation $AARL_0 = \tau$, where τ is some pre-specified target level for the ARL_0 (in the previous example, we have $\tau = 370.4$). In particular, practitioners adjust the control limits of a control chart, either through analytical or numerical methods, so that the unconditional performance of the control chart matches the target level τ . Many researchers, such as Jones (2002), Chakraborti (2006), Castagliola *et al.* (2009), and Goedhart *et al.* (2016), have embarked on

such a design, noting that the design indeed produces an expected in-control performance that is close to the desired level, given that a sufficient number of Phase-I samples is gathered. However, the idea of “average” performances quickly induces concerns in some practitioners, since practitioners are often interested only in the performance of their own control chart (and not everyone else’s control chart). Besides, as the practitioner-to-practitioner variability is still quite large under a modest number of Phase-I samples, it is likely that a practitioner would not obtain an ARL_0 value “close” enough to the desired value of 370.4. This calls for an alternative perspective, known as the *conditional perspective*. The conditional perspective aims to guarantee that each practitioner would have a high chance (say, 90%) of obtaining an ARL_0 performance that is greater than the desired value of 370.4 (recall from Remark 2.4.1 that we desire a large ARL_0 value). While the conditional perspective has gained much popularity over the past decade, there have still been several works that adopt the unconditional in-control metric matching design, see for example, Teoh *et al.* (2019), Khoo *et al.* (2019), Mim *et al.* (2022), Parvin *et al.* (2023), and Chong *et al.* (2024). We will provide a demonstration in Chapter 3 to explain why this method is less preferred nowadays.

2.5.4 *Guaranteed In-control Performance Design*

Before we formally define the framework of the guaranteed in-control performance design, let us consider the following scenario. Suppose that four practitioners, named A, B, C, and D, are responsible for monitoring the resistivity of a certain type of silicon wafer. Prior to Phase-II on-line process monitoring, each of the practitioners collect 200 individual Phase-I measurements from the same in-source in order to estimate the process mean and standard deviation. Their Phase-I parameter estimates are given as follows:

Practitioner A: *I obtained $\bar{X} = 4.3057$ and $S = 0.0637$.*

Practitioner B: *Mine were $\bar{X} = 4.3121$ and $S = 0.0629$.*

Practitioner C: *My samples yield $\bar{X} = 4.3019$ and $S = 0.0639$.*

Practitioner D: *I got $\bar{X} = 4.3114$ and $S = 0.0615$.*

Using their own (unique) sets of parameter estimates, the four practitioners A, B, C, and D deploy the same Phase-II control chart designed to produce an ARL_0 of 370.4 in the case where process parameters are known. Figure 2.6 shows the distribution of the ARL_0 values calculated for all possible parameter estimates. Note that the red dotted lines represent the conditional ARL_0 values obtained by Practitioners A, B, C, and D, whereas the red solid line represent the target level $ARL_0 = 370.4$.

From Figure 2.6, it is clear that Practitioners B and D have obtained ARL_0 performances that are much more satisfactory than the recommended level. This inference follows from Remark 2.4.1, which states that a practitioner always desire a large ARL_0 value to avoid frequent false alarms in SQC applications. On the other hand, Practitioners A and C are found to suffer from ARL_0 values that are shorter than the recommended value of 370.4. Given that Practitioner C obtains an ARL_0 value of 100. This implies that Practitioner C will be getting, on average, one false alarm in every 100 samples, which is more than three times as frequent as the recommended rate of one false alarm in every 370.4 samples. *This is a bad sign!*

The guaranteed in-control performance (GICP) design is introduced to tackle the aforementioned issue. The design involves adjusting (or correcting) the control limits of the control chart so that practitioners have a high chance of getting an ARL_0 performance exceeding the recommended threshold. Equivalently, the design promises a low risk of getting false alarm rates that exceed a certain recommended rate. Figure 2.7 shows the distribution of the ARL_0 values after the GICP correction is applied. From the graph, it is found that majority of the ARL_0 values now fall to the right of the target level of 370.4, and all the four practitioners

A, B, C, and D have attained conditional ARL_0 values that are at least as large as 370.4. This marks a sign of an improved in-control performance.

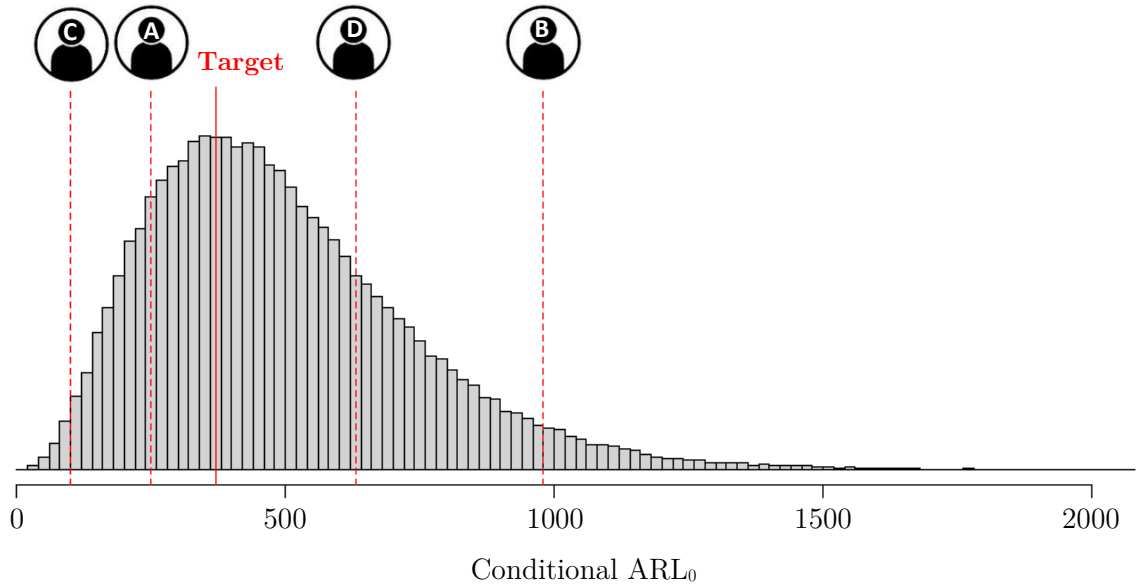


Figure 2.6 The distribution of ARL_0 values for various parameter estimates.

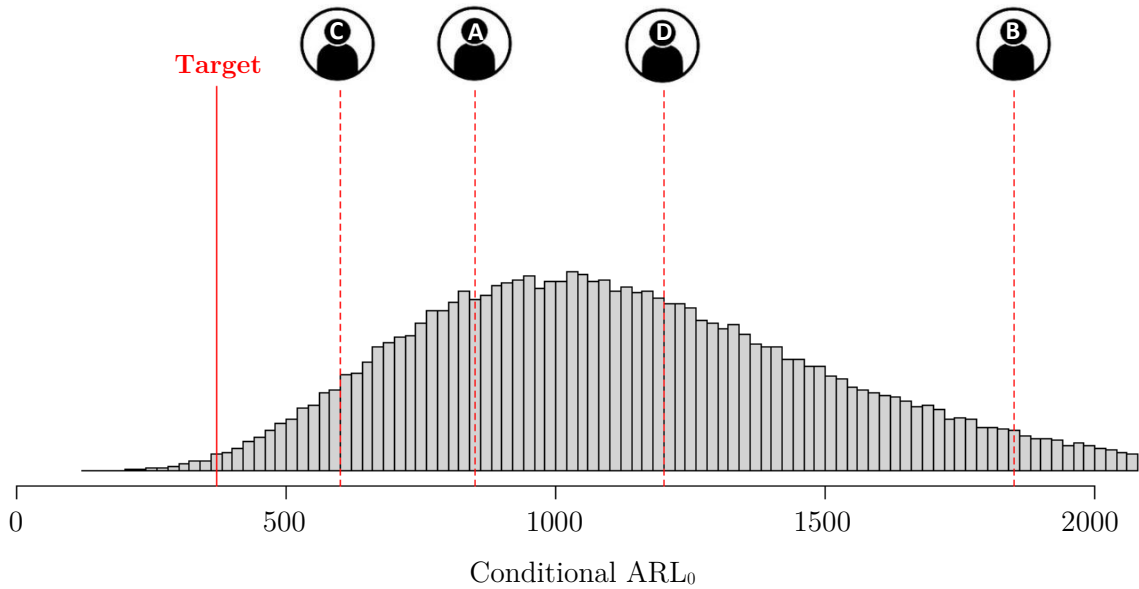


Figure 2.7 The distribution of ARL_0 values for various parameter estimates after the GICP design is applied.

Now, let us formulate the mathematical model that describes the GICP. Without loss of generality, suppose that the performance metric of interest is the ARL. The mathematical equation is given as

$$\Pr(\text{ARL}_0 \geq \tau) = 1 - p_e, \quad (2.46)$$

where ARL_0 is a function of the parameter estimate $\hat{\Theta}$, and τ is a pre-specified target level for the ARL_0 . Here, p_e is a user-specified risk probability, and its value is usually set to be a small positive number, e.g., 0.05 or 0.10. Note that Equation (2.46) is sometimes known as the exceedance probability criterion (EPC), since the expression on the left hand side is the probability of “exceeding” a certain threshold.

Albers and Kallenberg (2004) was among the first to develop the idea of correcting the control limits of the Shewhart $\bar{X}$ chart based on the EPC. They found out that, through correcting the control limits of the $\bar{X}$ chart, the problem with excessive false alarms can be effectively reduced, and the out-of-control sensitivity of the corrected $\bar{X}$ chart is slightly affected for large mean shift sizes. Gandy and Kvaløy (2013) then developed a general bootstrap theory for approximating the correction term for the control limits of the $\bar{X}$ chart. The approach involves creating a large number of bootstrap samples (at least 1,000) for the correction term, and subsequently calculating the mean of those samples to provide an estimate for the corrected control limits. The method was widely cited by researchers whose works span a multitude of control charts (e.g., the EWMA, CUSUM charts), see for example, Jones and Steiner (2012), Faraz *et al.* (2015), Saleh *et al.* (2015a), Saleh *et al.* (2016), Hany and Mahmoud (2016), and Khusna *et al.* (2020), and Aytacıoğlu *et al.* (2022).

Although the bootstrap method is proven effective in guaranteeing a satisfactory level of in-control performance, researchers have reported that the algorithm takes a large amount of time to complete and is very computer-intensive (Saleh *et al.*, 2015b, Diko *et al.*, 2019a). Furthermore, researchers have found that

the repeated application of the bootstrap approach can lead to varying control limits, hence it is often suggested to repeat the bootstrap method for at least 100 times, making the algorithm even more computationally expensive (Hu & Castagliola, 2017; Diko *et al.*, 2019a; Jardim *et al.*, 2020). To offer a more efficient alternative strategy, Diko *et al.* (2019a, 2019b) proposed adjusting the control limits of a control chart by constructing empirical distributions of the ARL_0 using an adequate number of simulation runs (they suggested 5000 runs). They claimed that the proposed method is more efficient than the bootstrap approach, and the results obtained from their method are comparable to some analytical results tabulated in the literature. However, it is found that the method relies heavily on the step size of the search, and an ill-chosen step size can often lead to very slow convergence. Recently, Capizzi and Masarotto (2020) developed a stochastic approximation algorithm for approximating the corrected control limits of a control chart based on the EPC. The algorithm is found to converge within 30,000 iterations, and it has a surprising resemblance with the Markov chain Monte Carlo algorithm, which is widely recognised for estimating the posterior distributions of intractable quantities. More details about the stochastic approximation algorithm will be given later in Chapter 3 of the thesis.

While most GICP designs involve computational methods to derive the corrected control limits, there have been quite a number of works devoted to the calculation of analytical control limits for the Shewhart control chart. For example, Goedhart *et al.* (2017b) derived the approximate correction term for the control limits of the Shewhart $\bar{X}$ chart under the GICP framework. They showed that the resulting conditional performance of the analytically corrected $\bar{X}$ chart has no real difference from that of the bootstrap-adjusted $\bar{X}$ chart, confirming the efficiencies of their approximating formulae. Jardim *et al.* (2019) subsequently derived the exact control limit of the $\bar{X}$ chart as a function of the conditional ARL_0 , which makes the statistical design of the $\bar{X}$ chart much more efficient than all the other

previous works. Similar developments for the Shewhart S and S^2 chart can be found in the works of Goedhart *et al.* (2017a), Faraz *et al.* (2018), and Sarmiento *et al.* (2022).

Despite the rising popularity of the GICP design, one of the biggest pitfalls of the GICP is that it causes the out-of-control performance of a control chart to deteriorate for small and moderate shift sizes. As researchers put it, the degradation in the out-of-control sensitivity is an inevitable price to pay in exchange for a large in-control ARL (Diko *et al.*, 2019a, 2019b; Jardim *et al.*, 2020; Sarmiento *et al.*, 2022), and this is consistent with the so-called “trade-off” between the in-control and out-of-control performances of a control chart. In Chapter 3, we will explain how we reduce this “trade-off” by marrying the GICP design with the concept of an optimisation design, and we call our proposed design the integrated optimal-GICP design.

2.6 Control Charts for Joint Monitoring of the Process Mean and Dispersion

An effective control scheme is one that possesses the ability to detect changes in both the location and scale parameters of a quality variable. There are two main types of control schemes dedicated for this purpose, i.e., multi-chart control schemes and single-chart control schemes. In the following subsections, we describe how and why each type of control scheme is derived, and provide some of their advantages and limitations within the SQC regimen. Note that throughout this section, we will be using the terms “dispersion”, “variance”, and “variability” interchangeably.

2.6.1 Multi-chart Control Schemes

Multi-chart schemes employ two or more individual control charts to simultaneously monitor the location and dispersion parameters of a process. While having many control charts is technically allowed in a multi-chart scheme, the most common type is still the two-chart scheme, which consists of one mean chart and one dispersion chart. According to McCracken and Chakraborti (2013), two-chart schemes offer a simple visual interpretation towards the status of a process. For instance, if a signal is produced by the mean chart, then there is most likely a shift in the process mean. On the other hand, if a signal is given by the dispersion chart, then the shift is most likely a shift in the process variance. Generally, two-chart schemes are preferred in many industrial quality control applications due to the familiarity of practitioners with the layout and interpretation of such schemes. The developments of many two-chart schemes were also justified on the grounds that the sample mean and the sample standard deviation (or the sample range) are statistically independent, given that data follow the normal distribution.

The simultaneous use of the Shewhart $\bar{X}$ and R charts for monitoring quality processes was first considered by Saniga (1977). To develop a joint scheme that minimises the expected cost, Saniga (1977) wrote an optimisation algorithm that searches the optimal control limits, sample size, and sampling interval for the joint $\bar{X}$ - R scheme. Many developments and extensions of the joint $\bar{X}$ - R scheme have since been contributed, see for example, the works of Rahim (1989), Saniga (1991), and Costa (1999). As stated in Section 2.5.2, the R statistic quickly loses its efficiency as the sample size increases. Therefore, many researchers have called for the development of the joint $\bar{X}$ - S scheme, or equivalently, the $\bar{X}$ - S^2 scheme, which monitors the sample standard deviation or the sample variance. Some of these works can be found in Raim *et al.* (1988), Wu and Wang (1997), and Wu *et al.* (2002). In certain applications, quality characteristics may have no basis for rational

subgrouping, hence it would only be possible to obtain individual measurements (i.e., $n = 1$) over fixed sampling times. In such cases, the Shewhart $\bar{X}$ and moving range (MR) charts are recommended, and the combined scheme is known as the $\bar{X}$ -MR scheme. Some discussions on the $\bar{X}$ -MR scheme can be found in Crowder (1987), Roes *et al.* (1993), Rigdon *et al.* (1994), and Amin and Ethridge (1998).

Referring to memory-type joint monitoring schemes, Gan (1995) developed a scheme that implements a two-sided EWMA mean chart and a two-sided EWMA variance chart simultaneously, known as the EWMA-EWMA (EE) scheme. They also considered a scheme that implements a two-sided CUSUM mean chart and a two-sided CUSUM variance chart, known as the CUSUM-CUSUM (CC) scheme. By comparing the performances of the EE and CC schemes, they deduced that the EE scheme is slightly more responsive than the CC scheme to small changes in the mean and a decrease in the variance. Generally, it was found that both the EE and CC schemes perform almost as well as each other in all circumstances. This finding was also reported by Reynolds and Stoumbos (2004a), who revealed a resemblance between the performances of the CUSUM multi-chart scheme and those of the EWMA multi-chart scheme. Other discussions on the performances of the EWMA and CUSUM multi-chart schemes can be found in the works of Stoumbos *et al.* (2003) and Reynolds and Stoumbos (2005).

While multi-chart schemes have been widely used in practice, there are some limitations with regard to their implementation and design. One of the major limitations of a multi-chart scheme is the high complexity associated with its design. Wu *et al.* (2010) attempted to develop an optimisation algorithm for the 3-CUSUM scheme, which consists of an upper one-sided CUSUM mean chart, a lower one-sided CUSUM mean chart, and an upper-sided CUSUM variance chart. They claimed that the optimal design for the three-CUSUM scheme is extremely arduous due to the large number of independent charting parameters required in the optimisation model, and one might as well resort to a single-chart scheme that is

slightly inferior in performance but is significantly easier to design. Besides, Cheng and Thaga (2006) inferred that the operation of multi-chart schemes often require a greater amount of effort (in terms of time and human resources) compared to single-chart schemes. As the design and operation of multi-chart schemes are relatively expensive, there are quite a limited number of developments on multi-chart schemes in the recent years, with some recent works including (but not limited to) Engmann and Han (2022) and Perry and Wang (2022).

Furthermore, it was found that the signals on the mean and/or variance chart can sometimes produce misleading conclusions about the status of a process. For example, Hawkins and Deng (2009) noticed that, although the $\bar{X}$ and S statistics are independent, a change in the variability of the process may cause the $\bar{X}$ chart to signal incorrectly. In particular, an increase in the process variance will likely lead to early signals from the $\bar{X}$ chart, which in turn leads to an incorrect conclusion of a mean shift. On the contrary, a decrease in the process variance will likely reduce the sensitivity of the $\bar{X}$ chart towards small mean shifts, which causes a delay in the response from the $\bar{X}$ chart. Note that in either case, the signals cannot be interpreted correctly, which renders the construction of a multi-chart scheme less meaningful. To that end, it seems that devising a scheme based on a single control chart could eliminate the hassle brought by a multi-chart scheme, while still achieving the goal of simultaneously monitoring the mean and dispersion of a process.

2.6.2 Single-chart Control Schemes

Single-chart schemes are control schemes that base their decisions on a single statistic (or a single control chart). Clearly, single-chart schemes are easier to operate compared to multi-chart schemes, since practitioners would only need to look at a single control chart and interpret the signals from it. Besides, single-chart

schemes are also easier to be designed than multi-chart schemes, as a typical single-chart scheme would involve a fewer number of charting parameters compared to schemes based on two separate charts. It is important to note that, when devising a single-chart control scheme, one needs to derive a statistic that captures the information from both the mean and variance. In the following texts, we will summarise some popular techniques used by researchers to achieve this purpose.

One of the most widely cited technique in the development of single-chart joint monitoring schemes is the maximum technique (Max). Essentially, the statistic of a Max-chart is developed by taking the maximum of two pivotal quantities, often transformed from the sample mean and sample variance (Chen & Cheng, 1998). Consider the following two pivotal quantities:

$$\frac{\bar{X}-\mu_0}{\frac{\sigma_0}{\sqrt{n}}} \quad \text{and} \quad \Phi^{-1}\left\{F_{\chi^2_{n-1}}\left[\frac{(n-1)S^2}{\sigma_0^2}\right]\right\},$$

where $\Phi^{-1}(\cdot)$ represents the inverse cumulative distribution function (cdf) of the standard normal distribution, and $F_{\chi^2_{n-1}}(\cdot)$ is the cdf of the chi-squared distribution with $n - 1$ degrees of freedom. It is immediate that both quantities above follow the standard normal distribution, given that the underlying data come from $\mathcal{N}(\mu_0, \sigma_0^2)$. Chen and Cheng (1998) proposed to monitor the maximum of the two quantities via an upper one-sided control chart, so that when the statistic goes above the upper control limit, we know we must have had an unusually large mean and/or variance or an unusually small mean and/or variance. By running some numerical results, Chen and Cheng (1998) showed that the Max chart is nearly as effective as the joint $\bar{X}$ - S scheme, and its implementation is a lot simpler than that of the Joint $\bar{X}$ - S scheme. The success of the Max chart has then motivated the inventions of other Max-type control schemes, see for example, the Max-EWMA chart (Chen *et al.*, 2001), the Max-CUSUM chart (Cheng & Thaga, 2010), and the multivariate versions of the Max-type control charts (Xie, 1999).

Apart from the Max function, the absolute value (ABS) function is also frequently used to incorporate the effects of mean and variance shifts in one charting statistic. One example is the single-chart EWMA scheme proposed by Domangue and Patch (1991), which describes an EWMA procedure used to monitor the statistic $|(X_i - \mu_0)/\sigma_0|$. Wu *et al.* (2010) subsequently developed an absolute value CUSUM (ABS-CUSUM) chart using the same recipe (i.e., monitoring the statistic $|(X_i - \mu_0)/\sigma_0|$ via a CUSUM procedure). They showed that the design of the ABS-CUSUM chart is a lot simpler than that of the 3-CUSUM scheme (see Section 2.6.1), and its performances are comparable to those of the 3-CUSUM scheme for a wide domain of shift sizes. Referring to developments of the SPRT chart, Ou *et al.* (2011a) developed the absolute value SPRT (ABS-SPRT) chart for joint monitoring of the mean and variance by simply monitoring the statistic $|(X_i - \mu_0)/\sigma_0|$ via a repeated sequence of SPRTs. They showed that the ABS-SPRT chart outperforms the ABS-CUSUM chart and the variable-sample-size version of the ABS-CUSUM chart in terms of the average detection speed.

Apart from the Max and ABS functions mentioned above, other methods include combining the mean and variance effects into one statistic via addition or multiplication. For instance, Wu and Tian (2006) proposed a chart for monitoring the weighted loss function, whose expression is given as

$$WL = \lambda_{WL}S^2 + (1 - \lambda_{WL})(\bar{X} - \mu_0)^2, \quad (2.47)$$

where $0 < \lambda_{WL} < 1$ is a weighting factor for the mean and variance losses. Wu and Tian (2005) subsequently proposed a new control chart for monitoring the quantity WL using a CUSUM procedure, which they call the weighted-loss CUSUM chart. Another method for combining the mean and variance effects is through the generalised likelihood ratio (GLR) statistic, which is expressed as a (slightly convoluted) sum of the pivotal quantities for the sample mean and sample variance (Hawkins & Deng, 2009), defined as

$$\text{GLR} = \left[\frac{\bar{X} - \mu_0}{\frac{\sigma_0}{\sqrt{n}}} \right]^2 + \frac{(n-1)S^2}{\sigma_0^2} + n \ln n - n - n \ln \left[\frac{(n-1)S^2}{\sigma_0^2} \right]. \quad (2.48)$$

Hawkins and Deng (2009) inferred that the GLR chart is a good alternative to the joint $\bar{X}$ - S chart in cases where the process variance may be decreasing. Some other developments and extensions can be found in the works of Zhang *et al.* (2010), Reynolds *et al.* (2013), and Song *et al.* (2018).

As the discerning reader might have noticed, one of the disadvantages of the single-chart control scheme is its inability to provide shift diagnosis. In other words, we would not be able to tell which quality of the process has changed upon observing an out-of-control signal, since both the mean and variance information has been merged into a single statistic. Nonetheless, as stated in Section 2.6.1, a multi-chart scheme may not always provide an accurate diagnosis of the nature of the shift occurred, and yet multi-chart schemes are significantly more complex in structure than single-chart schemes. Thus, we believe that single-chart schemes deserve more attention, and should be used more often in industrial practices.

2.7 Conclusions

In this chapter, the reader learns about the SPRT control chart and the foundational principles that make it successful. This knowledge will be carried forward to the subsequent chapters of this thesis, where we reveal some developments encompassing the SPRT chart. The performance measures introduced in this chapter (Section 2.4) will be used as the key performance criteria for assessing the performances of our proposed SPRT charts, as well as the objective functions of the statistical designs for those charts. In Chapter 3, we propose a design for the SPRT mean chart that couples the state-of-the-art guaranteed in-control performance (GICP) framework (see Section 2.5.4) and the concept of

optimisation design. In Chapter 4, we derive a new single-chart SPRT control scheme for simultaneous monitoring of the mean and variance of a normal process, echoing our statement at the end of Section 2.6.2.

3 The SPRT Chart with Estimated Process Parameters

3.1 Introduction

This chapter presents a thorough investigation of the SPRT chart with estimated process parameters, from both the theoretical and methodological perspectives. From the theoretical perspective, we provide the full derivation for the run length properties of the upper and lower one-sided SPRT charts with estimated process parameters. The derived properties include the conditional properties of the SPRT chart as functions of the Phase-I parameter estimates, as well as the unconditional properties “averaged” across all possible values of the parameter estimates (see Section 2.5.3).

From the methodological perspective, this chapter critically evaluates the three design methodologies currently adopted in research practice, i.e., the traditional design, unconditional metric matching design, and guaranteed in-control performance (GICP) design. The advantages and drawbacks of each method are revealed through their application to the SPRT control chart. It is important to note that the SPRT chart cited throughout this section refers to the SPRT chart for monitoring the mean detailed in Section 2.3.3, when the normal distribution is assumed.

3.2 The SPRT Chart with Known Process Parameters

This section provides derivations for the run length properties of the SPRT chart, i.e., the ASN, ARL, SDRL, ATS, SDTS, and ANOS (see Section 2.4 for their definitions), assuming that the process parameters μ_0 and σ_0 are known. Consider the upper one-sided SPRT chart with control statistic defined in Equation (2.27) where the value of the initial control statistic is set as zero. Let g and h be the

lower and upper control limits of the upper one-sided SPRT chart, respectively. In the existing literature, there are two methods for deriving the run length properties, i.e., through methods of integral equations or the Markov chain approach. To show that both methods lead to equivalent expressions, we first present the derivation through methods of integral equations. Then, we present the derivation through the more popular Markov chain approach. Note that we will be adopting the Markov chain approach in the derivations of the formulae in other chapters due to its learner-friendly interpretation.

Approach 1: The integral equation approach

Denote by $\text{ASN}(u)$ the ASN of a single SPRT when the initial control statistic takes the value u , i.e., $U_{i,0}^+ = u$. Then, $\text{ASN}(0)$ represents the ASN of a single SPRT given that the initial control statistic equals zero, i.e., $U_{i,0}^+ = 0$. Stoumbos and Reynolds (1997a) derived a Fredholm integral equation of the second kind for $\text{ASN}(0)$ as

$$\text{ASN}(0) = 1 + \int_g^h \text{ASN}(u) \cdot f_0(u) du. \quad (3.1)$$

The equation above can be explained via the law of total expectation: Since any SPRT must sample at least one observation, it follows that the minimum value of $\text{ASN}(0)$ is equal to one. The integral term on the right-hand side (RHS) of Equation (3.1) represents the expectation of the ASN, given that the SPRT takes the first observation but does not conclude, i.e., $U_{i,1}^+ \in (g, h)$. In this case, the expectation of the ASN can be expressed as the integral of the $\text{ASN}(u)$ multiplied by the one-step transition density $f_0(u)$ of the control statistic $U_{i,j}^+$ from the value 0 to an arbitrary value $u \in (g, h)$.

Denote by $OC(u)$ the operating characteristic (OC) of a single SPRT when the initial control statistic takes the value u , i.e., $U_{i,0}^+ = u$. Then, $OC(0)$ represents the OC of a single SPRT given that the initial control statistic equals zero, $U_{i,0}^+ = 0$. Using a similar approach, Stoumbos and Reynolds (1997a) provided an integral equation for the $OC(0)$ as

$$OC(0) = \Phi(g + \gamma - \delta) + \int_g^h OC(u) \cdot f_0(u) du, \quad (3.2)$$

where $\gamma > 0$ is the reference parameter of the SPRT chart, $\delta > 0$ is the mean shift size in units of the standard deviation, and $\Phi(\cdot)$ is the cdf of the standard normal distribution. The first term on the RHS of Equation (3.2) can be interpreted as the probability that the SPRT accepts the process as in-control immediately after the first observation is collected, which is expressed as $\Pr[(X_{i,1} - \mu_0)/\sigma_0 - \gamma < g]$. Using the fact that $X_{i,1} \sim \mathcal{N}(\mu_0 + \delta\sigma_0, \sigma_0^2)$, the probability can be simplified to $\Phi(g + \gamma - \delta)$. The second term on the RHS of Equation (3.2) represents the expectation of the OC, given that the SPRT does not terminate after the first observation. Similar to the case for the ASN, the expectation of the OC can be expressed as the integral of the $OC(u)$ multiplied by $f_0(u)$, integrated over the interval (g, h) .

Noting that Equations (3.1) and (3.2) do not have analytical solutions, Stoumbos and Reynolds (1997a) suggested applying Gaussian quadrature to approximate the integrals. In particular, Equations (3.1) and (3.2) can be rewritten as

$$ASN(0) = 1 + \sum_{a=1}^{\xi} \omega_a \cdot ASN(u_a) f_0(u_a) \quad (3.3)$$

and

$$OC(0) = \Phi(g + \gamma - \delta) + \sum_{a=1}^{\xi} \omega_a \cdot OC(u_a) f_0(u_a), \quad (3.4)$$

respectively, where ξ is a pre-defined number of quadrature nodes, $\{\omega_a\}_{a=1}^{\xi}$ is a set of quadrature weights, and $\{u_a\}_{a=1}^{\xi}$ is the set of quadrature nodes in the interval (g, h) . Stoumbos and Reynolds (1997a) found that Equations (3.3) and (3.4) can

be expressed as two systems of ξ equations, and subsequently solved via methods of matrices. However, it may not be too clear what the resulting matrices mean, and the approach can lead to some difficulty in the understanding of the underlying principles. The Markov chain approach, on the other hand, provides a more intuitive explanation to the calculation process, which makes the interpretation of the equations more insightful.

Approach 2: The Markov chain approach

Brook and Evans (1972) suggested to derive the run length properties of a control chart by considering a discrete-time Markov chain. The basic idea is as follows. First, the “acceptance” region (i.e., $(-\infty, g]$) and the “rejection” region (i.e., $[h, \infty)$) of the upper one-sided SPRT chart are treated as two absorbing states of a Markov chain. Next, the region (g, h) is partitioned into ξ subintervals, where each subinterval is treated as a transient state of the Markov chain (note that the same ξ that was previously used in methods of integral equations has been used here). Define the sequence of transient states $T_1, T_2, \dots, T_\xi$ as the intervals $(g, g + \Delta), (g + \Delta, g + 2\Delta), \dots, (g + (\xi - 1)\Delta, h)$, where $\Delta = (h - g)/\xi$ is the width of each state. The midpoint of state T_a is defined as

$$O_a = \frac{[g+(a-1)\Delta] + (g+a\Delta)}{2} = g + (a - 0.5)\Delta, \quad (3.5)$$

for $a = 1, 2, \dots, \xi$. We further define the two absorbing states as $T_{\xi+1} = (-\infty, g]$ and $T_{\xi+2} = [h, \infty)$. In canonical form, the full transition probability matrix of the absorbing Markov chain can be written as

$$\mathbf{P} = \begin{pmatrix} \mathbf{R} & \mathbf{Q} \\ \mathbf{0} & \mathbf{I}_2 \end{pmatrix} = \begin{pmatrix} r_{1,1} & r_{1,2} & \cdots & r_{1,\xi} & q_{1,(\xi+1)} & q_{1,(\xi+2)} \\ r_{2,1} & r_{2,2} & \cdots & r_{2,\xi} & q_{2,(\xi+1)} & q_{2,(\xi+2)} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ r_{\xi,1} & r_{\xi,2} & \cdots & r_{\xi,\xi} & q_{\xi,(\xi+1)} & q_{\xi,(\xi+2)} \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 \end{pmatrix}. \quad (3.6)$$

Here, $\mathbf{R}$ is the $\xi \times \xi$ matrix of transition probabilities between the transient states, $\mathbf{0}$ is a $2 \times \xi$ matrix with all zero entries, and $\mathbf{I}_2$ is the 2×2 identity matrix. The matrix $\mathbf{Q}$ is made up of two $\xi \times 1$ vectors, i.e., $\mathbf{q}_1$ and $\mathbf{q}_2$, each containing one-step transition probabilities from the transient states to the absorbing states $T_{\xi+1}$ and $T_{\xi+2}$, respectively.

Let $r_{a,b}$ denote the one-step transition probability that the control statistic transits from transient state T_a to another transient state T_b , for $a, b = 1, 2, \dots, \xi$. Let us also make the assumption that the control statistic lands on the midpoint of each transient state (this is approximately true for a very large ξ). The probability can be expressed as

$$\begin{aligned} r_{a,b} &= \Pr(U_{i,j}^+ \in T_b | U_{i,j-1}^+ \in T_a) \\ &\approx \Pr\left[\Delta \cdot (b-1) + g < O_a + \frac{X_{i,j} - \mu_0}{\sigma_0} - \gamma < \Delta \cdot b + g\right]. \end{aligned} \quad (3.7)$$

We now substitute Equation (3.5) into Equation (3.7) and simplify. With some slight abuse of notation, $r_{a,b}$ can be expressed as

$$\begin{aligned} r_{a,b} &= \Pr\left[\Delta \cdot (b-a-0.5) + \gamma < \frac{X_{i,j} - \mu_0}{\sigma_0} < \Delta \cdot (b-a+0.5) + \gamma\right] \\ &= \Pr\left[\Delta \cdot (b-a-0.5) + \gamma - \delta < \frac{X_{i,j} - (\mu_0 + \delta\sigma_0)}{\sigma_0} < \Delta \cdot (b-a+0.5) + \gamma - \delta\right] \\ &= \Phi[\Delta \cdot (b-a+0.5) + \gamma - \delta] - \Phi[\Delta \cdot (b-a-0.5) + \gamma - \delta], \end{aligned} \quad (3.8)$$

since $X_{i,j} \sim \mathcal{N}(\mu_0 + \delta\sigma_0, \sigma_0^2)$.

The entries of matrices $\mathbf{q}_1$ and $\mathbf{q}_2$, i.e., $q_{a,(\xi+1)}$ and $q_{a,(\xi+2)}$, can be derived in a way similar to the derivation for $r_{a,b}$ above. The formulae for $q_{a,(\xi+1)}$ is

$$\begin{aligned} q_{a,(\xi+1)} &= \Pr(U_{i,j}^+ \in T_{\xi+1} | U_{i,j-1}^+ \in T_a) \\ &\approx \Pr\left(O_a + \frac{X_{i,j} - \mu_0}{\sigma_0} - \gamma \leq g\right) \\ &= \Pr\left[g + \Delta \cdot (a-0.5) + \frac{X_{i,j} - \mu_0}{\sigma_0} - \gamma \leq g\right] \end{aligned}$$

$$\begin{aligned}
&= \Pr \left[\frac{X_{i,j} - (\mu_0 + \delta \sigma_0)}{\sigma_0} \leq \gamma + \Delta \cdot (0.5 - a) - \delta \right] \\
&= \Phi[\gamma + \Delta \cdot (0.5 - a) - \delta],
\end{aligned} \tag{3.9}$$

whereas the formulae for $q_{a,(\xi+2)}$ is

$$\begin{aligned}
q_{a,(\xi+2)} &= \Pr(U_{i,j}^+ \in T_{\xi+2} | U_{i,j-1}^+ \in T_a) \\
&\approx \Pr \left(O_a + \frac{X_{i,j} - \mu_0}{\sigma_0} - \gamma \geq h \right) \\
&= \Pr \left[g + \Delta \cdot (a - 0.5) + \frac{X_{i,j} - \mu_0}{\sigma_0} - \gamma \geq h \right] \\
&= \Pr \left[\frac{X_{i,j} - (\mu_0 + \delta \sigma_0)}{\sigma_0} \geq (h - g) + \gamma + \Delta \cdot (0.5 - a) - \delta \right] \\
&= 1 - \Phi[\gamma + \Delta \cdot (\xi + 0.5 - a) - \delta],
\end{aligned} \tag{3.10}$$

for $a = 1, 2, \dots, \xi$, since $h - g = \Delta \cdot \xi$.

The derivations of the ASN and OC of the SPRT chart involve computing the fundamental matrix, i.e., $(\mathbf{I}_\xi - \mathbf{R})^{-1}$, where $\mathbf{I}_\xi$ is a $\xi \times \xi$ identity matrix. Using the law of total expectation, $\text{ASN}(0)$ can be derived as (Ou *et al.*, 2010, 2012b)

$$\text{ASN}(0) = 1 + \mathbf{C}^\top (\mathbf{I}_\xi - \mathbf{R})^{-1} \mathbf{1}, \tag{3.11}$$

where $\mathbf{1}$ is a $\xi \times 1$ vector with entries filled with ones, and $\mathbf{C}$ is a $\xi \times 1$ vector containing one-step probabilities that the control statistic transits from the initial value $U_{i,0}^+ = 0$ to an arbitrary state T_a . Denote these one-step probabilities as c_a , for $a = 1, 2, \dots, \xi$, which has the expression given by

$$\begin{aligned}
c_a &= \Pr(U_{i,1}^+ \in T_a | U_{i,0}^+ = 0) \\
&= \Pr \left[g + \Delta \cdot (a - 1) < \frac{X_{i,1} - \mu_0}{\sigma_0} - \gamma < g + \Delta \cdot a \right] \\
&= \Phi(g + \Delta \cdot a + \gamma - \delta) - \Phi[g + \Delta \cdot (a - 1) + \gamma - \delta].
\end{aligned} \tag{3.12}$$

It is worth noting that the interpretation of Equation (3.11) is identical to that of Equation (3.3). Comparing both equations, one finds that the vector $\mathbf{C}$ is equivalent

to the set $\{\omega_a \cdot f_0(u_a)\}_{a=1}^\xi$, whereas $(\mathbf{I}_\xi - \mathbf{R})^{-1}\mathbf{1}$ is equivalent to the set $\{\text{ASN}(u_a)\}_{a=1}^\xi$ from the integral equations approach. As ω_a represents the set of quadrature weights, one may find it difficult to interpret the meaning of $\{\omega_a \cdot f_0(u_a)\}_{a=1}^\xi$ from the integral equations approach. However, with the Markov chain approach, it is easy to see that the set of values simply represent the transition probabilities from the initial state (where $U_{i,0}^+ = 0$) to the transient state T_a .

OC(0) can be derived in a similar fashion, which is given as (Ou *et al.*, 2010, 2012b)

$$\text{OC}(0) = P_0 + \mathbf{C}^\top (\mathbf{I}_\xi - \mathbf{R})^{-1} \mathbf{q}_1, \quad (3.13)$$

where $P_0 = \Phi(g + \gamma - \delta)$ is the probability that the control statistic transits to the acceptance region in the first step, as shown in Equation (3.2), whereas the entries of $\mathbf{C}$, $\mathbf{R}$, and $\mathbf{q}_1$ are given in Equations (3.12), (3.8), and (3.9), respectively. Note that the interpretation of Equation (3.13) is the same as that of Equation (3.4), with the vector $\mathbf{C}$ being equivalent to the set $\{\omega_a \cdot f_0(u_a)\}_{a=1}^\xi$ and $(\mathbf{I}_\xi - \mathbf{R})^{-1}\mathbf{q}_1$ being equivalent to the set $\{\text{OC}(u_a)\}_{a=1}^\xi$ from the integral equations approach.

In the remainder of this chapter, we will denote OC as OC(0) and ASN as ASN(0), which pertains to our assumption that the initial control statistic $U_{i,0} = 0$ for all integers $i \geq 1$. Now, let us derive the formulae of the ARL, SDRL, ATS, and SDTS of the SPRT chart. We first note that the run length, i.e., the number of SPRTs required to obtain a signal, is a geometric random variable with probability of success being the probability of rejecting H_0 , i.e., $\text{RL} \sim \text{Geometric}(1 - \text{OC})$. Hence, the ARL and SDRL of the SPRT chart are given as

$$\text{ARL} = \frac{1}{1 - \text{OC}}, \quad (3.14)$$

and

$$\text{SDRL} = \sqrt{\frac{\text{OC}}{[1 - \text{OC}]^2}}, \quad (3.15)$$

respectively. Note that Equations (3.14) and (3.15) apply in both cases under H_0 ($\delta = 0$) and under H_1 ($\delta \neq 0$).

Deriving the ATS and SDTS of the SPRT chart can be a little tricky. Let us denote OC_0 and OC_1 as the OC values evaluated under an in-control process and an out-of-control process, respectively. When the process is in-control, the in-control time to signal (TS_0) can be expressed as $TS_0 = d \times RL_0$ (see Equation (2.33)), where RL_0 is the in-control run length and $d > 0$ is a fixed sampling interval of the SPRT chart. It is easy to see that the ATS_0 and $SDTS_0$ of the SPRT chart can be expressed as

$$ATS_0 = d \times E[RL_0] = \frac{d}{1-OC_0}, \quad (3.16)$$

and

$$SDTS_0 = d \times \sqrt{\text{Var}[RL_0]} = d \sqrt{\frac{OC_0}{[1-OC_0]^2}}, \quad (3.17)$$

respectively.

When the process is out-of-control, recall from Section 2.4.1 that there are two ways to evaluate the run length properties of a control chart, i.e., via the zero-state mode or the steady-state mode (see Definitions 2.4.3 and 2.4.4). Under the zero state, we assume that the out-of-control event occurs at the outset, which means that the expressions for the ATS_1 and $SDTS_1$ will coincide with those in Equations (3.16) and (3.17), respectively. However, under the steady state, we assume that the SPRT chart has run for a long period of time, and the out-of-control event will likely occur at any point within a sampling interval. Figure 3.1 illustrates the situation where the SPRT chart (or any control chart) operates for a duration of κ sampling intervals, i.e., κd time units. The out-of-control event is found to occur at some time $(\kappa - 3)d < t < (\kappa - 2)d$ (presumably due to one of the machines failing). In this case, the run length is three, but the time to signal is not $3d$, since the duration from the out-of-control event to the next inspection time is

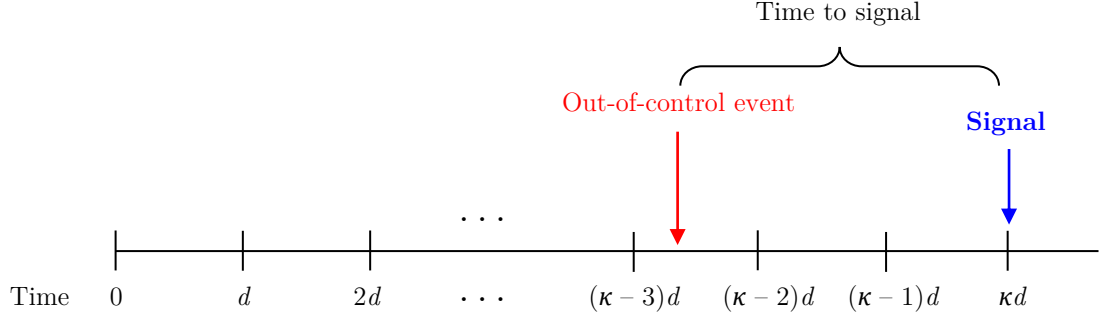


Figure 3.1 The visualisation of a single run of a control chart, with indications of the out-of-control event and signalling time.

not a full sampling interval. The time to signal should be expressed as $(2 + \mathcal{U})d$, where $\mathcal{U}$ is some random number that lies between 0 and 1 (inclusive).

It is clear that the steady-state out-of-control time to signal (TS_1) can be expressed as $TS_1 = d \times (RL_1 - 1 + \mathcal{U})$, where RL_1 is the out-of-control run length and $\mathcal{U}$ is a random number in $[0, 1]$. Several researchers, including Reynolds *et al.* (1990) and Stoumbos and Reynolds (1997a), stated that the variable $\mathcal{U}$ can be modelled using the standard uniform distribution, i.e., $\mathcal{U} \sim \text{Uniform}(0, 1)$, since there is often very little information about the distribution of times when a shift occurs. Also, it is reasonable to assume that RL and $\mathcal{U}$ are independent, since the time at which the shift occurs does not interfere with the time at which the control chart signals (Reynolds *et al.*, 1990; Stoumbos and Reynolds, 1997a; Ou *et al.*, 2012b). The steady-state ATS_1 and $SDTS_1$ of the SPRT chart are then derived as

$$ATS_1 = d \times E[RL_1 - 1 + \mathcal{U}] = d \left(\frac{1}{1-OC_1} - \frac{1}{2} \right) \quad (3.18)$$

and

$$\begin{aligned} SDTS_1 &= d \sqrt{\text{Var}[RL_1 - 1 + \mathcal{U}]} \\ &= d \sqrt{\text{Var}[RL_1] + \text{Var}[\mathcal{U}]} \\ &= d \sqrt{\frac{OC_1}{[1-OC_1]^2} + \frac{1}{12}}, \end{aligned} \quad (3.19)$$

respectively.

The ANOS of the SPRT chart can be calculated using the following formula (Stoumbos & Reynolds, 1997a; Godase & Mahadik, 2019)

$$\text{ANOS} = \text{ARL} \times \text{ASN}. \quad (3.20)$$

Note that Equation (3.20) is true under both H_0 and H_1 .

We also present the formulae for some measures of the overall performance of the SPRT chart. Here, we consider the AEQL and the EANOS metrics. Recall from Section 2.4.4 that, to evaluate the AEQL, one may select either the RL or the TS as the metric of evaluation. In this thesis, we select the TS since practitioners are often interested in the *length of time taken* to detect a fault or a shift in SQC applications. The formula for the AEQL is given as (Ou *et al.*, 2012b)

$$\text{AEQL} = \int_{\delta_{\min}}^{\delta_{\max}} \delta^2 \text{ATS}_1 f_{\delta}(\delta) d\delta = \frac{1}{\delta_{\max} - \delta_{\min}} \int_{\delta_{\min}}^{\delta_{\max}} \delta^2 \text{ATS}_1 d\delta, \quad (3.21)$$

where ATS_1 is a function of δ , $f_{\delta}(\delta)$ is the pdf of δ , and $(\delta_{\min}, \delta_{\max})$ are the minimum and maximum allowable mean shift sizes, respectively. Since there is insufficient information about the likelihood of δ , it is customary to set $f_{\delta}(\delta)$ as the pdf of a uniform distribution over a sensible range of δ , i.e., $\delta \in [\delta_{\min}, \delta_{\max}]$ (Wu *et al.*, 2010; Calzada & Scariano, 2013; Celano & Castagliola, 2016). The quadratic weights δ^2 used in Equation (3.21) indicates that larger weights are assigned to large mean shift sizes, whereas smaller weights are assigned to small mean shift sizes. As the AEQL amplifies the effect for larger mean shifts, it is suitable to be used as an optimisation criterion in the optimal design for the SPRT chart. This aligns with the fact that practitioners tend to have a greater interest in maximising the detection speed of control charts in relatively extreme cases.

The EANOS can be calculated via the following integral

$$\text{EANOS} = \int_{\delta_{\min}}^{\delta_{\max}} \text{ANOS}_1 f_{\delta}(\delta) d\delta = \frac{1}{\delta_{\max} - \delta_{\min}} \int_{\delta_{\min}}^{\delta_{\max}} \text{ANOS}_1 d\delta, \quad (3.22)$$

where ANOS_1 is a function of δ . Generally, a small EANOS implies that the control chart has a small ANOS_1 value along the shift range $[\delta_{\min}, \delta_{\max}]$. It should be noted that the ANOS evaluates the combined effect of the detection speed and the sampling efficiency of a control chart, rather than measuring the detection speed alone. Hence, the concept of quadratic loss proposed by Taguchi (1980) is not compatible with the ANOS criterion. For this reason, the EANOS metric, which assigns equal weights to ANOS_1 values along the range $[\delta_{\min}, \delta_{\max}]$, is used throughout the study.

The computer program for calculating the theoretical run length properties of the SPRT chart with known process parameters is available on the designated GitHub repository (labelled with A.1.1). To verify the accuracies of the formulae, we also provide a computer program based on Monte Carlo simulation (labelled with A.1.2). The reader should refer to the appendices for a complete guide on how to run the functions in the program.

It is important to note that the run length properties of the lower one-sided SPRT chart are identical to those of the upper one-sided SPRT chart derived above. For the purpose of demonstration, let us derive $r_{a,b}$ for the lower-sided case and show that its expression is indeed equivalent to that for the upper-sided case. Suppose now that we are interested in detecting a downward shift in the process mean, i.e., $\delta < 0$, and the control statistic of the SPRT chart is defined in Equation (2.28) with control limits $-g$ and $-h$. The region $(-h, -g)$ is partitioned into a sequence of transient states $T_1, T_2, \dots, T_{\xi}$ with intervals $(-g - \Delta, -g), (-g - 2\Delta, -g - \Delta), \dots, (-h, -g - (\xi - 1)\Delta)$, where the width of each interval remains as $\Delta = (h - g)/\xi$. The midpoint of each transient state is now

$$O_a = \frac{[-g - a\Delta] + [-g - (a-1)\Delta]}{2} = -g - (a - 0.5)\Delta, \quad (3.23)$$

for $a = 1, 2, \dots, \xi$. Denote by $T_{\xi+1}$ and $T_{\xi+2}$ the absorbing states $[-g, \infty)$ and $(-\infty, -h]$, respectively. The one-step transition probability $r_{a,b}$ is now expressed as

$$\begin{aligned}
r_{a,b} &= \Pr(U_{i,j}^- \in T_b | U_{i,j-1}^- \in T_a) \\
&\approx \Pr\left[-\Delta \cdot b - g < O_a + \frac{X_{i,j} - \mu_0}{\sigma_0} + \gamma < -\Delta \cdot (b-1) - g\right] \\
&= \Pr\left[-\Delta \cdot b - g < -g - \Delta \cdot (a-0.5) + \frac{X_{i,j} - \mu_0}{\sigma_0} + \gamma < -\Delta \cdot (b-1) - g\right] \\
&= \Pr\left[\Delta \cdot (a-b-0.5) - \gamma < \frac{X_{i,j} - \mu_0}{\sigma_0} < \Delta \cdot (a-b+0.5) - \gamma\right] \\
&= \Pr\left[\Delta \cdot (a-b-0.5) - \gamma - \delta < \frac{X_{i,j} - (\mu_0 + \delta\sigma_0)}{\sigma_0} < \Delta \cdot (a-b+0.5) - \gamma - \delta\right] \\
&= \Phi[\Delta \cdot (a-b+0.5) - \gamma - \delta] - \Phi[\Delta \cdot (a-b-0.5) - \gamma - \delta], \tag{3.24}
\end{aligned}$$

for $a, b = 1, 2, \dots, \xi$, where $\delta < 0$.

It is easy to see that, for fixed values of a, b, γ , and $|\delta|$, Equations (3.8) and (3.24) give the same values by the symmetric properties of the normal distribution. Hence, it suffices to compute the run length properties of the upper-sided SPRT chart and duplicate their values for the lower-sided counterpart.

3.3 The Proposed SPRT Chart with Estimated Process Parameters

In practice, the true values of μ_0 and σ_0 are rarely known. Hence, it is customary to estimate these process parameters from a set of reference data, known as Phase-I data. Suppose that a set of Phase-I individual observations of size m , i.e., $(Y_1, Y_2, \dots, Y_m)$, is independently sampled from an in-control process with $\mathcal{N}(\mu_0, \sigma_0^2)$. The common estimators for μ_0 and σ_0 are given as

$$\hat{\mu}_0 = \frac{1}{m} \sum_{\varsigma=1}^m Y_{\varsigma} \tag{3.25}$$

and

$$\hat{\sigma}_0 = \sqrt{\frac{1}{m-1} \sum_{\varsigma=1}^m (Y_{\varsigma} - \hat{\mu}_0)^2}, \tag{3.26}$$

respectively. Using the estimates given in Equations (3.25) and (3.26) in place of the true μ_0 and σ_0 , the upper- and lower-sided control statistics of the SPRT chart are updated as

$$\hat{U}_{i,j}^+ = \hat{U}_{i,j-1}^+ + \left(\frac{X_{i,j} - \hat{\mu}_0}{\hat{\sigma}_0} - \gamma \right) \quad (3.27)$$

and

$$\hat{U}_{i,j}^- = \hat{U}_{i,j-1}^- + \left(\frac{X_{i,j} - \hat{\mu}_0}{\hat{\sigma}_0} + \gamma \right). \quad (3.28)$$

respectively.

It is important to note that the estimators in Equations (3.25) and (3.26) are random variables of the Phase-I data. To facilitate the derivation of the run length properties under the estimated-parameter case, we will define the following pivotal quantities

$$W = \frac{\hat{\mu}_0 - \mu_0}{\frac{\sigma_0}{\sqrt{m}}} \quad (3.29)$$

and

$$V = \frac{\hat{\sigma}_0^2}{\sigma_0^2}, \quad (3.30)$$

whose distributions do not depend on the values of μ_0 and σ_0 . The choice of pivotal quantities can be justified based on two reasons. First, both W and V are either the original or transformed versions of well-established pivotal quantities for the normal distribution. Second, W and V are statistically independent, which is crucial for the evaluation of the unconditional properties of the SPRT chart with estimated process parameters later on.

It is known that the sample mean $\hat{\mu}_0$ follows the distribution $\mathcal{N}(\mu_0, \sigma_0^2/m)$. By standardising the variable, we have $(\hat{\mu}_0 - \mu_0)/(\sigma_0/\sqrt{m}) \sim \mathcal{N}(0, 1)$. Hence, the pivotal quantity W follows the standard normal distribution. It is also well-known that the quantity $(m-1)V^2 = (m-1)\hat{\sigma}_0^2/\sigma_0^2$ follows the chi-squared distribution

with $m - 1$ degrees of freedom. Let us define the variable $V' = (m - 1)V^2$. By transformation of variables, it can be shown that

$$\begin{aligned}
f_V(v) &= f_{\chi_{m-1}^2}(V') \cdot \left| \frac{dV'}{dV} \right| \\
&= f_{\chi_{m-1}^2}[(m-1)v^2] \cdot |2(m-1)v| \\
&= 2(m-1)v f_{\chi_{m-1}^2}[(m-1)v^2]
\end{aligned} \tag{3.31}$$

where $f_V(v)$ is the pdf of the pivotal quantity V , and $f_{\chi_{m-1}^2}(\cdot)$ is the pdf of the chi-squared distribution with $m - 1$ degrees of freedom. Note that when the process parameters are known, we have $W = 0$ and $V = 1$ unconditionally.

To evaluate the run length properties of the SPRT chart with estimated process parameters, it is necessary to include the effects of Phase-I parameter estimation (in the form of W and V) in the expressions defined in Section 3.2. Without loss of generality, consider the upper-sided case (i.e., $\delta > 0$) in the derivations. The new transition probability matrix $\hat{\mathbf{R}}$ has the following updated entries:

$$\begin{aligned}
\hat{r}_{a,b} &= \Pr(\hat{U}_{i,j}^+ \in T_b | \hat{U}_{i,j-1}^+ \in T_a) \\
&\approx \Pr\left[\Delta \cdot (b-1) + g < O_a + \frac{X_{i,j} - \hat{\mu}_0}{\hat{\sigma}_0} - \gamma < \Delta \cdot b + g\right] \\
&= \Pr\{\hat{\sigma}_0[\Delta \cdot (b-a-0.5) + \gamma] < X_{i,j} - \hat{\mu}_0 < \hat{\sigma}_0[\Delta \cdot (b-a+0.5) + \gamma]\} \\
&= \Pr\left\{\frac{\hat{\sigma}_0}{\sigma_0}[\Delta \cdot (b-a-0.5) + \gamma] < \frac{X_{i,j} - \hat{\mu}_0}{\sigma_0} < \frac{\hat{\sigma}_0}{\sigma_0}[\Delta \cdot (b-a+0.5) + \gamma]\right\} \\
&= \Pr\left\{V[\Delta \cdot (b-a-0.5) + \gamma] < \frac{X_{i,j} - (\mu_0 + \delta\sigma_0) - (\hat{\mu}_0 - \mu_0) + \delta\sigma_0}{\sigma_0} < V[\Delta \cdot (b-a+0.5) + \gamma]\right\} \\
&= \Pr\left\{V[\Delta \cdot (b-a-0.5) + \gamma] < \frac{X_{i,j} - (\mu_0 + \delta\sigma_0)}{\sigma_0} - \frac{\hat{\mu}_0 - \mu_0}{\sigma_0} + \delta < V[\Delta \cdot (b-a+0.5) + \gamma]\right\} \\
&= \Pr\left\{V[\Delta \cdot (b-a-0.5) + \gamma] < \frac{X_{i,j} - (\mu_0 + \delta\sigma_0)}{\sigma_0} - \frac{W}{\sqrt{m}} + \delta < V[\Delta \cdot (b-a+0.5) + \gamma]\right\} \\
&= \Pr\left\{V[\Delta \cdot (b-a-0.5) + \gamma] + \frac{W}{\sqrt{m}} - \delta < \frac{X_{i,j} - (\mu_0 + \delta\sigma_0)}{\sigma_0} < V[\Delta \cdot (b-a+0.5) + \gamma] + \frac{W}{\sqrt{m}} - \delta\right\} \\
&= \Phi\left\{V[\Delta \cdot (b-a+0.5) + \gamma] + \frac{W}{\sqrt{m}} - \delta\right\} - \Phi\left\{V[\Delta \cdot (b-a-0.5) + \gamma] + \frac{W}{\sqrt{m}} - \delta\right\}, \tag{3.32}
\end{aligned}$$

where W and V are given in Equations (3.29) and (3.30), respectively. On the other hand, by using the same approach as in the derivation of $\hat{r}_{a,b}$, the entries of the initial probability vector $\hat{\mathbf{C}}$ are updated as follows

$$\begin{aligned}
\hat{c}_a &= \Pr(\hat{U}_{i,1}^+ \in T_a | \hat{U}_{i,0}^+ = 0) \\
&= \Pr\left[g + \Delta \cdot (a - 1) < \frac{X_{i,1} - \hat{\mu}_0}{\hat{\sigma}_0} - \gamma < g + \Delta \cdot a\right] \\
&= \Pr\left\{V[g + \Delta \cdot (a - 1) + \gamma] < \frac{X_{i,1} - \hat{\mu}_0}{\sigma_0} < V[g + \Delta \cdot a + \gamma]\right\} \\
&= \Pr\left\{V[g + \Delta \cdot (a - 1) + \gamma] < \frac{X_{i,1} - (\mu_0 + \delta\sigma_0)}{\sigma_0} - \frac{W}{\sqrt{m}} + \delta < V[g + \Delta \cdot a + \gamma]\right\} \\
&= \Pr\left\{V[g + \Delta \cdot (a - 1) + \gamma] + \frac{W}{\sqrt{m}} - \delta < \frac{X_{i,1} - (\mu_0 + \delta\sigma_0)}{\sigma_0} < V[g + \Delta \cdot a + \gamma] + \frac{W}{\sqrt{m}} - \delta\right\} \\
&= \Phi\left\{V[g + \Delta \cdot a + \gamma] + \frac{W}{\sqrt{m}} - \delta\right\} - \Phi\left\{V[g + \Delta \cdot (a - 1) + \gamma] + \frac{W}{\sqrt{m}} - \delta\right\}. \quad (3.33)
\end{aligned}$$

The conditional ASN (CASN), which is expressed as a function of W and V , has the following expression

$$\text{CASN} = \mathbf{1} + \hat{\mathbf{C}}^\top (\mathbf{I}_\xi - \hat{\mathbf{R}})^{-1} \mathbf{1}, \quad (3.34)$$

where the matrices $\hat{\mathbf{R}}$ and $\hat{\mathbf{C}}$ can be computed using the formulae given in Equations (3.32) and (3.33), respectively.

The OC function of the SPRT chart also needs to be modified by taking into account the effect of parameter estimation. Referring to Equation (3.13), the term P_0 can be modified as

$$\begin{aligned}
\hat{P}_0 &= \Pr(\hat{U}_{i,1}^+ \in T_{\xi+1} | \hat{U}_{i,0}^+ = 0) \\
&= \Pr\left(\frac{X_{i,1} - \hat{\mu}_0}{\hat{\sigma}_0} - \gamma < g\right) \\
&= \Pr\left(\frac{X_{i,1} - \hat{\mu}_0}{\hat{\sigma}_0} < g + \gamma\right) \\
&= \Pr\left[\frac{X_{i,1} - \hat{\mu}_0}{\sigma_0} < V(g + \gamma)\right] \\
&= \Pr\left[\frac{X_{i,1} - (\mu_0 + \delta\sigma_0)}{\sigma_0} - \frac{W}{\sqrt{m}} + \delta < V(g + \gamma)\right]
\end{aligned}$$

$$= \Phi \left[V(g + \gamma) + \frac{w}{\sqrt{m}} - \delta \right], \quad (3.35)$$

whereas the entries $\hat{q}_{a,(\xi+1)}$ of the new probability vector $\hat{\mathbf{q}}_1$ are modified as

$$\begin{aligned} \hat{q}_{a,(\xi+1)} &= \Pr(\hat{U}_{i,j}^+ \in T_{\xi+1} | \hat{U}_{i,j}^+ \in T_a) \\ &\approx \Pr\left(O_a + \frac{X_{i,j} - \hat{\mu}_0}{\hat{\sigma}_0} - \gamma \leq g\right) \\ &= \Pr\left(g + \Delta \cdot (a - 0.5) + \frac{X_{i,j} - \hat{\mu}_0}{\hat{\sigma}_0} - \gamma \leq g\right) \\ &= \Pr\left[\frac{X_{i,j} - \hat{\mu}_0}{\hat{\sigma}_0} \leq \gamma + \Delta \cdot (0.5 - a)\right] \\ &= \Pr\left\{\frac{X_{i,j} - \hat{\mu}_0}{\sigma_0} \leq V[\gamma + \Delta \cdot (0.5 - a)]\right\} \\ &= \Pr\left\{\frac{X_{i,1} - (\mu_0 + \delta\sigma_0)}{\sigma_0} - \frac{w}{\sqrt{m}} + \delta \leq V[\gamma + \Delta \cdot (0.5 - a)]\right\} \\ &= \Phi\left\{V[\gamma + \Delta \cdot (0.5 - a)] + \frac{w}{\sqrt{m}} - \delta\right\}. \end{aligned} \quad (3.36)$$

Using Equations (3.35) and (3.36), the updated OC ($\widehat{\text{OC}}$) of the SPRT chart with estimated process parameters can be evaluated as follows:

$$\widehat{\text{OC}} = \hat{P}_0 + \hat{\mathbf{C}}^\top (\mathbf{I}_\xi - \hat{\mathbf{R}})^{-1} \hat{\mathbf{q}}_1. \quad (3.37)$$

When the process parameters are estimated, the ARL, ATS_0 , and ATS_1 of the SPRT chart should be replaced by their conditional versions, i.e., the conditional ARL (CARL), conditional ATS_0 (CATS₀), and conditional ATS_1 (CATS₁), respectively. The formulae are given as follows:

$$\text{CARL} = \frac{1}{1 - \widehat{\text{OC}}}, \quad (3.38)$$

$$\text{CATS}_0 = \frac{d}{1 - \widehat{\text{OC}}_0}, \quad (3.39)$$

and

$$\text{CATS}_1 = d \left(\frac{1}{1 - \widehat{\text{OC}}_1} - \frac{1}{2} \right), \quad (3.40)$$

where $\widehat{OC}_0$ and $\widehat{OC}_1$ are the in-control and out-of-control values of $\widehat{OC}$, respectively. The conditional ANOS (CANOS) of the SPRT chart with estimated process parameters has a similar expression as

$$CANOS = CARL \times CASN. \quad (3.41)$$

Note that Equations (3.38) and (3.41) are true under both H_0 and H_1 .

However, the derivations of the conditional SDRL (CSDRL), conditional SDTS₀ (CSDTS₀), and conditional SDTS₁ (CSDTS₁) are slightly different from those given in Section 3.2. To derive the formulae for the CSDRL, we need to first evaluate the first and second moments of RL. The first and second moments of RL are expressed as

$$E[RL] = CARL = \frac{1}{1-\widehat{OC}}$$

and

$$E[RL^2] = \frac{2-[1-\widehat{OC}]}{[1-\widehat{OC}]^2} = \frac{1+\widehat{OC}}{[1-\widehat{OC}]^2},$$

respectively, where $RL \sim \text{Geometric}(1 - \widehat{OC})$. The formula for the CSDRL can then be written as

$$CSDRL = \sqrt{E[RL^2] - (E[RL])^2} = \sqrt{\frac{1+\widehat{OC}}{[1-\widehat{OC}]^2} - CARL^2}, \quad (3.42)$$

and the formula is true under both H_0 and H_1 . Similarly, the CSDTS₀ can be derived as

$$CSDTS_0 = d \sqrt{\frac{1+\widehat{OC}_0}{[1-\widehat{OC}_0]^2} - CARL_0^2} = d \sqrt{\frac{1+\widehat{OC}_0}{[1-\widehat{OC}_0]^2} - \left(\frac{CATS_0}{d}\right)^2}, \quad (3.43)$$

where $CARL_0$ represents the in-control CARL. In the out-of-control situation, recall from Section 3.2 that the steady-state time to signal $TS_1 = d \times (RL_1 - 1 + \mathcal{U})$. The first and second moments of $(RL_1 - 1 + \mathcal{U})$ are

$$E[RL_1 - 1 + \mathfrak{U}] = \frac{1}{1-\widehat{OC}_1} - 1 + \frac{1}{2} = \frac{1}{1-\widehat{OC}_1} - \frac{1}{2}$$

and

$$\begin{aligned} E[(RL_1 - 1 + \mathfrak{U})^2] &= E[RL_1^2 + 2(\mathfrak{U} - 1)RL_1 + (\mathfrak{U} - 1)^2] \\ &= E[RL_1^2] + 2 \cdot E[\mathfrak{U} - 1] \cdot E[RL_1] + E[(\mathfrak{U} - 1)^2] \\ &= \frac{1+\widehat{OC}_1}{[1-\widehat{OC}_1]^2} + 2 \cdot \left(-\frac{1}{2}\right) \cdot \left(\frac{1}{1-\widehat{OC}_1}\right) + \frac{1}{3} \\ &= \frac{2\widehat{OC}_1}{[1-\widehat{OC}_1]^2} + \frac{1}{3}, \end{aligned}$$

respectively. Hence, the formula for the CSDTS₁ can be written as

$$\begin{aligned} \text{CSDTS}_1 &= d\sqrt{E[(RL_1 - 1 + \mathfrak{U})^2] - (E[RL_1 - 1 + \mathfrak{U}])^2} \\ &= d\sqrt{\frac{2\widehat{OC}_1}{[1-\widehat{OC}_1]^2} + \frac{1}{3} - \left(\frac{1}{1-\widehat{OC}_1} - \frac{1}{2}\right)^2} \\ &= d\sqrt{\frac{2\widehat{OC}_1}{[1-\widehat{OC}_1]^2} + \frac{1}{3} - \left(\frac{\text{CATS}_1}{d}\right)^2}. \end{aligned} \tag{3.44}$$

It is important to note that the conditional properties derived in Equations (3.34) and Equations (3.38) to (3.44) ultimately depend on the estimates $\hat{\mu}_0$ and $\hat{\sigma}_0$. If the values of $\hat{\mu}_0$ and $\hat{\sigma}_0$ are “close enough” to the true parameters μ_0 and σ_0 , one can expect the SPRT chart with estimated parameters to perform almost the same as the SPRT chart in the case where the process parameters are known. However, if the values of $\hat{\mu}_0$ and $\hat{\sigma}_0$ are far away from the true parameters μ_0 and σ_0 , it is likely that the conditional properties will deviate considerably from their nominal values in the known-parameter case. To measure the extent to which the performances differ across all possible estimates, it is important to assess the unconditional values of the run length properties (refer to Section 2.5.3 for the argument). The idea is simple, one simply evaluates the average values of the conditional properties across the domain of $\hat{\mu}_0$ and $\hat{\sigma}_0$, or equivalently, the domain of W and V . This can be done by integrating the conditional functions derived

earlier multiplied by the joint pdf of W and V . Thanks to the independence between W and V , the joint pdf of W and V is simply the product of the individual pdfs of W and V . Therefore, the average of the CASN (AASN), the average of the CARL (AARL), the average of the CATS (AATS), and the average of the CANOS (AANOS) are hence evaluated as

$$\text{AASN} = \mathbb{E}_{W,V}[\text{CASN}] = \int_0^\infty \int_{-\infty}^\infty \text{CASN} f_W(w) f_V(v) dw dv \quad (3.45)$$

$$\text{AARL} = \mathbb{E}_{W,V}[\text{CARL}] = \int_0^\infty \int_{-\infty}^\infty \text{CARL} f_W(w) f_V(v) dw dv \quad (3.46)$$

$$\text{AATS} = \mathbb{E}_{W,V}[\text{CATS}] = \int_0^\infty \int_{-\infty}^\infty \text{CATS} f_W(w) f_V(v) dw dv, \quad (3.47)$$

and

$$\text{AANOS} = \mathbb{E}_{W,V}[\text{CANOS}] = \int_0^\infty \int_{-\infty}^\infty \text{CANOS} f_W(w) f_V(v) dw dv, \quad (3.48)$$

respectively, where $f_W(w)$ and $f_V(v)$ represent the pdfs of the random variables W and V , respectively, and $\mathbb{E}_{W,V}[\cdot]$ represents the expectation with respect to W and V . Note that Equations (3.45) to (3.48) hold under both H_0 and H_1 .

Many researchers have discussed the importance of considering the practitioner-to-practitioner variability (see Section 2.5.1), which can be measured by the standard deviation of the CARL (SDARL), standard deviation of the CATS (SDATS), and the standard deviation of the CANOS (SDANOS). The formula for the SDARL is given as

$$\begin{aligned} \text{SDARL} &= \sqrt{\mathbb{E}_{W,V}[\text{CARL}^2] - (\mathbb{E}_{W,V}[\text{CARL}])^2} \\ &= \sqrt{\int_0^\infty \int_{-\infty}^\infty \text{CARL}^2 f_W(w) f_V(v) dw dv - \text{AARL}^2} \end{aligned} \quad (3.49)$$

Similarly, the SDATS and SDANOS can be calculated as

$$\text{SDATS} = \sqrt{\int_0^\infty \int_{-\infty}^\infty \text{CATS}^2 f_W(w) f_V(v) dw dv - \text{AATS}^2} \quad (3.50)$$

and

$$\text{SDANOS} = \sqrt{\int_0^\infty \int_{-\infty}^\infty \text{CANOS}^2 f_W(w) f_V(v) dw dv - \text{ANOS}^2}, \quad (3.51)$$

respectively. All these equations hold under both H_0 and H_1 .

To derive the unconditional value of the CSDRL, it is necessary to first evaluate the unconditional values of the quantities $E[\text{RL}]$ and $E[\text{RL}^2]$. The average standard deviation of the run length (ASDRL) is computed using the formula

$$\begin{aligned} \text{ASDRL} &= \sqrt{E_{W,V}\{E[\text{RL}^2]\} - (E_{W,V}\{E[\text{RL}]\})^2} \\ &= \sqrt{\int_0^\infty \int_{-\infty}^\infty \frac{1+\widehat{\text{OC}}}{[1-\widehat{\text{OC}}]^2} f_W(w) f_V(v) dw dv - \left(\int_0^\infty \int_{-\infty}^\infty \text{CARL} f_W(w) f_V(v) dw dv\right)^2} \\ &= \sqrt{\int_0^\infty \int_{-\infty}^\infty \frac{1+\widehat{\text{OC}}}{[1-\widehat{\text{OC}}]^2} f_W(w) f_V(v) dw dv - \text{AARL}^2}. \end{aligned} \quad (3.52)$$

Using the same method as in the derivation for the ASDRL, the formula for the in-control average standard deviation of the time to signal (ASDTS) (ASDTS_0) can be derived as follows:

$$\begin{aligned} \text{ASDTS}_0 &= d \sqrt{\int_0^\infty \int_{-\infty}^\infty \frac{1+\widehat{\text{OC}}_0}{[1-\widehat{\text{OC}}_0]^2} f_W(w) f_V(v) dw dv - \left(\int_0^\infty \int_{-\infty}^\infty \frac{\text{CATS}_0}{d} f_W(w) f_V(v) dw dv\right)^2} \\ &= d \sqrt{\int_0^\infty \int_{-\infty}^\infty \frac{1+\widehat{\text{OC}}_0}{[1-\widehat{\text{OC}}_0]^2} f_W(w) f_V(v) dw dv - \left(\frac{\text{AATS}_0}{d}\right)^2}, \end{aligned} \quad (3.53)$$

where AATS_0 represents the in-control AATS. Similarly, the out-of-control ASDTS (ASDTS_1) can be calculated as

$$\begin{aligned} \text{ASDTS}_1 &= d \sqrt{E_{W,V}\{E[(\text{RL}_1 - 1 + \mathfrak{U})^2]\} - (E_{W,V}\{E[\text{RL}_1 - 1 + \mathfrak{U}]\})^2} \\ &= d \sqrt{\int_0^\infty \int_{-\infty}^\infty \left(\frac{2\widehat{\text{OC}}_1}{[1-\widehat{\text{OC}}_1]^2} + \frac{1}{3}\right) f_W(w) f_V(v) dw dv - \left(\int_0^\infty \int_{-\infty}^\infty \frac{\text{CATS}_1}{d} f_W(w) f_V(v) dw dv\right)^2} \end{aligned}$$

$$= d \sqrt{\int_0^\infty \int_{-\infty}^\infty \frac{2\overline{OC}_1}{[1-\overline{OC}_1]^2} f_W(w)f_V(v) dw dv} + \frac{1}{3} - \left(\frac{\text{AATS}_1}{d}\right)^2 \quad (3.54)$$

where AATS_1 represents the out-of-control AATS.

It is important to note that the ASDRL and ASDTS have different interpretations from the SDARL and SDATS. The SDARL and SDATS measures the variability of the CARL and CATS values realised by different practitioners, hence it tells us how volatile are the average performances when different parameter estimates have been used. In the case where all practitioners use the same values of $\hat{\mu}_0$ and $\hat{\sigma}_0$, the SDARL and SDATS would be zero, since all practitioners share the same average performance and there would be no practitioner-to-practitioner variability. On the other hand, the ASDRL and ASDTS measure the variability of the run length values and the time to signal values obtained by the same practitioner over an infinite time horizon. It is unlikely that the ASDRL and ASDTS would attain the value zero, since there is always some fluctuation in the run length performances of a control chart. However, when the Phase-I sample size is “large enough”, the ASDRL and ASDTS should converge towards the SDRL and SDTS values in the known-parameter case.

Referring to the metrics for overall performance, the unconditional values for the AEQL and EANOS, known as the average of the AEQL (AAEQL) and the expected value of the AANOS (EAANOS), respectively, are derived. The AAEQL can be calculated as

$$\text{AAEQL} = \frac{1}{\delta_{\max} - \delta_{\min}} \int_0^\infty \int_{-\infty}^\infty \int_{\delta_{\min}}^{\delta_{\max}} \delta^2 \text{CATS} f_W(w)f_V(v) d\delta dw dv, \quad (3.55)$$

whereas the EAANOS can be calculated as

$$\text{EAANOS} = \frac{1}{\delta_{\max} - \delta_{\min}} \int_0^\infty \int_{-\infty}^\infty \int_{\delta_{\min}}^{\delta_{\max}} \text{CANOS} f_W(w)f_V(v) d\delta dw dv. \quad (3.56)$$

Note that all the integral expressions in Equations (3.45) - (3.56) can be approximated to a high degree of accuracy with the aid of the Gauss-Legendre Quadrature method. The computer program for calculating the unconditional properties of the SPRT chart with estimated process parameters is available on GitHub (labelled with A.2.1). To verify the accuracies of the formulae, a computer program based on Monte Carlo simulation is also provided (labelled with A.2.2).

3.4 An Evaluation of the Performances of the Traditional SPRT Chart with Estimated Process Parameters

In this section, we discuss the impact of Phase-I parameter estimation on the performances of the SPRT chart when the traditional design has been used. By traditional design, we refer to the design that treats the process parameters as if they were known (see Section 3.2). Several studies in the literature have pointed out that control charts designed under the traditional approach tend to be vulnerable to excessive false alarms when the process parameters are estimated (Quesenberry, 1993; Zhang *et al.*, 2014a; Saleh *et al.*, 2015a, 2015b). Besides, the perceived practitioner-to-practitioner variability leads to extreme difficulty in decision-making and subsequent efforts in process improvement. In this section, we investigate the extent to which the performances of the SPRT chart deteriorate by experimenting with various Phase-I sample sizes as well as mean shift sizes.

Prior to the investigation, it is necessary to outline the complete procedure of the traditional design approach. Note that in the traditional setting, it is implicitly assumed that the process parameters μ_0 and σ_0 are known. Therefore, all required metrics are calculated using the equations provided in Section 3.2. In the first step of the design, one needs to specify an objective function for the optimisation model. One example could be to minimise the AEQL of the SPRT

chart over the shift range $\delta \in [0.1, 2]$. Next, it is necessary to specify some input design parameters of the design (based on practical considerations and resources available), which will serve as the constraints of the aforementioned optimisation model. Some standard design parameters include the recommended ARL_0 , ATS_0 or $ANOS_0$, the sample size, and the sampling interval of a control chart. Finally, we specify a list of charting parameters to be tuned (or searched) in order to achieve the optimum objective function. The interested reader should refer to Haridy *et al.* (2024) for a complete review on control-chart-optimisation designs.

To demonstrate how the performances of the SPRT chart would change in cases where process parameters are estimated, we experiment with two different SPRT charts designed under the traditional approach based on

- (i) the optimisation of the AEQL (Section 3.4.1), and
- (ii) the optimisation of the EANOS (Section 3.4.2).

Note that both the AEQL and EANOS metrics provide measures for the overall performance of the SPRT chart over some specified shift range. Hence, the resulting optimal SPRT charts are expected to perform reasonably well across all mean shift sizes within their specified ranges.

3.4.1 Optimal Design Based on the AEQL

Recall from the introductory section that the AEQL-optimisation design is performed by first specifying an objective function, and then specifying some input specifications (which will act as the constraints to the optimisation model), followed by a list of charting parameters of the SPRT chart (which will act as the unknowns or variables to the optimisation model).

Design Specification

The user first specifies the following input parameters:

- (i) the recommended value of ATS_0 (τ_{ATS}),
- (ii) the minimum and maximum possible mean shift sizes ($\delta_{\min}$ and $\delta_{\max}$),
- (iii) the target inspection rate (IR),
- (iv) the minimum permissible sampling interval ($d_{\min}$), and
- (v) the maximum allowable ASN_0 ($\bar{n}_{\max}$).

The value of τ_{ATS} tells us how long we should expect to obtain the first false alarm. Generally, the value of τ_{ATS} is set to be a large number (> 100), and the choices vary for different practitioners. The values of $\delta_{\min}$ and $\delta_{\max}$ need to be specified as they are the arguments of the AEQL metric. Generally, it is advised to set $\delta_{\min}$ as a small number (say, 0.05 or 0.1) to avoid penalising extremely small shift sizes. For example, it may be of questionable value to detect a shift size of, say, $\delta = 0.1$, and smaller. In that case, the minimum shift size considered could be set as $\delta_{\min} = 0.1$. The same goes to the choice of $\delta_{\max}$, which is rather subjective.

The inspection rate is defined as the ratio of the ASN_0 to the sampling interval d of a control chart, i.e.,

$$IR = \frac{ASN_0}{d}.$$

The inspection rate describes the inverse relationship between the sample number and the sampling frequency in SQC applications. Setting a fixed IR implies that it would not be possible to increase the in-control average sample size without having to decrease the sampling frequency, and vice versa. This is consistent with the concept of resource allocation and management, which states that one cannot increase the number of observations inspected and increase the frequency of inspection simultaneously, since it would likely place a huge pressure on the

inspection system. Besides, a manufacturing process usually remains in-control for a long period of time, hence it is customary to set a fixed IR so as to maintain the workload and resources allocated for quality inspection.

It is worth noting that, for a fixed IR, it is possible to make both ASN_0 and d arbitrarily large or arbitrarily small. Hence, to avoid having unreasonably small values of d (i.e., sampling more often than we can afford), we impose a minimum sampling interval $d_{\min}$. Similarly, to avoid having unreasonably large ASN_0 values (i.e., sampling more observations than we can afford), we set a maximum value $\bar{n}_{\max}$. Note that the choices of those values depend on the capacity of the inspection system. For example, a practitioner may set an inspection rate of $IR = 5$ according to the advice of the resource management department. Due to constraints in labour force, the practitioner sets the minimum sampling interval as 15 minutes, and the average number of inspected units should not exceed 15 at any sampling time. These numbers could, of course, change according to the factory's policy and inspection capacity.

Variables Determination

Ou *et al.* (2012b) stated that, with a fixed inspection rate, practitioners may adopt either a small-sample-size-high-sampling-frequency or a large-sample-size-low-sampling-frequency strategy. However, in practice, practitioners often adopt a moderate-sample-size-moderate-sampling-frequency strategy, or a strategy that tends slightly towards either extreme (but not equal to the extremes). This leads to the following question: Would it be possible to seek a spot between smallest-sample-size-highest-sampling-frequency and largest-sample-size-lowest-sampling-frequency, such that the SPRT chart is optimised with respect to some performance metric? The answer is yes, and it could be achieved by developing an optimisation model for the design of the SPRT chart. Clearly, our objective function is to

minimise the AEQL (which is a loss function), and we have a number of fixed constraints to our model as discussed earlier (e.g., τ_{ATS} , $\delta_{\min}$, $\delta_{\max}$, IR , $d_{\min}$, $\bar{n}_{\max}$). In order to develop an optimisation design for the SPRT chart, we need to specify the design variables that will be searched (or tuned) during the design. The design variables, which are the charting parameters of the SPRT chart, are given as

- (i) the lower control limit g ,
- (ii) the upper control limit h ,
- (iii) the in-control average sample size ASN_0 ,
- (iv) the reference parameter γ , and
- (v) the sampling interval d .

Some researchers would argue that the ASN serves as a performance metric instead of a charting parameter. However, as stated by Ou *et al.* (2012b), the ASN_0 has a semantic meaning to the sample size in a fixed-sample-size control chart, and the sample size is a parameter of a control chart. Hence, in the AEQL-optimisation design, we perform a search in the space of the five charting parameters stated above, including the ASN_0 . Suppose $\bar{n}$ is the potential optimum value of the ASN_0 . The AEQL-optimisation model is outlined as

$$\underset{(\bar{n}, \gamma, d, g, h)}{\text{Minimise AEQL}}, \quad (3.57a)$$

subject to the constraints

$$\text{ATS}_0 = \tau_{\text{ATS}}, \quad (3.57b)$$

$$\text{ASN}_0 = \bar{n}, \quad (3.57c)$$

$$\bar{n} = \min\{\max\{1, \text{IR} \times d\}, \bar{n}_{\max}\}, \quad (3.57d)$$

and

$$d \geq d_{\min}, \quad (3.57e)$$

Note that Equation (3.57d) guarantees that the ASN_0 value is searched in the range $(1, \bar{n}_{\max})$.

It seems an intimidating task to run the algorithm above, since the minimisation problem involves searching an optimum solution in a five-dimensional parameter space (which is a really huge space). Note that the model defined in Equations (3.57a) to (3.57e) involves five variables (or unknowns) and three equality constraints. By the general implicit function theorem (where differentiability of functions is not necessary), the model consists of only two *free variables*. This means that we would only need to perform an exhaustive search on the parameter space of the two free variables, and perform a numerical search on the values of the remaining three dependent variables according to the three equality constraints. Ou *et al.* (2012b) suggested to make $\bar{n}$ and γ the free variables of the optimisation search. This suggestion stems from three main observations:

- First, the equality $IR = \bar{n}/d$ implies that either $\bar{n}$ or d is redundant, since we can always calculate the value of the other variable once we know the value of one of the variables.
- Second, recall from Section 2.3.3 that, given a deterministic mean shift size δ , the SPRT chart attains the best detection performance (in terms of the ARL) when γ is set to be equal to $|\delta|/2$. Although this is generally not true when minimising the AEQL, it is still worth citing the property so that we have an idea of the appropriate charting parameters to select as the free variables of our model.
- Third, Stoumbos and Reynolds (1997a) performed a sensitivity analysis on the performances of the SPRT chart when the values of g and h vary. They observed that, when the value of the ASN_0 changes (with the ATS_0 fixed), the value of g changes considerably, while the value of h changes by a smaller degree. On the other hand, when the value of the ATS_0 changes (with the ASN_0 fixed), the value of h is found to vary by a large degree, while the

value of g remains almost unchanged. Therefore, it seems natural to make g and h the dependent variables of the model, i.e., we can adjust (or tune) the values of g and h according to Equations (3.57c) and (3.57b), respectively, via some numerical root-finding methods.

The step-by-step procedure for determining the optimal charting parameters of the SPRT chart with known process parameters is given below:

Step 1: Specify the values of τ_{ATS} , IR , $\bar{n}_{\text{max}}$, δ_{min} , δ_{max} , and d_{min} .

Step 2: Perform a two-dimensional golden section search in the parameter space of $(\bar{n}, \gamma)$. To speed up the computation process, it is reasonable to restrict the searching ranges to $\bar{n} \in (\max\{1, \text{IR} \times d\}, \text{IR})$ and $\gamma \in (\delta_{\text{min}}/2, \delta_{\text{max}}/2)$. Note that the upper bound for $\bar{n}$ follows from an inference made by Ou *et al.* (2012b), which stated that the SPRT chart is most effective over a range of shift sizes when a small-sample-size-short-sampling-interval strategy is adopted. On the other hand, it is known from Section 2.3.3 that the value of γ that minimises the ARL_1 of the SPRT chart for any given δ is $\gamma = \delta/2$. Since the SPRT chart is to be optimised over the range of shifts $(\delta_{\text{min}}, \delta_{\text{max}})$, γ likely falls between the values $\delta_{\text{min}}/2$ and $\delta_{\text{max}}/2$.

Step 3: For each combination of $(\bar{n}, \gamma)$ searched in Step 2, calculate the value of d from the formula $d = \bar{n}/\text{IR}$. Adjust the values of g and h simultaneously using a hybrid Newton-Raphson method to satisfy both Equations (3.57b) and (3.57c). Terminate the search algorithm when the difference between successive values become less than 0.001. At the end of this step, all the possible combinations of $(\bar{n}, \gamma, d, g, h)$ satisfying constraints (3.57b) - (3.57e) are obtained when $\delta = 0$.

Step 4: Determine the combination $(\text{ASN}_0 = \bar{n}, \gamma, d, g, h)$ that produces the smallest AEQL value from all the possible combinations found in Steps 2 and 3.

Note that in the golden section search algorithm, the number of iterations to convergence is known upfront. Hence, one may prefer the golden section search algorithm over other standard optimisation algorithms (e.g., Nelder-Mead algorithm), whose speed of convergence is uncertain. The computer program for implementing the optimal design can be found on GitHub (labelled with A.3).

Table 3.1 tabulates the run length properties of the SPRT chart designed using the traditional approach based on the AEQL-optimisation model given in (3.57a) - (3.57e). Note that throughout this section, the results are only provided for a single set of design parameters, since the purpose of this section is to highlight the adverse effects of the traditional design on the performance of the SPRT chart, which have been well documented by many researchers for other types of control charts. For the purpose of illustration, the design parameters are selected as $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{min}} = 0.1$, and $\delta_{\text{max}} = 2.0$ in this example. The optimal charting parameters obtained from running the optimisation algorithm stated above are computed as $(\text{ASN}_0, \gamma, d, g, h) = (2.132, 0.306, 0.426, 0.317, 8.388)$. Note that these charting parameters are computed under the assumption that the process parameters are known. They are then used to produce results for the SPRT chart with estimated process parameters, i.e., to simulate the scenario where practitioners with different Phase-I samples use the same control chart designed under the traditional approach.

The run length properties are evaluated from two perspectives, i.e., conditional and unconditional. From the conditional perspective, the exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$, i.e., the probability that the CATS_0 exceeds the threshold τ_{ATS} , is calculated. This probability can be interpreted as the proportion of practitioners whose conditional in-control performance meets the minimum allowable standards, and we want this probability to be adequately high. The

Table 3.1 The exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$ and the unconditional metrics (AATS, ASDTS, SDATS) of the SPRT chart designed under the traditional approach based on the minimisation of the AEQL, when $\tau_{\text{ATS}} = 370.40$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$.

δ	$m = 50$			$m = 100$		
	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$		
	AATS	ASDTS	SDATS	AATS	ASDTS	SDATS
	48.35%			48.85%		
0.0	> 20,000	> 200,000	> 100,000	2,061.40	> 20,000	> 10,000
0.2	525.23	> 20,000	> 10,000	77.88	660.10	463.50
0.4	17.57	1,728.41	1,222.11	6.14	20.24	13.64
0.6	2.12	21.88	15.40	1.59	2.07	0.95
0.8	0.90	1.20	0.57	0.84	0.88	0.22
1.0	0.59	0.61	0.16	0.57	0.57	0.10
1.5	0.34	0.29	0.04	0.33	0.29	0.03
2.0	0.26	0.19	0.01	0.26	0.19	0.01
δ	$m = 200$			$m = 400$		
	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$		
	AATS	ASDTS	SDATS	AATS	ASDTS	SDATS
	49.20%			49.37%		
0.0	809.10	2,746.86	1,856.20	538.29	997.19	593.66
0.2	40.20	97.43	62.76	30.49	46.20	24.54
0.4	4.55	6.61	3.40	4.05	4.71	1.70
0.6	1.47	1.59	0.45	1.42	1.46	0.28
0.8	0.82	0.83	0.14	0.81	0.80	0.09
1.0	0.57	0.55	0.07	0.56	0.54	0.05
1.5	0.33	0.28	0.02	0.33	0.28	0.01
2.0	0.26	0.19	0.01	0.25	0.19	0.00
δ	$m = 600$			$m = 1000$		
	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$		
	AATS	ASDTS	SDATS	AATS	ASDTS	SDATS
	49.50%			49.59%		
0.0	473.56	720.82	384.40	428.54	555.21	249.79
0.2	28.00	36.79	16.87	26.22	30.84	11.48
0.4	3.91	4.29	1.26	3.81	4.01	0.91
0.6	1.40	1.43	0.22	1.39	1.40	0.16
0.8	0.81	0.80	0.08	0.80	0.79	0.06
1.0	0.56	0.54	0.04	0.56	0.54	0.03
1.5	0.33	0.28	0.01	0.33	0.28	0.01
2.0	0.25	0.19	0.00	0.25	0.19	0.00
δ	$m = 2000$			$m = +\infty$		
	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$		
	AATS	ASDTS	SDATS	ATS	SDTS	
	49.70%			100.00%		
0.0	398.19	454.80	155.65	370.46		370.24
0.2	24.99	27.09	7.40	23.85		23.85
0.4	3.73	3.83	0.61	3.66		3.66
0.6	1.38	1.38	0.11	1.38		1.37
0.8	0.80	0.79	0.04	0.80		0.78
1.0	0.56	0.53	0.02	0.56		0.53
1.5	0.33	0.28	0.01	0.33		0.28
2.0	0.25	0.19	0.00	0.25		0.19

exceedance probability is calculated by constructing an empirical distribution for the $CATS_0$ based on the simulation of many different parameter estimates from m Phase-I observations, say, 100,000 different parameter estimates. The computer program for computing the exceedance probability can be found on GitHub (labelled with A.2.2). From the unconditional perspective, both the in-control and out-of-control metrics averaged across all the possible parameter estimates, i.e., the AATS, ASDTS, and SDATS metrics, are evaluated. To observe how the performances of the SPRT chart with estimated process parameters vary for different Phase-I sample sizes, some results for $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$ are provided, where $m = +\infty$ corresponds to the case where process parameters are known. The unconditional metrics are evaluated for mean shift sizes $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$. It is important to note that, when $m = +\infty$, the SDATS is equal to zero, since the probability $\Pr(CATS_0 = 370.4) = 1$ in the case where process parameters are known.

From Table 3.1, it is observed that the exceedance probabilities fall below 50% for all Phase-I sample sizes, i.e., for m ranging from 50 to as large as 2000. This implies that, regardless of the size of the reference sample, practitioners have an at least 50% chance of obtaining a $CATS_0$ value shorter than the recommended value of 370.4. In other words, one out of two practitioners will likely obtain a false alarm rate greater than the recommended rate of $1/370.4 = 0.27$ per time unit on average. This is an alarming sign, since practitioners are susceptible to a high risk of reworking or scrapping nonconforming products under the traditional design of the SPRT chart with estimated process parameters, which in turn leads to potential economic losses.

Referring to the unconditional metrics, it is noticed that the $AATS_0$ and $ASDTS_0$ values (refer to rows corresponding to $\delta = 0$) vary drastically from the ATS_0 and $SDTS_0$ values obtained in the known-parameter case, especially when m is small. As m increases, the $AATS_0$ and $ASDTS_0$ values converge towards the

nominal ATS_0 and $SDTS_0$ values, agreeing closely with the law of large numbers. In the out-of-control case (refer to rows corresponding to $\delta \neq 0$), it is observed that the $AATS_1$ and $ASDTS_1$ deviate rather far away from the ATS_1 and $SDTS_1$ values for small shift sizes δ and Phase-I sample sizes m . For moderate and large shift sizes (i.e., $\delta \geq 0.6$), the disparity between the values become less visible for all Phase-I sample sizes.

It is also found that, for small Phase-I sample sizes, i.e., $m \leq 100$, the practitioner-to-practitioner variability is extremely large under the in-control case, as indicated by the in-control SDATS ($SDATS_0$) values. For m as large as 2000, the $SDATS_0$ value still amounts to around 40% of the τ_{ATS} , which signifies that there exists a large amount of variability among the in-control performances across practitioners (Zhang *et al.* (2014a) suggested that the $SDATS_0$ should be smaller than 10% of the τ_{ATS}). Combining both inferences from the conditional and unconditional perspectives, it is clear that the SPRT chart designed under the traditional approach is not suitable for use due to its unstable in-control performances and high false alarm rates. New design methodologies should be proposed to replace the traditional design.

3.4.2 Optimal Design Based on the EANOS

The EANOS optimisation model has a framework similar to the AEQL optimisation model. However, it involves a fewer number of design parameters and is generally easier to be implemented. Recall that the EANOS metric does not depend on the sampling interval d (see Equation (3.22)). Therefore, the specifications $d_{\min}$ and the inspection rate are not required in the EANOS optimisation model. It is also important to note that, since the inspection rate is not specified, it is no longer sensible to treat ASN_0 as a charting parameter in this design. Montgomery (2019) stated that the sensitivity of a control chart can be

increased by using a large sample size. If the value of ASN_0 is included in part of the charting parameters, then the optimisation algorithm may pursue an infinitely large ASN_0 , which is impractical in all settings. Hence, the ASN_0 is treated as a fixed input specification instead.

In the EANOS-optimisation design, we specify four input parameters, i.e., the recommended value of $ANOS_0$ (τ_{ANOS}), the desired value of ASN_0 ($\bar{n}$), minimum and maximum possible mean shift sizes ($\delta_{\min}$ and $\delta_{\max}$). Since $ANOS_0 = ARL_0 \times ASN_0$, the appropriate value for τ_{ANOS} can be rather subjective. For instance, if the practitioner desires an ARL_0 of 300 and an average sample size of five, then $\tau_{ANOS} = 1500$ could be used. Note that in the EANOS-optimisation model, the ASN_0 value is fixed and d is redundant. Hence, the only charting parameters that need to be tuned are γ , g , and h . The EANOS-optimisation model is given as

$$\underset{(\gamma, g, h)}{\text{Minimise EANOS}}, \quad (3.58a)$$

subject to the constraints

$$ANOS_0 = \tau_{ANOS} \quad (3.58b)$$

and

$$ASN_0 = \bar{n}. \quad (3.58c)$$

Similar to the previous argument in Section 3.4.1, the values of g and h are sensitive to changes in the ASN_0 and the $ANOS_0$, respectively, hence they are treated as the dependent variables of the optimisation model. Thus, the only independent variable to be searched in this model is γ . The step-by-step procedure for computing the optimal charting parameters of the SPRT chart is outlined as follows:

Step 1: Specify the values of τ_{ANOS} , $\bar{n}$, $\delta_{\min}$, and $\delta_{\max}$.

Step 2: This step is similar to Step 2 of the AEQL-optimisation algorithm, except that only a one-dimensional golden section search is performed on the parameter γ .

Step 3: For each γ searched in Step 2, calculate the values of g and h that satisfy both Equations (3.58b) and (3.58c). Terminate the search algorithm when the difference between successive values become less than 0.001. At the end of this step, all the possible combinations of (γ, g, h) are obtained when $\delta = 0$.

Step 4: Determine the combination (γ, g, h) that produces the smallest EANOS value from all the possible combinations found in Steps 2 and 3.

Note that the computer program for implementing this optimal design is available on GitHub (labelled with A.3).

Table 3.2 shows the run length properties of the SPRT chart designed using the EANOS-optimisation model stated above. The design parameters are chosen as $\tau_{\text{ANOS}} = 1500$, $\bar{n} = 5$, $\delta_{\min} = 0.1$, and $\delta_{\max} = 2.0$ for the purpose of illustration. By running the optimisation algorithm above, the optimal charting parameters are computed as $(\gamma, g, h) = (0.164, -0.122, 12.427)$, achieving a minimum EANOS of 33.889. Similar to Table 3.1, the run length performances of the SPRT chart are evaluated from both the conditional and unconditional perspectives, for $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$. From the conditional perspective, the exceedance probability considered is the probability that the in-control CANOS (CANOS_0) exceeds the threshold, i.e., $\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$. From the unconditional perspective, the AANOS and SDANOS metrics are evaluated to assess the central tendency and variability of the CANOS values across practitioners. Similar to Section 3.4.1, when $m = +\infty$, we have SDANOS equals zero, since the probability $\Pr(\text{CANOS}_0 = 1500) = 1$ in the case where process parameters are known.

Table 3.2 The exceedance probability $\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$ and the unconditional metrics (AANOS, SDANOS) of the SPRT chart designed under the traditional approach based on the minimisation of the EANOS, when $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = \bar{n} = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 2.0$, and $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$.

δ	$m = 50$		$m = 100$	
	$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$	
	AANOS	SDANOS	AANOS	SDANOS
	48.69%		49.10%	
0.0	> 100,000	> 1,000,000	> 10,000	> 100,000
0.2	5,427.70	> 500,000	329.86	3,220.44
0.4	96.11	> 10,000	57.15	44.12
0.6	32.17	35.32	29.99	8.04
0.8	21.16	6.01	20.63	3.75
1.0	16.00	3.40	15.80	2.27
1.5	10.07	1.51	10.04	1.05
2.0	7.41	0.93	7.41	0.65
δ	$m = 200$		$m = 400$	
	$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$	
	AANOS	SDANOS	AANOS	SDANOS
	49.38%		49.55%	
0.0	4,156.03	> 10,000	2,381.83	3,216.20
0.2	187.63	216.27	156.94	81.91
0.4	52.02	16.53	50.20	10.01
0.6	29.23	5.05	28.90	3.40
0.8	20.40	2.53	20.30	1.75
1.0	15.71	1.57	15.66	1.10
1.5	10.02	0.73	10.01	0.52
2.0	7.40	0.46	7.40	0.32
δ	$m = 600$		$m = 1000$	
	$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$	
	AANOS	SDANOS	AANOS	SDANOS
	49.66%		49.68%	
0.0	2,021.68	1,874.11	1,785.97	1,136.14
0.2	149.34	57.30	143.93	39.76
0.4	49.66	7.83	49.25	5.88
0.6	28.79	2.74	28.71	2.10
0.8	20.26	1.42	20.23	1.09
1.0	15.64	0.89	15.63	0.69
1.5	10.01	0.42	10.01	0.33
2.0	7.40	0.26	7.40	0.20
δ	$m = 2000$		$m = +\infty$	
	$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$	
	AANOS	SDANOS	ANOS	
	49.81%		100%	
0.0	1,633.97	677.17	1500	
0.2	140.22	26.06	136.77	
0.4	48.96	4.06	48.67	
0.6	28.65	1.47	28.59	
0.8	20.21	0.77	20.19	
1.0	15.62	0.49	15.62	
1.5	10.01	0.23	10.00	
2.0	7.40	0.14	7.40	

The findings from Table 3.2 are very similar to those from Table 3.1. In particular, the exceedance probabilities $\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$ are found to be lower than 50% even for a reasonably large Phase-I sample size, i.e., $m = 2000$. This implies that practitioners suffer an excessive number of false alarms, i.e., more than 50% of the CARL_0 values fall below the recommended level, since the ASN_0 value is fixed throughout the design. From the unconditional viewpoint, the in-control AANOS (AANOS_0) values lie quite far away from the nominal ANOS_0 value when m is moderately small ($m \leq 200$). When one increases the value of m , the AANOS_0 values tend to the nominal ANOS_0 value in the known-parameter case, whereas the in-control SDANOS (SDANOS_0) values converge slowly to the value zero. In the out-of-control case, however, the out-of-control AANOS (AANOS_1) values are quite consistent with the ANOS_1 values for moderate and large shift sizes (i.e., $\delta \geq 0.6$), even when the number of Phase-I samples is small. Nevertheless, practitioners are more concerned about the deteriorated performance of the SPRT chart in the in-control case, since an industrial process typically remains in the in-control state for an extended period of time, and excessive false alarms during this period can lower the factory's productivity and thus hinder business growth.

3.5 An Evaluation of the Performances of the SPRT Chart with Estimated Process Parameters Designed under the Unconditional Metric Matching Approach

This section discusses an alternative design for the SPRT chart with estimated process parameters, known as the unconditional metric matching approach. Recall from Section 2.5.3 that this method involves adjusting the control limits of the control chart so that its unconditional in-control metrics (e.g., in-control AARL, AATS, or AANOS) match their recommended values. In the context

of our proposed designs, practitioners would be required to equate $\text{AATS}_0 = \tau_{\text{ATS}}$ or $\text{AANOS}_0 = \tau_{\text{ANOS}}$ instead of $\text{ATS}_0 = \tau_{\text{ATS}}$ or $\text{ANOS}_0 = \tau_{\text{ANOS}}$ from Section 3.4. Note that the unconditional metric matching method is m -specific. In other words, the control limits of the SPRT chart adjusted for each value of m are unique.

In this section, we investigate two optimisation models based on the AAEQL and EAANOS under the unconditional metric matching approach. Note that, as opposed to the traditional design in Section 3.4, the unconditional metric matching approach treats the parameter estimates as the estimated process parameters, hence the AEQL and EANOS are replaced by their unconditional values AAEQL and EAANOS (averaged across all parameter estimates), respectively.

3.5.1 Optimal Design Based on the AAEQL

In the AAEQL-optimal design, we adopt the AATS-matching approach. Note that the optimisation framework is similar to those in Section 3.4.1. The AAEQL-optimisation model under the AATS-matching approach is given as follows:

$$\underset{(\bar{n}, \gamma, d, g, h)}{\text{Minimise AAEQL}}, \quad (3.59a)$$

subject to the constraints

$$\text{AATS}_0 = \tau_{\text{ATS}}, \quad (3.59b)$$

$$\text{AASN}_0 = \bar{n}, \quad (3.59c)$$

$$\bar{n} = \min\{\max\{1, \text{IR} \times d\}, \bar{n}_{\max}\}, \quad (3.59d)$$

and

$$d \geq d_{\min}, \quad (3.59e)$$

Note that constraints (3.59d) and (3.59e) coincide with constraints (3.57d) and (3.57e), respectively. In this model, the inspection rate IR is defined as $AASN_0/d$, where $AASN_0$ is the in-control AASN. The procedure for calculating the optimal charting parameters of the SPRT chart with estimated process parameters is outlined as follows:

Step 1: Specify the values of m , τ_{ATS} , IR, $\bar{n}_{\max}$, $\delta_{\min}$, $\delta_{\max}$, and $d_{\min}$.

Step 2: This step is similar to Step 2 of the AEQL-optimisation algorithm detailed in Section 3.4.1.

Step 3: This step is similar to Step 3 of the AEQL-optimisation algorithm detailed in Section 3.4.1. Note that constraints (3.57b) and (3.57c) should be replaced with constraints (3.59b) and (3.59c), respectively.

Step 4: Determine the combination ($AASN_0 = \bar{n}$, γ , d , g , h) that produces the smallest AAEQL value from all the possible combinations found in Steps 2 and 3.

The computer program for implementing the AAEQL-optimisation design based on the AATS-matching approach is available on GitHub (labelled with A.4.1).

In this illustration, the design parameters of the AAEQL-optimal SPRT chart are also selected as $\tau_{ATS} = 370.4$, IR = 5, $\bar{n}_{\max} = 15$, $d_{\min} = 0.2$, $\delta_{\min} = 0.1$, and $\delta_{\max} = 2.0$. These parameters are chosen to be the same as those in Section 3.4.1 in order to ensure that both results are comparable. Since the method is m -specific, there is a unique set of optimal charting parameters for each Phase-I sample size m . Table 3.3 tabulates the charting parameters and the resulting AAEQL values of the AAEQL-optimal SPRT chart designed using the AATS-matching approach, together with the exceedance probability $\Pr(CATS_0 \geq \tau_{ATS})$, AATS, ASDTS, and SDATS values, for $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$ and $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

Table 3.3 The optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$ and the resulting AAEQL value, the exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$, and the (AATS, ASDTS, SDATS) of the SPRT chart designed by minimising the AAEQL, when $\tau_{\text{ATS}} = 370.40$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$.

δ	$m = 50$			$m = 100$			$m = 200$			$m = 400$		
	(AASN ₀ , γ , d , g , h , AAEQL)			(AASN ₀ , γ , d , g , h , AAEQL)			(AASN ₀ , γ , d , g , h , AAEQL)			(AASN ₀ , γ , d , g , h , AAEQL)		
	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$		
	AATS	ASDTS	SDATS	AATS	ASDTS	SDATS	AATS	ASDTS	SDATS	AATS	ASDTS	SDATS
	(2.289, 0.250, 0.458, 0.388, 6.207, 0.805)			(2.241, 0.289, 0.448, 0.324, 6.896, 0.763)			(2.211, 0.303, 0.442, 0.296, 7.446, 0.733)			(2.180, 0.303, 0.436, 0.308, 7.899, 0.715)		
	14.01%			21.02%			27.75%			33.25%		
0.0	369.41	> 10,000	7,906.74	370.50	2,156.02	1,501.89	370.32	975.21	637.99	370.23	639.99	369.25
0.2	32.17	433.49	305.68	29.77	108.47	73.75	27.48	54.29	33.11	25.60	36.79	18.69
0.4	4.97	23.31	16.11	4.52	8.66	5.23	4.15	5.56	2.62	3.90	4.45	1.53
0.6	1.64	2.70	1.52	1.54	1.84	0.72	1.46	1.57	0.43	1.42	1.46	0.27
0.8	0.90	1.01	0.35	0.86	0.89	0.21	0.83	0.83	0.14	0.82	0.81	0.09
1.0	0.62	0.63	0.15	0.59	0.58	0.10	0.58	0.55	0.07	0.57	0.55	0.05
1.5	0.36	0.31	0.04	0.35	0.30	0.03	0.34	0.29	0.02	0.34	0.29	0.01
2.0	0.28	0.21	0.02	0.27	0.20	0.01	0.26	0.19	0.01	0.26	0.19	0.00

δ	$m = 600$			$m = 1000$			$m = 2000$			$m = +\infty$	
	(AASN ₀ , γ , d , g , h , AAEQL)			(AASN ₀ , γ , d , g , h , AAEQL)			(AASN ₀ , γ , d , g , h , AAEQL)			(ASN ₀ , γ , d , g , h , AEQL)	
	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$	
	AATS	ASDTS	SDATS	AATS	ASDTS	SDATS	AATS	ASDTS	SDATS	ATS	SDTS
	(2.170, 0.304, 0.434, 0.308, 8.050, 0.708)			(2.180, 0.304, 0.436, 0.299, 8.199, 0.703)			(2.179, 0.305, 0.436, 0.294, 8.292, 0.699)			(2.132, 0.306, 0.426, 0.317, 8.388, 0.694)	
	36.04%			38.99%			42.04%			100%	
0.0	369.83	544.42	282.65	370.28	473.86	209.28	369.88	421.08	142.58	370.46	370.24
0.2	25.01	32.07	14.19	24.40	28.45	10.34	24.04	26.01	7.01	23.85	23.85
0.4	3.82	4.16	1.18	3.73	3.92	0.87	3.68	3.77	0.59	3.66	3.66
0.6	1.40	1.43	0.22	1.39	1.39	0.16	1.38	1.37	0.11	1.38	1.37
0.8	0.81	0.80	0.08	0.81	0.79	0.06	0.80	0.78	0.04	0.80	0.78
1.0	0.57	0.54	0.04	0.56	0.54	0.03	0.56	0.53	0.02	0.56	0.53
1.5	0.33	0.28	0.01	0.33	0.28	0.01	0.33	0.28	0.01	0.33	0.28
2.0	0.26	0.19	0.00	0.26	0.19	0.00	0.26	0.19	0.00	0.25	0.19

Comparing the results between Tables 3.1 and 3.3, one finds that the exceedance probabilities $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$ of the SPRT chart drop by a considerable degree after the AATS-matching design is introduced. For instance, when $m = 50$, the exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$ of the optimal SPRT chart drops from 48.35% (see Table 3.1) to only 14.01% (see Table 3.3) as a result of the AATS-matching design. This implies that, under the AATS-matching design, practitioners have a higher risk of obtaining CATS_0 values that are below the minimum recommended level, indicating that the in-control performance has deteriorated further. The increase in the false alarm probability can be explained as follows. As the AATS_0 values of the traditional SPRT chart increase due to parameter estimation (see Table 3.1), the AATS-matching approach is introduced to bring the AATS_0 values back to the nominal level of 370.4 (i.e., τ_{ATS}). This is done by lowering the upper control limit h of the SPRT chart in order to reduce the large AATS_0 value. Referring to Table 3.3, one finds that the value of h decreases as m decreases, indicating that the AATS-matching mechanism responds to an increased “uncertainty” (due to small sample sizes) by lowering the upper control limit h . As a consequence, the control statistic has a higher chance of rising above h , hence leading to a higher false alarm probability on average.

From the unconditional viewpoint, it is already expected that the AATS_0 values will be very close to 370.4, since the aim of the design is to match the AATS_0 value to τ_{ATS} . For the other in-control metrics, one observes that the ASDTS_0 and SDATS_0 values of the SPRT chart designed under the AATS-matching approach are all smaller than those of the SPRT chart designed under the traditional approach. This is attributed to the lowered upper control limit h as a result of the design. In the out-of-control case, one also observes smaller values of AATS_1 and ASDTS_1 in Table 3.3 as compared to those in Table 3.1. This implies that the AATS_1 and ASDTS_1 values of the SPRT chart designed under the AATS-matching approach are closer to the ATS_1 and SDTS_1 values in the known-parameter case

(i.e., $m = +\infty$) compared to the SPRT chart designed under the traditional approach. Summing up the observations above, it seems that the AATS-matching approach does great at bringing the unconditional metrics closer to their values in the known-parameter case. However, the conditional in-control performance is substantially compromised, with a larger proportion of practitioners exposed to high risk of excessive false alarms.

3.5.2 Optimal Design Based on the EAANOS

In the EAANOS-optimal design, we adopt the AANOS-matching approach. Note that the optimisation framework is similar to those in Section 3.4.2. The objective function of the EAANOS-optimisation model is

$$\underset{(\gamma, g, h)}{\text{Minimise EAANOS}}, \quad (3.60a)$$

subject to the constraints

$$\text{AANOS}_0 = \tau_{\text{ANOS}} \quad (3.60b)$$

and

$$\text{AASN}_0 = \bar{n}. \quad (3.60c)$$

Note that, in all the ANOS-based and AANOS-based models, $\bar{n}$ should always be treated as a fixed design parameter instead of an adjustable parameter. The procedure for computing the optimal charting parameters of the SPRT chart with estimated process parameters is outlined as follows:

Step 1: Specify the values of m , τ_{ANOS} , $\bar{n}$, $\delta_{\min}$, and $\delta_{\max}$.

Step 2: This step is similar to Step 2 of the EANOS-optimisation algorithm detailed in Section 3.4.2.

Step 3: This step is similar to Step 3 of the EANOS-optimisation algorithm detailed in Section 3.4.2. Note that constraints (3.58b) and (3.58c) should be replaced with constraints (3.60b) and (3.60c), respectively.

Step 4: Determine the combination (γ, g, h) that produces the smallest EAANOS value from all the possible combinations found in Steps 2 and 3.

The computer program for implementing the optimal design based on the AANOS-matching approach is available on GitHub (labelled with A.4.2).

For the purpose of demonstration, the design parameters of the EAANOS-optimal design have been selected as $\tau_{\text{ANOS}} = 1500$, $\bar{n} = 5$, $\delta_{\min} = 0.1$, and $\delta_{\max} = 2.0$. The computed charting parameters (γ, g, h) , together with the minimum EAANOS values of the EAANOS-optimal SPRT chart, are given in Table 3.4. Besides, the conditional metric $\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$, as well as the unconditional metrics AANOS and SDANOS, are tabulated in Table 3.4, for $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$ and $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

Majority of the observations from Table 3.4 are found to coincide with those from Table 3.3. In particular, the exceedance probabilities $\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$ are found to deteriorate substantially, with the worst performance being observed when $m = 50$. It is perhaps quite surprising that, for m as large as 400, the exceedance probabilities are still below 30%. This shows that the unconditional metric matching approach is not suitable for Phase-II applications, especially when the amount of reference data available is modest. Referring to the unconditional metrics, all the AANOS_0 values are found to be very close to τ_{ANOS} , whereas the SDANOS_0 values show a decreasing trend for an increasingly large m . These observations are consistent with those made in Table 3.3. All in all, the unconditional metric matching approach is highly discommended due to its worsened conditional in-control performance. An alternative strategy should be

Table 3.4 The optimal charting parameters (γ, g, h) and the resulting EAANOS value, the exceedance probability $\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$, and the (AANOS, SDANOS) of the SPRT chart designed by minimising the EAANOS, when $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 2.0$, and $m \in \{50, 100, 200, 400, 600, 1000, 2000, +\infty\}$.

δ	$m = 50$		$m = 100$		$m = 200$		$m = 400$	
	$(\gamma, g, h, \text{EAANOS})$		$(\gamma, g, h, \text{EAANOS})$		$(\gamma, g, h, \text{EAANOS})$		$(\gamma, g, h, \text{EAANOS})$	
	$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$	
	AANOS	SDANOS	AANOS	SDANOS	AANOS	SDANOS	AANOS	SDANOS
	(0.010, 0.450, 10.528, 23.284)		(0.010, 0.669, 14.134, 26.650)		(0.086, 0.357, 13.556, 30.223)		(0.127, 0.122, 12.966, 32.067)	
	8.02%		11.97%		21.97%		29.93%	
0.0	1,499.78	> 100,000	1,499.72	> 10,000	1,503.68	5,294.73	1,500.69	1,926.45
0.2	86.58	1,129.77	89.83	172.61	111.53	79.57	124.18	54.18
0.4	30.83	23.70	38.97	12.07	43.83	10.42	46.19	8.14
0.6	19.55	5.73	25.54	4.87	27.32	4.01	27.93	3.05
0.8	14.52	3.07	19.09	2.79	19.91	2.21	20.04	1.64
1.0	11.60	2.02	15.27	1.87	15.69	1.44	15.64	1.05
1.5	7.79	1.03	10.23	0.96	10.30	0.71	10.15	0.51
2.0	5.91	0.67	7.74	0.63	7.71	0.46	7.55	0.32
δ	$m = 600$		$m = 1000$		$m = 2000$		$m = +\infty$	
	$(\gamma, g, h, \text{EAANOS})$		$(\gamma, g, h, \text{EAANOS})$		$(\gamma, g, h, \text{EAANOS})$		$(\gamma, g, h, \text{EAANOS})$	
	$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$		$\Pr(\text{CANOS}_0 \geq \tau_{\text{ANOS}})$	
	AANOS	SDANOS	AANOS	SDANOS	AANOS	SDANOS	ANOS	
	(0.139, 0.041, 12.782, 32.677)		(0.149, -0.024, 12.643, 33.164)		(0.157, -0.074, 12.525, 33.527)		(0.164, -0.122, 12.427, 33.889)	
	33.80%		37.42%		41.09%		100%	
0.0	1,498.80	1,342.91	1,499.66	935.23	1,502.29	616.40	1,500	
0.2	128.34	44.12	131.71	34.20	134.39	24.27	136.77	
0.4	47.00	6.85	47.67	5.43	48.18	3.91	48.67	
0.6	28.14	2.55	28.33	2.01	28.45	1.44	28.59	
0.8	20.09	1.36	20.13	1.06	20.16	0.76	20.19	
1.0	15.63	0.87	15.63	0.68	15.62	0.48	15.62	
1.5	10.10	0.42	10.06	0.32	10.03	0.23	10.00	
2.0	7.50	0.26	7.46	0.20	7.43	0.14	7.40	

proposed to address the issues arising from both the traditional design and the unconditional metric matching design.

3.6 Optimal Designs of the SPRT Chart with Estimated Process Parameters under the Guaranteed In-Control Performance Approach

The previous sections have discussed two different approaches for the optimal design of the SPRT chart with estimated process parameters, i.e., the traditional approach (see Section 3.4) and the unconditional metric matching approach (see Section 3.5). Both approaches are found to be unsatisfactory from the conditional perspective. The unconditional metric matching approach attempts to bring unconditional properties (e.g., AATS, ASDTS, AANOS) closer to their nominal values in the known-parameter case, however, at the expense of a higher false alarm probability. Therefore, this section aims to develop a design methodology that guarantees a satisfactory level of conditional in-control performance, with minimal compromise to the unconditional metrics of the SPRT chart in the out-of-control case. The methodology involves coupling the guaranteed in-control performance (GICP) approach with the concept of performance optimisation, known as the integrated optimal-GICP design (see Section 2.5.4).

In the following subsections, we present four optimisation models under the GICP approach based on the minimisation of the (i) AATS, (ii), AAEQL, (iii) AANOS, and (iv) EAANOS, respectively. Table 3.5 shows the objective each optimisation design aims to achieve, as well as the assumption that each optimisation design makes about the knowledge of shift size. For instance, the AATS- and AAEQL-optimisation designs aim to produce SPRT charts with the

Table 3.5 A two-way table showing the industrial objective each optimisation criterion aims to achieve, as well as the assumption each criterion makes about the shift sizes.

Assumption about shift sizes Industrial objective	Known/ Deterministic	Unknown/ Stochastic
Maximise detection speed	AATS	AAEQL
Maximise both detection speed and sampling efficiency	AANOS	EAANOS

shortest out-of-control detection time on average (see row 2 of Table 3.5). This is appropriate for a mass manufacturing application, e.g., monitoring the mean weight of canned chocolate beverages produced in a factory, since the production quantities are very large and all the inspected units are “reusable”. The AANOS- and EAANOS-optimisation designs, on the other hand, aim to produce SPRT charts with the smallest out-of-control detection time and the number of inspected units combined (see row 3 of Table 3.5). This is especially suitable for applications that involve destructive or expensive testing, e.g., durability testing for automobile airbags, since sampling incurs a high cost and practitioners generally want to reduce the number of units inspected.

Referring to the columns of Table 3.5, the AATS- and AANOS-optimisation designs make the assumption that the mean shift size δ is deterministic or known *a priori* (see column 2 of Table 3.5). According to Hawkins and Olwell (1998), the assumption of deterministic shift sizes is approximately valid when the practitioner has a specialised understanding of a certain production process. By citing an industrial example, they explained that it is possible to determine the size of shift in the mean emissions of a gasoline engine, given that there is sufficient historical evidence or a decent theory to support the computation. On the other hand, the AAEQL- and EAANOS-optimisation designs make the assumption that the mean

shift size δ is stochastic or unknown (see column 3 of Table 3.5). In other words, it is innately difficult to predict the exact shift to which a process will likely move (Calzada & Scariano, 2013; Chong et al., 2022), hence practitioners would be better off optimising the performance of the SPRT chart towards a shift range rather than a specific shift size.

Note that the frameworks for the optimal designs in this section (i.e., identifying the objectives, specifying the design parameters, determining the variables) are all identical to those in Sections 3.4 and 3.5. The computer programmes for implementing the aforementioned designs are available on GitHub (labelled with A.4.1 and A.4.2).

3.6.1 Optimal Design Based on the AATS

While researchers agree that process shift sizes tend to be unpredictable in practice (Wu *et al.*, 2010; Calzada & Scariano, 2013; Chong *et al.*, 2022), it is still worth pursuing a design based on the assumption that shift sizes are deterministic before extending it to cater the situation where shift sizes are unknown. Some insights gained from the preliminary design may also be helpful in simplifying sophisticated designs based on unknown shift sizes. In this section, we introduce the AATS-optimisation model for the SPRT chart with estimated process parameters based on the GICP approach.

The AATS-optimisation model for the SPRT chart with estimated process parameters is developed as follows:

$$\underset{(\bar{n}, \gamma, d, g, h)}{\text{Minimise AATS}_1}, \quad (3.61a)$$

subject to

$$\Pr(\text{CATS}_0 \geq (1 - \varepsilon)\tau_{\text{ATS}}) = 1 - p_e, \quad (3.61b)$$

and the same constraints given in Equations (3.59c), (3.59d), and (3.59e). Recall from Section 2.5.4 that Equation (3.61b) is called the EPC constraint. In Equation (3.61b), p_e is a user-defined risk probability, whereas ε is a user-defined tolerance term, whose value may be taken from the range $[0, 1)$. If $\varepsilon = 0$ is taken, then the EPC ensures that the CATS_0 must exceed the threshold τ_{ATS} with probability $1 - p_e$. If $\varepsilon > 0$ is taken, the EPC constraint is relaxed, and the threshold is lowered to some discounted level $(1 - \varepsilon)\tau_{\text{ATS}}$. This strategy has been recommended by several researchers, e.g., Diko *et al.* (2019a, 2019b), as it allows for some degree of flexibility in the original GICP design of control charts with estimated process parameters, thus making the design suitable for practitioners with different needs and in-control preferences.

The procedure for calculating the optimal charting parameters of the AATS-optimal SPRT chart with estimated process parameters designed under the GICP approach is outlined as follows:

Step 1: Specify the values of m , τ_{ATS} , IR , $\bar{n}_{\text{max}}$, δ_{opt} , d_{min} , p_e and ε . Here, δ_{opt} is the shift size for which a quick detection is required.

Step 2: According to Moustakides (1986) and Stoumbos and Reynolds (1997a), the value of γ that minimises the ARL_1 or the ATS_1 of the SPRT chart with known process parameters is equal to $\delta_{\text{opt}}/2$. While there is a reason to believe that this theorem can be extended to the estimated-parameter case, it is necessary to perform some checking in case the results go against with our intuition. Initiate the value of γ as $\delta_{\text{opt}}/2$. Increase or decrease the value of γ and proceed in the direction with a smaller AAQL. For each fixed value of γ , perform a one-dimensional golden section search in the space $\bar{n} \in (\max\{1, \text{IR} \times d\}, \bar{n}_{\text{max}})$.

Step 3: This step is similar to Step 3 of the AEQL-optimisation algorithm detailed in Section 3.4.1. In this step, the value of h is adjusted to satisfy constraint

(3.61b) by using a stochastic approximation algorithm. The value of g is then adjusted using the Newton-Raphson algorithm to satisfy constraint (3.59c). Both the values of g and h are adjusted in turns until they converge approximately.

Step 4: Determine the combination $(AASN_0 = \bar{n}, \gamma, d, g, h)$ that produces the smallest AAQL value from all the possible combinations found in Steps 2 and 3.

The computer program for implementing the AATS-optimal design based on the GICP approach is available on GitHub (labelled with A.4.1).

As stated in Section 2.5.4, we use the stochastic approximation algorithm proposed by Capizzi and Masarotto (2020) to calculate the value of h . However, we notice some limitations in the original algorithm (in terms of its computational efficiency), and hence we propose the following modified stochastic approximation algorithm:

Algorithm 1. A modified stochastic approximation algorithm

Inputs:

- m – Phase-I sample size
- τ_{ATS} – Recommended in-control ATS
- p_e – Error probability in the EPC constraint
- ε – Tolerance term in the EPC constraint
- h_{known} – The value of h that corresponds to the known-parameter case
- ∇ – Stochastic approximation gain
- $N_{\text{burn-in}}$ – Number of iterations in the burn-in period
- N_{final} – Number of iterations used to compute the final estimate for h

Computation:

- 1 Initialise a vector $\mathbf{h}$ of size N_{final} ;
- 2 Assign $h_0 \leftarrow 1.1 \times h_{\text{known}}$;
- 3 **For** iter = 1, ..., $N_{\text{burn-in}}, N_{\text{burn-in}} + 1, \dots, N_{\text{burn-in}} + N_{\text{final}}$ **do**:
- 4 Simulate $w_{\text{iter}} \sim \mathcal{N}(0, 1)$ and $v_{\text{iter}}^2 \sim \chi_{m-1}^2/(m-1)$;

```

5      Calculate the  $\text{CATS}_0$  value from Equations (3.40) using the simulated
      values of  $w_{\text{iter}}$  and  $v_{\text{iter}}$ ;
6      If  $\text{CATS}_0 < (1 - \varepsilon)\tau_{\text{ATS}}$  then:
7          Update  $h_{\text{iter}} \leftarrow \max\{h_{\text{known}}, h_{\text{iter}-1} - \nabla(p_e - 1)\}$ ;
8      else:
9          Update  $h_{\text{iter}} \leftarrow \max\{h_{\text{known}}, h_{\text{iter}-1} - \nabla(p_e)\}$ ;
10     If  $\text{iter} > N_{\text{burn-in}}$  then:
11         Store  $h_{\text{iter}}$  into the vector  $\mathbf{h}$ ;
12     End for;

```

Output:

Return the average value of all the N_{final} elements in the vector $\mathbf{h}$

Note: χ_{m-1}^2 represents the chi-square distribution with $m - 1$ degrees of freedom.

In the modified version of the algorithm (Algorithm 1), we have imposed the following two changes: First, Capizzi and Masarotto (2020) proposed to approximate the correction term for the control limit, i.e., approximate the value of $h - h_{\text{known}}$, and then add the correction term to h_{known} to determine the final value of h . However, we find that approximating the value of h directly would be much more efficient than approximating the correction term (and then applying the correction). Second, Capizzi and Masarotto (2020) proposed to simulate the value of CATS_0 in line 5 of Algorithm 1. However, we find it more efficient (and accurate) to calculate the theoretical CATS_0 value from Equation (3.40). These two changes have been incorporated in Algorithm 1.

Figure 3.2 shows the trace plot of a sequence of h values generated via Algorithm 1 using an initial value h_{known} that corresponds to the known-parameter case. The number of iterations performed in Figure 3.2 is 31,000. The first 1,000 values fall in the so-called “burn-in period”, and are hence discarded. The final value of h is estimated by taking the average of the remaining 30,000 values. Capizzi and Masarotto (2020) stated that, for an arbitrary stochastic gain ∇ , the stochastic

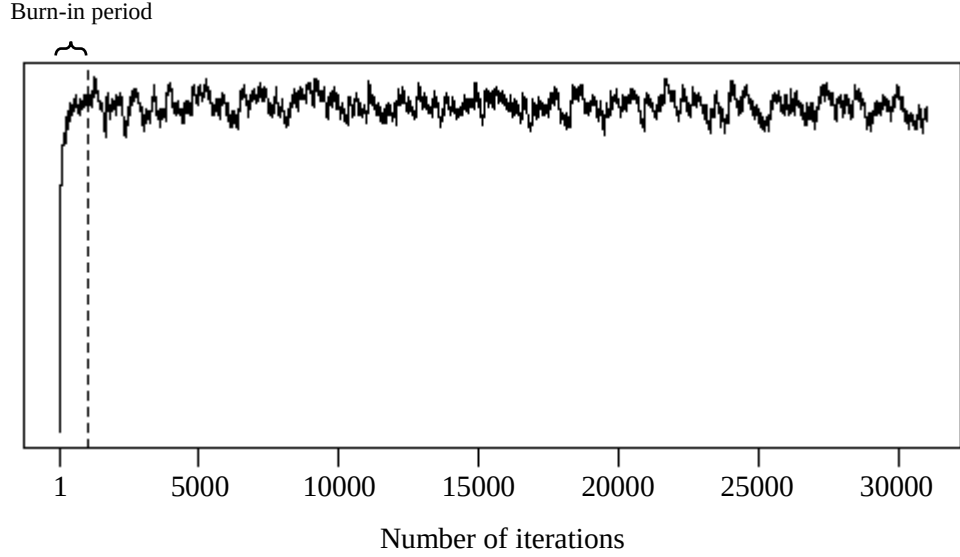


Figure 3.2 Trace plot of a sequence of h values generated from the modified stochastic approximation algorithm with 31,000 simulation runs.

approximation algorithm converges almost surely as the number of iterations tends to infinity. They cited the asymptotic results proven by Kushner and Yin (2003), stating that the approximation achieves a reasonable accuracy with 30,000 iterations. From Figure 3.2, it is indeed found that the algorithm converges very quickly.

There is another concern regarding the minimum Phase-I sample size that will be required for the integrated optimal-GICP design to be meaningful. Apparently, the optimisation design is not very meaningful when m is too small, since the unconditional properties of the SPRT chart can be extremely unstable when Phase-I samples are limited, especially when the shift sizes are also small. In the case where the unconditional properties are too volatile, the optimisation procedure is likely to get distorted, which can lead to an optimal solution that does not follow the “intended” trend (e.g., does not agree with the optimality theorem of the SPRT). To determine a suitable minimum Phase-I sample size for the integrated optimal-GICP design, we have conducted some investigations in regard to the unconditional performance of the SPRT chart with estimated process

Table 3.6 The adjusted control limits (g, h) of the SPRT chart designed under the EPC constraint, together with the AATS, ASDTS, and SDATS values, when $p_e = 0.05$, $\varepsilon = 0$, $\tau_{\text{ATS}} = 370.4$, $\text{AASN}_0 = 2.5$, $\gamma = 0.3$, $\text{IR} = 5$, $d = 0.5$, and $m \in \{50, 100, 150, 200, 250\}$.

(g, h)	$m = 50$	$m = 100$	$m = 150$	$m = 200$	$m = 250$
	(0.577, 29.695)	(0.299, 17.450)	(0.238, 14.764)	(0.211, 13.498)	(0.196, 12.734)
δ	AATS	AATS	AATS	AATS	AATS
	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS
	SDATS	SDATS	SDATS	SDATS	SDATS
0.0	> 10,000,000	> 1,000,000	> 200,000	> 60,000	> 10,000
	> 200,000,000	> 20,000,000	> 10,000,000	> 1,000,000	> 150,000
	> 100,000,000	> 10,000,000	> 2,000,000	> 600,000	> 100,000
0.1	> 5,000,000	> 500,000	> 10,000	4,536.58	1,787.43
	> 100,000,000	> 10,000,000	> 200,000	> 60,000	> 10,000
	> 50,000,000	> 5,000,000	> 100,000	> 45,000	8,543.64
0.2	> 1,000,000	> 100,000	1,523.29	343.02	171.90
	> 20,000,000	> 2,000,000	> 50,000	3,922.32	899.51
	> 10,000,000	> 1,000,000	> 20,000	2,762.88	624.33
0.3	> 500,000	4,408.78	93.81	34.69	22.80
	> 10,000,000	> 200,000	3,300.78	242.92	78.22
	> 5,000,000	> 100,000	2,333.06	170.01	52.91
0.5	> 100,000	6.36	2.92	2.53	2.38
	> 2,000,000	1,386.21	8.18	3.50	2.86
	> 1,000,000	980.19	5.41	1.72	1.13
0.7	> 10,000	1.26	1.14	1.09	1.07
	> 200,000	1.56	1.20	1.12	1.09
	> 100,000	0.67	0.30	0.23	0.20
1.0	0.89	0.66	0.62	0.60	0.59
	0.96	0.64	0.59	0.57	0.56
	0.27	0.11	0.08	0.07	0.06

parameters. Table 3.6 tabulates the unconditional metrics (AATS, ASDTS, SDATS) of the SPRT chart with estimated process parameters for $m \in \{50, 100, 150, 200, 250, 300\}$ and $\delta \in \{0.0, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$. For the purpose of exploration, the design specifications have been selected as $\tau_{\text{ATS}} = 370.4$, $\text{AASN}_0 = 2.5$, $\gamma = 0.3$, $\text{IR} = 5$, $d = 0.5$, $p_e = 0.05$, and $\varepsilon = 0$. Note that the control limits g and h are adjusted to satisfy Equations (3.59c) and (3.61b), i.e., the constraint $\text{AASN}_0 = \bar{n}$ and the EPC constraint, by using the Newton-Raphson algorithm and Algorithm 1, respectively. As an example, for $m = 150$, the GICP-adjusted control limits (g, h) are (0.238, 14.764), and the corresponding (AATS, ASDTS, SDATS) values are computed as (1.14, 1.20, 0.30) when $\delta = 0.7$.

From Table 3.6, it is clear that the performances of the SPRT chart with GICP-adjusted control limits are unstable when m is small. In particular, all the unconditional in-control metrics (i.e., $AATS_0$, $ASDTS_0$, and $SDATS_0$) exceed 10,000 for all values of m considered in Table 3.6, indicating that the in-control performance are extremely volatile when only a limited amount of reference data is available. While the in-control metrics are volatile, one is not too concerned as long as the conditional in-control performance is maintained at a satisfactory level, i.e., the exceedance probability $\Pr(CATS_0 \geq 370.4)$ is adequately high. Note that the proposed GICP design (with $p_e = 0.05$ and $\varepsilon = 0$) guarantees that 95% of the practitioners will obtain $CATS_0$ values exceeding 370.4. Hence, it really does not matter whether there is a large or small variation in the conditional in-control performances. Nonetheless, it is important to ensure that the out-of-control metrics (i.e., $AATS_1$, $ASDTS_1$, $SDATS_1$) do not increase beyond acceptable ranges, as large values of the unconditional out-of-control metrics can distort the mechanism of the optimisation algorithms (such as the AAEQL-optimal design). Referring to Table 3.6, we find that, for $m = 50$ (the smallest Phase-I sample size considered), the $AATS_1$ for $0.1 \leq \delta \leq 0.7$ are at least 10,000; whereas the $SDATS_1$ values exceed a striking value of 100,000. This implies that the out-of-control performances of the SPRT chart with process parameters estimated using $m = 50$ samples are extremely volatile, even for δ as large as 0.7. When m is increased from 50 through 100, the out-of-control performances improve slightly for $\delta = 0.5$, and when m is further increased to 150, the out-of-control performances seem to improve for $\delta = 0.3$. When $m = 200$ is used, the $AATS_1$, $ASDTS_1$, and $SDATS_1$ values seem to have become more stable for $\delta = 0.2$, and its out-of-control performances are arguably better than those for $m \leq 150$. Hence, in this thesis, we recommend using a bare minimum of $m = 200$ Phase-I samples to estimate the process parameters.

Tables 3.7 to 3.10 tabulate the optimal charting parameters of the AATS-optimal SPRT chart designed under the GICP approach for various combinations

Table 3.7 Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$ and the corresponding AATS_1 , ASDTS_1 , and SDATS_1 values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.05$, $\varepsilon = 0$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

δ_{opt}	$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = 2000$
	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)
	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)
	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$
0.2	(15.000, 0.400) (3.000, -5.147, 8.244) (277.92, 1,085.28, 741.82)	(15.000, 0.145) (3.000, -1.316, 26.17) (37.86, 496.98, 350.39)	(15.000, 0.100) (3, -0.649, 36.285) (16.08, 58.70, 39.93)	(15.000, 0.100) (3.000, -0.788, 28.765) (12.22, 14.53, 5.62)	(15.000, 0.100) (3.000, -0.884, 24.082) (10.68, 11.23, 2.60)
0.4	(8.713, 0.201) (1.743, -0.872, 22.988) (3.83, 17.58, 12.14)	(6.528, 0.200) (1.306, -0.564, 17.708) (3.02, 3.27, 0.95)	(6.597, 0.200) (1.319, -0.614, 16.169) (2.86, 2.96, 0.65)	(6.729, 0.202) (1.346, -0.687, 14.591) (2.75, 2.77, 0.45)	(6.682, 0.203) (1.336, -0.706, 13.362) (2.67, 2.65, 0.3)
0.6	(4.605, 0.300) (0.921, -0.608, 13.529) (1.40, 1.46, 0.40)	(4.40426, 0.300) (0.881, -0.591, 11.528) (1.30, 1.29, 0.22)	(4.370, 0.300) (0.874, -0.595, 10.870) (1.28, 1.25, 0.17)	(4.321, 0.300) (0.864, -0.591, 10.227) (1.25, 1.22, 0.13)	(4.320, 0.298) (0.864, -0.592, 9.688) (1.24, 1.19, 0.09)
0.8	(3.311, 0.400) (0.662, -0.542, 9.760) (0.78, 0.76, 0.14)	(3.219, 0.400) (0.644, -0.526, 8.689) (0.76, 0.72, 0.09)	(3.233, 0.400) (0.647, -0.541, 8.252) (0.75, 0.71, 0.07)	(3.156, 0.401) (0.631, -0.515, 7.879) (0.74, 0.70, 0.05)	(3.167, 0.400) (0.633, -0.524, 7.530) (0.74, 0.69, 0.04)
1.0	(2.591, 0.500) (0.518, -0.478, 7.709) (0.52, 0.49, 0.07)	(2.537, 0.500) (0.507, -0.461, 6.987) (0.51, 0.47, 0.05)	(2.511, 0.500) (0.502, -0.451, 6.701) (0.51, 0.46, 0.04)	(2.528, 0.500) (0.506, -0.467, 6.432) (0.50, 0.46, 0.03)	(2.514, 0.505) (0.503, -0.476, 6.135) (0.50, 0.46, 0.02)
1.5	(1.736, 0.750) (0.347, -0.353, 5.071) (0.27, 0.23, 0.02)	(1.736, 0.750) (0.347, -0.362, 4.675) (0.27, 0.23, 0.01)	(1.730, 0.750) (0.346, -0.359, 4.521) (0.27, 0.23, 0.01)	(1.723, 0.752) (0.345, -0.358, 4.368) (0.27, 0.23, 0.01)	(1.736, 0.748) (0.347, -0.367, 4.251) (0.26, 0.22, 0.01)
2.0	(1.398, 1.000) (0.28, -0.299, 3.734) (0.18, 0.14, 0.01)	(1.394, 1.000) (0.279, -0.300, 3.479) (0.18, 0.14, 0.01)	(1.394, 1.000) (0.279, -0.302, 3.375) (0.18, 0.14, 0.01)	(1.387, 1.005) (0.277, -0.299, 3.255) (0.18, 0.14, 0.00)	(1.391, 1.000) (0.278, -0.301, 3.177) (0.18, 0.14, 0.00)

Table 3.8 Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$ and the corresponding AATS_1 , ASDTS_1 , and SDATS_1 values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.05$, $\varepsilon = 0.2$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

δ_{opt}	$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = 2000$
	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)
	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)
	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$
0.2	(15.000, 0.380) (3.000, -4.851, 8.418) (215.870, 845.510, 578.060)	(15.000, 0.140) (3.000, -1.249, 25.650) (31.900, 335.070, 235.850)	(15.000, 0.100) (3, -0.674, 33.34) (15.41, 40.75, 26.69)	(14.500, 0.100) (2.900, -0.758, 26.763) (11.99, 14.07, 5.28)	(14.790, 0.100) (2.958, -0.885, 22.354) (10.47, 10.99, 2.50)
0.4	(7.500, 0.200) (1.500, -0.643, 21.876) (3.770, 12.670, 8.560)	(6.538, 0.200) (1.308, -0.572, 16.831) (3.00, 3.24, 0.94)	(6.713, 0.200) (1.343, -0.646, 15.283) (2.85, 2.94, 0.64)	(6.800, 0.200) (1.360, -0.694, 13.944) (2.73, 2.75, 0.44)	(6.785, 0.200) (1.357, -0.716, 12.842) (2.65, 2.63, 0.29)
0.6	(4.573, 0.300) (0.915, -0.600, 12.931) (1.39, 1.45, 0.40)	(4.385, 0.300) (0.877, -0.587, 11.063) (1.30, 1.29, 0.22)	(4.367, 0.300) (0.873, -0.597, 10.340) (1.27, 1.25, 0.17)	(4.35, 0.300) (0.870, -0.605, 9.738) (1.25, 1.21, 0.13)	(4.270, 0.300) (0.854, -0.588, 9.222) (1.23, 1.19, 0.09)
0.8	(3.318, 0.400) (0.664, -0.547, 9.362) (0.78, 0.76, 0.14)	(3.220, 0.400) (0.644, -0.528, 8.328) (0.76, 0.72, 0.09)	(3.231, 0.400) (0.646, -0.543, 7.912) (0.75, 0.71, 0.07)	(3.200, 0.400) (0.640, -0.535, 7.568) (0.74, 0.70, 0.05)	(3.155, 0.400) (0.631, -0.520, 7.221) (0.73, 0.69, 0.04)
1.0	(2.571, 0.500) (0.514, -0.466, 7.402) (0.52, 0.49, 0.07)	(2.536, 0.500) (0.507, -0.462, 6.714) (0.51, 0.47, 0.05)	(2.515, 0.500) (0.503, -0.455, 6.446) (0.50, 0.46, 0.04)	(2.520, 0.500) (0.504, -0.464, 6.185) (0.50, 0.46, 0.03)	(2.530, 0.500) (0.506, -0.474, 5.937) (0.50, 0.46, 0.02)
1.5	(1.730, 0.750) (0.346, -0.347, 4.876) (0.27, 0.23, 0.02)	(1.735, 0.750) (0.347, -0.362, 4.505) (0.27, 0.23, 0.01)	(1.731, 0.750) (0.346, -0.361, 4.359) (0.27, 0.23, 0.01)	(1.728, 0.750) (0.346, -0.361, 4.215) (0.27, 0.22, 0.01)	(1.734, 0.750) (0.347, -0.370, 4.080) (0.26, 0.22, 0.01)
2.0	(1.400, 1.000) (0.280, -0.304, 3.594) (0.18, 0.14, 0.01)	(1.397, 1.000) (0.279, -0.306, 3.346) (0.18, 0.14, 0.01)	(1.389, 1.000) (0.278, -0.294, 3.249) (0.18, 0.14, 0.01)	(1.390, 1.000) (0.278, -0.298, 3.151) (0.18, 0.14, 0.00)	(1.389, 1.000) (0.278, -0.298, 3.058) (0.18, 0.14, 0.00)

Table 3.9 Optimal charting parameters ($\text{AASN}_0, \gamma, d, g, h$) and the corresponding $\text{AATS}_1, \text{ASDTS}_1$, and SDATS_1 values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.10$, $\varepsilon = 0$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

δ_{opt}	$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = 2000$
	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)
	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)
	($\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1$)	($\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1$)	($\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1$)	($\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1$)	($\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1$)
0.2	(15.000, 0.300)	(15.000, 0.115)	(14.750, 0.100)	(15.000, 0.100)	(14.581, 0.100)
	(3.000, -3.631, 10.820)	(3.000, -0.865, 28.113)	(2.950, -0.702, 28.535)	(3.000, -0.831, 24.899)	(2.916, -0.868, 22.020)
	(156.330, 964.460, 672.960)	(22.970, 221.370, 155.690)	(14.31, 26.07, 15.43)	(11.73, 13.61, 4.95)	(10.43, 10.94, 2.48)
0.4	(7.500, 0.200)	(6.700, 0.200)	(6.732, 0.200)	(6.743, 0.200)	(6.800, 0.200)
	(1.500, -0.672, 18.642)	(1.340, -0.614, 15.805)	(1.346, -0.654, 14.755)	(1.349, -0.683, 13.801)	(1.360, -0.718, 12.957)
	(3.61, 6.91, 4.19)	(2.98, 3.20, 0.91)	(2.84, 2.92, 0.64)	(2.73, 2.75, 0.44)	(2.66, 2.63, 0.29)
0.6	(4.560, 0.300)	(4.370, 0.300)	(4.358, 0.300)	(4.320, 0.300)	(4.310, 0.300)
	(0.912, -0.602, 11.992)	(0.874, -0.584, 10.69)	(0.872, -0.595, 10.214)	(0.864, -0.594, 9.782)	(0.862, -0.600, 9.363)
	(1.39, 1.44, 0.39)	(1.30, 1.29, 0.22)	(1.27, 1.24, 0.17)	(1.25, 1.21, 0.13)	(1.24, 1.19, 0.09)
0.8	(3.281, 0.400)	(3.210, 0.400)	(3.199, 0.400)	(3.187, 0.400)	(3.215, 0.400)
	(0.656, -0.532, 8.936)	(0.642, -0.524, 8.183)	(0.640, -0.527, 7.891)	(0.637, -0.529, 7.619)	(0.643, -0.548, 7.344)
	(0.78, 0.76, 0.14)	(0.75, 0.72, 0.09)	(0.75, 0.71, 0.07)	(0.74, 0.7, 0.05)	(0.73, 0.69, 0.04)
1.0	(2.554, 0.500)	(2.530, 0.500)	(2.522, 0.500)	(2.516, 0.500)	(2.558, 0.500)
	(0.511, -0.457, 7.152)	(0.506, -0.459, 6.650)	(0.504, -0.459, 6.442)	(0.503, -0.461, 6.241)	(0.512, -0.491, 6.048)
	(0.52, 0.49, 0.07)	(0.51, 0.47, 0.05)	(0.50, 0.46, 0.04)	(0.50, 0.46, 0.03)	(0.50, 0.45, 0.02)
1.5	(1.735, 0.750)	(1.732, 0.750)	(1.729, 0.750)	(1.731, 0.750)	(1.730, 0.750)
	(0.347, -0.353, 4.782)	(0.346, -0.359, 4.499)	(0.346, -0.359, 4.386)	(0.346, -0.364, 4.270)	(0.346, -0.365, 4.167)
	(0.27, 0.23, 0.02)	(0.27, 0.23, 0.01)	(0.27, 0.23, 0.01)	(0.27, 0.22, 0.01)	(0.26, 0.22, 0.01)
2.0	(1.395, 1.000)	(1.392, 1.000)	(1.390, 1.000)	(1.392, 1.000)	(1.390, 1.000)
	(0.279, -0.295, 3.546)	(0.278, -0.297, 3.354)	(0.278, -0.296, 3.279)	(0.278, -0.301, 3.204)	(0.278, -0.299, 3.13)
	(0.18, 0.14, 0.01)	(0.18, 0.14, 0.01)	(0.18, 0.14, 0.01)	(0.18, 0.14, 0.00)	(0.18, 0.14, 0.00)

Table 3.10 Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$ and the corresponding AATS_1 , ASDTS_1 , and SDATS_1 values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.10$, $\varepsilon = 0.2$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

δ_{opt}	$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = 2000$
	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)	(AASN_0, γ)
	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)
	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$	$(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$
0.2	(15.000, 0.300) (3.000, -3.645, 10.292) (123.790, 648.180, 449.900)	(15.000, 0.110) (3.000, -0.805, 27.556) (20.680, 153.130, 107.290)	(15.000, 0.100) (3.000, -0.754, 26.463) (13.81, 22.58, 12.67)	(15.000, 0.100) (3.000, -0.856, 23.225) (11.46, 13.16, 4.66)	(15.000, 0.100) (3, -0.936, 20.664) (10.23, 10.7, 2.38)
0.4	(7.750, 0.200) (1.550, -0.731, 17.690) (3.57, 6.13, 3.55)	(6.663, 0.200) (1.333, -0.614, 14.977) (2.96, 3.17, 0.90)	(6.714, 0.200) (1.343, -0.658, 13.987) (2.82, 2.90, 0.63)	(6.683, 0.200) (1.337, -0.678, 13.125) (2.71, 2.73, 0.44)	(6.750, 0.200) (1.350, -0.714, 12.334) (2.64, 2.61, 0.29)
0.6	(4.453, 0.300) (0.891, -0.571, 11.408) (1.38, 1.44, 0.39)	(4.317, 0.300) (0.863, -0.570, 10.210) (1.29, 1.28, 0.22)	(4.335, 0.300) (0.867, -0.591, 9.792) (1.27, 1.24, 0.17)	(4.310, 0.300) (0.862, -0.595, 9.340) (1.25, 1.21, 0.13)	(4.322, 0.300) (0.864, -0.609, 8.947) (1.23, 1.19, 0.09)
0.8	(3.250, 0.400) (0.650, -0.520, 8.564) (0.78, 0.76, 0.14)	(3.238, 0.400) (0.648, -0.540, 7.832) (0.75, 0.72, 0.09)	(3.250, 0.400) (0.650, -0.554, 7.556) (0.75, 0.70, 0.07)	(3.253, 0.400) (0.651, -0.563, 7.292) (0.74, 0.69, 0.05)	(3.198, 0.400) (0.640, -0.543, 7.043) (0.73, 0.69, 0.04)
1.0	(2.543, 0.500) (0.509, -0.452, 6.877) (0.52, 0.49, 0.07)	(2.551, 0.500) (0.510, -0.474, 6.381) (0.51, 0.47, 0.05)	(2.529, 0.500) (0.506, -0.466, 6.188) (0.50, 0.46, 0.04)	(2.516, 0.500) (0.503, -0.462, 6.007) (0.50, 0.46, 0.03)	(2.537, 0.500) (0.507, -0.480, 5.811) (0.50, 0.45, 0.02)
1.5	(1.733, 0.750) (0.347, -0.351, 4.596) (0.27, 0.23, 0.02)	(1.730, 0.750) (0.346, -0.358, 4.323) (0.27, 0.23, 0.01)	(1.730, 0.750) (0.346, -0.361, 4.216) (0.27, 0.23, 0.01)	(1.732, 0.750) (0.346, -0.366, 4.111) (0.26, 0.22, 0.01)	(1.729, 0.750) (0.346, -0.365, 4.010) (0.26, 0.22, 0.01)
2.0	(1.395, 1.000) (0.279, -0.296, 3.408) (0.18, 0.14, 0.01)	(1.394, 1.000) (0.279, -0.301, 3.232) (0.18, 0.14, 0.01)	(1.394, 1.000) (0.279, -0.304, 3.152) (0.18, 0.14, 0.01)	(1.395, 1.000) (0.279, -0.308, 3.083) (0.18, 0.14, 0.00)	(1.391, 1.000) (0.278, -0.302, 3.013) (0.18, 0.14, 0.00)

of design specifications, as well as the corresponding $AATS_1$, $ASDTS_1$, and $SDATS_1$ values. In particular, the specifications $\tau_{ATS} = 370.4$, $m \in \{200, 400, 600, 1000, 2000\}$, $\delta_{opt} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, $p_e \in \{0.05, 0.10\}$, $\varepsilon \in \{0, 0.2\}$, $IR = 5$, and $d_{min} = 0.2$ have been chosen in this section. Note that the combination $(p_e, \varepsilon) = (0.05, 0)$ corresponds to the design with the strictest EPC, whereas the combination $(p_e, \varepsilon) = (0.10, 0.2)$ corresponds to the design with the least strict EPC. In practice, it is possible to pursue a spot in the middle, i.e., a moderately strict EPC, so that both the in-control and out-of-control performances can be preserved to some extent. With a 5% risk probability (i.e., $p_e = 0.05$), Tables 3.7 and 3.8 display results when 0% ($\varepsilon = 0$) and 20% ($\varepsilon = 0.2$) percentage differences are allowed, respectively. On the other hand, with a 10% risk probability (i.e., $p_e = 0.10$), Tables 3.9 and 3.10 display results when 0% ($\varepsilon = 0$) and 20% ($\varepsilon = 0.2$) percentage differences are applied, respectively. As a numeric example, when the input specifications are $m = 400$, $p_e = 0.05$, $\varepsilon = 0$, and $\delta_{opt} = 1.0$, the optimal charting parameters of the SPRT chart with estimated process parameters are read off as $(AASN_0, \gamma, d, g, h) = (2.537, 0.500, 0.507, -0.461, 6.987)$, and the resulting $AATS_1$, $ASDTS_1$, and $SDATS_1$ values are 0.51, 0.47, and 0.05, respectively (see Table 3.7). It is important to note that any attempt to reproduce the optimal charting parameters in Tables 3.7 to 3.10 can lead to slightly different results due to the nature of simulation-based computations. However, the minimum $AATS_1$ values achieved should be close to those tabulated in the tables provided.

From Tables 3.7 to 3.10, four key observations can be made:

- (i) For each δ_{opt} , the $(AATS_1, ASDTS_1, SDATS_1)$ values are found to decrease as m increases. This is consistent with the fact that the unconditional properties of the SPRT chart with estimated process parameters tend to those of the SPRT chart with known process parameters as more reference data are gathered. When the shift size of interest is small (i.e., $\delta_{opt} = 0.2$), there seem to be fairly large differences in the $(AATS_1, ASDTS_1, SDATS_1)$ values between

the SPRT chart designed with a small m and a large m . Nonetheless, when the shift size of interest grows larger (i.e., $\delta_{\text{opt}} \geq 0.6$), the performances become less volatile across different values of m .

- (ii) It is possible to improve the sensitivity of the optimal SPRT chart with GICP-adjusted limits by adopting a larger p_e and/or introducing a positive tolerance term ε . For instance, when $m = 1000$, $p_e = 0.05$, and $\delta_{\text{opt}} = 0.4$, the (AATS₁, ASDTS₁, SDATS₁) values decrease slightly from (2.75, 2.77, 0.45) to (2.73, 2.75, 0.44) when ε is adjusted from 0% to 20% (see Tables 3.7 and 3.8).
- (iii) Almost all the results shown in the tables agree with the optimality theorem of the SPRT, which states that the AATS₁ can be minimised by setting γ to $\delta_{\text{opt}}/2$, with the exception of the case where both m and δ_{opt} are small. In particular, we find that the optimal γ value of the SPRT chart falls around 0.3 to 0.4 when $m = 200$ and $\delta_{\text{opt}} = 0.2$ are assumed. This can be explained by the fact that, when m is small, there are considerable uncertainty and variability in the unconditional properties of the SPRT chart with estimated process parameters, which cause the optimisation algorithm to converge to a different solution. While the optimal solution is still plausible, practitioners are advised to collect a large number of Phase-I samples ($m \geq 400$) whenever possible.
- (iv) It is also noted that the optimal AASN₀ value increases as the shift size of interest δ_{opt} decreases. One possible explanation for the phenomenon is that, when a small δ_{opt} is expected, it is likely that the control statistic will go below the lower control limit g very frequently due to a small shift in the process mean. To maximise the detection effectiveness for a small shift, the algorithm responds by lowering the value of g so as to prevent the control statistic from going below g too frequently. As stated by Stoumbos and Reynolds (1997a), the AASN₀ value of the SPRT chart is mostly affected by changes in the value

of g , and the same goes to the converse. Hence, it makes sense that a large $AASN_0$ value is sought by the algorithm to achieve the best performance for a small δ_{opt} .

3.6.2 Optimal Design Based on the AAEQL

This section proposes an AAEQL-optimal design for the SPRT chart under the GICP framework. The AAEQL-optimisation model for the SPRT chart with estimated process parameters is given as follows:

$$\underset{(\bar{n}, \gamma, d, g, h)}{\text{Minimise AAEQL}}, \quad (3.62)$$

subject to the constraints detailed in Equations (3.61b), (3.59c), (3.59d), and (3.59e).

The procedure for calculating the charting parameters of the AAEQL-optimal SPRT chart with estimated process parameters designed under the GICP approach is outlined as follows:

Step 1: Specify the values of m , τ_{ATS} , IR , $\bar{n}_{\text{max}}$, δ_{min} , δ_{max} , d_{min} , p_e and ε .

Step 2: This step is similar to Step 2 of the AEQL-optimisation algorithm detailed in Section 3.4.1.

Step 3: This step is similar to Step 3 of the AATS-optimisation algorithm detailed in Section 3.6.1.

Step 4: Determine the combination ($AASN_0 = \bar{n}, \gamma, d, g, h$) that produces the smallest AAEQL value from all the possible combinations found in Steps 2 and 3.

The computer program for implementing the AAEQL-optimal design based on the GICP approach is available on GitHub (labelled with A.4.1).

Tables 3.11 to 3.14 tabulate the optimal charting parameters of the SPRT chart designed under the GICP approach, together with the minimum AAEQL value achieved by running the optimisation algorithm above. The unconditional

Table 3.11 Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$, the minimum AAEQL, and the resulting AATS, ASDTS, and SDATS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.05$, $\varepsilon = \{0.0, 0.2\}$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.

			$\delta = 0.0$	$\delta = 0.2$	$\delta = 0.4$	$\delta = 0.6$	$\delta = 0.8$	$\delta = 1.0$	$\delta = 1.5$	$\delta = 2.0$
			AATS	AATS	AATS	AATS	AATS	AATS	AATS	AATS
(AASN ₀ , γ , d)			ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS
m	ε	(g , h , AAEQL)	SDATS	SDATS	SDATS	SDATS	SDATS	SDATS	SDATS	SDATS
200	0.0	(2.250, 0.420, 0.450)	> 10,000	309.66	12.91	1.90	0.83	0.54	0.31	0.25
		(−0.034, 9.283, 2.111)	> 75,000	1,362.56	33.33	2.41	0.85	0.51	0.26	0.17
			> 50,000	938.27	21.72	1.06	0.18	0.07	0.02	0.00
	0.2	(2.250, 0.420, 0.450)	8,484.38	243.13	11.82	1.88	0.83	0.54	0.31	0.25
		(−0.035, 8.920, 1.784)	> 45,000	954.38	28.05	2.35	0.84	0.51	0.26	0.17
			> 30,000	652.58	17.99	1.01	0.18	0.07	0.02	0.00
400	0.0	(2.251, 0.275, 0.450)	5,914.80	70.30	4.27	1.47	0.87	0.61	0.36	0.27
		(0.395, 12.519, 0.983)	> 15,000	166.94	5.24	1.51	0.86	0.59	0.31	0.20
			> 10,000	107.06	2.15	0.27	0.10	0.05	0.01	0.01
	0.2	(2.251, 0.280, 0.450)	3,913.96	61.77	4.27	1.47	0.86	0.60	0.36	0.27
		(0.375, 11.780, 0.928)	> 10,000	133.91	5.21	1.50	0.85	0.58	0.30	0.20
			7,881.59	84.01	2.11	0.27	0.10	0.05	0.01	0.01
600	0.0	(2.250, 0.275, 0.450)	2,634.73	47.63	3.95	1.44	0.85	0.60	0.36	0.27
		(0.382, 11.721, 0.852)	5,538.31	77.53	4.40	1.46	0.84	0.58	0.30	0.20
			3,444.72	43.26	1.38	0.21	0.08	0.04	0.01	0.00
	0.2	(2.250, 0.280, 0.450)	1,833.65	42.83	3.95	1.43	0.84	0.60	0.35	0.27
		(0.362, 11.015, 0.821)	3,578.25	66.27	4.39	1.45	0.83	0.57	0.30	0.20
			2,172.83	35.75	1.37	0.21	0.08	0.04	0.01	0.00
1000	0.0	(2.250, 0.279, 0.450)	1,387.01	36.50	3.76	1.41	0.84	0.59	0.35	0.27
		(0.357, 10.876, 0.785)	2,084.03	46.87	3.98	1.41	0.82	0.56	0.30	0.20
			1,100.01	20.78	0.93	0.16	0.06	0.03	0.01	0.00
	0.2	(2.250, 0.275, 0.450)	1,054.39	32.30	3.68	1.40	0.84	0.59	0.35	0.27
		(0.368, 10.532, 0.765)	1,541.39	40.60	3.88	1.41	0.82	0.57	0.30	0.20
			795.19	17.39	0.88	0.15	0.06	0.03	0.01	0.00
2000	0.0	(2.249, 0.275, 0.450)	858.80	29.04	3.57	1.39	0.83	0.59	0.35	0.27
		(0.362, 10.379, 0.745)	1,040.87	32.41	3.65	1.38	0.81	0.56	0.30	0.20
			416.08	10.17	0.57	0.11	0.04	0.02	0.01	0.00
	0.2	(2.188, 0.290, 0.438)	641.58	28.30	3.70	1.39	0.82	0.58	0.34	0.26
		(0.345, 9.495, 0.730)	757.13	31.20	3.79	1.39	0.80	0.55	0.29	0.19
			284.51	9.29	0.61	0.11	0.04	0.02	0.01	0.00

Table 3.12 Optimal charting parameters (AASN_0 , γ , d , g , h), the minimum AAEQL, and the resulting AATS, ASDTS, and SDATS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.10$, $\varepsilon = \{0.0, 0.2\}$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 2.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.

m	ε	(AASN ₀ , γ , d) (g , h , AAEQL)	$\delta = 0.0$	$\delta = 0.2$	$\delta = 0.4$	$\delta = 0.6$	$\delta = 0.8$	$\delta = 1.0$	$\delta = 1.5$	$\delta = 2.0$
			AATS	AATS	AATS	AATS	AATS	AATS	AATS	AATS
			ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS
			SDATS	SDATS	SDATS	SDATS	SDATS	SDATS	SDATS	SDATS
200	0.0	(2.250, 0.380, 0.450)	7,169.16	180.04	8.86	1.70	0.82	0.55	0.32	0.26
		(0.080, 9.430, 1.476)	> 38,000	725.78	19.63	2.00	0.83	0.53	0.27	0.18
			> 25,000	497.16	12.39	0.76	0.16	0.07	0.02	0.01
	0.2	(2.124, 0.350, 0.425)	5,773.75	138.32	7.38	1.66	0.85	0.57	0.33	0.25
		(0.249, 9.753, 1.277)	> 30,000	550.74	15.04	1.88	0.86	0.55	0.28	0.18
			> 20,000	376.95	9.26	0.65	0.16	0.07	0.02	0.01
	0.0	(2.250, 0.275, 0.450)	3,044.05	54.09	4.16	1.47	0.86	0.61	0.36	0.27
		(0.391, 11.555, 0.890)	8,758.32	111.99	5.00	1.50	0.85	0.58	0.31	0.20
			5,807.03	69.34	1.96	0.27	0.10	0.05	0.01	0.01
400	0.0	(2.250, 0.280, 0.450)	2,085.48	47.92	4.14	1.46	0.85	0.60	0.35	0.27
		(0.370, 10.871, 0.851)	5,407.37	91.93	4.95	1.49	0.84	0.58	0.30	0.20
			3,527.84	55.48	1.91	0.27	0.10	0.05	0.01	0.01
	0.2	(2.250, 0.280, 0.450)	1,204.54	36.42	3.87	1.42	0.84	0.59	0.35	0.27
		(0.358, 10.362, 0.786)	2,187.60	53.42	4.27	1.44	0.83	0.57	0.30	0.20
			1,291.36	27.63	1.29	0.21	0.08	0.04	0.01	0.00
	0.0	(2.250, 0.260, 0.450)	1,780.31	37.56	3.74	1.44	0.87	0.62	0.36	0.28
		(0.430, 11.538, 0.810)	3,592.42	58.25	4.10	1.46	0.86	0.59	0.31	0.20
			2,206.44	31.48	1.20	0.20	0.08	0.04	0.01	0.00
600	0.0	(2.250, 0.280, 0.450)	1,204.54	36.42	3.87	1.42	0.84	0.59	0.35	0.27
		(0.358, 10.362, 0.786)	2,187.60	53.42	4.27	1.44	0.83	0.57	0.30	0.20
			1,291.36	27.63	1.29	0.21	0.08	0.04	0.01	0.00
	0.2	(2.250, 0.280, 0.450)	1,204.54	36.42	3.87	1.42	0.84	0.59	0.35	0.27
		(0.358, 10.362, 0.786)	2,187.60	53.42	4.27	1.44	0.83	0.57	0.30	0.20
			1,291.36	27.63	1.29	0.21	0.08	0.04	0.01	0.00
	0.0	(2.250, 0.279, 0.450)	1,042.82	32.84	3.71	1.40	0.83	0.59	0.35	0.27
		(0.354, 10.408, 0.765)	1,516.08	41.23	3.92	1.41	0.82	0.56	0.30	0.20
			778.29	17.62	0.90	0.16	0.06	0.03	0.01	0.00
1000	0.0	(2.250, 0.282, 0.450)	781.90	29.91	3.68	1.39	0.83	0.59	0.35	0.27
		(0.341, 9.860, 0.747)	1,097.47	36.74	3.88	1.40	0.81	0.56	0.30	0.20
			544.71	15.09	0.88	0.16	0.06	0.03	0.01	0.00
	0.2	(2.250, 0.282, 0.450)	781.90	29.91	3.68	1.39	0.83	0.59	0.35	0.27
		(0.341, 9.860, 0.747)	1,097.47	36.74	3.88	1.40	0.81	0.56	0.30	0.20
			544.71	15.09	0.88	0.16	0.06	0.03	0.01	0.00
	0.0	(2.250, 0.279, 0.450)	1,042.82	32.84	3.71	1.40	0.83	0.59	0.35	0.27
		(0.354, 10.408, 0.765)	1,516.08	41.23	3.92	1.41	0.82	0.56	0.30	0.20
			778.29	17.62	0.90	0.16	0.06	0.03	0.01	0.00
2000	0.0	(2.250, 0.275, 0.450)	711.97	27.14	3.53	1.38	0.83	0.59	0.35	0.27
		(0.359, 10.056, 0.735)	852.64	30.06	3.61	1.38	0.81	0.56	0.30	0.20
			331.97	9.15	0.56	0.11	0.04	0.02	0.01	0.00
	0.2	(2.188, 0.290, 0.438)	540.10	26.44	3.65	1.39	0.82	0.58	0.34	0.26
		(0.342, 9.213, 0.720)	631.08	28.98	3.74	1.38	0.80	0.55	0.29	0.19
			231.07	8.38	0.59	0.11	0.04	0.02	0.01	0.00
	0.0	(2.250, 0.275, 0.450)	711.97	27.14	3.53	1.38	0.83	0.59	0.35	0.27
		(0.359, 10.056, 0.735)	852.64	30.06	3.61	1.38	0.81	0.56	0.30	0.20
			331.97	9.15	0.56	0.11	0.04	0.02	0.01	0.00

properties, i.e., the AATS, ASDTS, and SDATS values, evaluated at various shift points along the range $[\delta_{\text{min}}, \delta_{\text{max}}]$ are provided in the tables. We have chosen the design specification as $\tau_{\text{ATS}} = 370.4$, $m \in \{200, 400, 600, 1000, 2000\}$, $p_e \in \{0.05, 0.10\}$, $\varepsilon \in \{0, 0.2\}$, $\text{IR} = 5$, and $d_{\text{min}} = 0.2$. In Tables 3.11 and 3.12, the range of mean shifts considered are $[\delta_{\text{min}}, \delta_{\text{max}}] = [0.1, 2.0]$, and the unconditional metrics are

Table 3.13 Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$, the minimum AAEQL, and the resulting AATS, ASDTS, and SDATS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.05$, $\varepsilon = \{0.0, 0.2\}$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 3.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.

m	ε	$(\text{AASN}_0, \gamma, d)$ (g, h, AAEQL)	$\delta = 0.0$	$\delta = 0.4$	$\delta = 0.7$	$\delta = 1.0$	$\delta = 1.5$	$\delta = 2.0$	$\delta = 2.5$	$\delta = 3.0$
			AATS	AATS	AATS	AATS	AATS	AATS	AATS	AATS
			ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS
			SDATS	SDATS	SDATS	SDATS	SDATS	SDATS	SDATS	SDATS
200	0.0	(1.650, 0.420, 0.330)	> 10,000	15.14	1.40	0.60	0.31	0.22	0.18	0.17
		(0.440, 9.148, 1.834)	> 50,000	37.04	1.55	0.60	0.28	0.17	0.12	0.10
			> 30,000	23.90	0.49	0.09	0.02	0.01	0.00	0.00
	0.2	(1.650, 0.380, 0.330)	> 10,000	11.57	1.35	0.62	0.32	0.22	0.19	0.17
		(0.540, 9.718, 1.615)	> 50,000	26.83	1.46	0.62	0.29	0.18	0.13	0.11
			> 30,000	17.12	0.41	0.09	0.02	0.01	0.00	0.00
400	0.0	(1.650, 0.280, 0.330)	4,799.39	5.49	1.31	0.68	0.35	0.24	0.19	0.17
		(0.795, 11.852, 1.110)	> 10,000	6.78	1.33	0.67	0.33	0.20	0.14	0.11
			9,679.81	2.81	0.21	0.07	0.02	0.01	0.00	0.00
	0.2	(1.650, 0.300, 0.330)	2,851.64	5.71	1.28	0.66	0.34	0.23	0.19	0.17
		(0.732, 10.618, 1.064)	7,259.38	7.08	1.31	0.65	0.32	0.19	0.14	0.11
			4,720.57	2.96	0.21	0.07	0.02	0.01	0.00	0.00
600	0.0	(1.660, 0.310, 0.332)	2,030.17	5.49	1.24	0.64	0.33	0.23	0.19	0.17
		(0.687, 10.213, 1.019)	3,721.53	6.27	1.26	0.63	0.31	0.19	0.13	0.11
			2,205.54	2.14	0.16	0.05	0.02	0.01	0.00	0.00
	0.2	(1.660, 0.310, 0.332)	1,468.81	5.36	1.24	0.63	0.33	0.23	0.19	0.17
		(0.684, 9.751, 0.992)	2,561.05	6.07	1.25	0.62	0.31	0.19	0.13	0.11
			1,483.58	2.01	0.16	0.05	0.02	0.01	0.00	0.00
1000	0.0	(1.670, 0.310, 0.334)	1,210.61	5.09	1.21	0.63	0.33	0.23	0.19	0.17
		(0.669, 9.683, 0.968)	1,702.91	5.45	1.22	0.61	0.30	0.19	0.13	0.11
			846.98	1.39	0.12	0.04	0.01	0.00	0.00	0.00
	0.2	(1.670, 0.310, 0.334)	905.78	4.97	1.21	0.62	0.33	0.23	0.19	0.17
		(0.666, 9.250, 0.949)	1,237.65	5.30	1.21	0.61	0.30	0.19	0.13	0.11
			596.51	1.31	0.12	0.04	0.01	0.00	0.00	0.00
2000	0.0	(1.650, 0.310, 0.330)	787.19	4.87	1.21	0.63	0.33	0.23	0.19	0.17
		(0.679, 9.166, 0.935)	922.39	5.02	1.21	0.61	0.30	0.19	0.13	0.11
			340.14	0.87	0.09	0.03	0.01	0.00	0.00	0.00
	0.2	(1.650, 0.310, 0.330)	609.51	4.77	1.21	0.62	0.33	0.23	0.19	0.17
		(0.676, 8.772, 0.921)	704.51	4.91	1.20	0.61	0.30	0.18	0.13	0.11
			250.04	0.83	0.09	0.03	0.01	0.00	0.00	0.00

evaluated at $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$. In Tables 3.13 and 3.14, the range of mean shifts considered are $[\delta_{\text{min}}, \delta_{\text{max}}] = [0.1, 3.0]$, and the unconditional metrics are evaluated at $\delta \in \{0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$. As a numeric example, when the input specifications are $m = 1000$, $p_e = 0.05$, $\varepsilon = 0$, $\delta_{\text{min}} = 0.1$, and $\delta_{\text{max}} = 2.0$, the optimal charting parameters of the SPRT chart with estimated

Table 3.14 Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$, the minimum AAEQL, and the resulting AATS, ASDTS, and SDATS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.10$, $\varepsilon = \{0.0, 0.2\}$, $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{min}} = 0.1$, $\delta_{\text{max}} = 3.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.

m	ε	$(\text{AASN}_0, \gamma, d)$ (g, h, AAEQL)	$\delta = 0.0$	$\delta = 0.4$	$\delta = 0.7$	$\delta = 1.0$	$\delta = 1.5$	$\delta = 2.0$	$\delta = 2.5$	$\delta = 3.0$
			AATS	AATS	AATS	AATS	AATS	AATS	AATS	AATS
			ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS	ASDTS
			SDATS	SDATS	SDATS	SDATS	SDATS	SDATS	SDATS	SDATS
200	0.0	(1.650, 0.360, 0.330)	7,177.45	9.70	1.33	0.63	0.32	0.23	0.19	0.17
		(0.591, 9.706, 1.408)	> 40,000	20.42	1.43	0.63	0.30	0.18	0.13	0.11
			> 20,000	12.71	0.38	0.09	0.03	0.01	0.00	0.00
	0.2	(1.650, 0.340, 0.330)	5,625.21	8.43	1.33	0.64	0.33	0.23	0.19	0.17
		(0.643, 9.814, 1.293)	> 30,000	16.5	1.41	0.64	0.31	0.19	0.13	0.11
			> 15,000	10.03	0.36	0.09	0.03	0.01	0.00	0.00
400	0.0	(1.650, 0.300, 0.330)	2,372.08	5.64	1.28	0.65	0.34	0.23	0.19	0.17
		(0.731, 10.364, 1.045)	5,797.07	6.93	1.30	0.65	0.31	0.19	0.14	0.11
			3,740.32	2.85	0.21	0.06	0.02	0.01	0.00	0.00
	0.2	(1.650, 0.300, 0.330)	1,690.56	5.49	1.27	0.65	0.34	0.23	0.19	0.17
		(0.728, 9.896, 1.013)	3,839.01	6.66	1.30	0.64	0.31	0.19	0.13	0.11
			2,437.27	2.66	0.21	0.06	0.02	0.01	0.00	0.00
600	0.0	(1.665, 0.310, 0.333)	1,400.73	5.32	1.23	0.63	0.33	0.23	0.19	0.17
		(0.679, 9.687, 0.988)	2,425.76	6.02	1.25	0.62	0.31	0.19	0.13	0.11
			1,400.49	1.99	0.16	0.05	0.02	0.01	0.00	0.00
	0.2	(1.665, 0.310, 0.333)	1,032.26	5.19	1.23	0.63	0.33	0.23	0.19	0.17
		(0.676, 9.251, 0.965)	1,707.88	5.82	1.24	0.62	0.30	0.19	0.13	0.11
			962.19	1.87	0.16	0.05	0.02	0.01	0.00	0.00
1000	0.0	(1.670, 0.310, 0.334)	936.66	4.98	1.21	0.62	0.33	0.23	0.19	0.17
		(0.666, 9.301, 0.951)	1,284.15	5.32	1.21	0.61	0.30	0.19	0.13	0.11
			621.3	1.32	0.12	0.04	0.01	0.00	0.00	0.00
	0.2	(1.670, 0.310, 0.334)	710.85	4.87	1.20	0.62	0.33	0.23	0.19	0.17
		(0.663, 8.889, 0.934)	948.79	5.17	1.21	0.61	0.30	0.19	0.13	0.11
			444.49	1.25	0.12	0.04	0.01	0.00	0.00	0.00
2000	0.0	(1.660, 0.310, 0.332)	669.63	4.78	1.20	0.62	0.33	0.23	0.19	0.17
		(0.668, 8.924, 0.926)	778.07	4.92	1.20	0.61	0.30	0.18	0.13	0.11
			280.37	0.84	0.08	0.03	0.01	0.00	0.00	0.00
	0.2	(1.660, 0.310, 0.332)	521.62	4.67	1.20	0.62	0.33	0.23	0.19	0.17
		(0.665, 8.540, 0.913)	598.27	4.80	1.19	0.61	0.30	0.18	0.13	0.11
			207.38	0.80	0.08	0.03	0.01	0.00	0.00	0.00

process parameters are $(\text{AASN}_0, \gamma, d, g, h) = (2.250, 0.279, 0.450, 0.357, 10.876)$ with a minimum AAEQL of 0.785, and the $(\text{AATS}_1, \text{ASDTS}_1, \text{SDATS}_1)$ values evaluated at $\delta = 0.8$ are (0.84, 0.82, 0.06) (see Table 3.11).

From Tables 3.11 to 3.14, we make the following four key observations:

- (i) Increasing the value of p_e and/or ε poses a favourable effect on the out-of-control performances of the SPRT chart with estimated process parameters. For instance, in Table 3.11, when $m = 600$ and $p_e = 0.05$ are used, the AAEQL value drops from 0.852 to 0.821 after a tolerance of 0.2 is applied, and the $(AATS_1, ASDTS_1, SDATS_1)$ values are found to decrease following the change. The improvement becomes much more pronounced when both m and δ are small.
- (ii) Increasing the value of $\delta_{\max}$ has the effect of shifting the sensitivity from small and moderate shift sizes to large shift sizes. For instance, when $m = 2000$, $p_e = 0.05$, and $\varepsilon = 0$, the $AATS_1$ value at $\delta = 0.4$ is found to increase from 3.57 to 4.87 when $\delta_{\max}$ is increased from 2 to 3 (see Tables 3.11 and 3.13). However, when $\delta = 2.0$, the $AATS_1$ value is found to drop from 0.27 to 0.23. Besides, the optimum AAEQL values are also found to increase slightly moving from $\delta_{\max} = 2.0$ to $\delta_{\max} = 3.0$.
- (iii) Generally, the upper control limit h of the SPRT chart drops as the Phase-I sample size increases. However, when $m = 200$, the values of h do not seem to follow the said trend, i.e., it is observed that the optimal γ values steer away from the values obtained in the other cases. Referring to Tables 3.11 and 3.12, we find that the optimal γ values obtained for $m = 200$ range from 0.350 to 0.420. However, for all other values of m (i.e., $m > 200$), the optimal γ values lie between 0.260 and 0.290. This phenomenon has been explained in Section 3.6.1, which states that the optimisation process likely gets distorted due to the unstable performances of the SPRT chart when a small m is used. However, since the results seem plausible, the optimisation design is still considered meaningful when $m = 200$.
- (iv) The $(AATS_1, ASDTS_1, SDATS_1)$ performances are almost identical for all the Phase-I sample sizes m when $\delta \geq 0.6$, even for m as small as 200. From Tables

3.11 to 3.14, it is observed that the $AATS_1$ values at $\delta \geq 1.0$ drop well below two times the sampling interval (i.e., $2d$), which indicates a satisfactory detection performance of the proposed optimal SPRT chart under the GICP framework. The resemblance between the performances of the SPRT chart with estimated process parameters and those with known process parameters for moderate and large shift sizes is undoubtedly a favourable sign to practitioners.

Apart from the unconditional metrics, we also conduct some investigations on the conditional performances of the GICP-adjusted SPRT chart in comparison to the SPRT chart designed under the traditional approach (see Section 3.4.1) and unconditional metric matching approach (see Section 3.5.1). For the purpose of illustration, the design specifications are chosen as $m = 200$, $\delta_{\min} = 0.1$, and $\delta_{\max} = 2.0$. We consider the following six designs:

Design 1: Traditional approach (refer to Table 3.1)

Design 2: AATS-matching approach (refer to Table 3.3)

Design 3: GICP approach with $p_e = 0.05$, and $\varepsilon = 0.0$ (refer to Table 3.11)

Design 4: GICP approach with $p_e = 0.05$, and $\varepsilon = 0.2$ (refer to Table 3.11)

Design 5: GICP approach with $p_e = 0.10$, and $\varepsilon = 0.0$ (refer to Table 3.12)

Design 6: GICP approach with $p_e = 0.10$, and $\varepsilon = 0.2$ (refer to Table 3.12)

Figure 3.3 displays boxplots of the in-control CATS ($CATS_0$) distributions generated from a large number of parameter estimates of $m = 200$ Phase-I samples. Each boxplot in Figure 3.3 corresponds to each of the six designs above. The vertical dotted line in Figure 3.3 shows the recommended threshold $\tau_{ATS} = 370.4$. It should be noted that the whiskers of each boxplot indicate the 5th and 95th quantiles of the CATS distributions, whereas the box shows the 25th, 50th, and 75th quantiles.

From Figure 3.3, it is found that the four SPRT charts designed under the GICP approach (Designs 3 to 6) exhibit a better in-control performance compared

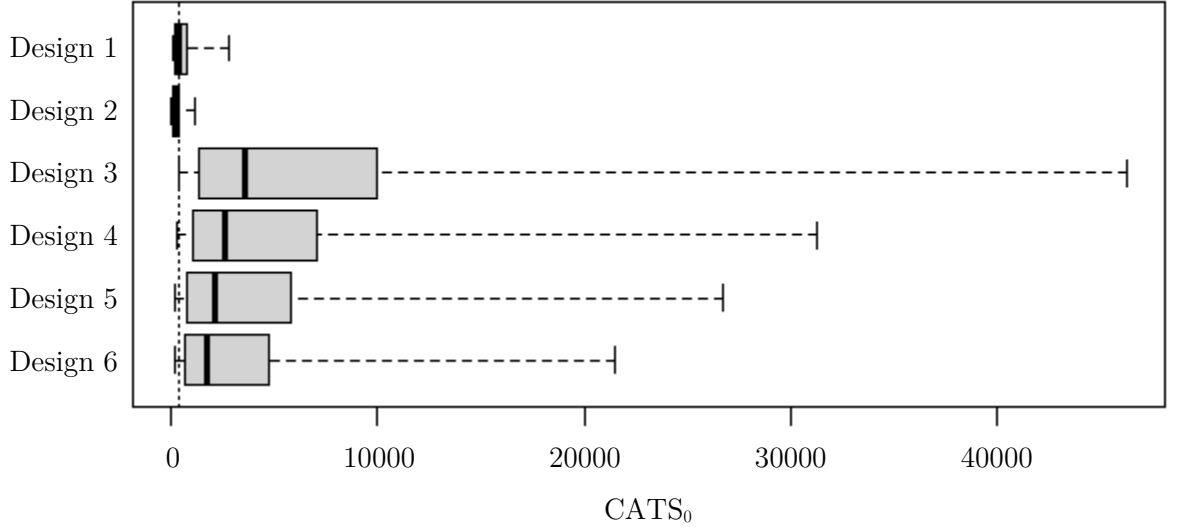


Figure 3.3 Boxplots of the $CATS_0$ distributions for the SPRT chart designed under the traditional approach, AATS-matching approach, and GICP approach with $p_e \in \{0.05, 0.10\}$ and $\varepsilon \in \{0.0, 0.2\}$.

to those designed under the traditional approach (Design 1) and AATS-matching approach (Design 2). As discussed in Sections 3.4 and 3.5, it is found that at least 50% of the $CATS_0$ from Design 1 and Design 2 fall below the recommended level τ_{ATS} , which indicates that practitioners are likely to suffer a large number of false alarms under Design 1 and Design 2. On the other hand, under Designs 3 to 6, the risk of high false alarm rates is effectively reduced to around 5% or 10%, by virtue of the EPC constraint (Equation (3.61b)) of the optimal design. Referring to the variability of the $CATS_0$ values, it is noticed that the $CATS_0$ distributions associated with the GICP design are much more volatile than those of the traditional and AATS-matching designs. This can be explained via the increase in the upper control limit h as a result of the EPC. While an increase in the practitioner-to-practitioner variability may seem concerning at first, one quickly realises that the majority of the $CATS_0$ values lie above the minimum acceptable threshold (i.e., τ_{ATS}), and a large $CATS_0$ value is very much indicative of a satisfactory in-control performance. To draw further inferences, it is necessary to consider the out-of-control performance of the SPRT chart under the GICP design.

Figure 3.4 displays boxplots of the out-of-control CATS ($CATS_1$) distributions for $\delta \in \{0.2, 0.4, 0.6, 0.8\}$. The boxplots in each subfigure (i.e., Figures 3.4(a), (b), (c), and (d)) correspond to the six designs aforementioned. From Figure 3.4, it is noticed that, when shift sizes are small ($\delta \leq 0.4$), all the GICP-adjusted SPRT charts (i.e., Designs 3 to 6) perform worse than the traditional SPRT chart (i.e., Design 1) and the AATS-matched SPRT chart (i.e., Design 2). This can be deduced from the extended whisker of the boxplots corresponding to Designs 3 to 6 in Figures 3.4(a) and 3.4(b). In fact, Saleh *et al.* (2015a) and Diko *et al.* (2019a, 2019b) have reported similar findings in their investigations of the EWMA and CUSUM charts with estimated process parameters. They agreed that the degradation of the out-of-control performance of a control chart is a necessary price to pay for the “guaranteed” in-control performance under the GICP framework. It is, however, possible to adopt a large value for p_e and/or introduce a tolerance term ε in the EPC to improve the out-of-control sensitivity. By carefully tuning p_e and/or ε , one can find a “nice spot” that balances both the in-control and out-of-control performances of the control chart. In Figure 3.4, it is clear that moving from Design 3 to Design 6 leads to a gradual improvement in the out-of-control performance of the SPRT chart, i.e., shorter whiskers in the boxplots.

It is also interesting to note that, for moderate mean shift sizes ($\delta \geq 0.6$), the performances among the six SPRT charts show very little disparities (see Figures 3.4(c) and 3.4(d)). Essentially, the SPRT chart with GICP-adjusted control limits is found to perform as well as the SPRT chart with unadjusted limits and AATS-matched limits, for $\delta \geq 0.6$. We attribute this to the AAEQL-optimisation design, which effectively reduces the $AATS_1$ values along the range $[\delta_{\min}, \delta_{\max}]$ based on the quadratic assignment of weights to the shift sizes. In existing literature, researchers (such as Saleh *et al.* (2015a) and Diko *et al.* (2019b)) have revealed that the EWMA and CUSUM charts only perform as well as their corresponding charts

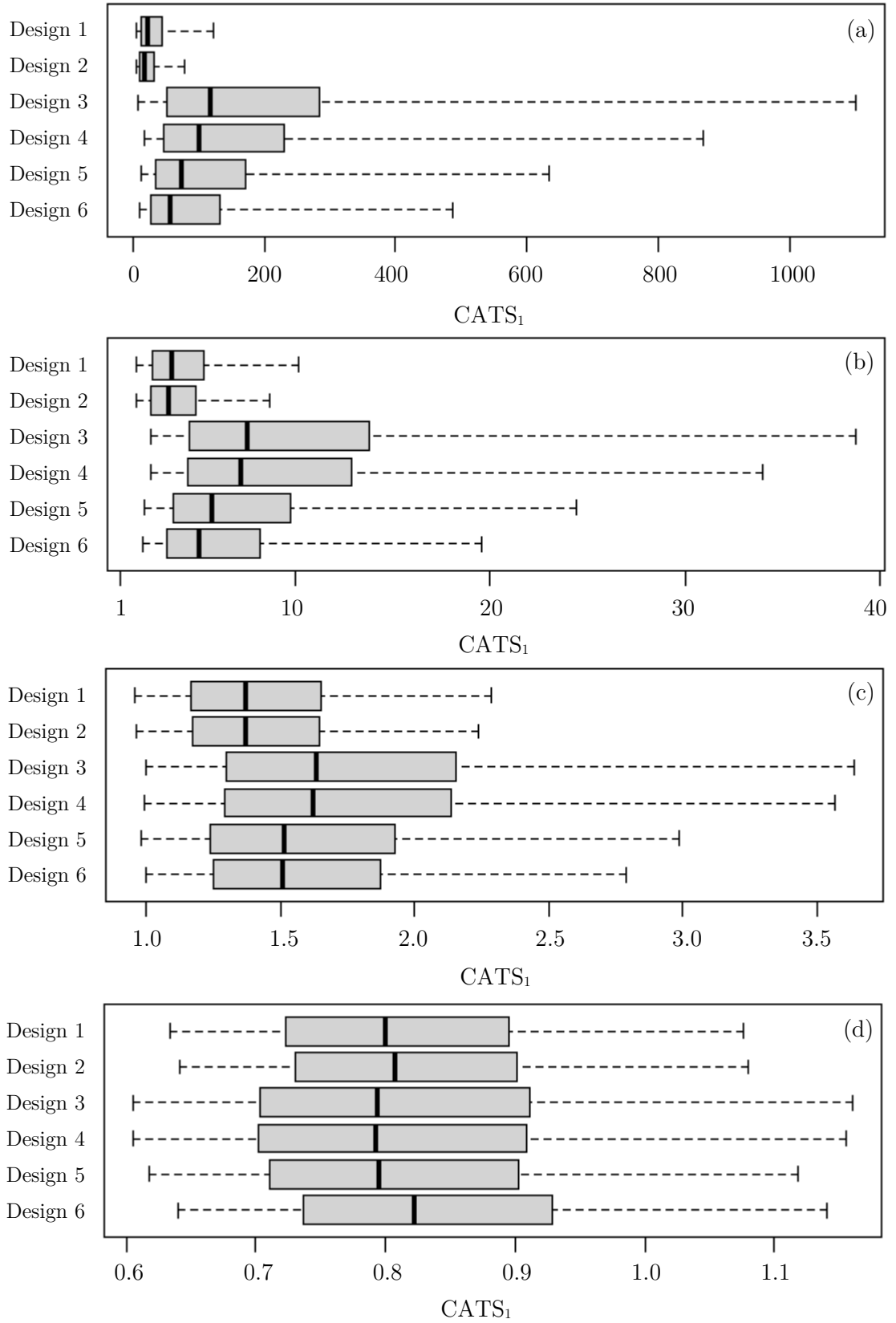


Figure 3.4 Boxplots of the $CATS_1$ distributions for the SPRT chart designed under the traditional approach, AATS-matching approach, and GICP approach with $p_e \in \{0.05, 0.10\}$ and $\varepsilon \in \{0.0, 0.2\}$ when (a) $\delta = 0.2$, (b) $\delta = 0.4$, (c) $\delta = 0.6$, and (d) $\delta = 0.8$.

in the known-parameter case for $\delta \geq 1.0$. Since our proposed SPRT charts (in Designs 3 to 6) perform well for $\delta \geq 0.6$, it is concluded that the optimisation design is effective in enhancing the robustness of the original GICP model.

3.6.3 Optimal Design Based on the AANOS

As stated in Section 2.4.3, the ANOS is an effective measure for assessing the combined effect of the ARL (i.e., detection speed) and the ASN (i.e., sampling efficiency) of variable-sample-size control charts, such as the SPRT chart. In this section, we introduce a model that minimises the $AANOS_1$ of the SPRT chart with estimated process parameters for a deterministic shift size based on the GICP approach.

The AANOS-optimisation model for the SPRT chart with estimated process parameters is developed as follows:

$$\underset{(\gamma, g, h)}{\text{Minimise}} AANOS_1, \quad (3.63a)$$

subject to

$$\Pr(CANOS_0 \geq (1 - \epsilon)\tau_{ANOS}) = 1 - p_e, \quad (3.63b)$$

and the constraint given in Equation (3.60c). It is important to note that Equation (3.63b) may lead to either one of the following conditions:

- (i) the $CARL_0$ is small but the in-control $CASN$ ($CASN_0$) is large,
- (ii) the $CARL_0$ is large but the $CASN_0$ is small, or
- (iii) both the $CARL_0$ and $CASN_0$ are large,

due to the fact that Equation (3.44) holds true. While a large $CARL_0$ value is desired, one may wish to avoid a large $CASN_0$ value in general, as a large $CASN_0$

can lead to an increase in the sampling costs. It is therefore clear that only condition (ii) is favourable. To ensure that condition (ii) is achieved, Equation (3.60c) (i.e., $\text{AASN}_0 = \bar{n}$) is imposed so that the major incidence of the EPC falls on the CARL_0 rather than the CASN_0 .

The procedure for computing the optimal charting parameters of the AANOS-optimal SPRT chart with estimated process parameters designed under the GICP approach is outlined as follows:

Step 1: Specify the values of m , τ_{ANOS} , $\bar{n}$, δ_{opt} , p_e , and ε .

Step 2: Initiate the value of γ as $\delta_{\text{opt}}/2$. Increase or decrease the value of γ in the range of $(0, +\infty)$ and proceed in the direction with a smaller AANOS_1 .

Step 3: This step is similar to Step 3 of the AATS-optimisation algorithm detailed in Section 3.6.1. In this step, the value of h is adjusted to satisfy constraint (3.63b) by using the modified stochastic approximation algorithm.

Step 4: Determine the combination (γ, g, h) that produces the smallest AANOS_1 value from all the possible combinations found in Steps 2 and 3.

Note that in Step 2, it is convenient to set $\gamma = \delta_{\text{opt}}/2$ as a starting point. This stems from the optimality theorem proven by Wald and Wolfowitz (1948), which states that the SPRT is the most efficient test among all sequential tests, i.e., the SPRT minimises the ASN_0 and ASN_1 simultaneously for any given δ_{opt} . In Step 3, we use Algorithm 1 outlined in Section 3.6.1 to compute the value of h . Note that the input τ_{ATS} of the algorithm should be replaced by τ_{ANOS} in this case. The computer program for implementing the AANOS-optimal design based on the GICP approach is available on GitHub (labelled with A.4.2).

Tables 3.15 to 3.18 tabulate the optimal charting parameters of the AANOS-optimal SPRT chart designed under the GICP approach for various combinations

Table 3.15 Optimal charting parameters (γ, g, h) and the corresponding AANOS₁ and SDANOS₁ values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.05$, $\varepsilon = 0$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

δ_{opt}	$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = 2000$
	γ	γ	γ	γ	γ
	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)
0.2	0.300 (-0.744, 11.882) (1126.34, 3338.34)	0.135 (0.228, 22.586) (336.5, 403.25)	0.130 (0.217, 20.941) (251.45, 133.98)	0.130 (0.176, 18.938) (204.88, 65.22)	0.120 (0.217, 18.281) (173.41, 32.77)
0.4	0.240 (-0.399, 15.200) (96.53, 59.79)	0.220 (-0.351, 14.107) (73.36, 19.51)	0.210 (-0.317, 13.733) (66.76, 13.00)	0.220 (-0.396, 12.355) (61.16, 8.98)	0.213 (-0.373, 11.947) (56.63, 5.54)
0.6	0.325 (-0.878, 10.894) (39.93, 11.07)	0.325 (-0.923, 9.637) (34.16, 5.97)	0.317 (-0.897, 9.408) (32.24, 4.37)	0.315 (-0.901, 8.993) (30.40, 3.09)	0.306 (-0.866, 8.815) (28.87, 2.00)
0.8	0.430 (-1.408, 7.995) (22.59, 4.71)	0.430 (-1.44, 7.243) (20.19, 2.80)	0.419 (-1.399, 7.153) (19.33, 2.10)	0.421 (-1.418, 6.819) (18.47, 1.54)	0.411 (-1.379, 6.704) (17.71, 1.01)
1.0	0.523 (-1.856, 6.405) (14.68, 2.53)	0.521 (-1.873, 5.877) (13.37, 1.56)	0.513 (-1.845, 5.768) (12.89, 1.20)	0.512 (-1.848, 5.56) (12.41, 0.88)	0.507 (-1.832, 5.412) (11.97, 0.59)
1.5	0.771 (-3.032, 4.012) (6.63, 0.88)	0.770 (-3.046, 3.723) (6.19, 0.57)	0.768 (-3.044, 3.610) (6.01, 0.45)	0.763 (-3.026, 3.524) (5.84, 0.33)	0.755 (-2.994, 3.459) (5.69, 0.22)
2.0	1.035 (-4.276, 2.765) (3.79, 0.43)	1.030 (-4.268, 2.593) (3.58, 0.28)	1.031 (-4.278, 2.511) (3.49, 0.22)	1.050 (-4.372, 2.384) (3.41, 0.17)	1.010 (-4.190, 2.428) (3.33, 0.11)

Table 3.16 Optimal charting parameters (γ, g, h) and the corresponding AANOS₁ and SDANOS₁ values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.05$, $\varepsilon = 0.2$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

δ_{opt}	$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = 2000$
	γ	γ	γ	γ	γ
	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)
0.2	0.285 (-0.666, 11.899) (867.46, 2320.70)	0.120 (0.336, 23.358) (293.90, 301.09)	0.126 (0.234, 20.007) (225.44, 109.51)	0.123 (0.214, 18.475) (186.34, 55.19)	0.112 (0.259, 17.905) (158.66, 28.17)
0.4	0.237 (-0.387, 14.561) (89.56, 50.95)	0.224 (-0.381, 13.135) (68.85, 18.12)	0.213 (-0.342, 12.831) (62.64, 12.08)	0.217 (-0.388, 11.841) (57.59, 8.23)	0.212 (-0.375, 11.375) (53.35, 5.12)
0.6	0.328 (-0.896, 10.239) (37.70, 10.38)	0.323 (-0.916, 9.230) (32.40, 5.57)	0.324 (-0.938, 8.777) (30.63, 4.17)	0.317 (-0.918, 8.526) (28.95, 2.92)	0.316 (-0.922, 8.181) (27.51, 1.92)
0.8	0.407 (-1.299, 8.107) (21.61, 4.24)	0.422 (-1.408, 7.037) (19.26, 2.60)	0.424 (-1.427, 6.726) (18.42, 2.01)	0.430 (-1.464, 6.370) (17.65, 1.48)	0.412 (-1.389, 6.387) (16.93, 0.96)
1.0	0.531 (-1.897, 6.016) (14.07, 2.45)	0.508 (-1.815, 5.781) (12.83, 1.46)	0.508 (-1.825, 5.560) (12.34, 1.13)	0.515 (-1.866, 5.275) (11.89, 0.85)	0.506 (-1.832, 5.180) (11.48, 0.56)
1.5	0.776 (-3.058, 3.794) (6.37, 0.85)	0.781 (-3.099, 3.496) (5.96, 0.55)	0.762 (-3.017, 3.479) (5.78, 0.42)	0.777 (-3.097, 3.288) (5.62, 0.32)	0.750 (-2.975, 3.323) (5.47, 0.21)
2.0	1.021 (-4.212, 2.678) (3.65, 0.40)	1.010 (-4.178, 2.527) (3.44, 0.26)	1.024 (-4.247, 2.410) (3.36, 0.21)	1.016 (-4.218, 2.362) (3.29, 0.16)	1.004 (-4.166, 2.328) (3.21, 0.11)

Table 3.17 Optimal charting parameters (γ, g, h) and the corresponding AANOS₁ and SDANOS₁ values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.10$, $\varepsilon = 0$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

δ_{opt}	$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = 2000$
	γ	γ	γ	γ	γ
	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)
0.2	0.230 (-0.354, 14.051) (674.74, 2280.24)	0.125 (0.281, 21.270) (267.68, 235.31)	0.115 (0.315, 20.926) (216.46, 99.10)	0.115 (0.275, 19.303) (184.67, 53.22)	0.110 (0.281, 18.466) (162.67, 28.97)
0.4	0.219 (-0.286, 14.793) (82.32, 40.86)	0.215 (-0.331, 13.292) (66.88, 16.94)	0.210 (-0.325, 12.883) (62.10, 11.84)	0.209 (-0.339, 12.323) (58.14, 8.17)	0.212 (-0.372, 11.625) (54.68, 5.28)
0.6	0.325 (-0.886, 9.977) (36.34, 9.80)	0.325 (-0.928, 9.105) (32.17, 5.54)	0.313 (-0.882, 9.066) (30.63, 4.07)	0.308 (-0.869, 8.867) (29.33, 2.92)	0.305 (-0.864, 8.633) (28.17, 1.93)
0.8	0.420 (-1.365, 7.633) (21.02, 4.23)	0.417 (-1.382, 7.126) (19.24, 2.58)	0.408 (-1.350, 7.042) (18.54, 1.96)	0.411 (-1.373, 6.768) (17.91, 1.46)	0.407 (-1.362, 6.624) (17.34, 0.98)
1.0	0.521 (-1.851, 6.031) (13.82, 2.35)	0.521 (-1.877, 5.619) (12.82, 1.49)	0.519 (-1.876, 5.486) (12.46, 1.16)	0.522 (-1.898, 5.301) (12.11, 0.87)	0.510 (-1.848, 5.280) (11.77, 0.58)
1.5	0.757 (-2.969, 3.880) (6.33, 0.82)	0.761 (-3.006, 3.630) (5.99, 0.54)	0.764 (-3.027, 3.521) (5.85, 0.43)	0.758 (-3.005, 3.468) (5.73, 0.32)	0.751 (-2.977, 3.420) (5.61, 0.22)
2.0	1.010 (-4.162, 2.711) (3.64, 0.4)	1.007 (-4.163, 2.571) (3.48, 0.26)	1.010 (-4.182, 2.502) (3.41, 0.21)	1.021 (-4.238, 2.414) (3.35, 0.16)	1.006 (-4.172, 2.401) (3.29, 0.11)

Table 3.18 Optimal charting parameters (γ, g, h) and the corresponding AANOS₁ and SDANOS₁ values of the SPRT chart with estimated process parameters designed under the GICP framework, when $p_e = 0.10$, $\varepsilon = 0.2$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $m \in \{200, 400, 600, 1000, 2000\}$, and $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$.

δ_{opt}	$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = 2000$
	γ	γ	γ	γ	γ
	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)	(g, h) (AANOS ₁ , SDANOS ₁)
0.2	0.219 (-0.297, 13.911) (536.48, 1546.39)	0.123 (0.281, 20.036) (237.16, 180.18)	0.107 (0.363, 20.530) (196.08, 81.93)	0.119 (0.229, 17.604) (168.49, 46.96)	0.112 (0.248, 17.090) (149.68, 25.86)
0.4	0.218 (-0.288, 14.041) (76.70, 36.04)	0.215 (-0.340, 12.583) (62.78, 15.55)	0.217 (-0.376, 11.840) (58.24, 11.08)	0.211 (-0.351, 12.145) (57.68, 8.13)	0.215 (-0.394, 10.945) (51.59, 4.95)
0.6	0.328 (-0.908, 9.387) (34.39, 9.19)	0.324 (-0.929, 8.693) (30.54, 4.11)	0.307 (-0.855, 8.812) (29.16, 3.78)	0.314 (-0.904, 8.309) (27.89, 2.78)	0.300 (-0.846, 8.340) (26.80, 1.80)
0.8	0.430 (-1.417, 7.090) (20.04, 4.08)	0.413 (-1.369, 6.848) (18.35, 1.95)	0.407 (-1.347, 6.744) (17.72, 1.86)	0.407 (-1.360, 6.520) (17.11, 1.37)	0.402 (-1.342, 6.409) (16.59, 0.92)
1.0	0.514 (-1.820, 5.850) (13.23, 2.21)	0.531 (-1.928, 5.260) (12.31, 1.17)	0.525 (-1.910, 5.171) (11.94, 1.12)	0.524 (-1.912, 5.041) (11.61, 0.84)	0.512 (-1.862, 5.025) (11.30, 0.56)
1.5	0.738 (-2.883, 3.820) (6.10, 0.77)	0.757 (-2.991, 3.488) (5.77, 0.42)	0.771 (-3.064, 3.325) (5.64, 0.42)	0.753 (-2.985, 3.338) (5.52, 0.31)	0.746 (-2.956, 3.296) (5.41, 0.21)
2.0	1.008 (-4.156, 2.585) (3.51, 0.38)	1.005 (-4.158, 2.455) (3.35, 0.25)	1.013 (-4.201, 2.374) (3.29, 0.20)	1.020 (-4.239, 2.298) (3.23, 0.15)	1.001 (-4.155, 2.307) (3.18, 0.10)

of design specifications, as well as the corresponding AANOS_1 and SDANOS_1 values. We choose the specification as $\tau_{\text{ANOS}} = 1500$, $m \in \{200, 400, 600, 1000, 2000\}$, $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, $p_e \in \{0.05, 0.10\}$, $\varepsilon \in \{0, 0.2\}$, and $\bar{n} = 5$. Using a risk probability of 5%, Tables 3.15 and 3.16 display results when 0% and 20% tolerances are applied to the threshold, respectively. On the other hand, using a risk probability of 10%, Tables 3.17 and 3.18 display results when 0% and 20% tolerances are applied, respectively. As a numeric example, when the input specifications are $m = 2000$, $p_e = 0.05$, $\varepsilon = 0$, and $\delta_{\text{opt}} = 2.0$, the optimal charting parameters of the SPRT chart with estimated process parameters are $(\gamma, g, h) = (1.010, -4.190, 2.428)$, and the resulting AANOS_1 and SDANOS_1 values are 3.33 and 0.11, respectively (see Table 3.15).

From Tables 3.15 to 3.18, we can draw observations similar to those found in Section 3.6.1. For each δ_{opt} , it is found that the $(\text{AANOS}_1, \text{SDANOS}_1)$ values decrease as the Phase-I sample size m increases. The differences are much more pronounced when a small δ_{opt} is assumed. For instance, when $p_e = 0.05$, $\varepsilon = 0$, and $\delta_{\text{opt}} = 0.2$, the optimum AANOS_1 values are achieved at 1126.34 and 173.41 for $m = 200$ and $m = 2000$, respectively (see Table 3.15). However, when $\delta_{\text{opt}} = 2.0$ (i.e., large shift size), the optimum AANOS_1 values become less distinct, reporting at 3.79 and 3.33 for $m = 200$ and $m = 2000$, respectively (see Table 3.15). Besides, the out-of-control performances of the SPRT chart can be improved by using a larger p_e and/or ε . For instance, when $m = 2000$ and $\delta_{\text{opt}} = 0.4$, the $(\text{AANOS}_1, \text{SDANOS}_1)$ values are reduced from (56.63, 5.54) (see Table 3.15) to (51.59, 4.95) (see Table 3.18) when (p_e, ε) are adjusted from (0.05, 0.0) to (0.10, 0.2). It is also worth noting that the optimal values of γ found are all approximately equal to $\delta_{\text{opt}}/2$, except for the case $m = 200$.

3.6.4 Optimal Design Based on the EAANOS

Pertaining to the fact that process shift sizes are almost always unknown, this section presents an optimisation model based on the EAANOS. The EAANOS-optimisation model for the SPRT chart with estimated process parameters is developed as follows:

$$\underset{(\gamma, g, h)}{\text{Minimise EAANOS}}, \quad (3.64)$$

subject to the constraints given in Equations (3.63b) and (3.60c).

The procedure for computing the optimal charting parameters of the EAANOS-optimal SPRT chart with estimated process parameters designed under the GICP approach is outlined as follows:

Step 1: Specify the values of m , τ_{ANOS} , $\bar{n}$, $\delta_{\min}$, $\delta_{\max}$, p_e , and ε .

Step 2: This step is similar to Step 2 of the EANOS-optimisation algorithm detailed in Section 3.4.2.

Step 3: This step is similar to Step 3 of the AANOS-optimisation algorithm detailed in Section 3.6.3.

Step 4: Determine the combination (γ, g, h) that produces the smallest EAANOS value from all the possible combinations found in Steps 2 and 3.

The computer program for implementing the EAANOS-optimal design based on the GICP approach is available on GitHub (labelled with A.4.2).

Tables 3.19 to 3.22 tabulate the charting parameters of the optimal SPRT chart designed under the GICP approach, the minimum EAANOS achieved, as well as the resulting AANOS_1 and SDANOS_1 values. For the purpose of illustration, the

Table 3.19 Optimal charting parameters (γ, g, h) , the minimum EAANOS, and the resulting AANOS and SDANOS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.05$, $\varepsilon \in \{0.0, 0.2\}$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 2.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.

m	ε	(γ, g) (h, EAANOS)	$\delta = 0.0$	$\delta = 0.2$	$\delta = 0.4$	$\delta = 0.6$	$\delta = 0.8$	$\delta = 1.0$	$\delta = 1.5$	$\delta = 2.0$
			AANOS	AANOS	AANOS	AANOS	AANOS	AANOS	AANOS	AANOS
			SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS
200	0.0	(0.422, -1.369)	> 20,000	1221.59	140.50	42.36	22.57	15.23	8.41	5.86
		(8.156, 204.644)	> 50,000	2105.36	119.19	15.45	4.62	2.20	0.76	0.42
	0.2	(0.420, -1.363)	> 12,000	957.19	125.25	39.84	21.51	14.57	8.07	5.63
		(7.810, 159.941)	> 30,000	1472.99	97.69	14.04	4.34	2.09	0.72	0.40
400	0.0	(0.222, -0.361)	> 15,000	388.06	73.34	37.27	24.99	18.82	11.68	8.52
		(13.997, 84.950)	> 35,000	363.16	19.61	5.11	2.39	1.43	0.63	0.39
	0.2	(0.226, -0.393)	> 10,000	335.04	68.48	34.87	23.36	17.58	10.91	7.95
		(12.968, 71.534)	> 20,000	277.36	18.10	4.80	2.24	1.34	0.59	0.36
600	0.0	(0.173, -0.087)	> 10,000	258.26	71.01	40.52	28.39	21.87	13.94	10.28
		(16.328, 63.344)	> 20,000	143.57	11.73	3.90	2.00	1.26	0.59	0.37
	0.2	(0.173, -0.096)	8286.72	237.53	64.63	36.27	25.22	19.35	12.28	9.04
		(15.354, 55.954)	>10,000	123.60	11.08	3.58	1.81	1.13	0.53	0.33
1000	0.0	(0.171, -0.102)	6103.45	216.69	63.04	35.75	24.95	19.17	12.18	8.97
		(15.254, 51.485)	5574.72	75.83	8.06	2.70	1.38	0.86	0.40	0.25
	0.2	(0.174, -0.131)	4213.97	195.92	58.81	33.39	23.29	17.90	11.38	8.38
		(14.146, 46.277)	3411.00	64.74	7.49	2.52	1.29	0.80	0.38	0.24
2000	0.0	(0.170, -0.117)	3526.97	183.91	57.92	33.24	23.28	17.92	11.41	8.42
		(14.248, 44.160)	1806.48	39.47	5.04	1.75	0.90	0.56	0.27	0.17
	0.2	(0.169, -0.122)	2664.62	167.70	54.61	31.50	22.10	17.03	10.86	8.02
		(13.523, 40.400)	1262.94	34.30	4.69	1.65	0.85	0.54	0.25	0.16

specifications chosen in the design are $\tau_{\text{ANOS}} = 1500$, $m \in \{200, 400, 600, 1000, 2000\}$, $p_e \in \{0.05, 0.10\}$, $\varepsilon \in \{0, 0.2\}$, and $\bar{n} = 5$. In Tables 3.19 and 3.20, the range of mean shifts considered are $[\delta_{\min}, \delta_{\max}] = [0.1, 2.0]$, and the unconditional metrics are evaluated at $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$. In Tables 3.21 and 3.22, the range of mean shifts considered are $[\delta_{\min}, \delta_{\max}] = [0.1, 3.0]$, and the unconditional metrics are evaluated at $\delta \in \{0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$. As a numeric example, when the input specifications are $m = 2000$, $p_e = 0.05$, $\varepsilon = 0.2$, $\delta_{\min} = 0.1$, and $\delta_{\max} = 2.0$, the optimal charting parameters of the SPRT chart with estimated process parameters are $(\gamma, g, h) = (0.169, -0.122, 13.523)$ with a minimum EAANOS

Table 3.20 Optimal charting parameters (γ, g, h) , the minimum EAANOS, and the resulting AANOS and SDANOS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.10$, $\varepsilon \in \{0.0, 0.2\}$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 2.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.

m	ε	(γ, g) (h, EAANOS)	$\delta = 0.0$	$\delta = 0.2$	$\delta = 0.4$	$\delta = 0.6$	$\delta = 0.8$	$\delta = 1.0$	$\delta = 1.5$	$\delta = 2.0$
			AANOS	AANOS	AANOS	AANOS	AANOS	AANOS	AANOS	AANOS
			SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS
200	0.0	(0.325, -0.886)	> 18,000	742.77	93.57	36.21	21.91	15.72	9.25	6.60
		(9.943, 136.471)	> 60,000	1395.95	61.73	9.75	3.63	1.95	0.77	0.45
	0.2	(0.315, -0.840)	> 10,000	592.12	84.54	34.31	21.08	15.21	9.02	6.46
		(9.758, 109.239)	> 40,000	997.10	51.03	8.86	3.41	1.85	0.74	0.43
400	0.0	(0.202, -0.253)	> 10,000	299.75	66.94	35.53	24.20	18.38	11.52	8.43
		(14.013, 66.780)	> 20,000	243.60	16.26	4.63	2.24	1.36	0.62	0.38
	0.2	(0.181, -0.132)	9092.65	253.01	63.75	35.11	24.27	18.56	11.74	8.63
		(14.544, 58.313)	> 15,000	187.92	14.24	4.36	2.17	1.34	0.62	0.39
600	0.0	(0.172, -0.092)	7608.12	230.96	63.74	35.85	24.96	19.16	12.17	8.96
		(15.213, 54.457)	>10,000	117.88	10.86	3.53	1.79	1.11	0.52	0.33
	0.2	(0.175, -0.122)	5122.47	207.96	59.55	33.54	23.34	17.92	11.38	8.38
		(14.136, 48.672)	6328.07	98.71	10.10	3.31	1.68	1.04	0.49	0.30
1000	0.0	(0.170, -0.105)	4362.88	195.62	59.27	33.81	23.63	18.18	11.57	8.53
		(14.450, 46.532)	3617.31	64.45	7.47	2.54	1.30	0.81	0.38	0.24
	0.2	(0.173, -0.133)	3197.13	178.97	55.69	31.80	22.23	17.10	10.88	8.03
		(13.502, 42.478)	2387.54	56.13	6.99	2.39	1.22	0.76	0.36	0.22
2000	0.0	(0.170, -0.125)	2921.18	173.27	55.67	32.03	22.45	17.29	11.02	8.13
		(13.719, 41.594)	1417.69	36.07	4.81	1.68	0.87	0.54	0.26	0.16
	0.2	(0.171, -0.141)	2207.80	158.10	52.33	30.21	21.20	16.34	10.42	7.69
		(12.913, 38.110)	985.47	31.42	4.48	1.58	0.82	0.51	0.24	0.15

of 40.400, and the AANOS_1 and SDANOS_1 values evaluated at $\delta = 1.0$ are 17.03 and 0.54, respectively (see Table 3.19).

From Tables 3.19 to 3.22, it is found that the minimum EAANOS value achieved decreases as m increases. This stems from the fact that the AANOS_1 values have decreased along the range as more Phase-I samples are collected. It is also observed that increasing the value of p_e and/or ε results in an improvement in the out-of-control performances, as indicated in previous sections. Unlike the AAEQL-optimal design, increasing the value of $\delta_{\max}$ (from 2 to 3) in the EAANOS-optimal design causes the minimum EAANOS to decrease. This may be explained by the

Table 3.21 Optimal charting parameters (γ, g, h) , the minimum EAANOS, and the resulting AANOS and SDANOS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.05$, $\varepsilon \in \{0.0, 0.2\}$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 3.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.

m	ε	(γ, g) (h, EAANOS)	$\delta = 0.0$	$\delta = 0.4$	$\delta = 0.7$	$\delta = 1.0$	$\delta = 1.5$	$\delta = 2.0$	$\delta = 2.5$	$\delta = 3.0$
			AANOS	AANOS	AANOS	AANOS	AANOS	AANOS	AANOS	AANOS
			SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS
200	0.0	(0.425, -1.383)	> 20,000	143.15	29.78	15.26	8.4	5.85	4.53	3.73
		(8.127, 138.354)	> 50,000	122.62	7.91	2.21	0.76	0.42	0.28	0.21
	0.2	(0.420, -1.363)	> 12,000	125.25	28.09	14.57	8.07	5.63	4.37	3.60
		(7.810, 106.061)	> 30,000	97.69	7.24	2.09	0.72	0.40	0.27	0.20
400	0.0	(0.225, -0.380)	> 15,000	73.4	29.72	18.66	11.57	8.43	6.67	5.54
		(13.817, 57.180)	> 35,000	19.84	3.35	1.42	0.63	0.39	0.27	0.21
	0.2	(0.225, -0.387)	> 10,000	68.66	28.09	17.67	10.97	8.00	6.33	5.27
		(13.050, 48.868)	> 20,000	18.10	3.15	1.34	0.6	0.36	0.26	0.20
600	0.0	(0.173, -0.087)	> 10,000	69.07	31.60	20.53	13.01	9.57	7.61	6.34
		(16.328, 43.587)	> 20,000	12.00	2.63	1.20	0.56	0.35	0.25	0.20
	0.2	(0.172, -0.090)	8249.89	64.59	29.78	19.39	12.31	9.06	7.20	6.01
		(15.400, 38.691)	> 10,000	11.04	2.46	1.13	0.53	0.33	0.24	0.19
1000	0.0	(0.170, -0.096)	6093.61	63.04	29.43	19.21	12.21	9.00	7.16	5.97
		(15.304, 35.889)	5582.16	8.03	1.87	0.86	0.40	0.25	0.18	0.14
	0.2	(0.172, -0.118)	4310.89	59.10	27.65	18.05	11.48	8.46	6.74	5.62
		(14.311, 32.640)	3537.03	7.48	1.75	0.81	0.38	0.24	0.17	0.13
2000	0.0	(0.170, -0.119)	3507.94	57.84	27.33	17.89	11.39	8.4	6.69	5.59
		(14.216, 31.037)	1792.44	5.04	1.21	0.56	0.26	0.17	0.12	0.09
	0.2	(0.171, -0.134)	2620.11	54.35	25.79	16.89	10.77	7.95	6.33	5.29
		(13.373, 28.487)	1229.3	4.69	1.14	0.53	0.25	0.16	0.11	0.09

fact that the EAANOS calculates a uniform average across the domain $[\delta_{\min}, \delta_{\max}]$, as opposed to the AAEQL, which calculates a quadratic weighted average across $[\delta_{\min}, \delta_{\max}]$. When the Phase-I sample sizes are moderate to large ($m \geq 600$), there is very little difference between the out-of-control performances of the SPRT chart with estimated process parameters for $\delta \geq 1.0$ and all combinations of p_e and ε . However, when m is small ($m = 200, 400$), the optimal charting parameters seem to deviate from the usual trend, leading to some discrepancies observed between the out-of-control performances of the SPRT chart for different values of δ .

Table 3.22 Optimal charting parameters (γ, g, h) , the minimum EAANOS, and the resulting AANOS and SDANOS values of the SPRT chart with estimated process parameters designed under the GICP framework for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, when $p_e = 0.10$, $\varepsilon \in \{0.0, 0.2\}$, $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = \bar{n} = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 3.0$, and $m \in \{200, 400, 600, 1000, 2000\}$.

m	ε	(γ, g) (h, EAANOS)	$\delta = 0.0$	$\delta = 0.4$	$\delta = 0.7$	$\delta = 1.0$	$\delta = 1.5$	$\delta = 2.0$	$\delta = 2.5$	$\delta = 3.0$
			AANOS	AANOS	AANOS	AANOS	AANOS	AANOS	AANOS	AANOS
			SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS	SDANOS
200	0.0	(0.335, -0.938)	> 15,000	95.72	27.20	15.50	9.07	6.46	5.06	4.19
		(9.647, 90.908)	> 50,000	64.36	5.67	1.95	0.76	0.44	0.30	0.23
	0.2	(0.315, -0.840)	> 10,000	84.69	26.15	15.23	9.02	6.46	5.07	4.21
		(9.769, 73.265)	> 40,000	51.20	5.17	1.85	0.74	0.43	0.30	0.23
400	0.0	(0.190, -0.178)	> 10,000	67.61	29.73	19.11	12.03	8.83	7.01	5.84
		(14.822, 45.835)	> 30,000	15.82	3.13	1.39	0.64	0.40	0.28	0.22
	0.2	(0.180, -0.060)	9299.64	63.94	28.82	18.66	11.80	8.68	6.90	5.75
		(15.373, 40.647)	> 20,000	14.25	2.98	1.34	0.62	0.39	0.28	0.22
600	0.0	(0.172, -0.092)	7608.12	63.74	29.42	19.16	12.17	8.96	7.12	5.94
		(15.213, 37.840)	>10,000	10.86	2.43	1.11	0.52	0.33	0.23	0.18
	0.2	(0.173, -0.109)	5180.75	59.67	27.67	18.03	11.46	8.44	6.72	5.61
		(14.264, 34.032)	6478.16	10.06	2.28	1.05	0.49	0.31	0.22	0.17
1000	0.0	(0.170, -0.105)	4362.88	59.27	27.82	18.18	11.57	8.53	6.79	5.67
		(14.450, 32.690)	3617.31	7.47	1.76	0.81	0.38	0.24	0.17	0.13
	0.2	(0.171, -0.121)	3235.53	55.86	26.32	17.22	10.96	8.09	6.44	5.38
		(13.633, 30.016)	2440.30	6.98	1.66	0.77	0.36	0.23	0.16	0.13
2000	0.0	(0.168, -0.112)	2921.18	55.67	26.5	17.38	11.08	8.18	6.52	5.44
		(13.825, 29.387)	1417.69	4.81	1.17	0.55	0.26	0.16	0.12	0.09
	0.2	(0.171, -0.141)	2207.80	52.33	24.92	16.34	10.42	7.69	6.13	5.13
		(12.913, 27.045)	985.47	4.48	1.10	0.51	0.24	0.15	0.11	0.09

Regarding the conditional performances of the SPRT chart, we produce boxplots of the conditional ANOS (CANOS) values. The specifications are set as $m = 200$, $\delta_{\min} = 0.1$, and $\delta_{\max} = 2.0$ for illustrative purposes. Figures 3.5 and 3.6 display boxplots of the CANOS distributions computed using the EAANOS-optimal SPRT chart with estimated process parameters. The naming convention of the design strategies is identical to those in Section 3.6.2, where Design 1 corresponds to the traditional approach (see Section 3.4.2 and Table 3.2), Design 2 corresponds to the AANOS-matching approach (see Section 3.5.2 and Table 3.4), and Designs 3 to 6 correspond to the GICP approach introduced in this section (see Tables 3.19 and 3.20). In Figure 3.5, the CANOS distributions correspond to the in-control case

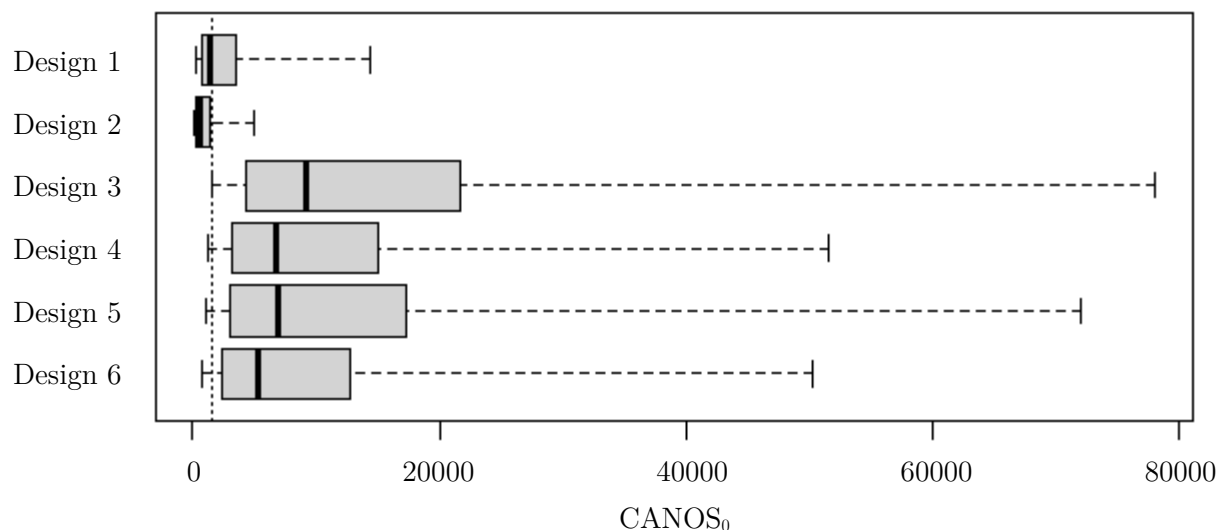


Figure 3.5 Boxplots of the $CANOS_0$ distributions for the SPRT chart designed under the traditional approach, AANOS-matching approach, and GICP approach with $p_e \in \{0.05, 0.10\}$ and $\varepsilon \in \{0.0, 0.2\}$.

($\delta = 0$), whereas in Figure 3.6, the $CANOS$ distributions correspond to the out-of-control case ($\delta = \{0.2, 0.4, 0.6, 0.8\}$). The vertical dotted line in Figure 3.5 indicates the recommended threshold $\tau_{ANOS} = 1500$.

From Figure 3.5, it is obvious that the in-control performance of the SPRT charts under Designs 1 and 2 are unsatisfactory, since at least half of the $CANOS_0$ values are smaller than the threshold 1500. After the GICP design is introduced (i.e., Designs 3 to 6), one observes substantial reduction in the number of high false alarm rates, although the improved in-control performance comes at the expense of an increased variability in the $CANOS_0$ distribution. Referring to Figure 3.6, it is found that the $CANOS_1$ distributions of the SPRT chart under Designs 3 to 6 resemble those of the SPRT chart under Designs 1 and 2 for $\delta \geq 0.8$ (see Figure 3.6(d)). This indicates that our proposed integrated optimal-GICP design is more effective than the standard GICP design practised by other researchers, who have reported better performances only for $\delta \geq 1.0$. Overall, the optimisation design is said to complement the GICP framework in reducing the trade-off between the in-control and out-of-control performances of the SPRT chart.

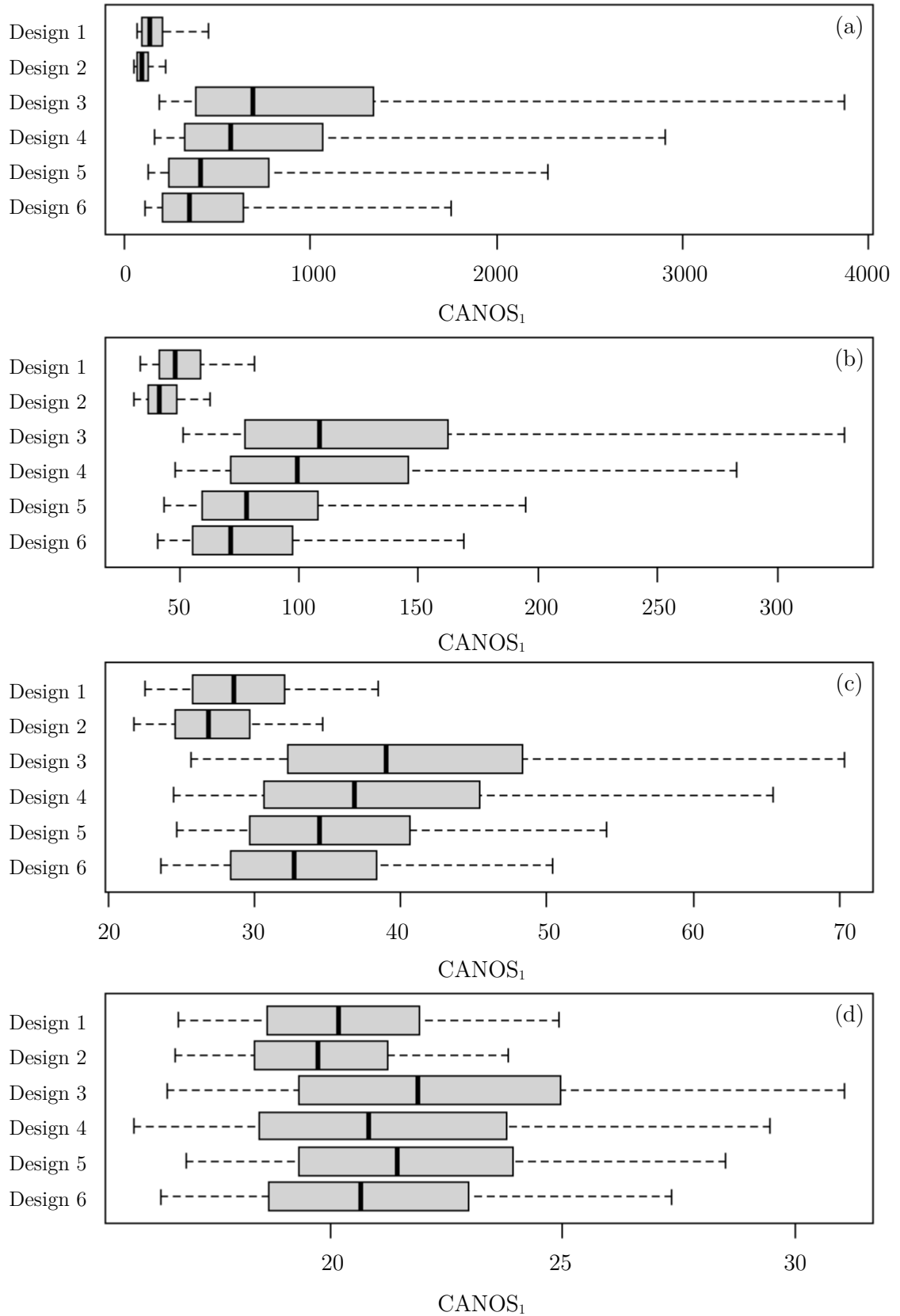


Figure 3.6 Boxplots of the $CANOS_1$ distributions for the SPRT chart designed under the traditional approach, AANOS-matching approach, and GICP approach with $p_e \in \{0.05, 0.10\}$ and $\varepsilon \in \{0.0, 0.2\}$, when (a) $\delta = 0.2$, (b) $\delta = 0.4$, (c) $\delta = 0.6$, and (d) $\delta = 0.8$.

3.7 A Comparison of the Performances between the SPRT Chart and the CUSUM Chart with Estimated Process Parameters

In this section, we compare the performances of the SPRT chart designed under the integrated optimal-GICP design to those of the CUSUM chart. It is worth noting that extensive comparative studies have been conducted by Stoumbos and Reynolds (1997a), Ou *et al.* (2012a, 2012c), all revealing that the SPRT chart has a superior detection performance over the Shewhart $\bar{X}$ and CUSUM charts. Using the AAELQ-optimal and EAANOS-optimal designs introduced in Section 3.6.2 and Section 3.6.4, this section aims to highlight the effectiveness of the GICP-adjusted optimal SPRT chart against its close competitor, i.e., the CUSUM chart.

To ensure that the comparison is fair, the CUSUM chart is designed with the same set of specifications, i.e., $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $d_{\min} = 0.2$, $\delta_{\min} = 0.1$, $\delta_{\max} = 2.0$, $m = 200$, $p_e = 0.05$, and $\varepsilon = 0$. Regarding the appropriate choice of sample size for the CUSUM chart, Ou *et al.* (2011a, 2012b) stated that the CUSUM chart with subgroup one (i.e., $n = 1$) has the best overall performance over a wide range of δ . Reynolds and Stoumbos (2004b) ascertained that the out-of-control performances of the CUSUM chart for different sample sizes n show very little difference when δ is small or moderate. However, when δ is large, the CUSUM chart is found to be the most effective when the sample size is set to be equal to one. Hence, in this comparison, the CUSUM chart for monitoring individual observations is used, and the complete details of the CUSUM chart can be found in Section 2.2.3.

The CUSUM chart has four charting parameters, i.e., control limit (H_{CUSUM}), reference value (γ_C), sample size (n), and sampling interval (d). Since we have $n = 1$ for the CUSUM chart with individual measurements, we must have $\text{ASN}_0 = 1$ (see Section 2.4.2) and $d = n/\text{IR} = 1/5 = 0.2$. This implies that the integrated optimal-GICP design for the CUSUM chart involves searching only two variables,

i.e., H_{CUSUM} and γ_C . Note that the control limit H_{CUSUM} can be adjusted to satisfy the EPC constraint (Equation (3.61b)) by Algorithm 1. This leaves γ_C as the only free variable in the optimisation design for the CUSUM chart with estimated process parameters. By running the model, the charting parameters of the AAEQL-optimal CUSUM chart with estimated process parameters are found as

$$\gamma_C = 0.416, H_{\text{CUSUM}} = 9.023, \text{ASN}_0 = 1, \text{ and } d = 0.2,$$

achieving a minimum AAEQL of 4.412. On the other hand, the charting parameters of the optimal SPRT chart can be quoted from Table 3.11 as $(\gamma, d, g, h) = (0.420, 0.450, -0.034, 9.283)$. Note that the minimum AAEQL achieved by the SPRT chart is 2.111, which is two times as small as that achieved by the optimal CUSUM chart.

Figures 3.7 and 3.8 show boxplots of the CATS distributions obtained from the AAEQL-optimal SPRT and CUSUM charts with charting parameters given in the preceding paragraph. The CATS distributions have been generated for the in-control case ($\delta = 0.0$) in Figure 3.7, and generated for four out-of-control cases ($\delta \in \{0.2, 0.4, 0.6, 0.8\}$) in Figure 3.8.

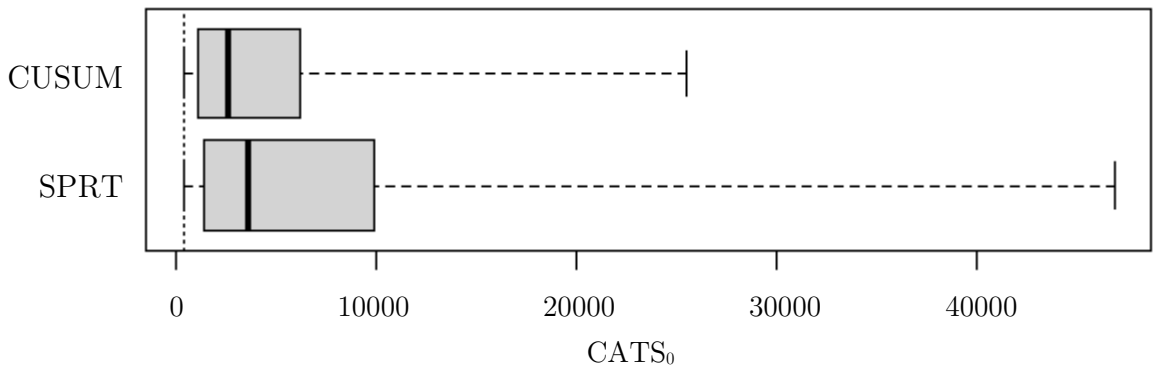


Figure 3.7 Boxplots of the CATS_0 distributions for the AAEQL-optimal CUSUM and SPRT charts designed under the GICP approach.

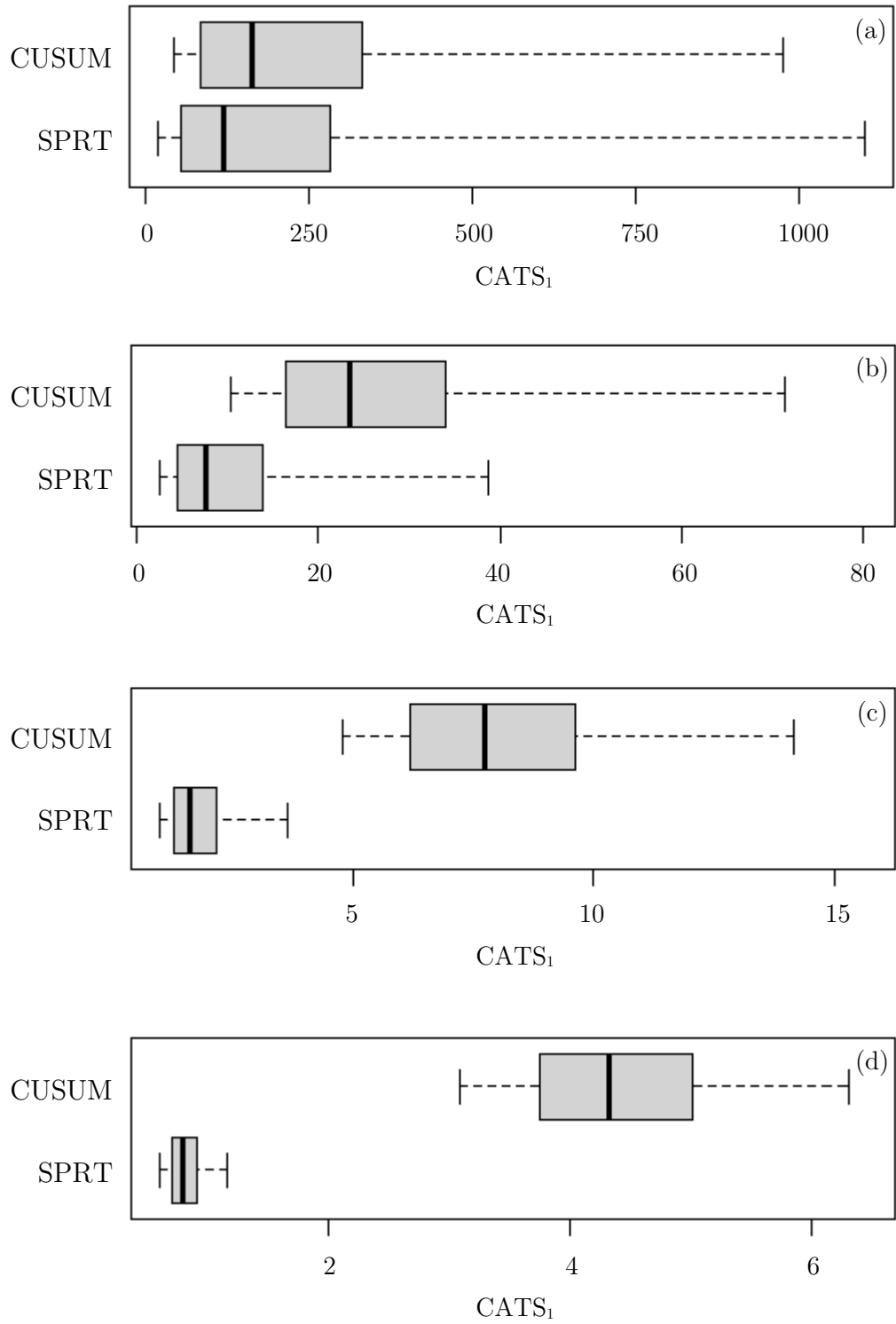


Figure 3.8 Boxplots of the $CATS_1$ distributions for the AAEL-optimal CUSUM and SPRT charts designed under the GICP approach when (a) $\delta = 0.2$, (b) $\delta = 0.4$, (c) $\delta = 0.6$, and (d) $\delta = 0.8$.

From Figure 3.7, both the SPRT and CUSUM charts perform equally well in terms of the conditional in-control performance, with the SPRT chart showing a more variable $CATS_0$ distribution compared to the CUSUM chart. However, the observation is not to be worried, since both the SPRT chart and CUSUM chart have achieved a satisfactory exceedance probability of 95%, i.e., 95% of the $CATS_0$ values are greater than the threshold τ_{ATS} , and by Remark 2.4.1, practitioners generally desire a large ATS_0 value. In the out-of-control case, one observes from Figure 3.8 that the variability of the $CATS_1$ distribution of SPRT chart decreases rapidly as δ increases from 0.2 to 0.8 (indicated by the shorter whiskers of the boxplots). In fact, the $CATS_1$ values of the SPRT chart decrease *faster* than those of the CUSUM chart as δ increases, indicating that SPRT chart outperforms the CUSUM chart in the case where both charts are designed under the integrated optimal-GICP model. It is therefore evident that the SPRT chart is favoured over the CUSUM chart from the aspect of detection sensitivity and speed.

Aside from the AAEQL-optimisation model, we also compare the performances of the SPRT and CUSUM control charts designed under the EAANOS-optimisation model. To ensure fairness in the comparisons, we fix the specifications as $\tau_{ANOS} = 1500$, $AASN_0 = \bar{n} = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 2.0$, $m = 200$, $p_e = 0.05$, and $\varepsilon = 0$ in both designs. Since $AASN_0 = 5$, the CUSUM chart must be operating on a sample size of $n = 5$ instead of $n = 1$ as in the AAEQL-optimal design. As d is redundant, the EAANOS-optimisation design involves searching only two charting parameters γ_C and H_{CUSUM} . By minimising the EAANOS value of the CUSUM chart with estimated process parameters, the charting parameters are calculated as

$$\gamma_C = 1.050 \text{ and } H_{CUSUM} = 2.811,$$

achieving a minimum EAANOS of 210.146. The reader may refer to the works of Saleh *et al.* (2016) and Diko *et al.* (2019b) for a full description of the CUSUM

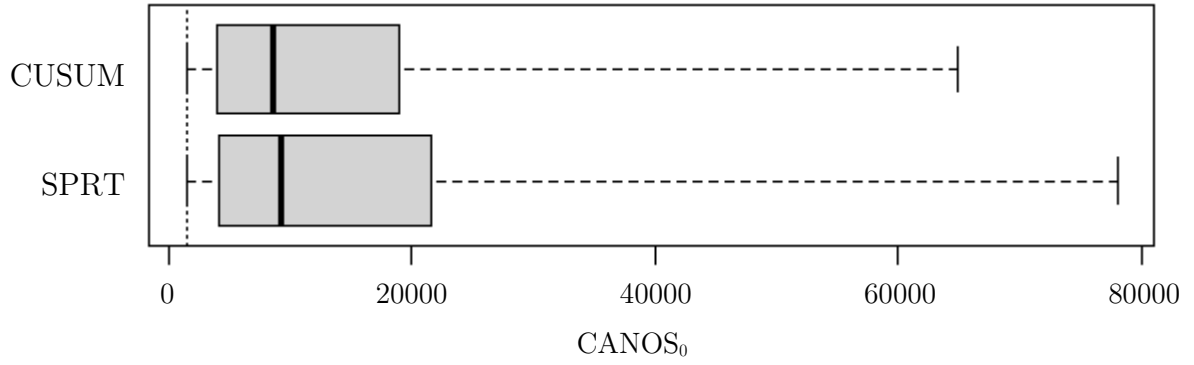


Figure 3.9 Boxplots of the $CANOS_0$ distributions for the EAANOS-optimal CUSUM and SPRT charts designed under the GICP approach.

chart with estimated process parameters. From Table 3.19, the charting parameters of the EAANOS-optimal SPRT chart are obtained as $(\gamma, g, h) = (0.422, -1.369, 8.156)$, with a minimum EAANOS equal to 204.644. Note that the minimum EAANOS achieved by the SPRT chart is lower than that of the CUSUM charts.

Figures 3.9 and 3.10 contain boxplots of the $CANOS$ distributions generated using the EAANOS-optimal SPRT and CUSUM charts with charting parameters specified above. The $CANOS$ distributions have been illustrated for the in-control case ($\delta = 0.0$) in Figure 3.9, and four out-of-control cases ($\delta \in \{0.2, 0.4, 0.6, 0.8\}$) in Figure 3.10. From Figure 3.9, it is again evident that the GICP-adjusted SPRT chart has a more variable $CANOS_0$ distribution compared to those of the GICP-adjusted CUSUM chart. In the out-of-control case, however, one notices that both the GICP-adjusted SPRT and CUSUM charts perform equally well for all shift sizes, although it may be argued that the SPRT chart has a slightly lower median $CANOS_1$ than the CUSUM chart (see Figure 3.10). It is also observed that the GICP-adjusted SPRT chart has a less variable $CANOS_1$ distribution compared to that of the CUSUM chart, as shown by the shorter whiskers in the boxplots. This implies that, although the SPRT chart has a more volatile in-control performance, it is more robust than the CUSUM chart against adverse effects of the GICP design in terms of the out-of-control sensitivity. To conclude, the SPRT chart is preferred over the CUSUM chart due to its superior detection sensitivity.

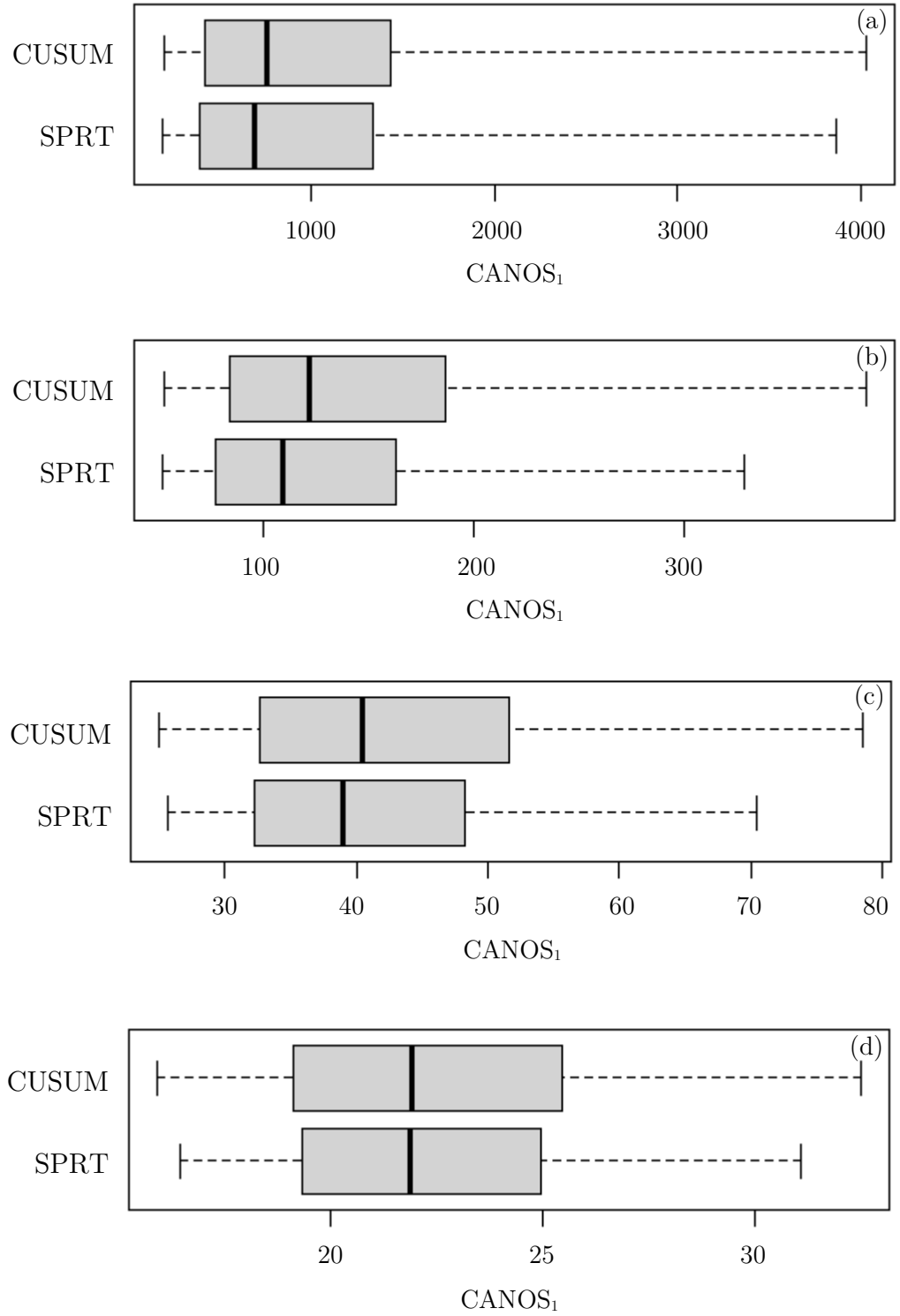


Figure 3.10 Boxplots of the $CANOS_1$ distributions for the EAANOS-optimal CUSUM and SPRT charts designed under the GICP approach when (a) $\delta = 0.2$, (b) $\delta = 0.4$, (c) $\delta = 0.6$, and (d) $\delta = 0.8$.

3.8 Real Industrial Examples

In this section, two industrial examples are provided to illustrate the implementation of the optimal SPRT chart with estimated process parameters proposed in Sections 3.6.2 and 3.6.4. In Section 3.8.1, an application for monitoring the mean resistivity of epitaxial wafer substrates is demonstrated using the AAQL-optimal SPRT chart with GICP-adjusted limits. In Section 3.8.2, an application for monitoring the mean loop height of a wire bonding process is demonstrated using the EAANOS-optimal SPRT chart with GICP adjusted limits.

3.8.1 Epitaxial Process of Silicon Wafers

In the manufacturing industry, silicon epitaxial wafers are commonly used in part of the fabrication of integrated circuits. Meticulous control of the epitaxial resistivity is essential to ensure that the quality of the end products is satisfactory. During the epitaxial process, the silicon layers may have a higher resistivity in the case where a low N-type dopant is deposited, and a lower resistivity in the case where a high N-type dopant is deposited. Therefore, it is crucial to monitor the epitaxial resistivity of silicon wafers using both an upper-sided and a lower-sided SPRT chart to detect potential shifts in either direction.

The raw data used in this study have been obtained from an N-type epitaxial process. Prior to the implementation of the Phase-II SPRT chart, it is necessary to first collect a set of Phase-I data for the purpose of parameter estimation. In this example, $m = 200$ Phase-I observations ($Y_1, Y_2, \dots, Y_{200}$) have been collected. Note that the resistivity measurements are expressed in units of ohm-centimetre (ohm-cm). The Phase-I samples are carefully examined using the average moving range (AMR) control chart for individual measurements. Roes *et al.* (1993) stated that using a single AMR chart is sufficient for checking the stability of individual

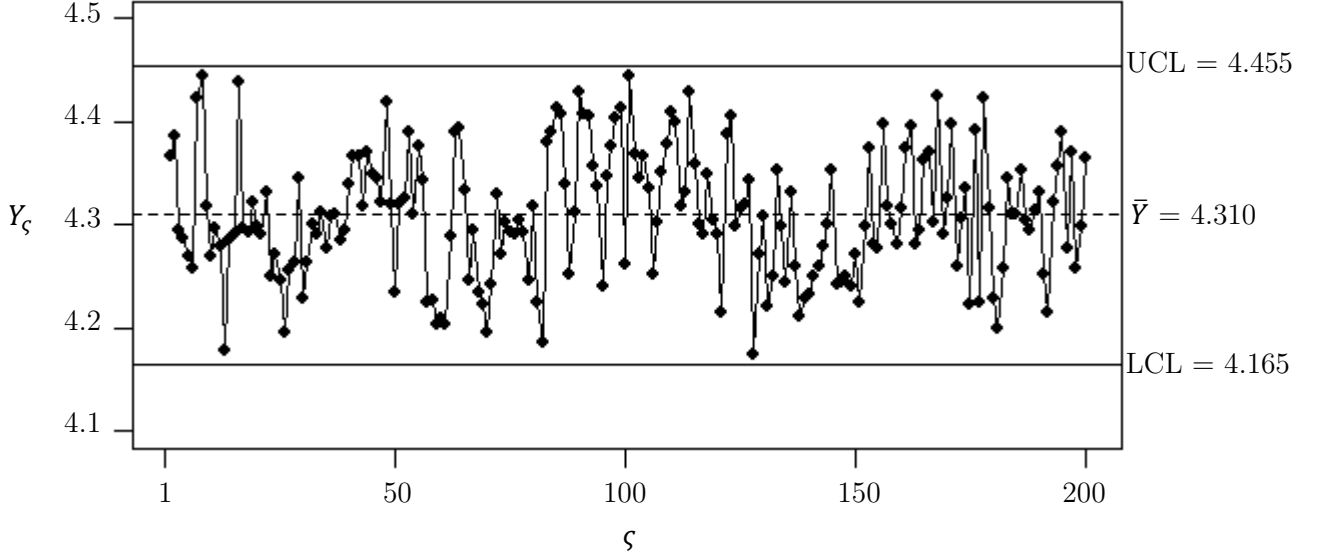


Figure 3.11 The Phase-I AMR chart for $m = 200$ individual resistivity measurements.

measurements with respect to the mean and variability, since the moving range chart adds no real advantage to the sensitivity of the scheme. Therefore, in this illustration, only a single AMR chart is used in the Phase-I analysis.

Figure 3.11 shows the AMR control chart for monitoring the 200 Phase-I measurements. The upper (UCL) and lower (LCL) control limits of the AMR chart are determined by $\bar{Y} \pm (3\sqrt{\pi}/2) \overline{MR} = (4.165, 4.455)$, where $\bar{Y} = \sum_{\varsigma=1}^{200} Y_{\varsigma} / 200 = 4.310$ ohm-cm is the sample mean and $\overline{MR} = \sum_{\varsigma=2}^{200} |Y_{\varsigma} - Y_{\varsigma-1}| / 199 = 0.055$ ohm-cm is the sample average of the moving ranges. From Figure 3.11, it is clear that all the Phase-I measurements come from an in-control source. Hence, the Phase-I estimates $\hat{\mu}_0$ and $\hat{\sigma}_0$ of the epitaxial resistivity can be computed from Equations (3.25) and (3.26) as 4.310 ohm-cm and 0.061 ohm-cm, respectively.

Suppose that a wafer manufacturing company is currently inspecting five silicon wafers every three hours. Letting $d = 1$ correspond to a duration of three hours, the inspection rate practised by the manufacturing company is calculated as $IR = 5/1 = 5$. Using the specifications $\tau_{ATS} = 370.4$, $IR = 5$, $d_{\min} = 0.2$, $\delta_{\min} = 0.1$, $\delta_{\max} = 2.0$, $p_e = 0.05$ and $\varepsilon = 0$, the charting parameters of the upper one-sided

AAEQL-optimal SPRT chart with GICP-adjusted control limits can be quoted from Table 3.11 as $(\text{AASN}_0, \gamma, d, g, h) = (2.250, 0.420, 0.450, -0.034, 9.283)$. The new sampling interval of the optimal SPRT chart with adjusted control limits is $d = 0.450$, which is equivalent to 81 minutes. For the convenience of execution, the sampling interval is rounded to 80 minutes in this example.

Recall from Remark 2.3.1 that, under the same in-control conditions, the lower-sided counterpart of the SPRT chart would be symmetric to the upper-sided counterpart, i.e., the acceptance limit and rejection limit are $-g$ and $-h$, respectively. As a general practice, we treat both an upper-sided mean shift and a lower-sided mean shift as being equally likely to occur. This practice is common in other control chart applications, see for example, the Shewhart $\bar{X}$ chart and the EWMA chart, which have UCL and LCL symmetric about the centre line. Note, however, that this design is generally not applicable when there is strong evidence of a very skewed data (which is outside the scope of this thesis).

The decision rule of the upper- and lower-sided SPRT charts are described as follows: At any sampling point i ($i = 1, 2, \dots$), for $j = 1, 2, \dots$,

- sampling is terminated and the process is declared as in-control, if
 - both $\hat{U}_{i,j}^+ \in (-\infty, g]$ and $\hat{U}_{i,j}^- \in [-g, +\infty)$, or
 - $\hat{U}_{i,j}^+ \in (-\infty, g]$, given that $\hat{U}_{i,t}^- \in [-g, +\infty)$ at some $t < j$, or
 - $\hat{U}_{i,j}^- \in [-g, +\infty)$, given that $\hat{U}_{i,t}^+ \in (-\infty, g]$ at some $t < j$.
- sampling is terminated and the process is declared as out-of-control, if
 - $\hat{U}_{i,j}^+ \in [h, +\infty)$, regardless of the value of $\hat{U}_{i,j}^-$, or
 - $\hat{U}_{i,j}^- \in (-\infty, -h]$, regardless of the value of $\hat{U}_{i,j}^+$,
- sampling continues as long as none of the conditions above are fulfilled.

Table 3.23 shows the recursively computed upper- and lower-sided Phase-II control statistics of the AAEQL-optimal SPRT chart. Note that the boldfaced

Table 3.23 Phase-II summary statistics for monitoring resistivity measurements using the AAEQL-optimal SPRT chart with GICP-adjusted control limits.

Sampling time	Sample index, i	Sample number, j	$\hat{U}_{i,j}^+$	$\hat{U}_{i,j}^-$	Conclusion
08:00	—	—	—	—	—
09:20	1	1	−0.827	0.013	*
		2	*	1.724	In-control
10:40	2	1	−0.030	0.810	*
		2	−0.581	*	In-control
12:00	3	1	−0.764	0.076	In-control
13:20	4	1	0.218	1.058	*
		2	1.161	*	*
		3	3.189	*	*
		4	2.795	*	*
		5	4.781	*	*
		6	4.003	*	*
		7	5.040	*	*
		8	7.969	*	*
		9	8.314	*	*
		10	8.025	*	*
		11	8.397	*	*
		12	10.554	*	Out-of-control

NOTE: The boldfaced values in **blue** indicate that the statistic falls in the acceptance region, whereas the boldfaced value in **red** indicates that the statistic falls in the rejection region.

values in blue represent the control statistics that fall in the acceptance region, whereas the boldfaced values in red represent the control statistics that fall in the rejection region. Two plots depicting the implementation of the SPRT chart are given in Figure 3.12, with the vertical and horizontal axes revealing the control statistics and precise timestamps, respectively. Note that the first point of each SPRT represents the initial value of the control statistic, i.e., $\hat{U}_{i,0}^+ = \hat{U}_{i,0}^- = 0$, whereas the subsequent points represent the control statistic computed from Equations (3.27) and (3.28) until the SPRT reaches a conclusion. At time 08:00, the SPRT chart is first initialised with an 80-minute period for machinery warmup.

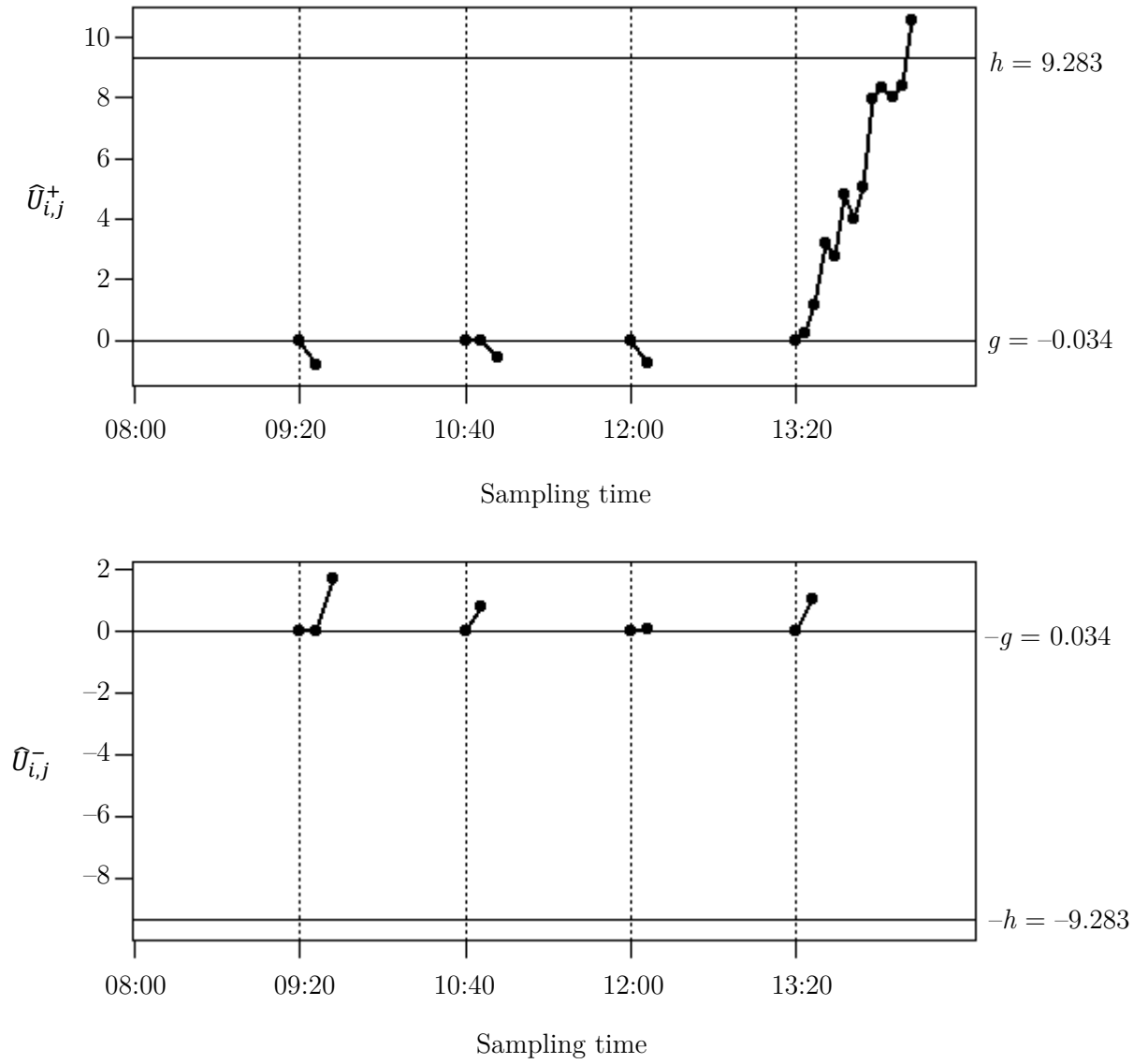


Figure 3.12 The Phase-II upper- and lower-sided SPRT charts for monitoring resistivity measurements of epitaxial wafers.

In the following three sampling points (i.e., 09:20, 10:40, 12:00), the upper- and lower-sided SPRT charts both indicate that the process is in-control. At time 13:20, the fourth SPRT samples a total number of 12 observations before emitting an out-of-control signal on the upper-sided chart, suggesting that there is a probable upward shift in the mean of the epitaxial resistivity. The alarm prompts practitioners to swiftly conduct an investigation on the issue, and the time to signal is reported to be less than 5 hours and 20 minutes.

3.8.2 Wire Bonding Process

Wire bonding is a crucial process in the semiconductor industry for establishing interconnections between substrates and integrated circuits. An incorrect loop geometry can lead to wire loop heights that are either too high or too low, which in turn affect the reliability of the bonding process. It is, hence, important to monitor the loop height measurements of the wire bonding process to ensure that the wire formation is optimum.

Similar to Section 3.8.1, $m = 200$ Phase-I loop height measurements ($Y_1, \dots, Y_{200}$) have been collected for the purpose of Phase-I parameter estimation. The loop height readings are measured in one-thousandth of an inch (mils). To verify the in-control status of the Phase-I measurements, the Phase-I AMR chart have been used. The UCL and LCL of the AMR chart are computed as $\bar{Y} \pm (3\sqrt{\pi}/2) \overline{MR} = (6.382, 8.878)$, where $\bar{Y} = \sum_{\zeta=1}^{200} Y_{\zeta}/200 = 7.630$ mils and $\overline{MR} = \sum_{\zeta=2}^{200} |Y_{\zeta} - Y_{\zeta-1}|/199 = 0.469$ mils. Figure 3.13 displays the Phase-I AMR chart for monitoring the 200 Phase-I loop height measurements. From the figure, it is clear that all the samples

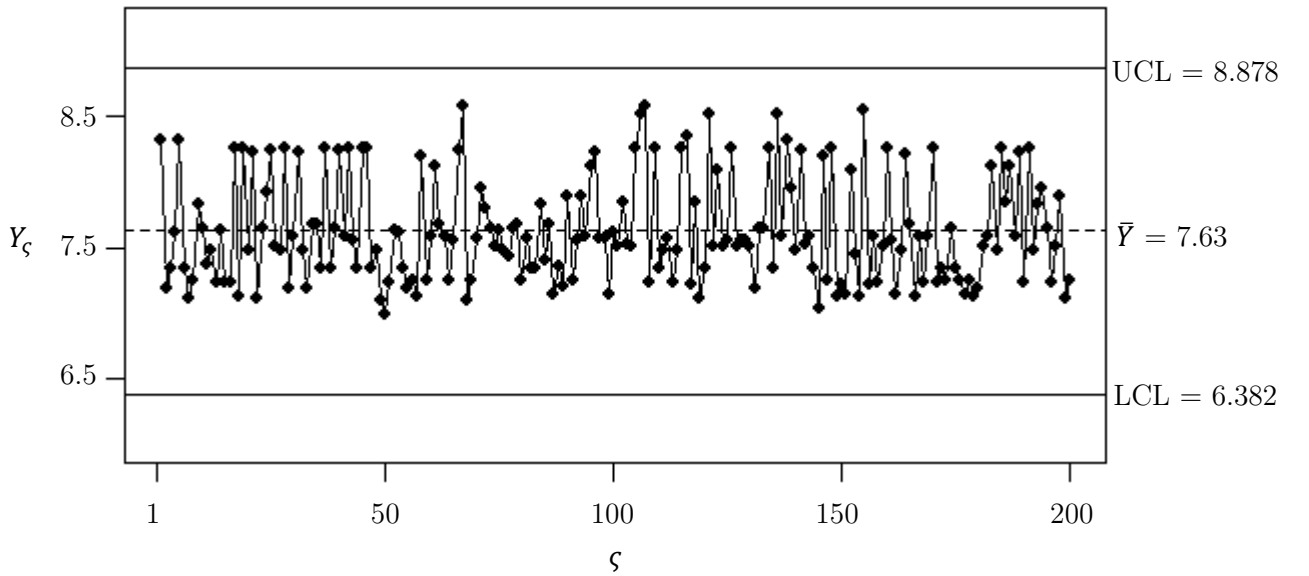


Figure 3.13 The Phase-I AMR chart for $m = 200$ individual loop height measurements.

are in-control, and the process parameters are estimated from Equations (3.25) and (3.26) as $\hat{\mu}_0 = 7.630$ mils and $\hat{\sigma}_0 = 0.402$ mils, respectively.

Suppose that the semiconductor manufacturing company expects to sample five loop height measurements on average at each sampling point. Using the specifications $\tau_{\text{ANOS}} = 1500$, $\text{AASN}_0 = 5$, $\delta_{\min} = 0.1$, $\delta_{\max} = 2.0$, $p_e = 0.05$ and $\varepsilon = 0$, the charting parameters of the EAANOS-optimal upper one-sided SPRT chart with GICP-adjusted control limits can be quoted from Table 3.19 as $(\gamma, g, h) = (0.422, -1.369, 8.156)$. Note that the lower one-sided SPRT chart is symmetric to its upper-sided counterpart. Since the sampling interval is not relevant in this design, the sample index i is referred to instead of the sampling time.

Table 3.24 shows the upper-sided and lower-sided Phase-II control statistics of the EAANOS-optimal SPRT chart. The boldfaced values in blue represent the control statistics that fall in the acceptance region, whereas the boldfaced values in red represent the control statistics that fall in the rejection region. The upper-sided and lower-sided SPRT chart are depicted in Figure 3.14. From the figure, one observes that the loop height measurements are declared as in-control in the first five SPRTs. The numbers of measurements taken in both the upper- and lower-sided SPRT charts are equal to 10, 4, 5, 4, and 6, for the first five sampling points, respectively. At the sixth sampling point, the upper-sided control statistic rises above h after 13 measurements are sought, which lead to an out-of-control alarm being produced. Investigative plans are initiated immediately to determine the cause of the shift. The number of observations to signal is then reported as $10 + 4 + 5 + 4 + 6 + 13 = 42$.

Table 3.24 Phase-II summary statistics for monitoring loop height measurements using the EAANOS-optimal SPRT chart with GICP-adjusted control limits.

Sample index, i	Sample number, j	$\hat{U}_{i,j}^+$	$\hat{U}_{i,j}^-$	Conclusion
1	1	-1.368	0.524	*
	2	-2.339	0.651	*
	3	*	0.902	*
	4	*	-0.315	*
	5	*	0.309	*
	6	*	0.435	*
	7	*	0.014	*
	8	*	0.663	*
	9	*	-0.380	*
	10	*	-1.622	In-control
2	1	-0.647	-0.197	*
	2	-1.070	-0.618	*
	3	-0.946	-1.586	*
	4	-2.364	*	In-control
3	1	-0.373	-0.471	*
	2	-1.244	-0.444	*
	3	-1.618	-0.914	*
	4	*	-1.012	*
	5	*	-1.458	In-control
4	1	-0.672	-0.172	*
	2	-1.642	-0.046	*
	3	*	-0.467	*
	4	*	-1.435	In-control
5	1	0.995	-1.839	*
	2	-0.100	*	*
	3	1.218	*	*
	4	0.696	*	*
	5	0.297	*	*
	6	-1.444	*	In-control
6	1	2.114	-2.958	*
	2	2.661	*	*
	3	3.507	*	*
	4	5.621	*	*
	5	6.367	*	*
	6	5.994	*	*
	7	5.198	*	*
	8	4.551	*	*
	9	3.953	*	*
	10	3.157	*	*
	11	5.023	*	*
	12	6.142	*	*
	13	8.207	*	Out-of-control

NOTE: The boldfaced values in **blue** indicate that the statistic falls in the acceptance region, whereas the boldfaced value in **red** indicates that the statistic falls in the rejection region.

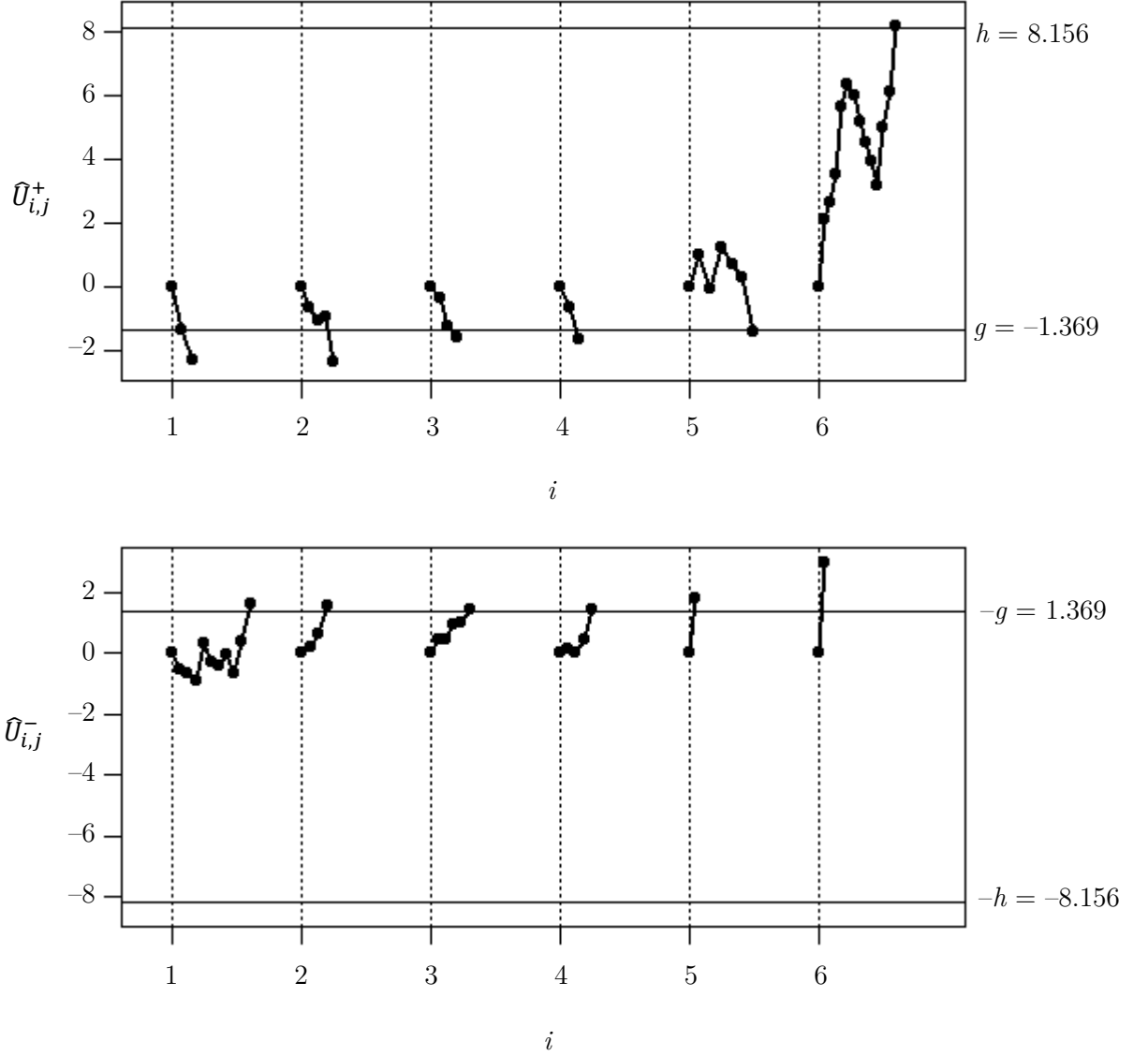


Figure 3.14 The Phase-II upper- and lower-sided SPRT charts for monitoring loop height measurements of the wire bonding process.

3.9 Conclusions

This chapter details the full theoretical framework for the SPRT chart in cases where process parameters are known and estimated. The Markov chain approach has been used to derive the essential run length properties of the SPRT

chart, such as the ASN, ARL, SDRL, ATS, SDTS, ANOS, etc. We reveal that, when process parameters are estimated, these aforementioned properties become conditional functions of the parameter estimates. It is useful to quantify the average values of these properties across various different parameter estimates, thus giving rise to *unconditional* properties such as the AASN, AARL, ASDRL, AATS, ASDTS, AANOS, etc. To measure practitioner-to-practitioner variability, we suggest calculating the standard deviation of the conditional properties, i.e., SDARL, SDATS, SDANOS.

To develop enhanced designs for the SPRT chart with estimated process parameters, we propose the use of four different performance metrics, i.e., the AATS, AAEQL, AANOS, and EAANOS, as the objective function for the optimisation model. We have provided explanations to why these metrics have been chosen in the thesis, and how they are relevant in achieving the appropriate industrial goals. In this chapter, we propose a design that marries the state-of-the-art guaranteed in-control performance (GICP) approach with the concept of optimisation, which we refer to as the integrated optimal-GICP design. The proposed design is shown to effectively mitigate issues with excessive false alarms that are present in other older models (i.e., the traditional design and the unconditional metric matching design). While many researchers have reported considerable deterioration in the out-of-control performance of GICP-designed control charts for $\delta < 1$, our proposed integrated optimal-GICP design has only seen concern-worthy deteriorations for $\delta < 0.6$. This proves that the integrated optimal-GICP design is very much robust against trade-offs between the in-control and out-of-control performances of control charts due to the GICP alone. To demonstrate the ease of implementation of the SPRT chart in SQC applications, we have illustrated two real applications involving an epitaxial process and a wire pull test. All the charting parameters tabulated in this chapter are ready to use and can be reproduced via the computer programmes published on GitHub.

4 The SPRT Chart for Joint Monitoring of Process Mean and Dispersion

4.1 Introduction

In this chapter, we develop a new single-chart SPRT scheme, which we name as the omnibus SPRT (OSPRT) scheme, for joint monitoring of both the mean and dispersion of a process. The word “omnibus” has the meaning of *collective*, *comprehensive*, and *inclusive*. As stated in Sections 2.3.3 and 2.3.4, the SPRT mean chart and SPRT dispersion chart have been constructed through first principles by Stoumbos and Reynolds (1997a) and Godase and Mahadik (2019), respectively. However, the literature lacks a well-justified SPRT chart for joint monitoring of both the mean and dispersion of a process, noting that the only single-chart joint monitoring SPRT scheme, i.e., the ABS-SPRT chart, is simply an “absolute-value” version of the SPRT mean chart. In this chapter, we derive the full theoretical framework of the OSPRT chart from first principles, and subsequently derive the run length properties of the OSPRT chart through the Markov chain approach (see Section 3.2). We also compare the performances of our proposed OSPRT chart to those of the ABS-SPRT chart, as well as other joint monitoring control charts from the same family (i.e., CUSUM charts). Two real applications of the OSPRT chart with industrial data are subsequently presented in the final section of this chapter.

4.2 The Proposed Omnibus SPRT (OSPRT) Chart

4.2.1 Theoretical Framework and Working Mechanism of the OSPRT

Chart

Let X denote the quality characteristic of a normally distributed process with unknown mean μ and unknown standard deviation σ . In the process control regimen, practitioners are often interested in testing the hypotheses $H_0: (\mu, \sigma) = (\mu_0, \sigma_0)$ against $H_1: (\mu, \sigma) = (\mu_1, \sigma_1)$, where (μ_0, σ_0) is the vector of the target mean and standard deviation of the process, and (μ_1, σ_1) is the vector of the shifted mean and standard deviation. To formulate a more tractable problem, it is conventional to express the alternative hypothesis as $H_1: (\mu, \sigma) = (\mu_0 + \delta\sigma_0, \eta\sigma_0)$, where δ and η represent the magnitudes of the mean and standard deviation shifts, respectively. Intuitively, δ can be thought of as a “drift” trigger of a control chart, and η can be interpreted as a “scale” trigger.

Wald (1947) defines the statistic of the SPRT as the log-likelihood ratio of a sequence of observed data evaluated under H_1 and H_0 . Suppose that the observations $X_1, X_2, \dots$ are taken sequentially from a process. Recall from Section 2.3.1 that, for an integer $j \geq 1$, the log-likelihood ratio for the sample $(X_1, \dots, X_j)$ has the expression

$$L_j = \ln \left[\prod_{j'=1}^j \frac{f(X_{j'}|H_1)}{f(X_{j'}|H_0)} \right] = \sum_{j'=1}^j \ln \left[\frac{f(X_{j'}|H_1)}{f(X_{j'}|H_0)} \right], \quad (4.1)$$

where $f(x|H_0)$ and $f(x|H_1)$ are the pdfs of the quality variable X evaluated under the hypotheses H_0 and H_1 , respectively. Since X follows the normal distribution, it is convenient to write Equation (4.1) as

$$L_j = \sum_{j'=1}^j \ln \left\{ \frac{\frac{1}{\sqrt{2\pi}\sigma_1} \exp \left[-\frac{(X_{j'} - \mu_1)^2}{2\sigma_1^2} \right]}{\frac{1}{\sqrt{2\pi}\sigma_0} \exp \left[-\frac{(X_{j'} - \mu_0)^2}{2\sigma_0^2} \right]} \right\}. \quad (4.2)$$

Note that some algebraic effort is required to express Equation (4.2) in a neat, elegant form. By using the properties of a logarithmic function and factorisation techniques, L_j can be simplified as

$$\begin{aligned}
L_j &= \sum_{j'=1}^j \ln \left\{ \frac{\frac{1}{\sqrt{2\pi}\sigma_1} \exp \left[-\frac{(X_{j'} - \mu_1)^2}{2\sigma_1^2} \right]}{\frac{1}{\sqrt{2\pi}\sigma_0} \exp \left[-\frac{(X_{j'} - \mu_0)^2}{2\sigma_0^2} \right]} \right\} \\
&= \sum_{j'=1}^j \left\{ \ln \left(\frac{\sigma_0}{\sigma_1} \right) + \frac{(X_{j'} - \mu_0)^2}{2\sigma_0^2} - \frac{(X_{j'} - \mu_1)^2}{2\sigma_1^2} \right\} \\
&= \sum_{j'=1}^j \left\{ \ln \left(\frac{\sigma_0}{\sigma_1} \right) + \frac{(\sigma_1 X_{j'} - \sigma_1 \mu_0)^2 - (\sigma_0 X_{j'} - \sigma_0 \mu_1)^2}{2\sigma_0^2 \sigma_1^2} \right\} \\
&= \sum_{j'=1}^j \left\{ \ln \left(\frac{\sigma_0}{\sigma_1} \right) + \frac{[(\sigma_1 - \sigma_0)X_{j'} + \sigma_0 \mu_1 - \sigma_1 \mu_0][(\sigma_1 + \sigma_0)X_{j'} - (\sigma_0 \mu_1 + \sigma_1 \mu_0)]}{2\sigma_0^2 \sigma_1^2} \right\} \\
&= \sum_{j'=1}^j \left\{ \ln \left(\frac{\sigma_0}{\sigma_1} \right) + \frac{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}{2\sigma_0^2 \sigma_1^2} \left(X_{j'} + \frac{\sigma_0 \mu_1 - \sigma_1 \mu_0}{\sigma_1 - \sigma_0} \right) \left(X_{j'} - \frac{\sigma_0 \mu_1 + \sigma_1 \mu_0}{\sigma_1 + \sigma_0} \right) \right\} \\
&= \sum_{j'=1}^j \left\{ \ln \left(\frac{\sigma_0}{\sigma_1} \right) + \frac{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}{2\sigma_0^2 \sigma_1^2} \left[X_{j'} - \mu_0 + \frac{(\mu_1 - \mu_0)\sigma_0}{\sigma_1 - \sigma_0} \right] \left[X_{j'} - \mu_0 - \frac{(\mu_1 - \mu_0)\sigma_0}{\sigma_1 + \sigma_0} \right] \right\} \\
&= \sum_{j'=1}^j \left\{ \ln \left(\frac{\sigma_0}{\sigma_1} \right) + \frac{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}{2\sigma_1^2} \left(\frac{X_{j'} - \mu_0}{\sigma_0} + \frac{\mu_1 - \mu_0}{\sigma_1 - \sigma_0} \right) \left(\frac{X_{j'} - \mu_0}{\sigma_0} - \frac{\mu_1 - \mu_0}{\sigma_1 + \sigma_0} \right) \right\}. \tag{4.3}
\end{aligned}$$

Define a standardised variable $Z_{j'} = (X_{j'} - \mu_0)/\sigma_0$. By completing the square and factoring, Equation (4.3) can be expressed as

$$\begin{aligned}
L_j &= \sum_{j'=1}^j \left\{ \ln \left(\frac{\sigma_0}{\sigma_1} \right) + \frac{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}{2\sigma_1^2} \left(Z_{j'} + \frac{\mu_1 - \mu_0}{\sigma_1 - \sigma_0} \right) \left(Z_{j'} - \frac{\mu_1 - \mu_0}{\sigma_1 + \sigma_0} \right) \right\} \\
&= \sum_{j'=1}^j \left\{ \ln \left(\frac{\sigma_0}{\sigma_1} \right) + \frac{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}{2\sigma_1^2} \left[\left(Z_{j'} + \frac{\sigma_0(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 - \left(\frac{\sigma_1(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 \right] \right\} \\
&= \frac{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}{2\sigma_1^2} \sum_{j'=1}^j \left\{ \left(Z_{j'} + \frac{\sigma_0(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 - \left[\frac{\sigma_1(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right]^2 - \right. \\
&\quad \left. \frac{2\sigma_1^2(\ln(\sigma_1/\sigma_0))}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right\}. \tag{4.4}
\end{aligned}$$

It may not be immediately obvious that Equation (4.4) is the most simplified form of the log-likelihood function L_j . By defining the following constants

$$c = \frac{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}{2\sigma_1^2}, \quad (4.5a)$$

$$k = \frac{\sigma_0(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}, \quad (4.5b)$$

and

$$\gamma = \left[\frac{\sigma_1(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right]^2 + \frac{2\sigma_1^2(\ln \sigma_1 - \ln \sigma_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}, \quad (4.5c)$$

Equation (4.4) can be written as

$$L_j = c \sum_{j'=1}^j \left[(Z_{j'} + k)^2 - \gamma \right]. \quad (4.6)$$

To express c , k , and γ in terms of δ and η only, we substitute $\mu_1 = \mu_0 + \delta\sigma_0$ and $\sigma_1 = \eta\sigma_0$ into Equations (4.5a) to (4.5c) to obtain the following:

$$c = \frac{1}{2} \left(1 - \frac{1}{\eta^2} \right), \quad (4.7a)$$

$$k = \frac{\delta}{\eta^2 - 1}, \quad (4.7b)$$

and

$$\gamma = \left(\frac{\delta\eta}{\eta^2 - 1} \right)^2 + \frac{2\eta^2 \ln \eta}{\eta^2 - 1}. \quad (4.7c)$$

Note that the constants represented by Equations (4.7a) to (4.7c) have rather meaningful interpretations. For instance, if one expects an increase in the process variance, then the constant c takes a positive value, hence the control procedure takes positive sums of $(Z_{j'} + k)^2 - \gamma$ for $j' = 1, \dots, j$. However, if the process variance is expected to decrease, the constant c becomes negative, and hence the procedure takes negative sums of $(Z_{j'} + k)^2 - \gamma$. Generally, the sign of c plays a major role in determining how the data should be accumulated (see Mahadik *et al.*, 2021). However, its value has little significance on the construction of the OSPRT procedure, and may be omitted without affecting the operation of the control chart.

Suppose now that the sequences of observations $(X_{1,1}, \dots, X_{1,N_1}), (X_{2,1}, \dots, X_{2,N_2}), \dots$ are taken at the sampling points $i = 1, 2, \dots$. Assume further that the measurements are sampled independently within a negligible time. Define the standardised measurement $Z_{i,j} = (X_{i,j} - \mu_0)/\sigma_0$. The control statistic, $U_{i,j}$, of the OSPRT chart is calculated recursively as

$$U_{i,j} = U_{i,j-1} + \text{sgn}(\eta - 1) \left[(Z_{i,j} + k)^2 - \gamma \right], \quad (4.8)$$

for $i = 1, 2, \dots, j = 1, 2, \dots, N_i$, where N_i is the sample number of the i -th SPRT, and $U_{i,0} = 0$ is the initial value of the control statistic. Here, $\text{sgn}(\cdot)$ is the sign function, and k and γ are known as the reference parameters of the OSPRT chart, whose values can be calculated from Equations (4.7b) and (4.7c), respectively. It is clear that $\gamma > 0$ for all values of $(\delta, \eta) \neq (0, 1)$, whereas the sign of k depends on the signs of δ and η . When $\eta > 1$, k shares the same sign as δ , i.e., $k\delta > 0$; when $0 < \eta < 1$, k takes the opposite sign of δ , i.e., $k\delta < 0$. Suppose that an upper-sided variance shift (i.e., $\eta > 1$) is of interest, the control statistics for the OSPRT chart for detecting an increasing mean shift ($U_{i,j}^{++}$) and a decreasing mean shift ($U_{i,j}^{-+}$) are given as

$$U_{i,j}^{++} = U_{i,j-1}^{++} + (Z_{i,j} + |k|)^2 - \gamma \quad (4.9a)$$

and

$$U_{i,j}^{-+} = U_{i,j-1}^{-+} + (Z_{i,j} - |k|)^2 - \gamma, \quad (4.9b)$$

respectively. If a lower-sided variance shift (i.e., $0 < \eta < 1$) is of interest instead, the control statistics for the OSPRT chart for detecting an increasing mean shift ($U_{i,j}^{+-}$) and a decreasing mean shift ($U_{i,j}^{--}$) are given as

$$U_{i,j}^{+-} = U_{i,j-1}^{+-} - (Z_{i,j} - |k|)^2 + \gamma \quad (4.10a)$$

and

$$U_{i,j}^{--} = U_{i,j-1}^{--} - (Z_{i,j} + |k|)^2 + \gamma, \quad (4.10b)$$

respectively.

It is generally agreed that monitoring of the mean in both directions (upward and downward) is crucial, as a mean shift in either direction always indicates a compromised quality. In most cases, an upper-sided shift and a lower-sided shift would have the same effects on the mean chart due to symmetry, hence all derivations associated with the upper-sided mean chart can be easily extended to the lower-sided counterpart. Concerning shifts in the variance, many researchers have emphasised the importance of detecting an increasing shift in the variance over a decreasing shift in the variance, since an inflated process variance is usually more indicative of a deteriorated product quality. A decrease in the process variance, on the other hand, reveals an improvement in the process stability and would require minimal (or no) signals from the control chart. Similar remarks have been made by Reynolds and Stoumbos (1998) in their article, where they claimed that the primary goal of the SPRT proportion chart is to detect an increase in the proportion defect rather than to detect a decrease in the proportion defect. Hence, this chapter only discusses the OSPRT chart for monitoring an increase in the variance (i.e., $\eta > 1$), i.e., the OSPRT chart with control statistics defined in Equations (4.9a) and (4.9b), although the OSPRT chart for a lower-sided variance could be constructed using a similar fashion. Note that from now onwards, the term “OSPRT chart” refers to the “OSPRT chart for monitoring an upper-sided variance shift”.

The OSPRT chart has two control limits g and h . It is worth noting that both the OSPRT chart for detecting an increase in the mean and for detecting a decrease in the mean have the same orientation, i.e., both the control statistics defined in Equations (4.9a) and (4.9b) can be plotted on the same control chart. To elaborate, the OSPRT chart accepts the process as in-control if $U_{i,j}^{++}$ (or $U_{i,j}^{--}$)

falls in the region $(-\infty, g]$. If $U_{i,j}^{++}$ (or $U_{i,j}^{-+}$) falls in the region $[h, \infty)$, the process is said to be out-of-control with a potential upward (or downward) mean shift and/or increasing shift in the variance. If $U_{i,j}^{++}$ (or $U_{i,j}^{-+}$) lies within the region (g, h) , further observations are required until the OSPRT reaches a decision.

4.2.2 Run Length Properties of the OSPRT Chart

This section aims to derive the run length properties of the OSPRT chart for detecting an upper-sided shift in the mean (i.e., $U_{i,j}^{++}$), which can be easily extended to the lower-sided counterpart by the symmetric property of the mean chart (by Remark 2.3.1). The computer programmes for computing the run length properties of the OSPRT chart through theoretical formulae and Monte Carlo simulation are available on GitHub, labelled with B.1.1 and B.1.2, respectively. The run length properties of the OSPRT chart are derived through the construction of an absorbing Markov chain (see Section 3.2).

Suppose that the region (g, h) of the OSPRT chart is partitioned into ξ subintervals, each representing a transient state T_a , for $a = 1, 2, \dots, \xi$. Define further the acceptance state $T_{\xi+1} = (-\infty, g]$ and the rejection state $T_{\xi+2} = [h, \infty)$. Let O_a be the midpoint of state T_a . It is clear that

$$O_a = g + (a - 0.5)\Delta, \quad (4.11)$$

where $\Delta = (h - g)/\xi$ is the width of each transient state. In canonical form, the full transition probability matrix of the absorbing Markov chain can be written as

$$\mathbf{P} = \begin{pmatrix} \mathbf{R} & \mathbf{Q} \\ \mathbf{0} & \mathbf{I}_2 \end{pmatrix} = \begin{pmatrix} r_{1,1} & r_{1,2} & \cdots & r_{1,\xi} & q_{1,(\xi+1)} & q_{1,(\xi+2)} \\ r_{2,1} & r_{2,2} & \cdots & r_{2,\xi} & q_{2,(\xi+1)} & q_{2,(\xi+2)} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ r_{\xi,1} & r_{\xi,2} & \cdots & r_{\xi,\xi} & q_{\xi,(\xi+1)} & q_{\xi,(\xi+2)} \\ 0 & 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 \end{pmatrix}, \quad (4.12)$$

where $\mathbf{R}$ is the $\xi \times \xi$ matrix of transition probabilities between the transient states, and $\mathbf{Q}$ is made up of two vectors $\mathbf{q}_1$ and $\mathbf{q}_2$, each containing one-step transition probabilities from the transient states to the absorbing states $T_{\xi+1}$ and $T_{\xi+2}$, respectively. Note that these definitions coincide with those in Section 3.2.

To construct the transition probability matrix $\mathbf{R}$ between the transient states, it is necessary to derive an expression for the one-step probability that the control statistic transits from transient state T_a to transient state T_b . The probability, denoted as $r_{a,b}$, is derived as follows:

$$\begin{aligned}
r_{a,b} &= \Pr(U_{i,j}^{++} \in T_b \mid U_{i,j-1}^{++} \in T_a) \\
&\approx \Pr[g + \Delta(b-1) \leq O_a + (Z_{i,j} + |k|)^2 - \gamma \leq g + \Delta b] \\
&= \Pr[g + \Delta(b-1) \leq g + (a-0.5)\Delta + (Z_{i,j} + k)^2 - \gamma \leq g + \Delta b] \\
&= \Pr[\Delta(b-a-0.5) + \gamma \leq (Z_{i,j} + k)^2 \leq \Delta(b-a+0.5) + \gamma] \\
&= \Pr\left[\frac{\Delta(b-a-0.5)+\gamma}{\eta^2} \leq \left(\frac{Z_{i,j}+k}{\eta}\right)^2 \leq \frac{\Delta(b-a+0.5)+\gamma}{\eta^2}\right], \tag{4.13}
\end{aligned}$$

for $a, b = 1, 2, \dots, \xi$. Note that $|k| = k$ follows from the fact that $k > 0$ when both $\delta > 0$ and $\eta > 1$. In the case where $\delta < 0$ and $\eta > 1$ (i.e., $k < 0$), the control statistic $U_{i,j}^{++} = (Z_{i,j} - |k|)^2 - \gamma = [Z_{i,j} - (-k)]^2 - \gamma = (Z_{i,j} + k)^2 - \gamma$, which leads to the same final expression in Equation (4.13). It should also be noted that, under the alternative hypothesis, the standardised quality characteristic $Z_{i,j} = (X_{i,j} - \mu_0)/\sigma_0$ follows the normal distribution $\mathcal{N}(\delta, \eta^2)$. Defining a linear transformation $Z' = (Z_{i,j} + k)/\eta$, it is easy to show that $Z' \sim \mathcal{N}((\delta + k)/\eta, 1)$ by properties of the normal distribution. It follows that $(Z')^2$ follows a non-central chi-squared distribution with one degree of freedom and non-centrality parameter $[(\delta + k)/\eta]^2$. Hence, Equation (4.13) can be evaluated as

$$r_{a,b} = F_{\chi_1^2} \left[\frac{\Delta(b-a+0.5)+\gamma}{\eta^2} \mid \left(\frac{\delta+k}{\eta} \right)^2 \right] - F_{\chi_1^2} \left[\frac{\Delta(b-a-0.5)+\gamma}{\eta^2} \mid \left(\frac{\delta+k}{\eta} \right)^2 \right], \tag{4.14}$$

where $F_{\chi^2_1}\{\cdot \mid [(\delta + k)/\eta]^2\}$ is the cdf of the non-central chi-squared distribution with one degree of freedom and non-centrality parameter $[(\delta + k)/\eta]^2$.

Instead of performing the computation from Equation (4.14), some researchers may prefer solving the inequality in Equation (4.13) for $Z_{i,j}$, i.e., isolate $Z_{i,j}$ in the inequality so that the final expression becomes $\Pr(\dots \leq Z_{i,j} \leq \dots)$. The reason for such a pursuit is that most programming languages calculates the cdf of the normal distribution (a lot) faster than the cdf of the non-central chi-squared distribution. Hence, from the aspect of computational efficiency, it seems that performing the calculation using functions of the normal distribution would be more favourable, even though the resulting formula may appear extremely messy and cumbersome. For completeness, the formula for computing $r_{a,b}$ using the cdf of the standard normal distribution is given in Appendix A. Note that the derivation can be easily extended to other formulaic expressions involved in the calculation of the run length properties of the OSPRT chart. The computer programmes for computing the run length properties of the OSPRT chart are provided under the two different methods, i.e., the “chi-squared distribution” method (using the equations shown in this section) and the “normal distribution” method (using the equations shown in Appendix A). Note that the latter is recommended for code optimisation.

Next, define a $\xi \times 1$ vector of initial probabilities, $\mathbf{C}$, that contains one-step transition probabilities from the initial state to an arbitrary transient state T_a . The entries of $\mathbf{C}$, c_a , is derived as

$$\begin{aligned}
c'_a &= \Pr(U_{i,1}^{++} \in T_a \mid U_{i,0}^{++} = 0) \\
&= \Pr\left[g + \Delta(a - 1) \leq (Z_{i,1} + k)^2 - \gamma \leq g + \Delta a\right] \\
&= \Pr\left[g + \Delta(a - 1) + \gamma \leq (Z_{i,1} + k)^2 \leq g + \Delta a + \gamma\right] \\
&= \Pr\left[\frac{g + \Delta(a - 1) + \gamma}{\eta^2} \leq \left(\frac{Z_{i,1} + k}{\eta}\right)^2 \leq \frac{g + \Delta a + \gamma}{\eta^2}\right]
\end{aligned}$$

$$= F_{\chi_1^2} \left[\frac{g+\Delta a+\gamma}{\eta^2} \left| \left(\frac{\delta+k}{\eta} \right)^2 \right| \right] - F_{\chi_1^2} \left[\frac{g+\Delta(a-1)+\gamma}{\eta^2} \left| \left(\frac{\delta+k}{\eta} \right)^2 \right| \right]. \quad (4.15)$$

The ASN of the OSPRT chart, which represents the expected number of visits between the transient states before $U_{i,j}^{++}$ falls into one of the absorbing states, can be computed as

$$\text{ASN} = \mathbf{1} + \mathbf{C}'^\top (\mathbf{I}_\xi - \mathbf{R}')^{-1} \mathbf{1}, \quad (4.16)$$

where $\mathbf{I}_\xi$ is a $\xi \times \xi$ identity matrix and $\mathbf{1}$ is a $\xi \times 1$ vector with entries filled with ones.

To calculate the ARL and SDRL of the OSPRT chart, it is necessary to first evaluate the operating characteristic function OC of a single OSPRT. According to the law of total probability, the OC can be computed as the following sum

$$\text{OC} = P_0 + \mathbf{C}'^\top (\mathbf{I}_\xi - \mathbf{R}')^{-1} \mathbf{q}_1, \quad (4.17)$$

where P_0 is the one-step transition probability from the initial state to the state $T_{\xi+1}$, and $\mathbf{q}_1$ contains one-step transition probabilities from an arbitrary state T_a to the acceptance state $T_{\xi+1}$. The probability P_0 is derived as

$$\begin{aligned} P'_0 &= \Pr(U_{i,1}^{++} \in T_{\xi+1} \mid U_{i,0}^{++} = 0) \\ &= \Pr \left[(Z_{i,1} + k)^2 - \gamma \leq g \right] \\ &= \Pr \left[\left(\frac{Z_{i,1} + k}{\eta} \right)^2 \leq \frac{g+\gamma}{\eta^2} \right] \\ &= F_{\chi_1^2} \left[\frac{g+\gamma}{\eta^2} \left| \left(\frac{\delta+k}{\eta} \right)^2 \right| \right], \end{aligned} \quad (4.18)$$

and the entries of vector $\mathbf{q}_1$, $q_{a,(\xi+1)}$, can be derived as follows:

$$\begin{aligned} q'_{a,(\xi+1)} &= \Pr(U_{i,j}^{++} \in T_{\xi+1} \mid U_{i,j-1}^{++} \in T_a) \\ &\approx \Pr \left[O_a + (Z_{i,j} + k)^2 - \gamma \leq g \right] \end{aligned}$$

$$\begin{aligned}
&= \Pr \left[g + (a - 0.5)\Delta + (Z_{i,j} + k)^2 - \gamma \leq g \right] \\
&= \Pr \left[\left(\frac{Z_{i,j} + k}{\eta} \right)^2 \leq \frac{\Delta(0.5 - a) + \gamma}{\eta^2} \right] \\
&= F_{\chi_1^2} \left[\frac{\Delta(0.5 - a) + \gamma}{\eta^2} \left| \left(\frac{\delta + k}{\eta} \right)^2 \right. \right].
\end{aligned} \tag{4.19}$$

Note that the formulae for computing c_a , P_0 , and $q_{a,(\xi+1)}$ using the cdf of the standard normal distribution are given in Appendix A.

Recall from Section 3.2 (see Chapter 3) that the run length of the OSPRT chart follows a geometric distribution with probability equal to $1 - \text{OC}$. Hence, the ARL and SDRL of the OSPRT chart can be evaluated as

$$\text{ARL} = \frac{1}{1 - \text{OC}} \tag{4.20}$$

and

$$\text{SDRL} = \sqrt{\frac{\text{OC}}{[1 - \text{OC}]^2}}. \tag{4.21}$$

Note that both Equations (4.20) and (4.21) are true under H_0 and H_1 .

When the process is in-control, the ATS_0 and SDTS_0 are calculated as

$$\text{ATS}_0 = \frac{d}{1 - \text{OC}_0}, \tag{4.22}$$

and

$$\text{SDTS}_0 = d \sqrt{\frac{\text{OC}_0}{[1 - \text{OC}_0]^2}}, \tag{4.23}$$

respectively, where OC_0 is calculated by substituting $(\delta, \eta) = (0, 1)$ into Equation (4.17). When the process is out-of-control, we follow the argument in Section 3.2 regarding the steady-state time to signal of the SPRT chart. The steady-state ATS_1 and SDTS_1 of the SPRT chart are calculated as

$$\text{ATS}_1 = d \left(\frac{1}{1 - \text{OC}_1} - \frac{1}{2} \right) \tag{4.24}$$

and

$$\text{SDTS}_1 = d \sqrt{\frac{\text{OC}_1}{[1-\text{OC}_1]^2} + \frac{1}{12}}, \quad (4.25)$$

respectively.

As stated in Section 3.2 of Chapter 3, the ANOS of the OSPRT chart can be computed as the product of the ARL and the ASN, i.e.,

$$\text{ANOS} = \text{ARL} \times \text{ASN}, \quad (4.26)$$

where the values of ARL and ASN are given in Equations (4.20) and (4.16), respectively. Note that Equation (4.26) is true under H_0 and H_1 .

The formulae for two measures of the overall performance of the OSPRT chart, i.e., the AEQL and the EANOS, are also presented in this section. It should be noted that, for joint monitoring control charts, the set of shift sizes considered is no longer an interval of values, but rather a two-dimensional set of values. Specifically, the set of shift sizes considered is a (rectangular) domain formed by $[\delta_{\min}, \delta_{\max}] \times [\eta_{\min}, \eta_{\max}]$. Here, $\delta_{\min}$ and $\delta_{\max}$ represent the minimum and maximum allowable mean shift sizes, whereas $\eta_{\min}$ and $\eta_{\max}$ represent the minimum and maximum allowable standard deviation shift sizes. The formula for the AEQL is given as

$$\begin{aligned} \text{AEQL} &= \int_{\eta_{\min}}^{\eta_{\max}} \int_{\delta_{\min}}^{\delta_{\max}} (\delta^2 + \eta^2 - 1) \text{ATS}_1 f_{\delta}(\delta) f_{\eta}(\eta) d\delta d\eta \\ &= \frac{1}{(\delta_{\max} - \delta_{\min})(\eta_{\max} - \eta_{\min})} \int_{\eta_{\min}}^{\eta_{\max}} \int_{\delta_{\min}}^{\delta_{\max}} (\delta^2 + \eta^2 - 1) \text{ATS}_1 d\delta d\eta, \end{aligned} \quad (4.27)$$

where ATS_1 is a function of both δ and η , and both $f_{\delta}(\delta)$ and $f_{\eta}(\eta)$ are taken to be the pdfs of uniform distributions over the range $[\delta_{\min}, \delta_{\max}]$ and $[\eta_{\min}, \eta_{\max}]$, respectively. Note that the weight $(\delta^2 + \eta^2 - 1)$ was introduced by Ou *et al.* (2011a, 2012a) as a variant to the original quadratic weight δ^2 used with the mean chart. On the other hand, the EANOS can be computed using the formula

$$\begin{aligned}
\text{EANOS} &= \int_{\eta_{\min}}^{\eta_{\max}} \int_{\delta_{\min}}^{\delta_{\max}} \text{ANOS}_1 f_{\delta}(\delta) f_{\eta}(\eta) d\delta d\eta \\
&= \frac{1}{(\delta_{\max} - \delta_{\min})(\eta_{\max} - \eta_{\min})} \int_{\eta_{\min}}^{\eta_{\max}} \int_{\delta_{\min}}^{\delta_{\max}} \text{ANOS}_1 d\delta d\eta, \tag{4.28}
\end{aligned}$$

where ANOS_1 is a function of both $\boldsymbol{\delta}$ and $\boldsymbol{\eta}$. Note that the ANOS is not suitable to be modelled using the Taguchi's quadratic loss function, since the ANOS measures the combined effect of both the run length and the sampling efficiency. Hence, the ordinary average value is used to evaluate the overall performance of the OSPRT chart in terms of the ANOS_1 . The computer program for calculating the theoretical run length properties of the OSPRT chart is available on GitHub (labelled with B.1.1). To verify the accuracies of the formulae, a computer program based on Monte Carlo simulation is also provided (labelled with B.1.2).

4.3 Optimal Designs of the OSPRT Chart

This section describes four optimisation models for the OSPRT chart based on the minimisation of the (i) ATS, (ii) AEQL, (iii) ANOS, and (iv) EANOS metrics, respectively. Recall from Section 3.6 that the ATS- and ANOS-optimal designs are implemented based on the assumption that the shift sizes are known *a priori*, and they are outlined in Sections 4.3.1 and 4.3.3, respectively. The AEQL- and EANOS-optimal designs, on the other hand, are implemented assuming that the shift sizes are unknown, and they are outlined in Sections 4.3.2 and 4.3.4, respectively. Both the ATS- and AEQL-optimal designs aim to minimise the out-of-control average detection time of the OSPRT chart, making them suitable for general applications in mass manufacturing. On the other hand, the ANOS- and EANOS-optimal designs aim to minimise both the out-of-control average detection time as well as the average sample size, which make them useful in applications that involve high-cost testing. The computer programmes for implementing the aforementioned designs are available on GitHub. The ATS- and AEQL-

optimisation programmes are labelled with B.2.1, whereas the ANOS- and EANOS- optimisation programmes are labelled with B.2.2.

4.3.1 Optimal Design Based on the ATS

This section outlines an optimisation algorithm for adjusting the charting parameters of the OSPRT chart that yields the minimum ATS_1 at a specified shift point (δ, η) . Prior to the optimisation design, it is necessary to list out the charting parameters and input specifications for the OSPRT chart. The proposed design requires six fixed input variables, i.e.,

- (i) the recommended value of ATS_0 (τ_{ATS}),
- (ii) the target mean shift size δ_{opt} and standard deviation shift size η_{opt} ,
- (iii) the target inspection rate (IR),
- (iv) the minimum permissible sampling interval (d_{min}), and
- (v) the maximum allowable ASN_0 ($\bar{n}_{max}$).

Recall from Section 3.4.1 that the appropriate choice of τ_{ATS} is quite subjective, but the general recommendation is to choose a value > 100 . The values of δ_{opt} and η_{opt} can be specified based on the practitioners' knowledge about a specific process, or based on some historical evidence of the process going out of control. In the context of resource management, the sampling frequency and sample size cannot be increased concurrently for a fixed amount of resources (e.g., labour, machinery). In other words, if practitioners decide to inspect a greater number of measurements, they should collect the samples less frequently to avoid placing excessive pressure on the inspection system, and vice versa. The constraint described above is achieved by imposing the inspection rate IR, and its value should be chosen carefully based on the capacity of the inspection system. Meanwhile, $\bar{n}_{max}$ is set to avoid an extremely large sample size being searched by the optimisation algorithm. d_{min} is

also set to avoid an impractically small sampling interval being obtained during the optimisation effort.

Unlike the SPRT mean chart, the OSPRT chart has six charting parameters:

- (i) the control limits g and h ,
- (ii) the in-control average sample size ASN_0 ,
- (iii) the reference parameters k and γ , and
- (iv) the sampling interval d .

whose values can be tuned to achieve an optimum level of performance. Recall from Section 3.4.1 that, although the ASN_0 acts as a performance metric of the OSPRT chart, it is semantically related to the sample size (or sample number) of a fixed-sample-size control chart. Hence, the ASN_0 can be treated as an adjustable parameter in the optimisation search, given that some constraint is imposed throughout the design (e.g., the ratio of ASN_0 to d).

The ATS-optimisation model for the OSPRT chart is outlined as follows:

$$\underset{(\bar{n}, k, \gamma, d, g, h)}{\text{Minimise}} \text{ATS}_1, \quad (4.29a)$$

subject to

$$\text{ATS}_0 = \tau_{\text{ATS}}, \quad (4.29b)$$

$$\text{ASN}_0 = \bar{n}, \quad (4.29c)$$

$$\bar{n} = \min\{\max\{1, \text{IR} \times d\}, \bar{n}_{\max}\}, \quad (4.29d)$$

and

$$d \geq d_{\min}. \quad (4.29e)$$

Note that all discussions in Section 3.4.1 regarding free variables and dependent variables of the optimisation model also apply to the optimisation design of the OSPRT chart. In particular, since the value of d depends on the ASN_0 , we only need to perform an independent search on $\bar{n}$, and compute d using the equality $d = \bar{n}/IR$. Besides, according to Reynolds and Stoumbos (1997a) and Ou *et al.* (2012b), the values of g and h can be adjusted via numerical root-finding methods to satisfy constraints (4.29c) and (4.29b), respectively. This implies that the values of g and h need not be searched independently via extensive search methods. This leaves the optimisation model described in Equations (4.29a) to (4.29e) with only three independent search parameters, i.e., $\bar{n}$, k , and γ (recall from Section 3.4.1 that the optimal design for the SPRT mean chart requires two independent search parameters $\bar{n}$ and γ). While the optimisation model has been reduced substantially, it is still rather time-consuming to perform an exhaustive search in the three-dimensional parameter space of $(\bar{n}, k, \gamma)$ on a standard computer. It turns out that, by virtue of the optimality of the SPRT, the three-parameter model described above can be reduced to a single-parameter model. In the following paragraph, we introduce a proposition and a corollary regarding the optimality of the OSPRT chart.

Proposition 4.1. For a fixed ARL_0 and $ASN_0 \in (1, +\infty)$, the OSPRT scheme that minimises the ARL_1 at the shift point (δ, η) has the control statistic defined in Equation (4.8) and reference parameters defined in Equations (4.7b) and (4.7c).

Proof. Moustakides (1986) proved that the Page’s procedure (Page, 1954), or more commonly known as the CUSUM procedure, is exactly optimal for detecting changes in distributions in terms of the average delay to detection. As the CUSUM is a repeated sequence of Wald’s SPRT, it follows that the optimality properties of the CUSUM can be extended to the SPRT (Tartakovsky *et al.*, 2015).

A more formal proof can be constructed by connecting the SPRT to the Neyman-Pearson (NP) lemma, which proves the existence of the most powerful test among all statistical tests with a common Type-I error. This follows from the fact that the SPRT relies on the calculation of likelihood ratios to derive a conclusion, but instead of evaluating the likelihood ratio for a fixed number of samples, it does so in a sequential manner. Suppose that we are interested in testing the hypotheses $H_0: (\mu, \sigma) = (\mu_0, \sigma_0)$ versus $H_1: (\mu, \sigma) = (\mu_1, \sigma_1)$, where $\mu_1 > \mu_0$ and $\sigma_1 > \sigma_0$ without loss of generality. It should be noted that minimising the ARL_1 of the OSPRT chart is equivalent to maximising the quantity $(1 - OC)$, i.e., *maximising the statistical power of a single OSPRT*. According to the NP lemma, the most powerful test (i.e., the test with the highest statistical power) has the critical region $\{\mathbf{X}_n : L(\mathbf{X}_n) > c_R\}$, where $\mathbf{X}_n = (X_1, X_2, \dots, X_n)$ is a sample of n observations, $L(\cdot)$ is the likelihood function, and $c_R > 0$ is a critical value. The likelihood function for a fixed-sample-size (FSS) likelihood ratio test (with sample size n) can be evaluated as

$$\begin{aligned}
L(\mathbf{X}_n) &= \prod_{j=1}^n \left[\frac{\frac{1}{\sqrt{2\pi}\sigma_1} \exp\left\{-\frac{(X_j - \mu_1)^2}{2\sigma_1^2}\right\}}{\frac{1}{\sqrt{2\pi}\sigma_0} \exp\left\{-\frac{(X_j - \mu_0)^2}{2\sigma_0^2}\right\}} \right] \\
&= \left(\frac{\sigma_0}{\sigma_1}\right)^n \exp\left\{\sum_{j=1}^n \left[\frac{(X_j - \mu_0)^2}{2\sigma_0^2} - \frac{(X_j - \mu_1)^2}{2\sigma_1^2}\right]\right\} \\
&= \left(\frac{\sigma_0}{\sigma_1}\right)^n \exp\left\{\frac{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}{2\sigma_1^2} \sum_{j=1}^n \left[\left(\frac{X_j - \mu_0}{\sigma_0} + \frac{\sigma_0(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}\right)^2 - \left(\frac{\sigma_1(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}\right)^2\right]\right\}
\end{aligned}$$

Note that the expression above has been simplified in a similar way as in Equations (4.3) and (4.4). By algebra, the criterion for rejection $L(\mathbf{X}_n) > c_R$ can be re-expressed as

$$\begin{aligned}
\left(\frac{\sigma_0}{\sigma_1}\right)^n \exp\left\{\frac{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}{2\sigma_1^2} \sum_{j=1}^n \left[\left(\frac{X_j - \mu_0}{\sigma_0} + \frac{\sigma_0(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}\right)^2 - \left(\frac{\sigma_1(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}\right)^2\right]\right\} &> c_R \\
\sum_{j=1}^n \left(\frac{X_j - \mu_0}{\sigma_0} + \frac{\sigma_0(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}\right)^2 &> \tilde{c}_R,
\end{aligned}$$

where

$$\tilde{c}_R = n \left(\frac{\sigma_1(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 + \frac{2\sigma_1^2 [\ln c_R + n \ln(\frac{\sigma_1}{\sigma_0})]}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)}.$$

It is possible to determine the rejection threshold $\tilde{c}_R$ by specifying a desired Type-I error α for the FSS likelihood ratio test. Under the null hypothesis H_0 , it is not difficult to show that the statistic of the critical region has a non-central chi-squared distribution with n degrees of freedom and non-centrality parameter $\sigma_0^2(\mu_1 - \mu_0)^2/[(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)]^2$. Hence, the rejection threshold $\tilde{c}_R$ can be computed by solving the equation

$$F_{\chi_n^2} \left[\tilde{c}_R \mid \frac{\sigma_0^2(\mu_1 - \mu_0)^2}{(\sigma_1 - \sigma_0)^2(\sigma_1 + \sigma_0)^2} \right] = 1 - \alpha.$$

Under H_1 , the (highest) statistical power achieved by the FSS likelihood ratio test is determined by the following expression

$$\begin{aligned} & \Pr \left[\sum_{j=1}^n \left(\frac{X_j - \mu_0}{\sigma_0} + \frac{\sigma_0(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 > \tilde{c}_R \right] \\ &= \Pr \left[\sum_{j=1}^n \left(\frac{X_j - \mu_1}{\sigma_0} + \frac{\mu_1 - \mu_0}{\sigma_0} + \frac{\sigma_0(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 > \tilde{c}_R \right] \\ &= \Pr \left[\sum_{j=1}^n \left(\frac{\sigma_1}{\sigma_0} \cdot \frac{X_j - \mu_1}{\sigma_1} + \frac{(\mu_1 - \mu_0)(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0) + \sigma_0^2(\mu_1 - \mu_0)}{\sigma_0(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 > \tilde{c}_R \right] \\ &= \Pr \left[\sum_{j=1}^n \left(\frac{\sigma_1}{\sigma_0} \cdot \frac{X_j - \mu_1}{\sigma_1} + \frac{\sigma_1^2(\mu_1 - \mu_0)}{\sigma_0(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 > \tilde{c}_R \right] \\ &= \Pr \left[\sum_{j=1}^n \left(\frac{\sigma_1}{\sigma_0} \right)^2 \left(\frac{X_j - \mu_1}{\sigma_1} + \frac{\sigma_1(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 > \tilde{c}_R \right] \\ &= \Pr \left[\sum_{j=1}^n \left(\frac{X_j - \mu_1}{\sigma_1} + \frac{\sigma_1(\mu_1 - \mu_0)}{(\sigma_1 - \sigma_0)(\sigma_1 + \sigma_0)} \right)^2 > \left(\frac{\sigma_0}{\sigma_1} \right)^2 \tilde{c}_R \right] \\ &= 1 - F_{\chi_n^2} \left[\left(\frac{\sigma_0}{\sigma_1} \right)^2 \tilde{c}_R \mid \frac{\sigma_1^2(\mu_1 - \mu_0)^2}{(\sigma_1 - \sigma_0)^2(\sigma_1 + \sigma_0)^2} \right]. \end{aligned}$$

Note that the OSPRT has the same rejection criterion as the FSS likelihood ratio test derived above. It is however worth noting that, since the rejection threshold $\tilde{c}_R$ depends on μ_1 and σ_1 , the likelihood ratio test is *not* the uniformly most powerful test for a composite hypothesis $H_1: (\mu, \sigma) > (\mu_0, \sigma_0)$. Likewise, the OSPRT is also not the uniformly most powerful sequential test.

Note that the OSPRT chart that achieves the minimum ARL_1 for a fixed pair of (δ, η) has reference parameters k and γ that depend on (δ, η) . To see this, note that for a specified (δ, η) , we can calculate the values of k and γ using Equations (4.7b) and (4.7c), respectively. Likewise, the OSPRT chart with a fixed k and γ is said to achieve the minimum ARL_1 for some shift size (δ^*, η^*) . We want to show that the shift size (δ^*, η^*) is in fact unique. We first rewrite Equation (4.7b) as $\delta = k(\eta^2 - 1)$, and substitute it into Equation (4.7c) as

$$\gamma = k^2 \eta^2 + \frac{2\eta^2 \ln \eta}{\eta^2 - 1}. \quad (4.30)$$

It is not difficult to show that the expression on the RHS of Equation (4.30) is monotone increasing in $\eta \in (0, 1) \cup (1, +\infty)$, for all $k \in \mathbb{R}$. As $\eta \rightarrow 0^+$, the expression on the RHS tends to zero. It follows that Equation (4.30) has one and only one real solution η^* for any $\gamma > 0$. It follows that the solution δ^* is also unique. $\square$

Corollary 4.1. For a fixed ATS_0 and $ASN_0 \in (1, +\infty)$, the OSPRT scheme that minimises the ATS_1 value at the shift point (δ, η) has the same form as that in Proposition 4.1.

Proof. From Equations (4.22) and (4.24), the ATS_0 and ATS_1 can be expressed as $d \times ARL_0$ and $d \times (ARL_1 - 1/2)$, respectively. Since both the ATS_0 and ATS_1 are linear functions of the ARL_0 and ARL_1 (with a constant slope and intercept term), the minimisation problem coincides with that of Proposition 4.1 and the result follows. $\square$

By using Corollary 4.1, the step-by-step procedure for determining the charting parameters of the ATS-optimal OSPRT chart is given as follows:

Step 1: Specify the values of τ_{ATS} , IR , $\bar{n}_{\max}$, $d_{\min}$, δ_{opt} , and η_{opt} . Here, δ_{opt} and η_{opt} are the mean and standard deviation shift sizes for which a quick detection is required.

Step 2: Using the shift sizes $(\delta_{\text{opt}}, \eta_{\text{opt}})$ specified in Step 1, calculate the optimal values of k and γ from Equations (4.7b) and (4.7c), respectively. Perform a one-dimensional golden section search on the parameter space of $\bar{n}$ by satisfying Equation (4.29d) and (4.29e).

Step 3: For each combination of $(\bar{n}, k, \gamma)$ searched in Step 2, calculate the value of d from the formula $d = \bar{n}/\text{IR}$. Adjust the values of g and h simultaneously using a hybrid Newton-Raphson method to satisfy both Equations (4.29b) and (4.29c). Terminate the search algorithm when the difference between successive values become less than 0.001. At the end of this step, all the possible combinations of $(\bar{n}, k, \gamma, d, g, h)$ satisfying constraints (4.29b) - (4.29e) are obtained when $\delta = 0$.

Step 4: Determine the combination $(\text{ASN}_0 = \bar{n}, k, \gamma, d, g, h)$ that produces the smallest ATS_1 value from all the possible combinations found in Steps 2 and 3.

The computer program for implementing the ATS-optimal design for the OSPRT chart is available on GitHub (labelled with B.2.1).

Tables 4.1 and 4.2 present the optimal charting parameters $(\text{ASN}_0, k, \gamma, d, g, h)$ of the ATS-optimal OSPRT chart, and the resulting $(\text{ATS}_1, \text{SDTS}_1)$ values for $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$. The input specifications are chosen as $\tau_{\text{ATS}} = 370.4$, $\text{IR} \in \{3, 5\}$, $\bar{n}_{\text{max}} = 15$, and $d_{\text{min}} = 0.2$. In Table 4.1, an inspection rate of $\text{IR} = 3$ has been used, whereas in Table 4.2, an inspection rate of $\text{IR} = 5$ has been used. As a numeric example, when $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (1.0, 1.4)$ and $\text{IR} = 5$, the optimal OSPRT chart has charting parameters $(\text{ASN}_0, k, \gamma, d, g, h) = (1.957, 1.042, 3.501, 0.391, -1.257, 21.255)$ and yields $(\text{ATS}_1, \text{SDTS}_1) = (0.607, 0.585)$ (see Table 4.2). All the results tabulated in Tables 4.1 and 4.2 have been verified with Monte Carlo simulations.

Table 4.1 Optimal charting parameters ($\text{ASN}_0, k, \gamma, d, g, h$), and the corresponding ($\text{ATS}_1, \text{SDTS}_1$) values of the OSPRT chart, when $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 3$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta_{\text{opt}} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.

δ_{opt}	η_{opt}				
	1.2	1.4	1.6	1.8	2.0
	(ASN_0, k, γ) (d, g, h) ($\text{ATS}_1, \text{SDTS}_1$)	(ASN_0, k, γ) (d, g, h) ($\text{ATS}_1, \text{SDTS}_1$)	(ASN_0, k, γ) (d, g, h) ($\text{ATS}_1, \text{SDTS}_1$)	(ASN_0, k, γ) (d, g, h) ($\text{ATS}_1, \text{SDTS}_1$)	(ASN_0, k, γ) (d, g, h) ($\text{ATS}_1, \text{SDTS}_1$)
0.2	(4.282, 0.455, 1.491) (1.427, -0.749, 22.893) (7.018, 6.994)	(2.886, 0.208, 1.459) (0.962, -0.657, 16.363) (3.169, 3.145)	(2.291, 0.128, 1.585) (0.764, -0.670, 13.953) (1.914, 1.888)	(1.970, 0.089, 1.726) (0.657, -0.691, 12.630) (1.359, 1.332)	(1.772, 0.067, 1.866) (0.591, -0.716, 11.762) (1.059, 1.031)
0.4	(3.441, 0.909, 2.383) (1.147, -1.053, 26.118) (3.762, 3.732)	(2.605, 0.417, 1.714) (0.868, -0.742, 17.023) (2.471, 2.445)	(2.177, 0.256, 1.711) (0.726, -0.710, 14.131) (1.696, 1.670)	(1.914, 0.179, 1.804) (0.638, -0.714, 12.687) (1.269, 1.242)	(1.741, 0.134, 1.918) (0.580, -0.726, 11.787) (1.015, 0.987)
0.6	(2.959, 1.364, 3.871) (0.986, -1.542, 28.750) (2.157, 2.119)	(2.331, 0.625, 2.140) (0.777, -0.889, 17.783) (1.815, 1.787)	(2.035, 0.385, 1.921) (0.678, -0.770, 14.418) (1.429, 1.402)	(1.837, 0.268, 1.933) (0.612, -0.751, 12.779) (1.145, 1.117)	(1.695, 0.200, 2.008) (0.565, -0.753, 11.804) (0.949, 0.921)
0.8	(2.568, 1.818, 5.954) (0.856, -2.021, 30.596) (1.379, 1.334)	(2.116, 0.833, 2.735) (0.705, -1.067, 18.538) (1.333, 1.302)	(1.896, 0.513, 2.216) (0.632, -0.856, 14.724) (1.174, 1.145)	(1.751, 0.357, 2.114) (0.584, -0.804, 12.894) (1.008, 0.979)	(1.641, 0.267, 2.133) (0.547, -0.789, 11.843) (0.871, 0.842)
1.0	(2.242, 2.273, 8.631) (0.747, -2.401, 31.950) (0.964, 0.915)	(1.947, 1.042, 3.501) (0.649, -1.257, 19.194) (1.005, 0.969)	(1.775, 0.641, 2.595) (0.592, -0.960, 15.028) (0.959, 0.929)	(1.666, 0.446, 2.346) (0.555, -0.864, 13.030) (0.876, 0.846)	(1.583, 0.333, 2.293) (0.528, -0.831, 11.886) (0.789, 0.759)
1.5	(1.708, 3.409, 17.929) (0.569, -3.138, 33.691) (0.513, 0.457)	(1.632, 1.563, 6.159) (0.544, -1.717, 20.190) (0.570, 0.525)	(1.548, 0.961, 3.909) (0.516, -1.254, 15.622) (0.604, 0.566)	(1.489, 0.669, 3.152) (0.496, -1.053, 13.356) (0.612, 0.577)	(1.446, 0.500, 2.848) (0.482, -0.958, 12.032) (0.600, 0.567)
2.0	(1.416, 4.545, 30.945) (0.472, -3.614, 34.156) (0.344, 0.285)	(1.413, 2.083, 9.881) (0.471, -2.065, 20.519) (0.384, 0.333)	(1.389, 1.282, 5.750) (0.463, -1.529, 15.900) (0.419, 0.374)	(1.360, 0.893, 4.283) (0.453, -1.257, 13.550) (0.444, 0.404)	(1.337, 0.666, 3.625) (0.446, -1.112, 12.121) (0.459, 0.421)

Table 4.2 Optimal charting parameters ($ASN_0, k, \gamma, d, g, h$), and the corresponding ($ATS_1, SDTS_1$) values of the OSPRT chart, when $\tau_{ATS} = 370.4$, $IR = 5$, $\bar{n}_{\max} = 15$, $d_{\min} = 0.2$, $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta_{\text{opt}} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.

δ_{opt}	η_{opt}				
	1.2	1.4	1.6	1.8	2.0
	(ASN_0, k, γ) (d, g, h) ($ATS_1, SDTS_1$)	(ASN_0, k, γ) (d, g, h) ($ATS_1, SDTS_1$)	(ASN_0, k, γ) (d, g, h) ($ATS_1, SDTS_1$)	(ASN_0, k, γ) (d, g, h) ($ATS_1, SDTS_1$)	(ASN_0, k, γ) (d, g, h) ($ATS_1, SDTS_1$)
0.2	(4.358, 0.455, 1.491) (0.872, -0.729, 26.055) (4.351, 4.337)	(2.923, 0.208, 1.459) (0.585, -0.650, 18.405) (1.934, 1.919)	(2.312, 0.128, 1.585) (0.462, -0.662, 15.617) (1.161, 1.146)	(1.983, 0.089, 1.726) (0.397, -0.687, 14.091) (0.822, 0.806)	(1.781, 0.067, 1.866) (0.356, -0.715, 13.104) (0.640, 0.623)
0.4	(3.518, 0.909, 2.383) (0.704, -1.083, 29.281) (2.302, 2.284)	(2.632, 0.417, 1.714) (0.526, -0.738, 19.063) (1.503, 1.488)	(2.193, 0.256, 1.711) (0.439, -0.709, 15.770) (1.028, 1.012)	(1.926, 0.179, 1.804) (0.385, -0.711, 14.146) (0.767, 0.751)	(1.750, 0.134, 1.918) (0.350, -0.727, 13.129) (0.613, 0.596)
0.6	(2.983, 1.364, 3.871) (0.597, -1.533, 32.014) (1.310, 1.287)	(2.349, 0.625, 2.140) (0.470, -0.886, 19.826) (1.101, 1.084)	(2.048, 0.385, 1.921) (0.410, -0.772, 16.059) (0.865, 0.848)	(1.847, 0.268, 1.933) (0.369, -0.751, 14.233) (0.692, 0.675)	(1.703, 0.200, 2.008) (0.341, -0.753, 13.146) (0.573, 0.556)
0.8	(2.584, 1.818, 5.954) (0.517, -2.016, 33.883) (0.835, 0.807)	(2.130, 0.833, 2.735) (0.426, -1.063, 20.599) (0.807, 0.788)	(1.907, 0.513, 2.216) (0.381, -0.859, 16.370) (0.709, 0.692)	(1.759, 0.357, 2.114) (0.352, -0.802, 14.352) (0.609, 0.591)	(1.648, 0.267, 2.133) (0.330, -0.786, 13.190) (0.526, 0.508)
1.0	(2.257, 2.273, 8.631) (0.451, -2.414, 35.254) (0.582, 0.552)	(1.957, 1.042, 3.501) (0.391, -1.257, 21.255) (0.607, 0.585)	(1.784, 0.641, 2.595) (0.357, -0.960, 16.687) (0.579, 0.561)	(1.674, 0.446, 2.346) (0.335, -0.864, 14.493) (0.528, 0.510)	(1.589, 0.333, 2.293) (0.318, -0.830, 13.232) (0.476, 0.458)
1.5	(1.713, 3.409, 17.929) (0.343, -3.140, 37.013) (0.309, 0.275)	(1.637, 1.563, 6.159) (0.327, -1.716, 22.267) (0.343, 0.316)	(1.553, 0.961, 3.909) (0.311, -1.254, 17.290) (0.364, 0.341)	(1.494, 0.669, 3.152) (0.299, -1.054, 14.826) (0.369, 0.348)	(1.450, 0.500, 2.848) (0.290, -0.956, 13.386) (0.361, 0.341)
2.0	(1.422, 4.545, 30.945) (0.284, -3.678, 37.478) (0.207, 0.171)	(1.416, 2.083, 9.881) (0.283, -2.066, 22.604) (0.231, 0.200)	(1.392, 1.282, 5.750) (0.278, -1.527, 17.580) (0.252, 0.225)	(1.363, 0.893, 4.283) (0.273, -1.255, 15.030) (0.267, 0.243)	(1.340, 0.666, 3.625) (0.268, -1.110, 13.482) (0.276, 0.253)

From Tables 4.1 and 4.2, the $(\text{ATS}_1, \text{SDTS}_1)$ values exhibit a general downward trend as δ_{opt} and/or η_{opt} increase. However, for some fixed value of δ_{opt} (i.e., $\delta_{\text{opt}} \geq 1.0$), we notice a “camel hump” pattern in the $(\text{ATS}_1, \text{SDTS}_1)$ values as η_{opt} is increased from 1.2 through 2.0 (which can be counter-intuitive to some). For instance, when $\delta_{\text{opt}} = 1.5$, the $(\text{ATS}_1, \text{SDTS}_1)$ values are found to increase as η increases from 1.2 to 1.8, and then decrease when η expands beyond 1.8 (see Tables 4.1 and 4.2). A visualisation of the “camel hump” pattern is provided in Figure 4.1. From Figure 4.1, we observe that, for $\delta_{\text{opt}} \in \{0.8, 1.0, 1.5\}$, the ATS_1 value seems to increase slightly before it drops back down as η_{opt} increases. When $\delta_{\text{opt}} = 2.0$, the ATS_1 value is found to increase continuously in $1.2 \leq \eta_{\text{opt}} \leq 2.0$, but will presumably fall back down as we increase η_{opt} beyond 2.0. In fact, similar observations have been reported by other researchers working on statistical designs of joint monitoring

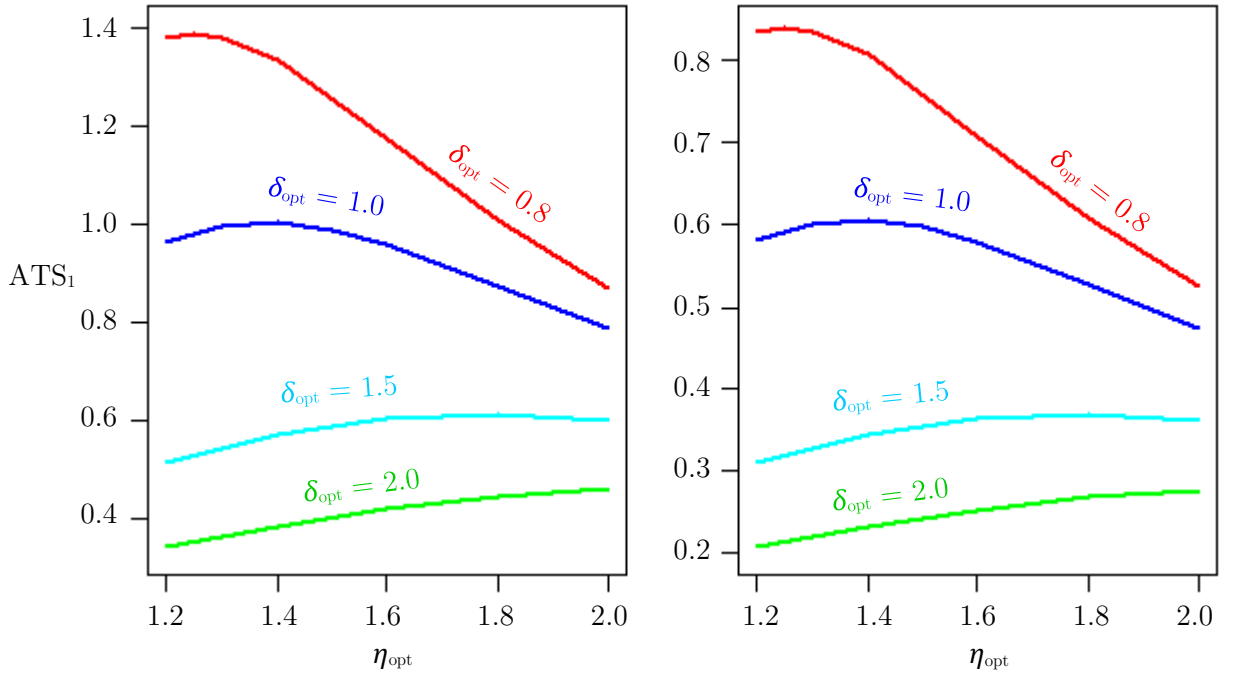


Figure 4.1 The optimum ATS_1 values achieved by the OSPRT chart for $\delta_{\text{opt}} \in \{0.8, 1.0, 1.5, 2.0\}$ against the standard deviation shift size $1.2 \leq \eta_{\text{opt}} \leq 2.0$, when $\text{IR} = 3$ (left) and $\text{IR} = 5$ (right).

control charts, see for example, Wu and Tian (2005), Zhang *et al.* (2010), Ou *et al.* (2011a), and Khusna *et al.* (2019), One possible explanation for the phenomenon could be, as the variability of the observations increases (i.e., $\eta > 1$), the observations are more likely to scatter high above or below the shifted mean, resulting in a “near zero” effect on the control statistics of the control chart in the short run. To illustrate this situation, let us consider two normal distributions, each with $(\delta, \eta) = (2.0, 1.0)$ and $(\delta, \eta) = (2.0, 2.0)$, respectively. It is clear that observations from the former distribution are more likely to cluster around the shifted mean, which would then lead to a sustained rise in the control statistic. On the contrary, observations from the latter distribution are more likely to deviate far above or below the shifted mean, which in turn causes a “delay” in the reaction of the control chart. Although this phenomenon can be slightly counter-intuitive, one quickly realises that it does not lead to too much concern as the $(ATS_1, SDTS_1)$ have attained relatively small values for $\delta_{\text{opt}} \geq 1.0$.

From the tables, we also notice that the optimal ASN_0 value decreases as the shift sizes $(\delta_{\text{opt}}, \eta_{\text{opt}})$ increase, and vice versa. For example, the optimal ASN_0 values are found to decrease persistently from around 4.3 to 1.4 as $(\delta_{\text{opt}}, \eta_{\text{opt}})$ increase (see Tables 4.1 and 4.2). Besides, when the inspection rate is adjusted from three to five, the optimal ASN_0 value is found to stay about the same, whereas the sampling interval d is shortened by nearly 40%. For instance, when $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (0.4, 1.4)$, increasing IR from three to five leads to a fall in the value of d from 2.471 to 1.503, while only a slight increase in the optimal ASN_0 value from 2.605 to 2.632 is observed. This implies that an increase in the inspection capacity almost always leads to a higher sampling frequency rather than an increased sample size of the OSPRT chart (i.e., small-sample-size-high-sampling-frequency strategy). In fact, Ou *et al.* (2012b) showed that, given a fixed inspection rate IR, the optimisation algorithm tends to seek a combination with a small ASN_0 and a small d in order to achieve the best detection effectiveness for the SPRT chart. It is, however,

important to ensure that the sampling interval sought does not go below $d_{\min}$ to avoid impractically high sampling frequencies.

4.3.2 Optimal Design Based on the AEQL

In practice, the assumption that shift sizes are deterministic is almost always invalid. Therefore, it is wise to consider an optimal design based on an aggregated metric of the ATS_1 , i.e., the AEQL, over the domain of shifts $[\delta_{\min}, \delta_{\max}] \times [\eta_{\min}, \eta_{\max}]$. The AEQL-optimisation model for the OSPRT chart has the objective function

$$\underset{(\bar{n}, k, \gamma, d, g, h)}{\text{Minimise AEQL}}, \quad (4.31)$$

subject to the same constraints in Equations (4.29b) - (4.29e). The step-by-step procedure for determining the charting parameters of an AEQL-optimal OSPRT chart is described as follows:

Step 1: Specify the values of τ_{ATS} , IR, $\bar{n}_{\max}$, $d_{\min}$, $\delta_{\min}$, $\delta_{\max}$, $\eta_{\min}$, and $\eta_{\max}$.

Step 2: Search the value of $\bar{n}$ using the golden section search algorithm as in Step 2 of the ATS-optimisation model. As there are no formulae to compute the optimal values of k and γ , perform a two-dimensional golden section search on the parameter space of k and γ by restricting the search intervals to $(k_{\min}, k_{\max})$ and $(\gamma_{\min}, \gamma_{\max})$, respectively, where $(k_{\min}, k_{\max})$ and $(\gamma_{\min}, \gamma_{\max})$ are suitable lower and upper bounds for k and γ , respectively.

Step 3: This step is similar to Step 3 of the ATS-optimisation algorithm detailed in Section 4.3.1.

Step 4: Determine the combination ($ASN_0 = \bar{n}$, k , γ , d , g , h) that produces the smallest AEQL value from all the possible combinations found in Steps 2 and 3.

The computer program for implementing the AEQL-optimal design for the OSPRT chart is available on GitHub (labelled with B.2.1).

Tables 4.3 and 4.4 tabulate the optimal charting parameters (ASN_0 , k , γ , d , g , h) of the AEQL-optimal OSPRT chart, and the corresponding minimum AEQL values for $\delta_{\max}$, $\eta_{\max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$. The input specifications of the design are chosen as $\tau_{ATS} = 370.4$, $IR \in \{3, 5\}$, $\bar{n}_{\max} = 15$, $d_{\min} = 0.2$, $\delta_{\min} = 0.05$, and $\eta_{\min} = 1.05$. In Table 4.3, an inspection rate of $IR = 3$ has been used, whereas in Table 4.4, an inspection rate of $IR = 5$ has been used. As a numeric example, when $(\delta_{\max}, \eta_{\max}) = (2.50, 3.00)$ and $IR = 3$, the optimal OSPRT chart has charting parameters $(ASN_0, k, \gamma, d, g, h) = (1.471, 0.373, 2.198, 0.490, -0.165, 12.779)$, and the resulting minimum AEQL value is 3.607 (see Table 4.3). It is worth noting that, by substituting $(k, \gamma) = (0.373, 2.198)$ into Equations (4.7b) and (4.7c), the corresponding shift sizes would be calculated as $\delta = 0.886$ and $\eta = 1.837$. This implies that the AEQL-optimal OSPRT chart with $(\delta_{\max}, \eta_{\max}) = (2.50, 3.00)$ is also optimum (in terms of the ATS_1) for detecting the deterministic shift sizes $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (0.886, 1.837)$.

From both tables, one observes that the minimum AEQL value generally increases as the values of $\delta_{\max}$ and $\eta_{\max}$ increase, with some exceptions noticed for $\delta_{\max} = 2.25$ and $\eta_{\max} \leq 2.50$. In particular, for $\eta_{\max} \in \{2.00, 2.25, 2.50\}$, the AEQL values are found to drop when $\delta_{\max}$ is increased from 2.00 to 2.25, and then increase when $\delta_{\max}$ rises above 2.25. For instance, referring to Table 4.3, when $\eta_{\max} = 2.25$ and $IR = 3$, the minimum AEQL value decreases from 3.014 to 3.008 when $\delta_{\max}$ is adjusted from 2.00 to 2.25, and then increases persistently thereafter. This slightly unusual trend may be attributed to the quadratic weights used in the formula for

Table 4.3 Optimal charting parameters ($\text{ASN}_0, k, \gamma, d, g, h$), and the corresponding optimum AEQL value of the OSPRT chart, when $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 3$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{max}} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$, and $\eta_{\text{max}} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$.

δ_{max}	η_{max}				
	2.00	2.25	2.50	2.75	3.00
	(ASN_0, k, γ)	(ASN_0, k, γ)	(ASN_0, k, γ)	(ASN_0, k, γ)	(ASN_0, k, γ)
	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)
	AEQL	AEQL	AEQL	AEQL	AEQL
2.00	(1.799, 0.506, 2.028) (0.600, -0.174, 15.965) 2.824	(1.715, 0.455, 2.034) (0.572, -0.211, 14.899) 3.014	(1.640, 0.415, 2.052) (0.547, -0.214, 14.097) 3.203	(1.589, 0.379, 2.062) (0.530, -0.212, 13.522) 3.393	(1.537, 0.355, 2.098) (0.512, -0.215, 13.012) 3.587
2.25	(1.725, 0.523, 2.088) (0.575, -0.090, 15.846) 2.816	(1.652, 0.468, 2.087) (0.551, -0.136, 14.776) 3.008	(1.593, 0.428, 2.108) (0.531, -0.175, 13.983) 3.200	(1.546, 0.391, 2.117) (0.515, -0.178, 13.393) 3.394	(1.505, 0.364, 2.147) (0.502, -0.200, 12.888) 3.591
2.50	(1.656, 0.535, 2.138) (0.552, 0.015, 15.730) 2.822	(1.597, 0.482, 2.144) (0.532, -0.067, 14.671) 3.016	(1.548, 0.439, 2.161) (0.516, -0.123, 13.873) 3.211	(1.509, 0.401, 2.166) (0.503, -0.139, 13.283) 3.408	(1.471, 0.373, 2.198) (0.490, -0.165, 12.779) 3.607
2.75	(1.593, 0.543, 2.177) (0.531, 0.135, 15.614) 2.841	(1.546, 0.492, 2.188) (0.515, 0.025, 14.576) 3.035	(1.505, 0.447, 2.202) (0.502, -0.046, 13.776) 3.232	(1.467, 0.413, 2.231) (0.489, -0.098, 13.149) 3.432	(1.438, 0.381, 2.243) (0.479, -0.117, 12.680) 3.634
3.00	(1.539, 0.555, 2.226) (0.513, 0.248, 15.529) 2.871	(1.499, 0.499, 2.227) (0.500, 0.128, 14.481) 3.066	(1.461, 0.457, 2.248) (0.487, 0.047, 13.697) 3.264	(1.432, 0.423, 2.281) (0.477, -0.035, 13.068) 3.466	(1.407, 0.389, 2.288) (0.469, -0.061, 12.596) 3.671

Table 4.4 Optimal charting parameters ($\text{ASN}_0, k, \gamma, d, g, h$), and the corresponding optimum AEQL value of the OSPRT chart, when $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{max}} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$, and $\eta_{\text{max}} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$.

δ_{max}	η_{max}				
	2.00	2.25	2.50	2.75	3.00
	(ASN_0, k, γ)	(ASN_0, k, γ)	(ASN_0, k, γ)	(ASN_0, k, γ)	(ASN_0, k, γ)
	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)	(d, g, h)
	AEQL	AEQL	AEQL	AEQL	AEQL
2.00	(1.818, 0.513, 1.997) (0.364, -0.065, 18.233) 1.740	(1.732, 0.455, 1.984) (0.346, -0.079, 17.015) 1.853	(1.660, 0.420, 2.008) (0.332, -0.107, 16.097) 1.966	(1.609, 0.385, 2.017) (0.322, -0.115, 15.419) 2.080	(1.552, 0.358, 2.038) (0.310, -0.079, 14.877) 2.196
2.25	(1.741, 0.530, 2.056) (0.348, 0.027, 18.094) 1.734	(1.672, 0.471, 2.040) (0.334, -0.018, 16.869) 1.848	(1.609, 0.433, 2.064) (0.322, -0.063, 15.942) 1.963	(1.557, 0.398, 2.074) (0.311, -0.059, 15.267) 2.079	(1.521, 0.369, 2.092) (0.304, -0.079, 14.715) 2.197
2.50	(1.678, 0.546, 2.113) (0.336, 0.106, 17.971) 1.736	(1.613, 0.482, 2.087) (0.323, 0.069, 16.742) 1.851	(1.561, 0.443, 2.112) (0.312, 0.001, 15.810) 1.968	(1.518, 0.407, 2.120) (0.304, -0.014, 15.134) 2.086	(1.483, 0.383, 2.165) (0.297, -0.077, 14.522) 2.206
2.75	(1.613, 0.543, 2.121) (0.323, 0.266, 17.831) 1.746	(1.562, 0.496, 2.140) (0.312, 0.149, 16.630) 1.862	(1.516, 0.455, 2.162) (0.303, 0.073, 15.697) 1.980	(1.481, 0.416, 2.167) (0.296, 0.040, 15.014) 2.099	(1.449, 0.386, 2.183) (0.290, 0.018, 14.468) 2.220
3.00	(1.557, 0.558, 2.173) (0.311, 0.381, 17.739) 1.764	(1.511, 0.505, 2.180) (0.302, 0.262, 16.522) 1.880	(1.481, 0.460, 2.187) (0.296, 0.164, 15.635) 1.998	(1.445, 0.425, 2.211) (0.289, 0.104, 14.909) 2.119	(1.419, 0.394, 2.227) (0.284, 0.067, 14.363) 2.241

the AEQL (see Equation (4.27)). Besides, it is generally observed that a larger domain of shift sizes specified tends to lead to a smaller d being produced, thus resulting in smaller ASN_0 values. Increasing the value of IR also has the effect of reducing the sampling interval d by a large degree (but not the ASN_0), agreeing with most of the inferences made in Section 4.3.1.

To assess how the AEQL-optimal OSPRT chart performs at various shift sizes, we tabulate the ATS and SDTS values of the OSPRT charts designed using the specifications $IR = 5$ and $(\delta_{\max}, \eta_{\max}) \in \{(2.0, 2.0), (3.0, 3.0)\}$ in Tables 4.5 and 4.6, respectively. In Table 4.5, the (ATS, SDTS) values are evaluated for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$, whereas in Table 4.6, the (ATS, SDTS) values are evaluated for $\delta \in \{0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta \in \{1.0, 1.4, 1.7, 2.0, 2.5, 3.0\}$. To produce the results in Table 4.5, the

Table 4.5 (ATS, SDTS) values of the AEQL-optimal OSPRT chart designed using $\tau_{ATS} = 370.4$, $IR = 5$, $\bar{n}_{\max} = 15$, $d_{\min} = 0.2$, $\delta_{\max} = 2.00$, and $\eta_{\max} = 2.00$, evaluated at $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$.

δ	η					
	1.0	1.2	1.4	1.6	1.8	2.0
	ATS SDTS	ATS SDTS	ATS SDTS	ATS SDTS	ATS SDTS	ATS SDTS
0.0	370.40 370.22	18.47 18.47	3.70 3.70	1.66 1.65	1.07 1.06	0.81 0.79
0.2	67.05 67.05	7.34 7.34	2.42 2.41	1.35 1.34	0.95 0.94	0.75 0.74
0.4	12.16 12.16	3.16 3.16	1.61 1.60	1.09 1.08	0.84 0.82	0.69 0.68
0.6	3.08 3.08	1.63 1.63	1.13 1.12	0.88 0.87	0.73 0.72	0.63 0.61
0.8	1.29 1.28	1.01 1.00	0.84 0.83	0.72 0.71	0.64 0.62	0.57 0.55
1.0	0.75 0.74	0.70 0.69	0.65 0.64	0.61 0.59	0.56 0.54	0.52 0.50
1.5	0.35 0.32	0.39 0.36	0.41 0.38	0.41 0.39	0.42 0.39	0.41 0.38
2.0	0.24 0.19	0.27 0.23	0.29 0.25	0.31 0.27	0.33 0.29	0.33 0.30

Table 4.6 (ATS, SDTS) values of the AEQL-optimal OSPRT chart designed using $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{max}} = 3.00$, and $\eta_{\text{max}} = 3.00$, evaluated at $\delta \in \{0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.4, 1.7, 2.0, 2.5, 3.0\}$.

δ	η					
	1.0	1.4	1.7	2.0	2.5	3.0
	ATS	ATS	ATS	ATS	ATS	ATS
	SDTS	SDTS	SDTS	SDTS	SDTS	SDTS
0.0	370.40	4.86	1.40	0.77	0.48	0.37
	370.22	4.86	1.39	0.77	0.46	0.35
0.4	28.78	2.14	1.00	0.66	0.45	0.36
	28.78	2.14	1.00	0.65	0.43	0.34
0.7	4.19	1.17	0.75	0.57	0.42	0.35
	4.19	1.16	0.75	0.56	0.40	0.32
1.0	1.08	0.71	0.57	0.48	0.39	0.33
	1.08	0.70	0.56	0.47	0.37	0.31
1.5	0.37	0.39	0.39	0.37	0.33	0.30
	0.35	0.38	0.37	0.35	0.31	0.28
2.0	0.22	0.26	0.28	0.29	0.29	0.28
	0.19	0.24	0.26	0.27	0.26	0.25
2.5	0.17	0.20	0.22	0.24	0.25	0.25
	0.12	0.17	0.19	0.21	0.22	0.22
3.0	0.15	0.17	0.19	0.20	0.22	0.23
	0.10	0.13	0.15	0.17	0.19	0.20

optimal charting parameters of the OSPRT chart are taken from Table 4.4 (the cell corresponding to $(\delta_{\text{max}}, \eta_{\text{max}}) = (2.0, 2.0)$) as $(\text{ASN}_0, k, \gamma, d, g, h) = (1.818, 0.513, 1.997, 0.364, -0.065, 18.233)$. To produce the results in Table 4.6, the optimal charting parameters of the OSPRT chart are taken from Table 4.4 (the cell corresponding to $(\delta_{\text{max}}, \eta_{\text{max}}) = (3.0, 3.0)$) as $(\text{ASN}_0, k, \gamma, d, g, h) = (1.419, 0.394, 2.227, 0.284, 0.067, 14.363)$.

From both tables, it is found that the $(\text{ATS}_1, \text{SDTS}_1)$ values decrease rapidly as the shift sizes δ and η increase, signifying that the AEQL-optimisation design is generally effective for almost all shift sizes within the domain $[\delta_{\text{min}}, \delta_{\text{max}}] \times [\eta_{\text{min}}, \eta_{\text{max}}]$. However, for large mean shift sizes, i.e., $\delta \geq 1.5$, the sensitivity of the OSPRT chart is found to decrease slightly as η increases. For instance, when $\delta = 2.5$ and $(\delta_{\text{max}}, \eta_{\text{max}}) = (3.0, 3.0)$, the $(\text{ATS}_1, \text{SDTS}_1)$ values are found to increase

continuously as η increases from 1.0 to 2.0 (see Table 4.6). An explanation for this observation has been given in Section 3.6.1, citing several research works on joint monitoring control schemes which have shared the same observation. The unusual pattern is, however, not to be worried about because the $(\text{ATS}_1, \text{SDTS}_1)$ have attained very small values when $\delta \geq 1.5$.

4.3.3 Optimal Design Based on the ANOS

This section presents an optimisation framework for determining the charting parameters of the OSPRT chart that achieves the minimum ANOS_1 at a deterministic shift point (δ, η) . Unlike the ATS-optimal and AEQL-optimal designs in the previous sections, the ANOS-optimal design only involves searching four charting parameters, i.e., k , γ , g , and h , of the OSPRT chart. As the sampling interval d is not involved in the calculation of the ANOS, it is not necessary to specify its value. Also recall from Section 3.6.3 of Chapter 3 that the value of the ASN_0 should be fixed for all ANOS-based designs to avoid unjustifiable comparisons between control charts. Hence, in the ANOS-optimal design, only four input specifications are required, i.e., the recommended value of ANOS_0 (τ_{ANOS}), the desired ASN_0 value ($\bar{n}$), and the target shift sizes δ_{opt} and η_{opt} .

The ANOS-optimisation model for the OSPRT chart is outlined as follows:

$$\underset{(k, \gamma, g, h)}{\text{Minimise}} \text{ANOS}_1, \quad (4.32a)$$

subject to

$$\text{ANOS}_0 = \tau_{\text{ANOS}} \quad (4.32b)$$

and Equation (4.29c). Before we move to the procedure of implementing the ANOS-optimisation design, let us consider the following corollary.

Corollary 4.2. For a fixed ARL_0 and $ASN_0 \in (1, +\infty)$, the OSPRT scheme that minimises the $ANOS_1$ value at the shift point (δ, η) has the same form as that in Proposition 4.1.

Proof. Noting that $ANOS_1 = ARL_1 \times ASN_1$ holds by Equation (4.26), our proof involves showing that the OSPRT scheme with prescribed reference parameters achieves both a minimum ARL_1 and a minimum ASN_1 . By Proposition 4.1, we conclude that the OSPRT scheme indeed minimises the ARL_1 at any specified shift point (δ, η) , when the reference parameters k and γ are set to be equal to Equations (4.7b) and (4.7c), respectively. To conclude that the OSPRT also minimises the ASN_1 , we cite Wald's optimality theorem, which proves that the SPRT minimises both the ASN_0 and ASN_1 for any simple hypothesis test. As the proof of Wald's optimality theorem is non-trivial, we refer the interested reader to the full paper by Wald and Wolfowitz (1948). $\square$

By citing Corollary 4.2, the step-by-step procedure for determining the charting parameters of the ANOS-optimal OSPRT chart is given as follows:

Step 1: Specify the values of τ_{ANOS} , $\bar{n}$, δ_{opt} , and η_{opt} .

Step 2: Using the shift sizes $(\delta_{opt}, \eta_{opt})$ specified in Step 1, calculate the optimal values of k and γ from Equations (4.7b) and (4.7c), respectively.

Step 3: For each combination of (k, γ) searched in Step 2, adjust the values of g and h simultaneously using a hybrid Newton-Raphson method to satisfy both Equations (4.32b) and (4.29c). Terminate the search algorithm when the difference between successive values become less than 0.001.

Step 4: Determine the combination (k, γ, g, h) that produces the smallest $ANOS_1$ value from all the possible combinations found in Steps 2 and 3.

Note that, by virtue of Corollary 4.2, the ANOS-optimisation algorithm can be completed in a very short amount of time since it involves no independent parameter searches. The computer program for implementing the ANOS-optimal design for the OSPRT chart is available on GitHub (labelled with B.2.2).

Tables 4.7 and 4.8 show the charting parameters (k, γ, g, h) of the ANOS-optimal OSPRT chart, together with the minimum ARL_1 and $ANOS_1$ values achieved for $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$. The input specifications are selected as $\tau_{\text{ANOS}} = 1500$ and $ASN_0 \in \{3, 5\}$. In Table 4.7, the target ASN_0 value is set to be equal to 3, whereas in Table 4.8, the target ASN_0 value is set to be equal to 5. As a numeric example, when $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (0.8, 1.8)$ and $ASN_0 = 3$, the optimal OSPRT chart has charting parameters $(k, \gamma, g, h) = (0.357, 2.114, -2.076, 13.217)$, yielding $(ARL_1, ANOS_1) = (1.571, 7.486)$ (see Table 4.7). All the results tabulated in Tables 4.7 and 4.8 have been verified with Monte Carlo simulations.

From both tables, we notice that the $(ARL_1, ANOS_1)$ values exhibit decreasing trends as the optimum shift sizes δ_{opt} and η_{opt} increase. When δ_{opt} is moderate and large, i.e., $\delta_{\text{opt}} \geq 1.0$, the “camel hump” pattern (which has appeared in Sections 4.3.1 and 4.3.2) can be found in the ARL_1 performance of the OSPRT chart. However, the “camel hump” pattern is not found in the $ANOS_1$ performance of the optimal OSPRT chart. As δ_{opt} and η_{opt} increase, the optimum $ANOS_1$ values are found to drop gradually, agreeing with the concept that the number of measurements inspected is inversely related to the magnitude of a shift. It is also found that, by adjusting the ASN_0 from 3 to 5, the minimum ARL_1 value achieved is reduced by up to 38%. However, no significant reduction in the $ANOS_1$ values is observed following the adjustment. For instance, when $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (0.2, 1.2)$, the ARL_1 can be reduced from 7.763 (see Table 4.7) to 4.804 (see Table 4.8) through an adjustment of the ASN_0 from 3 to 5, while the $ANOS_1$ value has only dropped

Table 4.7 Optimal charting parameters (k, γ, g, h) , and the corresponding $(\text{ARL}_1, \text{ANOS}_1)$ values of the OSPRT chart, when $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 3$, $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta_{\text{opt}} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.

δ_{opt}	η_{opt}				
	1.2	1.4	1.6	1.8	2.0
	(k, γ)	(k, γ)	(k, γ)	(k, γ)	(k, γ)
	(g, h)	(g, h)	(g, h)	(g, h)	(g, h)
	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$
0.2	(0.455, 1.491)	(0.208, 1.459)	(0.128, 1.585)	(0.089, 1.726)	(0.067, 1.866)
	(0.122, 24.812)	(−0.724, 17.521)	(−1.231, 14.728)	(−1.576, 13.153)	(−1.825, 12.080)
	(7.763, 58.050)	(3.704, 26.414)	(2.448, 15.402)	(1.907, 10.432)	(1.620, 7.760)
0.4	(0.909, 2.383)	(0.417, 1.714)	(0.256, 1.711)	(0.179, 1.804)	(0.133, 1.920)
	(−0.562, 28.060)	(−1.075, 18.083)	(−1.416, 14.852)	(−1.684, 13.163)	(−1.889, 12.074)
	(4.321, 34.143)	(2.999, 20.724)	(2.234, 13.532)	(1.821, 9.650)	(1.578, 7.375)
0.6	(1.364, 3.871)	(0.625, 2.140)	(0.385, 1.921)	(0.268, 1.933)	(0.200, 2.008)
	(−1.583, 30.623)	(−1.599, 18.758)	(−1.705, 15.025)	(−1.856, 13.182)	(−1.995, 12.047)
	(2.674, 21.286)	(2.346, 15.502)	(1.974, 11.318)	(1.701, 8.597)	(1.516, 6.819)
0.8	(1.818, 5.954)	(0.833, 2.735)	(0.513, 2.216)	(0.357, 2.114)	(0.267, 2.133)
	(−2.988, 32.259)	(−2.272, 19.352)	(−2.078, 15.213)	(−2.076, 13.217)	(−2.131, 12.023)
	(1.894, 14.403)	(1.874, 11.660)	(1.728, 9.281)	(1.571, 7.486)	(1.443, 6.179)
1.0	(2.273, 8.631)	(1.042, 3.501)	(0.641, 2.595)	(0.446, 2.346)	(0.333, 2.293)
	(−4.799, 33.179)	(−3.101, 19.761)	(−2.523, 15.374)	(−2.338, 13.249)	(−2.391, 11.993)
	(1.493, 10.391)	(1.558, 8.975)	(1.524, 7.603)	(1.447, 6.450)	(1.366, 5.529)
1.5	(3.409, 17.929)	(1.563, 6.159)	(0.962, 3.909)	(0.670, 3.153)	(0.500, 2.848)
	(−11.187, 33.498)	(−5.887, 20.064)	(−4.067, 15.521)	(−3.685, 13.242)	(−3.578, 11.878)
	(1.109, 5.514)	(1.170, 5.196)	(1.203, 4.827)	(1.211, 4.459)	(1.202, 4.112)
2.0	(4.545, 30.945)	(2.083, 9.881)	(1.282, 5.750)	(0.893, 4.283)	(0.667, 3.626)
	(−20.354, 32.442)	(−9.847, 19.625)	(−6.957, 15.261)	(−5.754, 13.023)	(−5.146, 11.658)
	(1.019, 3.431)	(1.043, 3.389)	(1.067, 3.309)	(1.086, 3.210)	(1.097, 3.101)

Table 4.8 Optimal charting parameters (k, γ, g, h) , and the corresponding $(\text{ARL}_1, \text{ANOS}_1)$ values of the OSPRT chart, when $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 5$, $\delta_{\text{opt}} \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta_{\text{opt}} \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.

δ_{opt}	η_{opt}				
	1.2	1.4	1.6	1.8	2.0
	(k, γ)	(k, γ)	(k, γ)	(k, γ)	(k, γ)
	(g, h)	(g, h)	(g, h)	(g, h)	(g, h)
	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$
0.2	(0.455, 1.491)	(0.208, 1.459)	(0.128, 1.585)	(0.089, 1.726)	(0.067, 1.866)
	(-1.024, 24.585)	(-1.458, 17.115)	(-2.293, 14.243)	(-2.994, 12.563)	(-3.588, 11.418)
	(4.804, 58.032)	(2.452, 26.364)	(1.707, 15.343)	(1.400, 10.363)	(1.246, 7.684)
0.4	(0.909, 2.383)	(0.417, 1.714)	(0.256, 1.711)	(0.179, 1.804)	(0.133, 1.920)
	(-2.093, 27.553)	(-2.009, 17.637)	(-2.643, 14.311)	(-3.223, 12.538)	(-3.733, 11.389)
	(2.807, 34.093)	(2.031, 20.669)	(1.584, 13.470)	(1.353, 9.579)	(1.224, 7.298)
0.6	(1.364, 3.871)	(0.625, 2.140)	(0.385, 1.921)	(0.268, 1.933)	(0.200, 2.008)
	(-3.739, 29.719)	(-2.981, 18.125)	(-3.219, 14.387)	(-3.597, 12.508)	(-3.998, 11.326)
	(1.844, 21.204)	(1.649, 15.435)	(1.437, 11.247)	(1.289, 8.522)	(1.193, 6.740)
0.8	(1.818, 5.954)	(0.833, 2.735)	(0.513, 2.216)	(0.357, 2.114)	(0.267, 2.133)
	(-6.080, 30.912)	(-4.270, 18.498)	(-3.976, 14.464)	(-4.092, 12.474)	(-4.439, 11.250)
	(1.397, 14.287)	(1.382, 11.573)	(1.303, 9.200)	(1.221, 7.406)	(1.157, 6.098)
1.0	(2.273, 8.631)	(1.042, 3.501)	(0.641, 2.595)	(0.446, 2.346)	(0.333, 2.293)
	(-9.563, 31.360)	(-5.857, 18.702)	(-4.929, 14.491)	(-4.749, 12.418)	(-5.006, 11.161)
	(1.183, 10.237)	(1.214, 8.866)	(1.197, 7.510)	(1.159, 6.363)	(1.122, 5.444)
1.5	(3.409, 17.929)	(1.563, 6.159)	(0.962, 3.909)	(0.670, 3.153)	(0.500, 2.848)
	(-21.745, 30.707)	(-11.376, 18.460)	(-8.393, 14.287)	(-7.330, 12.160)	(-6.936, 10.870)
	(1.020, 5.290)	(1.039, 5.037)	(1.052, 4.703)	(1.056, 4.354)	(1.053, 4.019)
2.0	(4.545, 30.945)	(2.083, 9.881)	(1.282, 5.750)	(0.893, 4.283)	(0.667, 3.626)
	(-39.061, 29.210)	(-19.300, 17.682)	(-13.408, 13.757)	(-10.829, 11.736)	(-9.530, 10.493)
	(1.001, 3.220)	(1.004, 3.220)	(1.009, 3.174)	(1.014, 3.098)	(1.018, 3.005)

slightly from 58.050 (see Table 4.7) to 58.032 (see Table 4.8). This implies that increasing the inspection capacity almost always leads to a shorter detection time (a higher detection speed), but not a smaller number of inspected units (a higher savings in sampling costs).

4.3.4 *Optimal Design Based on the EANOS*

We would like to devise a model that minimises the out-of-control ANOS averaged across a possible domain of shift sizes, assuming that there is no way to determine the actual shift size of a process. The EANOS-optimisation model for the OSPRT chart is outlined as follows:

$$\underset{(k, \gamma, g, h)}{\text{Minimise EANOS}}, \quad (4.33)$$

subject to the constraints defined in Equations (4.32b) and (4.29c). Note that the design strategy is similar to the ANOS-optimisation model detailed in Section 4.3.3. The step-by-step procedure for determining the charting parameters of the EANOS-optimal OSPRT chart is given as follows:

Step 1: Specify the values of τ_{ANOS} , $\bar{n}$, $\delta_{\min}$, $\delta_{\max}$, $\eta_{\min}$, and $\eta_{\max}$.

Step 2: Perform a two-dimensional golden section search on the parameter space of k and γ by restricting the search intervals to $(k_{\min}, k_{\max})$ and $(\gamma_{\min}, \gamma_{\max})$, respectively, where $(k_{\min}, k_{\max})$ and $(\gamma_{\min}, \gamma_{\max})$ are suitable lower and upper bounds for k and γ , respectively.

Step 3: This step is similar to Step 3 of the ANOS-optimisation algorithm detailed in Section 4.3.3.

Step 4: Determine the combination (k, γ, g, h) that produces the smallest EANOS value from all the possible combinations found in Steps 2 and 3.

The computer program for implementing the EANOS-optimal design for the OSPRT chart is available on GitHub (labelled with B.2.2).

Tables 4.9 and 4.10 display the charting parameters (k, γ, g, h) of the EANOS-optimal OSPRT chart, together with the corresponding minimum EANOS achieved for $\delta_{\max}, \eta_{\max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$. The design specifications chosen are set as $\tau_{\text{ANOS}} = 1500$, $\delta_{\min} = 0.05$, $\eta_{\min} = 1.05$, and $\text{ASN}_0 \in \{3, 5\}$. As a numeric example, when $(\delta_{\max}, \eta_{\max}) = (2.00, 2.50)$ and $\text{ASN}_0 = 3$, the charting parameters of the EANOS-optimal OSPRT chart are $(k, \gamma, g, h) = (0.485, 1.775, -1.025, 19.218)$, and the minimum EANOS value achieved is 10.887 (see Table 4.9). By substituting $k = 0.485$ and $\gamma = 1.775$ into Equations (4.7b) and (4.7c), respectively, the optimum shift sizes for detection are calculated as $\delta = 0.413$ and $\eta = 1.361$ by solving a system of two equations. This implies that the EANOS-optimal OSPRT chart with $(\delta_{\max}, \eta_{\max}) = (2.00, 2.50)$ is also optimum (in terms of the ARL_1 and ANOS_1) for the deterministic shift sizes $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (0.413, 1.361)$.

Table 4.9 Optimal charting parameters (k, γ, g, h) , and the corresponding optimum EANOS value of the OSPRT chart, when $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 3$, $\delta_{\max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$, and $\eta_{\max} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$.

$\delta_{\max}$	$\eta_{\max}$				
	2.00	2.25	2.50	2.75	3.00
	(k, γ)	(k, γ)	(k, γ)	(k, γ)	(k, γ)
	(g, h)	(g, h)	(g, h)	(g, h)	(g, h)
	EANOS	EANOS	EANOS	EANOS	EANOS
2.00	(0.531, 1.775) (-0.808, 20.890) 13.752	(0.510, 1.788) (-0.965, 19.845) 12.116	(0.485, 1.775) (-1.025, 19.218) 10.887	(0.473, 1.778) (-1.089, 18.766) 9.922	(0.460, 1.778) (-1.132, 18.394) 9.143
2.25	(0.535, 1.805) (-0.916, 20.438) 12.620	(0.510, 1.803) (-1.011, 19.630) 11.156	(0.490, 1.798) (-1.090, 18.993) 10.054	(0.475, 1.800) (-1.148, 18.526) 9.187	(0.464, 1.798) (-1.181, 18.229) 8.485
2.50	(0.538, 1.820) (-0.954, 20.302) 11.667	(0.513, 1.815) (-1.053, 19.485) 10.343	(0.493, 1.816) (-1.135, 18.834) 9.345	(0.475, 1.810) (-1.175, 18.403) 8.559	(0.463, 1.811) (-1.231, 17.997) 7.923
2.75	(0.539, 1.832) (-0.987, 20.157) 10.852	(0.513, 1.825) (-1.088, 19.325) 9.645	(0.493, 1.828) (-1.168, 18.680) 8.734	(0.477, 1.824) (-1.224, 18.219) 8.017	(0.464, 1.827) (-1.274, 17.823) 7.435
3.00	(0.540, 1.841) (-1.010, 20.062) 10.148	(0.513, 1.836) (-1.123, 19.167) 9.039	(0.495, 1.835) (-1.187, 18.629) 8.202	(0.478, 1.837) (-1.260, 18.076) 7.543	(0.465, 1.842) (-1.311, 17.676) 7.008

Table 4.10 Optimal charting parameters (k, γ, g, h) , and the corresponding optimum EANOS value of the OSPRT chart, when $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 5$, $\delta_{\text{max}} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$, and $\eta_{\text{max}} \in \{2.00, 2.25, 2.50, 2.75, 3.00\}$.

	η_{max}				
	2.00	2.25	2.50	2.75	3.00
	(k, γ)	(k, γ)	(k, γ)	(k, γ)	(k, γ)
	(g, h)	(g, h)	(g, h)	(g, h)	(g, h)
δ_{max}	EANOS	EANOS	EANOS	EANOS	EANOS
2.00	(0.537, 1.799) (-1.795, 20.187) 13.700	(0.510, 1.789) (-1.883, 19.307) 12.069	(0.490, 1.785) (-1.985, 18.717) 10.841	(0.473, 1.781) (-2.074, 18.231) 9.878	(0.461, 1.780) (-2.154, 17.856) 9.100
2.25	(0.532, 1.800) (-1.800, 19.947) 12.570	(0.511, 1.805) (-1.964, 19.085) 11.109	(0.493, 1.805) (-2.085, 18.485) 10.008	(0.476, 1.801) (-2.181, 17.991) 9.143	(0.464, 1.804) (-2.262, 17.605) 8.443
2.50	(0.539, 1.822) (-1.868, 19.783) 11.617	(0.514, 1.825) (-2.042, 18.892) 10.296	(0.493, 1.818) (-2.169, 18.260) 9.299	(0.478, 1.821) (-2.271, 17.771) 8.515	(0.466, 1.825) (-2.348, 17.395) 7.880
2.75	(0.540, 1.833) (-1.924, 19.633) 10.804	(0.513, 1.829) (-2.087, 18.765) 9.598	(0.495, 1.831) (-2.227, 18.131) 8.689	(0.480, 1.835) (-2.326, 17.648) 7.973	(0.466, 1.830) (-2.394, 17.291) 7.393
3.00	(0.540, 1.845) (-1.986, 19.447) 10.100	(0.514, 1.838) (-2.142, 18.633) 8.993	(0.495, 1.845) (-2.291, 17.970) 8.157	(0.479, 1.842) (-2.375, 17.511) 7.500	(0.470, 1.850) (-2.482, 17.123) 6.966

From Tables 4.9 and 4.10, one notices that the optimum EANOS value decreases as the values of δ_{max} and η_{max} increase. This can be explained by the fact that, when the domain increases in size (i.e., δ_{max} and η_{max} increase), the EANOS metric includes a wider set of smaller ANOS_1 values (calculated at large shift sizes), which causes the average value to be “diluted” in general. We also notice that changing the value of the ASN_0 imposes little (or no) change on the optimal charting parameters as well as the overall EANOS value. It is interesting to note that the optimal value of k is mostly affected by the value of η_{max} , whereas the optimal value of γ is mostly affected by the value of δ_{max} . For instance, for $\eta_{\text{max}} = 2.00$, the optimal values of k obtained are within the range 0.530 to 0.540, whereas for $\eta_{\text{max}} = 3.00$, the optimal values of k obtained fall to around 0.460 to 0.470. This observation suggests that, when a wider range of standard deviation shifts is considered, one should adopt a smaller value of k in the OSPRT chart. On the

other hand, when a wider range of mean shifts is considered, one should adopt a larger value of γ .

To observe how the EANOS-optimal OSPRT chart performs for various shift sizes, we tabulate both the ARL and ANOS values of the OSPRT chart designed using $ASN_0 = 5$, $(\delta_{\max}, \eta_{\max}) \in \{(2.0, 2.0), (3.0, 3.0)\}$ in Tables 4.11 and 4.12, respectively. In Table 4.11, the (ARL, ANOS) values are evaluated for $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$, whereas in Table 4.12, the (ARL, ANOS) values are evaluated for $\delta \in \{0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta \in \{1.0, 1.4, 1.7, 2.0, 2.5, 3.0\}$. To produce the results in Table 4.11, the optimal charting parameters of the OSPRT chart are taken from Table 4.10 (the cell corresponding to $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$) as $(k, \gamma, g, h) = (0.537, 1.799, -1.795, 20.187)$. On the other hand, to produce the results in Table

Table 4.11 (ARL, ANOS) values of the EANOS-optimal OSPRT chart designed using $\tau_{\text{ANOS}} = 1500$, $ASN_0 = 5$, $\delta_{\max} = 2.00$, and $\eta_{\max} = 2.00$, evaluated at $\delta \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$, and $\eta \in \{1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$.

δ	η					
	1.0	1.2	1.4	1.6	1.8	2.0
	ARL	ARL	ARL	ARL	ARL	ARL
	ANOS	ANOS	ANOS	ANOS	ANOS	ANOS
0.0	300	13.07	3.53	2.14	1.69	1.48
	1500	120.13	38.53	20.86	13.95	10.38
0.2	40.00	5.55	2.63	1.90	1.59	1.43
	309.60	61.95	28.28	17.49	12.44	9.58
0.4	7.09	2.97	2.06	1.69	1.49	1.37
	87.48	35.45	20.88	14.45	10.91	8.71
0.6	2.64	2.00	1.70	1.52	1.40	1.32
	37.79	22.69	15.78	11.92	9.49	7.84
0.8	1.67	1.56	1.47	1.39	1.32	1.27
	21.66	15.83	12.25	9.89	8.23	7.02
1.0	1.32	1.33	1.31	1.29	1.25	1.22
	14.45	11.74	9.76	8.28	7.15	6.27
1.5	1.06	1.09	1.12	1.13	1.13	1.13
	7.20	6.61	6.06	5.57	5.13	4.75
2.0	1.01	1.02	1.04	1.05	1.07	1.07
	4.49	4.34	4.17	4.00	3.84	3.68

Table 4.12 (ARL, ANOS) values of the EANOS-optimal OSPRT chart designed using $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 5$, $\delta_{\text{max}} = 3.00$, and $\eta_{\text{max}} = 3.00$, evaluated at $\delta \in \{0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.4, 1.7, 2.0, 2.5, 3.0\}$.

δ	η					
	1.0	1.4	1.7	2.0	2.5	3.0
	ARL	ARL	ARL	ARL	ARL	ARL
	ANOS	ANOS	ANOS	ANOS	ANOS	ANOS
0.0	300	3.58	1.76	1.39	1.19	1.11
	1500	38.02	15.72	9.58	5.78	4.20
0.4	10.37	2.04	1.50	1.30	1.16	1.10
	110.41	20.76	11.83	8.13	5.33	4.01
0.7	2.22	1.53	1.34	1.24	1.14	1.09
	30.89	13.58	9.27	6.96	4.90	3.82
1.0	1.32	1.27	1.22	1.18	1.12	1.08
	14.69	9.42	7.28	5.89	4.45	3.59
1.5	1.05	1.09	1.10	1.10	1.08	1.07
	6.90	5.75	5.04	4.47	3.73	3.20
2.0	1.01	1.03	1.04	1.05	1.05	1.05
	4.21	3.91	3.68	3.46	3.12	2.82
2.5	1.00	1.01	1.02	1.02	1.03	1.03
	2.93	2.88	2.82	2.76	2.63	2.49
3.0	1.00	1.00	1.01	1.01	1.02	1.02
	2.23	2.25	2.26	2.27	2.25	2.20

4.12, the optimal charting parameters of the OSPRT chart are taken from Table 4.10 (the cell corresponding to $(\delta_{\text{max}}, \eta_{\text{max}}) = (3.0, 3.0)$) as $(k, \gamma, g, h) = (0.470, 1.850, -2.482, 17.123)$.

From both tables, the ARL_1 and ANOS_1 values are found to decrease rapidly as the shift sizes δ and η increase, confirming that the EANOS-optimal design is effective in boosting the sensitivity of the OSPRT chart for a wide range of shifts. Some unusual pattern, i.e., the “camel hump” pattern, can be observed in both the ARL_1 and ANOS_1 values for large mean shift size, i.e., $\delta \geq 1.0$. However, the pattern only affects ARL_1 and ANOS_1 values that are less than 2, which is too small to provoke any concern.

4.4 A Brief Review on Some Joint Monitoring Control Schemes

This section provides a brief review on four joint monitoring control charts included in the comparative study of this chapter (Section 4.5), i.e., the Shewhart $\bar{X}$ - S chart, the weighted-loss CUSUM (WLC) chart, the variable-sample-size (VSS) WLC chart, and the absolute-value SPRT (ABS-SPRT) chart. The Shewhart $\bar{X}$ - S chart is a classical joint monitoring control scheme that operates on two control charts, i.e., the $\bar{X}$ chart for monitoring process mean and S chart for monitoring process variability (see Section 2.2.1). Despite being a multi-chart scheme, the Shewhart $\bar{X}$ - S chart has a surprisingly small number of charting parameters, making it favourable from the aspects of design and operation. Multiple research works have cited the Shewhart $\bar{X}$ - S chart as a classical benchmark in their comparative studies, see for example, Wu and Tian (2005, 2006), Haridy *et al.* (2014), and Quinino *et al.* (2020). The WLC chart, on the other hand, is also a widely cited CUSUM chart for joint monitoring of the mean and dispersion. The WLC chart has received quite some attention due to its strong association with quality engineering as well as its high detection sensitivity. Since the OSPRT chart is a VSS control chart by construction, it seems natural to include the VSS WLC chart in the comparative study due to the similarity in their sampling structures. Finally, the ABS-SPRT chart has been chosen in this study because it is the only single-chart SPRT scheme for jointly monitoring the mean and dispersion of a process.

Note that this section only discusses the properties of the upper-sided component of each control chart mentioned in the preceding paragraph. As stated in Section 3.2 and Section 4.2.1, the run length properties associated with a lower-sided mean component are identical to those associated with an upper-sided mean component due to symmetry. Hence, it suffices to compare the performances among the upper-sided mean components of the control charts and provide a general conclusion (which will also apply to the lower-sided mean component). Concerning

the process variability, it has been stated in Section 4.2.1 that an increase in the process variability is much more hazardous (and worthy of attention) than a decrease in the process variability. Besides, the WLC, VSS WLC, and ABS-SPRT charts are designed in a way such that they are insensitive to decreases in the variance. Hence, throughout this chapter, only the upper-sided mean and variance components of each control chart are considered in the performance comparisons.

4.4.1 The Shewhart $\bar{X}$ - S Chart

Let us provide a quick recap of Section 2.2.1 here. Suppose that a sequence of n measurements $(X_{1,1}, X_{1,2}, \dots, X_{1,n}), (X_{2,1}, X_{2,2}, \dots, X_{2,n}), \dots$ are taken from a normal distribution $\mathcal{N}(\mu_0, \sigma_0^2)$ at the sampling points $i = 1, 2, \dots$. The control statistics of the $\bar{X}$ chart and S chart are computed as the average $\bar{X}_i$ and standard deviation S_i of the i -th sample, respectively, using the formulae

$$\bar{X}_i = \frac{1}{n} \sum_{j'=1}^n X_{i,j'} \quad (4.34)$$

and

$$S_i = \sqrt{\frac{1}{n-1} \sum_{j'=1}^n (X_{i,j'} - \bar{X}_i)^2}. \quad (4.35)$$

In an upper-sided $\bar{X}$ chart, the UCL is calculated as $\text{UCL}_{\bar{X}} = \mu_0 + L_{\bar{X}}\sigma_0/\sqrt{n}$, where $L_{\bar{X}} > 0$ is the control limit multiplier for the $\bar{X}$ chart. In an upper-sided S chart, the UCL is given as $\text{UCL}_S = c_4\sigma_0 + L_S\sigma_0\sqrt{1 - c_4^2}$, where $L_S > 0$ is the control limit multiplier for the S chart and c_4 is a bias adjustment factor. The decision mechanism of the Shewhart $\bar{X}$ - S chart is detailed as follows:

- If both $\bar{X}_i \leq \text{UCL}_{\bar{X}}$ and $S_i \leq \text{UCL}_S$, the process is indicated as in-control at the i -th sampling point.

- If $\bar{X}_i > \text{UCL}_{\bar{X}}$ and/or $S_i > \text{UCL}_S$, the process is indicated as out-of-control at the i -th sampling point, and correction measures must be taken to restore the in-control status of the process.

Assuming that μ_0 and σ_0 are known, the Shewhart $\bar{X}$ - S chart has four charting parameters, i.e., the sample size (n), sampling interval (d), upper control limit of the $\bar{X}$ chart ($\text{UCL}_{\bar{X}}$), and upper control limit of the S chart (UCL_S). In the optimal design of the $\bar{X}$ - S chart, the algorithm searches a minimum point in the space of the objective function by tuning the values of the four charting parameters, subject to some practical constraints. Note that the $\bar{X}$ - S chart would have only three charting parameters when the sampling interval d is redundant.

4.4.2 The Weighted Loss CUSUM (WLC) and Variable-sample-size

WLC Charts

The weighted-loss CUSUM (WLC) chart was proposed by Wu and Tian (2005) in an attempt to improve the detection performance of the original weighted-loss control chart (Wu & Tian, 2006). As a recap to what we have seen in Section 2.6.2, the weighted-loss function has the following formula:

$$\text{WL}_i = \lambda_{\text{WL}} S_i^2 + (1 - \lambda_{\text{WL}})(\bar{X}_i - \mu_0)^2, \quad (4.36)$$

where $\bar{X}_i$ and S_i can be computed from Equations (4.34) and (4.35), respectively, and $\lambda_{\text{WL}} \in (0, 1)$ is a weighting factor for the mean and variance losses. Wu and Tian (2005) suggested to monitor the weighted-loss function using a CUSUM procedure, and the control statistic is computed recursively as

$$\text{WLC}_i = \max(0, \text{WLC}_{i-1} + \text{WL}_i - \gamma_{\text{WLC}}), \quad (4.37)$$

for $i = 1, 2, \dots$, where the initial control statistic $WLC_0 = 0$, and $\gamma_{WLC} > 0$ is the reference parameter of the WLC chart.

Similar to the ordinary CUSUM chart, the WLC chart operates on an upper control limit H_{WLC} . The process is indicated as in-control as long as the statistic WLC_i lies inside the region $[0, H_{WLC}]$. If the statistic $WLC_i > H_{WLC}$ at the sampling point i , an out-of-control signal is produced and immediate actions are taken to investigate the cause(s) of shift.

Assuming that μ_0 and σ_0 are known, the WLC chart has five charting parameters, i.e., the sample size (n), sampling interval (d), weighting factor (λ_{WL}), reference parameter (γ_{WLC}), and upper control limit (H_{WLC}). In the optimal design of the WLC chart, the algorithm searches a minimum objective function by tuning the values of the five charting parameters. When the sampling interval is not involved in the design, the WLC chart would have only four charting parameters.

An improved variant of the WLC chart is the variable-sample-size (VSS) WLC chart proposed by Zhang and Wu (2007). The VSS WLC adopts two different sample sizes n_1 and n_2 (with $n_1 < n_2$) when calculating the sample statistics from Equations (4.34) and (4.35). The smaller sample size n_1 , also known as the relax sample size, is used when the process measurements are relatively stable. When the measurements become less stable, the larger sample size n_2 , also known as the alert sample size, will be sought in the subsequent sampling point. In particular, the VSS WLC chart is divided into three main charting regions, i.e., the safe region $[0, W_{WLC}]$, the warning region $(W_{WLC}, H_{WLC}]$, and the out-of-control region $(H_{WLC}, +\infty)$, where W_{WLC} is the warning limit of the VSS WLC chart. When the control statistic falls in the safe region, n_1 observations will be taken at the next sampling point. When the control statistic falls in the warning region, n_2 observations will be taken at the subsequent sampling point. A visualisation of the VSS WLC chart is given in Figure 4.2.

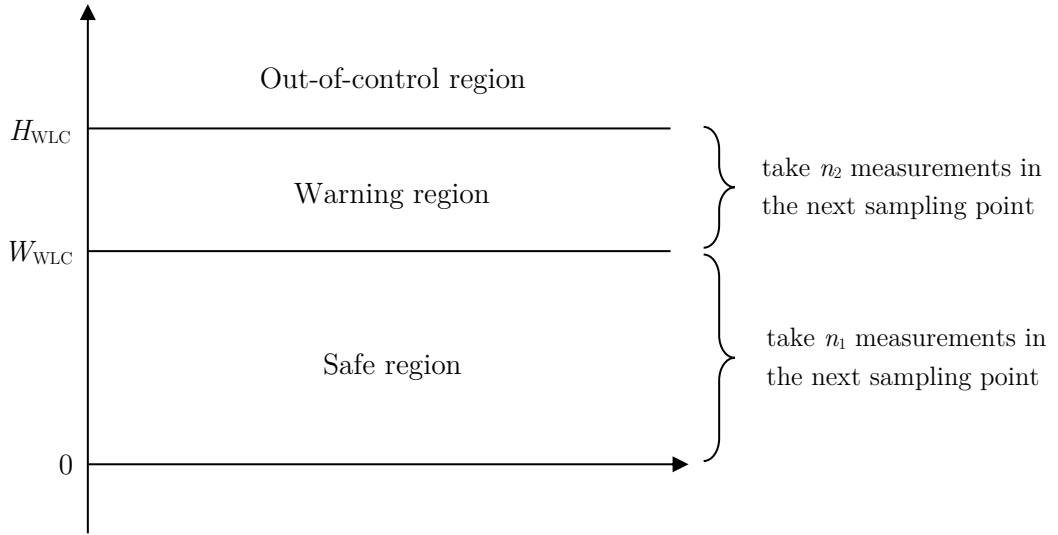


Figure 4.2 A visualisation of the VSS WLC control chart and its sampling mechanism.

Zhang and Wu (2007) showed that, by introducing the VSS feature, the overall sensitivity of the WLC chart can be boosted by almost 15%. However, this comes at the expense of an increased number of charting parameters. The VSS WLC chart now has seven charting parameters, i.e., two distinct sample sizes (n_1 and n_2), sampling interval (d), weighting factor (λ_{WL}), reference parameter (γ_{WLC}), upper warning limit (W_{WLC}), and upper control limit (H_{WLC}). All the seven parameters need to be searched in the optimal design for the VSS WLC chart.

4.4.3 The Absolute-value SPRT Chart

Ou *et al.* (2011a) proposed the ABS-SPRT chart for monitoring changes in both the process mean and variance by applying the absolute value function to the SPRT mean chart. The control statistic of the ABS-SPRT chart is computed recursively as

$$U_{i,j}^{ABS} = U_{i,j-1}^{ABS} + |Z_{i,j}| - \gamma_{ABS}, \quad (4.38)$$

where $U_{i,j}^{\text{ABS}} = 0$, $Z_{ij} = (X_{ij} - \mu_0)/\sigma_0$ is the standardised measurement, and $\gamma_{\text{ABS}} > 0$ is the reference parameter of the ABS-SPRT chart. Note that Ou *et al.* (2011a) has not provided an expression for γ_{ABS} in terms of the shift sizes δ and η (in fact, they were not able to provide any mathematical justification for the value of γ_{ABS} , since the ABS-SPRT chart was not derived from first principles).

Similar to the OSPRT chart (and any other SPRT charts), the ABS-SPRT chart has two control limits g_{ABS} and h_{ABS} . If the control statistic $U_{i,j}^{\text{ABS}}$ goes below g_{ABS} , sampling is terminated, and the process is declared as in-control. If the control statistic $U_{i,j}^{\text{ABS}}$ rises above h_{ABS} , sampling is terminated, and the process is declared as out-of-control. Investigations are carried out immediately to eliminate the root cause(s) of the shift. If the control statistic $U_{i,j}^{\text{ABS}}$ falls in the region between g_{ABS} and h_{ABS} , sampling continues, and no conclusion can be drawn about the process.

Note that the optimal design for the ABS-SPRT chart is identical to that for the SPRT mean chart given in Section 3.4. In particular, all five charting parameters of the ABS-SPRT chart (i.e., ASN_0 , d , g_{ABS} , h_{ABS} , γ_{ABS}) are tuned to achieve a minimum objective function specified by the user (e.g., the AEQL). The reader should refer to Section 3.4 for the full implementation of the optimisation design.

4.5 A Comparison of the Performances between the OSPRT Chart and the Four Competing Charts

This section presents a thorough comparative analysis of the performances of the OSPRT chart and the four joint monitoring control schemes described in Section 4.4, i.e., the Shewhart $\bar{X}$ - S chart, WLC chart, VSS WLC chart, and ABS-

SPRT chart. To demonstrate the effectiveness of the OSPRT chart, the comparisons are made with respect to all four optimisation designs outlined in Section 4.3, i.e., the ATS-optimal, AEQL-optimal, ANOS-optimal, and EANOS-optimal designs. The results associated with the ATS-optimal and AEQL-optimal designs are presented in Section 4.5.1, whereas those associated with the ANOS-optimal and EANOS-optimal designs are presented in Section 4.5.2.

4.5.1 Comparative Analyses Based on the ATS-optimal and AEQL-optimal Designs

This section compares the performances of the optimal OSPRT chart with the Shewhart $\bar{X}$ - S , WLC, VSS WLC, and ABS-SPRT charts designed under the ATS-optimisation and AEQL-optimisation models. To ensure that the comparison is fair, we fix the design specifications for all the five control charts as $\tau_{\text{ATS}} = 370.4$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, and $\text{IR} = 5$. For the ease of demonstration, we also assume that $\mu_0 = 0$ and $\sigma_0 = 1$. The optimisation models used to design the control charts can be found in Sections 4.3.1 and 4.3.2, bearing in mind that each control chart has a different set of charting parameters (or variables) to be tuned in the optimisation algorithm.

Since the Shewhart $\bar{X}$ - S and WLC charts are fixed-sample-size (FSS) control charts, it follows that their ASN_0 values are equal to their sample sizes n (see Section 2.4.2). For the VSS WLC chart, the ASN_0 can be calculated as a weighted average of n_1 and n_2 , and the interested reader should find the formula given in Zhang and Wu (2007). Note that each control chart is designed such that their performances are optimised with respect to upper-sided mean and variance shifts only, and the constraints of the optimisation problem are set to be the same as those described in Sections 4.3.1 and 4.3.2.

We first consider the ATS-optimisation design. To demonstrate the effectiveness of the OSPRT chart over other control charts, we consider nine different pairs of shift sizes constructed using $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$. The shift sizes are deliberately chosen to include one representative from each of the “small”, “moderate”, and “large” categories. For example, $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (1.0, 2.0)$ corresponds to the combination of a moderate mean shift and a large standard deviation shift, whereas $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (2.0, 1.2)$ corresponds the combination of a large mean shift and a small standard deviation shift. Note that the definition of a small (or moderate/ large) shift size can be quite subjective, and the values provided herein can be considered as reasonable representations for each category (for example, we avoid using $\delta_{\text{opt}} = 0.2$ to represent a “small” shift size, since some practitioners may consider $\delta < 0.2$ as being unimportant). Table 4.13 tabulates the charting parameters of the Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts, optimised for the nine shift sizes stated above. As a numeric example, when $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (0.4, 1.6)$ is chosen, the resulting optimal WLC chart has charting parameters $n = 15$, $\lambda_{\text{WL}} = 0.398$, $\gamma_{\text{WLC}} = 0.660$, $d = 3.000$, and $H_{\text{WLC}} = 0.276$.

To contrast the performances of the five control charts, we tabulate their resulting $(\text{ATS}_1, \text{SDTS}_1)$ values evaluated at the shift points $(\delta_{\text{opt}}, \eta_{\text{opt}})$ in Table 4.14. For example, from the previous example, the optimal WLC chart produces a minimum ATS_1 value of 2.569 and an SDTS_1 value of 2.108 (see Table 4.14). To make the performance ranking among the five control charts much more prominent, we have shaded the table cells using a spectrum of blue tones. In particular, we shade the cell with the smallest (best) ATS_1 value of each row in the lightest blue, and shade the cell with the largest (worst) ATS_1 value of each row in the darkest blue. Cells with the lowest ATS_1 values are also bolded to assist the reader in identifying the best-performing chart for each category.

Table 4.13 Optimal charting parameters of the ATS-optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts, optimised for $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$, when $\tau_{\text{ATS}} = 370.4$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, and $\text{IR} = 5$.

δ_{opt}	η_{opt}	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
		$(n, \text{UCL}_{\bar{X}}, \text{UCL}_S)$ d	$(n, \lambda_{\text{WLC}}, \gamma_{\text{WLC}})$ (d, H_{WLC})	$(n_1, n_2, \lambda, \gamma_{\text{WLC}})$ $(d, W_{\text{WLC}}, H_{\text{WLC}})$	$(\text{ASN}_0, \gamma_{\text{ABS}})$ $(d, g_{\text{ABS}}, h_{\text{ABS}})$	$(\text{ASN}_0, k, \gamma)$ (d, g, h)
0.4	1.2	(15, 0.640, 1.580)	(15, 0.244, 0.357)	(9, 15, 0.272, 0.397)	(4.234, 0.897)	(3.518, 0.909, 2.383)
		3.000	(3.000, 0.424)	(2.016, 0.297, 0.778)	(0.847, -0.045, 8.505)	(0.704, -1.083, 29.281)
	1.6	(15, 0.757, 1.477)	(15, 0.398, 0.660)	(5, 11, 0.427, 0.702)	(2.296, 1.015)	(2.193, 0.256, 1.711)
		3.000	(3.000, 0.276)	(1.120, 0.571, 1.420)	(0.459, -0.026, 5.327)	(0.439, -0.709, 15.770)
	2.0	(9, 1.059, 1.676)	(6, 0.437, 0.936)	(3, 5, 0.398, 0.854)	(1.794, 1.112)	(1.750, 0.134, 1.918)
		1.800	(1.200, 0.879)	(0.640, 0.807, 2.434)	(0.359, -0.020, 4.203)	(0.350, -0.727, 13.129)
1.0	1.2	(12, 0.718, 1.912)	(13, 0.165, 0.403)	(11, 15, 0.174, 0.370)	(2.541, 1.000)	(2.257, 2.273, 8.631)
		2.400	(2.600, 0.266)	(2.280, 0.070, 0.408)	(0.508, -0.076, 5.568)	(0.451, -2.414, 35.254)
	1.6	(10, 0.857, 1.700)	(8, 0.320, 0.700)	(4, 7, 0.339, 0.692)	(1.926, 1.079)	(1.784, 0.641, 2.595)
		2.000	(1.600, 0.520)	(0.860, 0.560, 1.638)	(0.385, -0.023, 4.518)	(0.357, -0.960, 16.687)
	2.0	(7, 1.121, 1.847)	(4, 0.377, 1.009)	(3, 5, 0.379, 0.878)	(1.649, 1.159)	(1.589, 0.333, 2.293)
		1.400	(0.800, 1.381)	(0.640, 0.657, 2.255)	(0.330, -0.017, 3.833)	(0.318, -0.830, 13.232)
2.0	1.2	(4, 1.427, 3.543)	(4, 0.151, 0.992)	(3, 5, 0.144, 0.843)	(1.523, 1.284)	(1.422, 4.545, 30.945)
		0.800	(0.800, 1.232)	(0.624, 0.526, 2.182)	(0.305, -0.148, 3.123)	(0.284, -3.678, 37.478)
	1.6	(4, 1.438, 2.604)	(4, 0.265, 1.064)	(2, 4, 0.248, 0.939)	(1.450, 1.277)	(1.392, 1.282, 5.750)
		0.800	(0.800, 1.097)	(0.440, 1.024, 3.639)	(0.290, -0.053, 3.160)	(0.278, -1.527, 17.580)
	2.0	(3, 1.743, 2.820)	(3, 0.310, 1.262)	(2, 3, 0.283, 1.075)	(1.388, 1.304)	(1.340, 0.666, 3.625)
		0.600	(0.600, 1.697)	(0.420, 0.821, 3.551)	(0.278, -0.022, 3.042)	(0.268, -1.110, 13.482)

Table 4.14 (ATS₁, SDTS₁) values of the ATS-optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts, optimised for $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$, when $\tau_{\text{ATS}} = 370.4$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, and IR = 5.

		Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
δ_{opt}	η_{opt}	(ATS ₁ , SDTS ₁)	(ATS ₁ , SDTS ₁)	(ATS ₁ , SDTS ₁)	(ATS ₁ , SDTS ₁)	(ATS ₁ , SDTS ₁)
0.4	1.2	(10.371, 10.298)	(7.752, 6.026)	(7.349, 5.475)	(4.337, 4.323)	(2.302, 2.284)
	1.6	(2.866, 2.591)	(2.569, 2.108)	(2.379, 1.760)	(1.166, 1.151)	(1.028, 1.012)
	2.0	(1.532, 1.344)	(1.372, 1.148)	(1.270, 0.980)	(0.652, 0.635)	(0.613, 0.596)
1.0	1.2	(1.826, 1.541)	(1.894, 1.497)	(1.865, 1.450)	(1.219, 1.202)	(0.582, 0.552)
	1.6	(1.693, 1.483)	(1.514, 1.258)	(1.450, 1.107)	(0.748, 0.731)	(0.579, 0.561)
	2.0	(1.207, 1.063)	(1.068, 0.899)	(0.997, 0.784)	(0.529, 0.512)	(0.476, 0.458)
2.0	1.2	(0.564, 0.459)	(0.599, 0.468)	(0.593, 0.444)	(0.296, 0.269)	(0.207, 0.171)
	1.6	(0.639, 0.549)	(0.628, 0.508)	(0.605, 0.461)	(0.325, 0.303)	(0.252, 0.225)
	2.0	(0.631, 0.581)	(0.590, 0.499)	(0.559, 0.450)	(0.320, 0.299)	(0.276, 0.253)

From Table 4.14, it is quite clear that the OSPRT chart outperforms the Shewhart $\bar{X}$ - S , WLC, VSS WLC, and ABS-SPRT charts for all magnitudes of joint shift sizes. The disparity between the performances of the OSPRT chart and the other four control charts is particularly obvious when both δ_{opt} and η_{opt} are small. Specifically, the OSPRT chart is found to be at least three times as sensitive as the Shewhart $\bar{X}$ - S , WLC, and VSS WLC charts, and twice as sensitive as the ABS-SPRT chart, when $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (0.4, 1.2)$ (see Table 4.14). Although the disparity is found to close off as the shift sizes get larger, the OSPRT chart remains the best performer among all the five control charts. Overall, the OSPRT chart is found to be at least twice as responsive as the Shewhart $\bar{X}$ - S , WLC, and VSS WLC charts towards all process shift sizes, and it outperforms the ABS-SPRT chart by percentages that range from around 6.4% to 109.5%.

Referring the sampling efficiency, one observes that the OSPRT chart adopts the smallest ASN₀ value among all the control charts. In particular, the optimal OSPRT chart requires only around 1.340 to 3.518 observations at each sampling point on average when the process is in-control (see Table 4.13). These numbers are substantially smaller than those of the Shewhart $\bar{X}$ - S , WLC, and VSS WLC

charts, whose sample sizes reach up to 15 when both δ_{opt} and η_{opt} are small. While the OSPRT chart has a smaller ASN_0 than the Shewhart $\bar{X}$ - S and WLC charts, its ASN_0 value is not too different from that of the ABS-SPRT chart. This is consistent with the inference made by Ou *et al.* (2012b), which states that SPRT-based control charts tend to seek a “small-sample-size-high-sampling-frequency” strategy in any attempt to minimise its average detection time.

It is also important to note that, by virtue of Proposition 4.1 and Corollary 4.1, the ATS-optimal design of the OSPRT chart is implemented by simply performing a one-dimensional search on the independent parameter ASN_0 . For the other four control charts, we would require at least two independent searches to complete the optimal design with respect to the ATS_1 (with the VSS WLC chart being the most demanding). For example, for the Shewhart $\bar{X}$ - S chart, the ATS-optimisation model contains four variables (i.e., four charting parameters) and two equality constraints (4.29b) and (4.29d) (note that Equation (4.29c) is redundant because we have $\text{ASN}_0 = n$ for an FSS control chart). By the general implicit function theorem, the optimisation model for the ATS-optimal Shewhart $\bar{X}$ - S chart contains $4 - 2 = 2$ free variables. The same argument applies to the WLC, VSS WLC, and ABS-SPRT charts, and the reader is directed to the works of Wu and Tian (2006), Zhang and Wu (2007), and Ou *et al.* (2011a), respectively, for more details on their optimal designs. It is therefore evident that the OSPRT chart not only renders the best performance, but it also possesses a low design complexity.

Now, we compare the performance of the OSPRT chart to those of the other four control charts under the AEQL-optimisation design described in Section 4.3.2. In our illustration, we select the design specifications as $\tau_{\text{ATS}} = 370.4$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, and $\text{IR} = 5$, and $(\delta_{\text{max}}, \eta_{\text{max}}) \in \{(2.0, 2.0), (3.0, 3.0)\}$. Note that all the design procedures carried out are identical to the ATS-optimisation design discussed earlier in this section, with the only difference being the objective function used by the models. Table 4.15 tabulate the charting parameters of the Shewhart

$\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed under the AEQL-optimal design, together with the minimum AEQL values achieved. From Table 4.15, it is evident that the OSPRT chart achieves the smallest AEQL value among all charts, signifying that it is the overall best-performing chart, in both cases where $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$ and $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$. However, we are interested in understanding the performance of the OSPRT chart (relative to other control charts) in various different shift patterns, in particular, when (i) only the process mean increases ($\delta > 0$ and $\eta = 1$), (ii) only the process standard deviation increases ($\delta = 0$ and $\eta > 1$), and (iii) both the process mean and standard deviation increase ($\delta > 0$ and $\eta > 1$).

Table 4.15 Optimal charting parameters of the Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS SPRT, and OSPRT charts designed for $\tau_{\text{ATS}} = 370.4$, $\bar{n}_{\max} = 15$, $d_{\min} = 0.2$, $\text{IR} = 5$, and $(\delta_{\max}, \eta_{\max}) \in \{(2.0, 2.0), (3.0, 3.0)\}$.

Charts	Charting parameters					AEQL
	Sample size(s) or ASN_0	Sampling interval	Control limits & warning limits	Reference parameter(s)	Weighting Factor	
$(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$						
$\bar{X}$ - S	$n = 9$	$d = 1.800$	$\text{UCL}_{\bar{X}} = 0.930,$ $\text{UCL}_S = 1.734$	—	—	5.061
WLC	$n = 6$	$d = 1.200$	$H_{\text{WLC}} = 1.085$	$\gamma_{\text{WLC}} = 0.630$	$\lambda_{\text{WL}} = 0.321$	4.335
VSS WLC	$n_1 = 5,$ $n_2 = 9$	$d = 1.080$	$H_{\text{WLC}} = 1.497,$ $W_{\text{WLC}} = 0.692$	$\gamma_{\text{WLC}} = 0.542$	$\lambda_{\text{WL}} = 0.312$	4.047
ABS-SPRT	$\text{ASN}_0 = 1.917$	$d = 0.383$	$g = 0.265,$ $h = 5.823$	$\gamma_{\text{ABS}} = 0.984$	—	2.257
SPRT	$\text{ASN}_0 = 1.818$	$d = 0.364$	$g = -0.065,$ $h = 18.233$	$k = 0.513,$ $\gamma = 1.997$	—	1.740
$(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$						
$\bar{X}$ - S	$n = 4$	$d = 0.800$	$\text{UCL}_{\bar{X}} = 1.508,$ $\text{UCL}_S = 2.347$	—	—	5.432
WLC	$n = 2$	$d = 0.400$	$H_{\text{WLC}} = 4.172$	$\gamma_{\text{WLC}} = 1.119$	$\lambda_{\text{WL}} = 0.277$	4.636
VSS WLC	$n_1 = 3,$ $n_2 = 5$	$d = 0.640$	$H_{\text{WLC}} = 2.425,$ $W_{\text{WLC}} = 0.877$	$\gamma_{\text{WLC}} = 0.752$	$\lambda_{\text{WL}} = 0.329$	4.431
ABS-SPRT	$\text{ASN}_0 = 1.452$	$d = 0.290$	$g_{\text{ABS}} = 0.311,$ $h_{\text{ABS}} = 4.306$	$\gamma_{\text{ABS}} = 1.096$	—	2.540
SPRT	$\text{ASN}_0 = 1.419$	$d = 0.284$	$g = 0.067,$ $h = 14.363$	$k = 0.394,$ $\gamma = 2.227$	—	2.241

Case (i): Only the process mean increases

Tables 4.16 and 4.17 compare the $(\text{ATS}_1, \text{SDTS}_1)$ values of the AEQL-optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$ and $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, respectively. In Table 4.16, the results are presented for $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ (since $\delta_{\max} = 2.0$), whereas in Table 4.17, the results are presented for $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ (since $\delta_{\max} = 3.0$). A similar colour coding based on blue shades has been applied to each table. From both Tables 4.16 and 4.17, it is obvious that the

Table 4.16 $(\text{ATS}_1, \text{SDTS}_1)$ values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta = 1$.

	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
δ	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$
0.2	(108.63, 108.62)	(171.31, 171.17)	(162.01, 161.63)	(230.09, 230.09)	(67.05, 67.05)
0.4	(30.17, 30.16)	(43.90, 42.83)	(35.68, 33.77)	(68.77, 68.77)	(12.16, 12.16)
0.6	(10.16, 10.13)	(13.40, 12.06)	(10.43, 8.44)	(15.70, 15.70)	(3.08, 3.08)
0.8	(4.25, 4.19)	(5.72, 4.60)	(4.76, 3.42)	(4.32, 4.32)	(1.29, 1.28)
1.0	(2.18, 2.06)	(3.13, 2.33)	(2.77, 1.90)	(1.78, 1.77)	(0.75, 0.74)
1.5	(0.98, 0.65)	(1.17, 0.87)	(1.15, 0.81)	(0.57, 0.55)	(0.35, 0.32)
2.0	(0.90, 0.52)	(0.69, 0.47)	(0.68, 0.49)	(0.32, 0.28)	(0.24, 0.19)

Table 4.17 $(\text{ATS}_1, \text{SDTS}_1)$ values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta = 1$.

	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
δ	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$
0.4	(55.95, 55.95)	(91.69, 91.29)	(67.71, 67.03)	(92.52, 92.52)	(28.78, 28.78)
0.7	(14.46, 14.46)	(18.67, 17.81)	(11.05, 9.66)	(13.64, 13.64)	(4.19, 4.19)
1.0	(4.75, 4.74)	(5.43, 4.54)	(3.55, 2.55)	(2.57, 2.57)	(1.08, 1.08)
1.5	(1.22, 1.18)	(1.63, 1.12)	(1.27, 0.84)	(0.57, 0.55)	(0.37, 0.35)
2.0	(0.56, 0.45)	(0.80, 0.52)	(0.68, 0.47)	(0.28, 0.26)	(0.22, 0.19)
2.5	(0.42, 0.26)	(0.47, 0.32)	(0.44, 0.32)	(0.20, 0.16)	(0.17, 0.12)
3.0	(0.40, 0.23)	(0.32, 0.23)	(0.35, 0.23)	(0.16, 0.11)	(0.15, 0.10)

OSPRT chart outperforms the other four control charts in terms of both the ATS_1 and $SDTS_1$. In particular, for small and moderate values of δ , the OSPRT chart is found to be at least twice as sensitive as the other four control charts. For instance, from Table 4.16, the ratios between the ATS_1 values of the Shewhart $\bar{X}$ - S , WLC, VSS WLC, and ABS-SPRT charts and that of the OSPRT chart can be calculated as 2.5, 3.6, 2.9, and 5.7, respectively, when $\delta = 0.4$. It is perhaps quite surprising that the ABS-SPRT chart is a rather bad performer when δ is small, and it shows a worse performance than even the Shewhart $\bar{X}$ - S chart for $\delta \leq 0.6$. For larger mean shift sizes, the OSPRT chart remains more powerful than the other four control charts. However, the disparity between the performances of the OSPRT chart and the ABS-SPRT chart is found to close off as δ increases.

Case (ii): Only the process standard deviation increases

Tables 4.18 and 4.19 compare the $(ATS_1, SDTS_1)$ values of the five AEQL-optimal control charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$ and $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, respectively. In Table 4.18, the results are presented for $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$ (since $\eta_{\max} = 2.0$), whereas in Table 4.19, the results are presented for $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$ (since $\eta_{\max} = 3.0$). From both Tables 4.18 and 4.19, it is perhaps a little surprising that the OSPRT chart is no longer the best performer among all control charts. It turns out that the ABS-SPRT chart is the best performer among the control charts, showing slightly lower ATS_1 and $SDTS_1$

Table 4.18 $(ATS_1, SDTS_1)$ values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta = 0$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.

	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
η	$(ATS_1, SDTS_1)$	$(ATS_1, SDTS_1)$	$(ATS_1, SDTS_1)$	$(ATS_1, SDTS_1)$	$(ATS_1, SDTS_1)$
1.2	(41.00, 40.99)	(20.89, 19.37)	(17.46, 15.09)	(11.17, 11.17)	(18.47, 18.47)
1.4	(10.41, 10.38)	(6.14, 4.97)	(5.38, 3.93)	(2.55, 2.54)	(3.70, 3.70)
1.6	(4.43, 4.37)	(3.24, 2.49)	(2.97, 2.12)	(1.32, 1.31)	(1.66, 1.65)
1.8	(2.55, 2.44)	(2.14, 1.64)	(2.01, 1.46)	(0.91, 0.90)	(1.07, 1.06)
2.0	(1.77, 1.61)	(1.59, 1.23)	(1.52, 1.12)	(0.72, 0.70)	(0.81, 0.79)

Table 4.19 $(\text{ATS}_1, \text{SDTS}_1)$ values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta = 0$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$.

	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
η	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$
1.4	(14.74, 14.73)	(6.81, 6.10)	(5.87, 4.84)	(3.59, 3.59)	(4.86, 4.86)
1.7	(4.62, 4.61)	(2.58, 2.11)	(2.31, 1.75)	(1.18, 1.18)	(1.40, 1.39)
2.0	(2.29, 2.27)	(1.51, 1.21)	(1.39, 1.06)	(0.70, 0.69)	(0.77, 0.77)
2.5	(1.16, 1.11)	(0.88, 0.72)	(0.84, 0.66)	(0.45, 0.44)	(0.48, 0.46)
3.0	(0.80, 0.73)	(0.63, 0.52)	(0.63, 0.50)	(0.35, 0.33)	(0.37, 0.35)

performances compared to the OSPRT chart for all values of η . Similar to case (i), as η increases, the disparity between the performances of the OSPRT and ABS-SPRT charts is found to diminish gradually. Although the OSPRT chart is slightly inferior to the ABS-SPRT chart, its detection speed is still significantly higher than those of the Shewhart $\bar{X}$ - S , WLC, and VSS WLC control charts. Based on our observations from case (i) and case (ii), it seems that the optimal ABS-SPRT chart is better-suited for detecting a shift in the variance only, whereas the OSPRT chart has a more balanced performance in terms of mean shift and variance shift detection.

Case (iii): Both the process mean and standard deviation increase

Tables 4.20 and 4.21 compare the $(\text{ATS}_1, \text{SDTS}_1)$ performances of the five control charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$ and $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, respectively. In Table 4.20, the results are presented for $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$, whereas in Table 4.21, the results are presented for $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$. From both Tables 4.20 and 4.21, we observe that, for (almost) all magnitudes of joint shifts, the OSPRT chart stands out as the best-performing control chart among all its competitors. In particular, the OSPRT chart is found to perform slightly better than the ABS-SPRT chart, and it is at least 1.5 times as responsive

Table 4.20 $(\text{ATS}_1, \text{SDTS}_1)$ values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.

δ	η	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
		$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$
0.2	1.2	(26.34, 26.33)	(16.59, 15.14)	(13.75, 11.54)	(9.33, 9.33)	(7.34, 7.34)
	1.4	(8.58, 8.55)	(5.64, 4.54)	(4.95, 3.61)	(2.39, 2.38)	(2.42, 2.41)
	1.6	(3.98, 3.91)	(3.11, 2.39)	(2.85, 2.04)	(1.28, 1.27)	(1.35, 1.34)
	1.8	(2.38, 2.27)	(2.09, 1.60)	(1.96, 1.43)	(0.90, 0.89)	(0.95, 0.94)
	2.0	(1.69, 1.53)	(1.56, 1.21)	(1.49, 1.11)	(0.71, 0.70)	(0.75, 0.74)
0.4	1.2	(13.77, 13.76)	(9.97, 8.70)	(8.29, 6.52)	(5.88, 5.88)	(3.16, 3.16)
	1.4	(6.32, 6.28)	(4.53, 3.61)	(4.02, 2.93)	(2.01, 2.00)	(1.61, 1.60)
	1.6	(3.38, 3.30)	(2.76, 2.13)	(2.54, 1.84)	(1.18, 1.17)	(1.09, 1.08)
	1.8	(2.17, 2.04)	(1.94, 1.50)	(1.83, 1.34)	(0.86, 0.85)	(0.84, 0.82)
	2.0	(1.60, 1.42)	(1.49, 1.16)	(1.43, 1.06)	(0.69, 0.68)	(0.69, 0.68)
0.6	1.2	(6.89, 6.85)	(5.80, 4.78)	(4.97, 3.69)	(3.32, 3.32)	(1.63, 1.63)
	1.4	(4.31, 4.25)	(3.41, 2.68)	(3.08, 2.24)	(1.59, 1.58)	(1.13, 1.12)
	1.6	(2.74, 2.64)	(2.33, 1.81)	(2.17, 1.58)	(1.05, 1.04)	(0.88, 0.87)
	1.8	(1.92, 1.77)	(1.74, 1.36)	(1.65, 1.22)	(0.80, 0.79)	(0.73, 0.72)
	2.0	(1.48, 1.29)	(1.39, 1.09)	(1.33, 1.00)	(0.66, 0.64)	(0.63, 0.61)
0.8	1.2	(3.68, 3.60)	(3.60, 2.83)	(3.19, 2.30)	(1.95, 1.94)	(1.01, 1.00)
	1.4	(2.89, 2.79)	(2.54, 1.98)	(2.33, 1.70)	(1.23, 1.22)	(0.84, 0.83)
	1.6	(2.16, 2.03)	(1.92, 1.50)	(1.80, 1.33)	(0.91, 0.90)	(0.72, 0.71)
	1.8	(1.66, 1.49)	(1.53, 1.20)	(1.46, 1.09)	(0.73, 0.72)	(0.64, 0.62)
	2.0	(1.36, 1.15)	(1.27, 0.99)	(1.22, 0.92)	(0.62, 0.60)	(0.57, 0.55)
1.0	1.2	(2.18, 2.06)	(2.40, 1.83)	(2.19, 1.57)	(1.25, 1.24)	(0.70, 0.69)
	1.4	(2.00, 1.86)	(1.91, 1.49)	(1.79, 1.31)	(0.96, 0.94)	(0.65, 0.64)
	1.6	(1.71, 1.54)	(1.57, 1.23)	(1.49, 1.12)	(0.78, 0.76)	(0.61, 0.59)
	1.8	(1.44, 1.24)	(1.32, 1.04)	(1.27, 0.96)	(0.66, 0.65)	(0.56, 0.54)
	2.0	(1.25, 1.01)	(1.14, 0.90)	(1.11, 0.84)	(0.58, 0.56)	(0.52, 0.50)
1.5	1.2	(1.05, 0.74)	(1.12, 0.85)	(1.09, 0.80)	(0.58, 0.56)	(0.39, 0.36)
	1.4	(1.09, 0.81)	(1.06, 0.82)	(1.03, 0.77)	(0.56, 0.54)	(0.41, 0.38)
	1.6	(1.10, 0.81)	(0.99, 0.77)	(0.97, 0.74)	(0.54, 0.51)	(0.41, 0.39)
	1.8	(1.07, 0.78)	(0.93, 0.72)	(0.91, 0.70)	(0.51, 0.48)	(0.42, 0.39)
	2.0	(1.03, 0.73)	(0.88, 0.68)	(0.86, 0.65)	(0.48, 0.45)	(0.41, 0.38)
2.0	1.2	(0.91, 0.53)	(0.71, 0.50)	(0.70, 0.50)	(0.36, 0.32)	(0.27, 0.23)
	1.4	(0.92, 0.55)	(0.72, 0.51)	(0.71, 0.51)	(0.38, 0.35)	(0.29, 0.25)
	1.6	(0.93, 0.57)	(0.73, 0.52)	(0.71, 0.52)	(0.39, 0.36)	(0.31, 0.27)
	1.8	(0.93, 0.58)	(0.72, 0.52)	(0.70, 0.51)	(0.39, 0.36)	(0.33, 0.29)
	2.0	(0.94, 0.58)	(0.72, 0.51)	(0.69, 0.51)	(0.39, 0.36)	(0.33, 0.30)

as the Shewhart $\bar{X}$ - S , WLC, and VSS WLC charts in detecting simultaneous shifts in the mean and variance. When the joint shift sizes are very large, e.g., $\delta > 2.0$ and $\eta > 2.0$, the ABS-SPRT chart performs almost similarly to the OSPRT chart.

Table 4.21 $(\text{ATS}_1, \text{SDTS}_1)$ values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$.

δ	η	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
		$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$	$(\text{ATS}_1, \text{SDTS}_1)$
0.4	1.4	(8.26, 8.25)	(5.19, 4.54)	(4.35, 3.46)	(2.68, 2.68)	(2.14, 2.14)
	1.7	(3.41, 3.39)	(2.30, 1.88)	(2.06, 1.56)	(1.06, 1.06)	(1.00, 1.00)
	2.0	(1.90, 1.87)	(1.42, 1.14)	(1.31, 1.00)	(0.67, 0.66)	(0.66, 0.65)
	2.5	(1.05, 0.99)	(0.86, 0.70)	(0.82, 0.65)	(0.44, 0.43)	(0.45, 0.43)
	3.0	(0.75, 0.67)	(0.62, 0.51)	(0.62, 0.49)	(0.35, 0.33)	(0.36, 0.34)
0.7	1.4	(4.69, 4.68)	(3.44, 2.88)	(2.84, 2.17)	(1.69, 1.68)	(1.17, 1.16)
	1.7	(2.51, 2.48)	(1.89, 1.53)	(1.68, 1.28)	(0.88, 0.87)	(0.75, 0.75)
	2.0	(1.57, 1.54)	(1.27, 1.02)	(1.17, 0.90)	(0.61, 0.60)	(0.57, 0.56)
	2.5	(0.95, 0.89)	(0.81, 0.66)	(0.78, 0.61)	(0.43, 0.41)	(0.42, 0.40)
	3.0	(0.71, 0.63)	(0.60, 0.50)	(0.60, 0.48)	(0.34, 0.32)	(0.35, 0.32)
1.0	1.4	(2.64, 2.62)	(2.23, 1.79)	(1.86, 1.38)	(1.03, 1.03)	(0.71, 0.70)
	1.7	(1.78, 1.75)	(1.49, 1.19)	(1.32, 1.01)	(0.70, 0.69)	(0.57, 0.56)
	2.0	(1.28, 1.23)	(1.09, 0.88)	(1.01, 0.78)	(0.54, 0.52)	(0.48, 0.47)
	2.5	(0.85, 0.79)	(0.75, 0.61)	(0.73, 0.57)	(0.40, 0.38)	(0.39, 0.37)
	3.0	(0.66, 0.58)	(0.57, 0.48)	(0.58, 0.46)	(0.33, 0.31)	(0.33, 0.31)
1.5	1.4	(1.16, 1.11)	(1.19, 0.90)	(1.02, 0.76)	(0.53, 0.52)	(0.39, 0.38)
	1.7	(1.03, 0.98)	(0.97, 0.77)	(0.88, 0.68)	(0.47, 0.46)	(0.39, 0.37)
	2.0	(0.89, 0.83)	(0.82, 0.66)	(0.77, 0.60)	(0.42, 0.41)	(0.37, 0.35)
	2.5	(0.71, 0.63)	(0.63, 0.52)	(0.63, 0.49)	(0.36, 0.34)	(0.33, 0.31)
	3.0	(0.60, 0.50)	(0.52, 0.43)	(0.53, 0.42)	(0.31, 0.29)	(0.30, 0.28)
2.0	1.4	(0.64, 0.55)	(0.72, 0.54)	(0.65, 0.48)	(0.33, 0.31)	(0.26, 0.24)
	1.7	(0.66, 0.57)	(0.66, 0.52)	(0.62, 0.48)	(0.34, 0.32)	(0.28, 0.26)
	2.0	(0.64, 0.56)	(0.61, 0.49)	(0.59, 0.46)	(0.33, 0.31)	(0.29, 0.27)
	2.5	(0.59, 0.49)	(0.52, 0.43)	(0.53, 0.42)	(0.31, 0.29)	(0.29, 0.26)
	3.0	(0.54, 0.43)	(0.46, 0.38)	(0.48, 0.37)	(0.29, 0.26)	(0.28, 0.25)
2.5	1.4	(0.47, 0.33)	(0.48, 0.36)	(0.46, 0.34)	(0.24, 0.21)	(0.20, 0.17)
	1.7	(0.50, 0.37)	(0.47, 0.37)	(0.47, 0.35)	(0.26, 0.23)	(0.22, 0.19)
	2.0	(0.51, 0.39)	(0.46, 0.37)	(0.47, 0.36)	(0.27, 0.24)	(0.24, 0.21)
	2.5	(0.51, 0.39)	(0.43, 0.36)	(0.46, 0.35)	(0.27, 0.24)	(0.25, 0.22)
	3.0	(0.49, 0.36)	(0.40, 0.33)	(0.44, 0.33)	(0.26, 0.24)	(0.25, 0.22)
3.0	1.4	(0.41, 0.25)	(0.34, 0.26)	(0.37, 0.26)	(0.19, 0.15)	(0.17, 0.13)
	1.7	(0.43, 0.28)	(0.35, 0.28)	(0.39, 0.28)	(0.21, 0.17)	(0.19, 0.15)
	2.0	(0.44, 0.30)	(0.36, 0.29)	(0.40, 0.29)	(0.23, 0.19)	(0.20, 0.17)
	2.5	(0.45, 0.32)	(0.36, 0.29)	(0.40, 0.30)	(0.24, 0.21)	(0.22, 0.19)
	3.0	(0.45, 0.31)	(0.35, 0.29)	(0.40, 0.29)	(0.24, 0.21)	(0.23, 0.20)

4.5.2 Comparative Analyses Based on the ANOS-optimal and EANOS-optimal Designs

In this section, we compare the performances of the optimal OSPRT chart with the Shewhart $\bar{X}$ - S , WLC, VSS WLC, and ABS-SPRT charts designed under the ANOS-optimisation and EANOS-optimisation models. Note that the ANOS provides a combined measure for both the detection speed and sampling efficiency of a control chart. Generally, practitioners desire a small out-of-control ANOS (ANOS_1), since a small ANOS_1 indicates that only a few number of units need to be inspected on average before an out-of-control signal is produced. In the case where sampling cost is high, the control chart with the lowest ANOS_1 would be preferred since it could lead to potential savings in sampling and inspection costs in the long run.

To ensure a fair comparison, we fix the design specifications for the five control charts as $\tau_{\text{ANOS}} = 1500$ and $\text{ASN}_0 = 5$ (note that the sampling interval d is not relevant in all ANOS-based statistical designs). For the ease of demonstration, we again take $\mu_0 = 0$ and $\sigma_0 = 1$, without loss of generality. Since the ASN_0 is fixed at five, it follows that the sample sizes for the Shewhart $\bar{X}$ - S and WLC charts are both $n = 5$. Assuming that shift sizes are deterministic, we compute the charting parameters of the Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts using the ANOS-optimisation model described in Section 4.3.3. Note once again that each control chart has a different set of charting parameters, and hence the optimum objective function should be searched in their respective parameter space. Table 4.22 shows the optimal charting parameters of the five control charts optimised for nine pairs of shift sizes constructed by $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$. As a numeric example, when $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (1.0, 2.0)$ is chosen, the optimal Shewhart $\bar{X}$ - S chart has charting parameters $\text{UCL}_{\bar{X}} = 1.329$, $\text{UCL}_S = 2.067$, and $n = 5$.

Table 4.22 Optimal charting parameters of the ATS-optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts, optimised for $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$, when $\tau_{\text{ANOS}} = 1500$ and $\text{ASN}_0 = 5$.

δ_{opt}	η_{opt}	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
		$\text{UCL}_{\bar{X}}$	$(\lambda_{\text{WL}}, \gamma_{\text{WLC}})$	$(n_1, n_2, \lambda_{\text{WL}}, \gamma_{\text{WLC}})$	γ_{ABS}	(k, γ)
		UCL_S	H_{WLC}	$(W_{\text{WLC}}, H_{\text{WLC}})$	$(g_{\text{ABS}}, h_{\text{ABS}})$	(g, h)
0.4	1.2	1.229	(0.295, 0.515)	(4, 6, 0.299, 0.485)	0.917	(0.909, 2.383)
		2.283	2.041	(0.364, 2.205)	(-0.308, 7.265)	(-2.093, 27.553)
	1.6	1.367	(0.406, 0.775)	(4, 6, 0.410, 0.664)	1.078	(0.256, 1.711)
		2.043	1.383	(0.010, 1.787)	(-1.049, 4.055)	(-2.643, 14.311)
	2.0	1.422	(0.428, 0.950)	(4, 6, 0.425, 0.680)	1.208	(0.133, 1.920)
		2.021	1.090	(0.010, 1.823)	(-1.670, 2.967)	(-3.733, 11.389)
1.0	1.2	1.214	(0.189, 0.543)	(4, 6, 0.180, 0.429)	1.041	(2.273, 8.631)
		2.734	1.320	(0.048, 1.697)	(-0.903, 4.514)	(-9.563, 31.360)
	1.6	1.264	(0.325, 0.775)	(4, 6, 0.320, 0.574)	1.164	(0.641, 2.595)
		2.151	1.063	(0.013, 1.625)	(-1.462, 3.268)	(-4.929, 14.491)
	2.0	1.329	(0.381, 0.950)	(4, 6, 0.370, 0.625)	1.258	(0.333, 2.293)
		2.067	0.930	(0.010, 1.694)	(-1.900, 2.683)	(-5.006, 11.161)
2.0	1.2	1.213	(0.153, 0.949)	(4, 6, 0.130, 0.396)	1.330	(4.545, 30.945)
		3.078	0.708	(0.010, 1.717)	(-2.226, 2.355)	(-39.061, 29.210)
	1.6	1.227	(0.263, 1.002)	(4, 6, 0.245, 0.503)	1.362	(1.282, 5.750)
		2.303	0.654	(0.010, 1.585)	(-2.371, 2.234)	(-13.408, 13.757)
	2.0	1.265	(0.322, 1.094)	(4, 6, 0.310, 0.566)	1.396	(0.667, 3.626)
		2.149	0.620	(0.010, 1.606)	(-2.524, 2.117)	(-9.530, 10.493)

Table 4.23 (ARL₁, ANOS₁) values of the ANOS-optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts, optimised for $\delta_{\text{opt}} \in \{0.4, 1.0, 2.0\}$ and $\eta_{\text{opt}} \in \{1.2, 1.6, 2.0\}$, when $\tau_{\text{ANOS}} = 1500$ and $\text{ASN}_0 = 5$.

		Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
δ_{opt}	η_{opt}	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)
0.4	1.2	(14.984, 74.921)	(8.971, 44.846)	(8.302, 46.808)	(4.847, 58.899)	(2.807, 34.093)
	1.6	(4.214, 21.068)	(3.060, 15.298)	(3.001, 16.633)	(1.742, 16.007)	(1.584, 13.470)
	2.0	(2.122, 10.608)	(1.828, 9.141)	(1.877, 10.136)	(1.268, 8.172)	(1.224, 7.298)
1.0	1.2	(2.894, 14.468)	(2.691, 13.453)	(2.631, 14.506)	(1.782, 18.472)	(1.183, 10.237)
	1.6	(2.293, 11.466)	(2.003, 10.017)	(2.026, 11.007)	(1.350, 9.954)	(1.197, 7.510)
	2.0	(1.681, 8.406)	(1.539, 7.697)	(1.595, 8.502)	(1.168, 6.383)	(1.122, 5.444)
2.0	1.2	(1.077, 5.384)	(1.102, 5.511)	(1.142, 5.884)	(1.022, 4.428)	(1.001, 3.220)
	1.6	(1.147, 5.737)	(1.141, 5.703)	(1.181, 6.099)	(1.037, 4.040)	(1.009, 3.174)
	2.0	(1.160, 5.801)	(1.134, 5.672)	(1.174, 6.055)	(1.036, 3.571)	(1.018, 3.005)

To compare the performances among the five ANOS-optimal control charts, we tabulate their (ARL₁, ANOS₁) values at each shift point (δ_{opt} , η_{opt}) in Table 4.23. It should be noted that an effective control chart would have both a low ARL₁ value and a low ANOS₁ value, both representing a high detection speed and a high inspection efficiency of the control chart, respectively. From the previous example, the optimal Shewhart $\bar{X}$ - S chart has (ARL₁, ANOS₁) values equal to (1.681, 8.406) (see Table 4.23). To assist the reader in identifying the best performing chart, we have shaded the table cells according to the sizes of the ANOS₁ values of each row. That is, the cell with the smallest ANOS₁ value of each row is shaded in the lightest blue, while the cell with the largest ANOS₁ value of each row is shaded in the darkest blue. Note that our main metric of evaluation in this section is the ANOS₁, i.e., we favour the control chart that has the smallest ANOS₁ value, even if it means that the control chart has a larger ARL₁ value compared to other contestants. This is because a small ARL₁ value does not necessarily warrant a small number of observations inspected – the control chart that signals within two sampling periods could end up sampling 100 units of products. Note, however, that a control chart would be ideal if it achieves both the lowest ARL₁ and ANOS₁ values among all contestants.

From Table 4.23, it is clear that the OSPRT chart outperforms the Shewhart $\bar{X}$ - S , WLC, VSS WLC, and ABS-SPRT charts for all small, moderate, and large sizes of process shifts, in terms of both the ARL_1 and $ANOS_1$. In particular, the OSPRT chart is found to have significantly lower ARL_1 values compared to the other four control charts, especially when shift sizes are small. For instance, when $(\delta_{\text{opt}}, \eta_{\text{opt}}) = (0.4, 1.2)$, the ARL_1 value of the OSPRT chart ($ARL_1 = 2.807$) is just about one half of that of the ABS-SPRT chart ($ARL_1 = 4.847$), about one-third of those of the WLC ($ARL_1 = 8.971$) and VSS WLC ($ARL_1 = 8.302$) charts, and about one-fifth of that of the Shewhart $\bar{X}$ - S chart ($ARL_1 = 14.984$). It is therefore evident that the OSPRT chart has a relative advantage over the other four control charts in terms of detection speed, making it appealing to applications where a high precision is required (e.g., power, medical, and aeronautical industries). In terms of the $ANOS_1$ performance, one observes that the OSPRT chart remains the most efficient among all the control charts, i.e., it has the smallest $ANOS_1$ value across all magnitudes of shifts. Although the disparities between the $ANOS_1$ performances of the OSPRT chart and those of the other control charts are not as prominent as those observed in the ARL_1 performances, the OSPRT chart is still the best performer in terms of its sensitivity and out-of-control sampling efficiency. It should be noted that, by Proposition 4.1 and Corollary 4.2, the OSPRT chart with prescribed reference parameters in Equations (4.7b) and (4.7c) is optimum with respect to both the ARL_1 and $ANOS_1$ for any deterministic shift size $(\delta_{\text{opt}}, \eta_{\text{opt}})$. This implies that the ANOS-optimal design for the OSPRT chart requires no parameter searches and is very much design-friendly.

Assuming that shift sizes are unpredictable, we now compare the performance of the OSPRT chart with those of the four control charts designed under the EANOS-optimisation model described in Section 4.3.4. In our illustration, we select the design specifications as $\tau_{\text{ANOS}} = 1500$, $ASN_0 = 5$, and $(\delta_{\text{max}}, \eta_{\text{max}}) \in$

Table 4.24 Optimal charting parameters of the Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS SPRT, and OSPRT charts designed for $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 5$, and $(\delta_{\text{max}}, \eta_{\text{max}}) \in \{(2.0, 2.0), (3.0, 3.0)\}$.

Charts	Charting parameters			EANOS
	Control limits & warning limits	Reference parameter(s)	Weighting Factor	
$(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$				
$\bar{X}$ - S	$\text{UCL}_{\bar{X}} = 1.268, \text{UCL}_S = 2.143$	—	—	23.258
WLC	$H_{\text{WLC}} = 1.858$	$\gamma_{\text{WLC}} = 0.546$	$\lambda_{\text{WL}} = 0.307$	16.774
VSS WLC	$H_{\text{WLC}} = 2.052, W_{\text{WLC}} = 0.263$	$\gamma_{\text{WLC}} = 0.510$	$\lambda_{\text{WL}} = 0.313$	17.455
ABS-SPRT	$g = -0.623, h = 5.605$	$\gamma_{\text{ABS}} = 0.977$	—	20.058
SPRT	$g = -1.795, h = 20.187$	$k = 0.537, \gamma = 1.799$	—	13.700
$(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$				
$\bar{X}$ - S	$\text{UCL}_{\bar{X}} = 1.274, \text{UCL}_S = 2.131$	—	—	11.621
WLC	$H_{\text{WLC}} = 1.659$	$\gamma_{\text{WLC}} = 0.594$	$\lambda_{\text{WL}} = 0.325$	9.580
VSS WLC	$H_{\text{WLC}} = 1.976, W_{\text{WLC}} = 0.214$	$\gamma_{\text{WLC}} = 0.523$	$\lambda_{\text{WL}} = 0.319$	9.975
ABS-SPRT	$g_{\text{ABS}} = -0.960, h_{\text{ABS}} = 4.329$	$\gamma_{\text{ABS}} = 1.055$	—	9.581
SPRT	$g = -2.482, h = 17.123$	$k = 0.470, \gamma = 1.850$	—	6.966

$\{(2.0, 2.0), (3.0, 3.0)\}$. Table 4.24 tabulates the charting parameters of the EANOS-optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts, as well as their resulting minimum EANOS values. From the table, the OSPRT chart is found to achieve the smallest EANOS value among all control charts, proving its exceptional capability in both shift detection and cost saving. To assess the performance of the EANOS-optimal OSPRT chart for various shift sizes, we present discussions based on the three different scenarios, i.e., when (i) only the process mean increases, (ii) only the process standard deviation increases, and (iii) both the process mean and standard deviation increase.

Case (i): Only the process mean increases

Tables 4.25 and 4.26 compare the $(\text{ARL}_1, \text{ANOS}_1)$ values of the EANOS-optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\text{max}}, \eta_{\text{max}}) = (2.0, 2.0)$ and $(\delta_{\text{max}}, \eta_{\text{max}}) = (3.0, 3.0)$, respectively. In Table 4.25, the metrics are calculated for $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ (since $\delta_{\text{max}} = 2.0$), whereas in Table 4.26, the metrics are calculated for $\delta \in \{0.4, 0.7, 1.0, 1.5,$

Table 4.25 (ARL₁, ANOS₁) values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta = 1$.

	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
δ	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)
0.2	(105.23, 526.13)	(131.11, 655.57)	(125.11, 648.86)	(186.75, 1008.00)	(40.00, 309.60)
0.4	(36.83, 184.15)	(32.68, 163.41)	(29.29, 161.24)	(56.06, 382.23)	(7.09, 87.48)
0.6	(14.58, 72.88)	(11.49, 57.43)	(10.43, 58.68)	(13.23, 127.53)	(2.64, 37.79)
0.8	(6.73, 33.66)	(5.86, 29.29)	(5.48, 30.76)	(4.17, 51.74)	(1.67, 21.66)
1.0	(3.63, 18.16)	(3.69, 18.43)	(3.51, 19.59)	(2.16, 27.19)	(1.32, 14.45)
1.5	(1.43, 7.16)	(1.80, 9.02)	(1.77, 9.57)	(1.23, 10.55)	(1.06, 7.20)
2.0	(1.05, 5.27)	(1.22, 6.10)	(1.22, 6.40)	(1.06, 6.19)	(1.01, 4.49)

Table 4.26 (ARL₁, ANOS₁) values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta = 1$.

	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
δ	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)
0.4	(37.81, 189.05)	(37.58, 187.88)	(30.80, 169.15)	(70.06, 438.13)	(10.37, 110.41)
0.7	(9.93, 49.66)	(8.45, 42.26)	(7.44, 41.91)	(9.73, 92.89)	(2.22, 30.89)
1.0	(3.69, 18.46)	(3.74, 18.70)	(3.53, 19.70)	(2.35, 28.45)	(1.32, 14.69)
1.5	(1.44, 7.21)	(1.78, 8.92)	(1.76, 9.56)	(1.16, 9.60)	(1.05, 6.90)
2.0	(1.06, 5.28)	(1.21, 6.03)	(1.22, 6.38)	(1.03, 5.36)	(1.01, 4.21)
2.5	(1.00, 5.02)	(1.03, 5.14)	(1.04, 5.30)	(1.01, 3.69)	(1.00, 2.93)
3.0	(1.00, 5.00)	(1.00, 5.01)	(1.00, 5.08)	(1.00, 2.85)	(1.00, 2.23)

2.0, 2.5, 3.0} (since $\delta_{\max} = 3.0$). A similar colour coding based on blue shades has been applied to each table. Note that the colour of each cell is determined based on the size of the ANOS₁ values instead of the ARL₁ values. From both Tables 4.25 and 4.26, the OSPRT chart clearly outperforms the other four control charts in terms of both the ARL₁ and ANOS₁. In particular, we observe an overwhelming disparity between the ARL₁ performance of the OSPRT chart and those of the other four control charts, and their differences can reach up to five times for a small mean shift size. Regarding the ANOS₁ performances, the OSPRT chart defeats the ABS-SPRT chart substantially (> 2 folds) when δ is small and moderate ($\delta \leq 1.0$), and wins over all other control charts by a modest degree when δ is large ($\delta > 1.0$)

(with the exception of $\delta = 1.5$ in Table 4.25). As a numeric example, referring to Table 4.18, the ANOS_1 value of the OSPRT chart ($= 14.69$) is about twice as small as that of the ABS-SPRT chart ($= 28.45$) for $\delta = 1.0$.

Case (ii): Only the process standard deviation increases

Tables 4.27 and 4.28 compare the $(\text{ARL}_1, \text{ANOS}_1)$ values of the five EANOS-optimal control charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$ and $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, respectively. In Table 4.27, the metrics are evaluated for $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$ (since $\eta_{\max} = 2.0$), whereas in Table 4.28, the metrics are evaluated for $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$ (since $\eta_{\max} = 3.0$). From the tables, it seems that none of the control charts is consistently the best performer in terms of the ANOS_1 metric. In particular, when $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$ is used, the lowest ANOS_1 is found to be achieved by the WLC chart for $1.2 \leq \eta \leq 1.6$, whereas for $\eta \geq 1.8$, the

Table 4.27 $(\text{ARL}_1, \text{ANOS}_1)$ values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta = 0$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.

	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
η	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$
1.2	(46.55, 232.74)	(15.85, 79.26)	(14.44, 81.55)	(9.83, 97.49)	(13.07, 120.13)
1.4	(13.75, 68.74)	(6.14, 30.71)	(5.77, 32.52)	(2.88, 32.63)	(3.53, 38.53)
1.6	(6.24, 31.20)	(3.78, 18.88)	(3.59, 20.03)	(1.89, 18.67)	(2.14, 20.86)
1.8	(3.71, 18.53)	(2.76, 13.79)	(2.64, 14.55)	(1.56, 13.10)	(1.69, 13.95)
2.0	(2.60, 13.02)	(2.21, 11.03)	(2.12, 11.57)	(1.41, 10.15)	(1.48, 10.38)

Table 4.28 $(\text{ARL}_1, \text{ANOS}_1)$ values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta = 0$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$.

	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
η	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$	$(\text{ARL}_1, \text{ANOS}_1)$
1.4	(13.42, 67.08)	(6.02, 30.08)	(5.72, 32.29)	(3.24, 33.87)	(3.58, 38.02)
1.7	(4.60, 23.01)	(3.04, 15.21)	(2.99, 16.61)	(1.63, 14.47)	(1.76, 15.72)
2.0	(2.57, 12.86)	(2.11, 10.53)	(2.09, 11.41)	(1.31, 9.15)	(1.39, 9.58)
2.5	(1.59, 7.96)	(1.50, 7.49)	(1.50, 7.98)	(1.15, 5.81)	(1.19, 5.78)
3.0	(1.29, 6.43)	(1.25, 6.27)	(1.26, 6.60)	(1.10, 4.37)	(1.11, 4.20)

lowest ANOS_1 is achieved by the ABS-SPRT chart. When $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$ is used, the OSPRT chart is found to achieve the smallest ANOS_1 among all control charts, for a moderately large standard deviation shift size. In terms of the ARL_1 performance, the ABS-SPRT chart is still the best performing control chart among all competing charts (see Section 4.5.1), although it could be argued that the OSPRT chart performs nearly as well as the ABS-SPRT chart for moderately large standard deviation shift sizes, i.e., $1.4 \leq \eta \leq 2.0$. When η is very large (i.e., $\eta > 2.0$), the OSPRT chart becomes slightly more efficient than the ABS-SPRT chart (see Table 4.28).

Case (iii): Both the process mean and standard deviation increase

Tables 4.29 and 4.30 compare the $(\text{ATS}_1, \text{SDTS}_1)$ performances of the five control charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$ and $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, respectively. In Table 4.29, the metrics are evaluated for $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$, whereas in Table 4.30, the metrics are evaluated for $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$. From both Tables 4.20 and 4.21, we find that the OSPRT chart remains the best performer among all its competitors for all magnitudes of shift sizes. It is worth noting that, even though the ABS-SPRT chart has one of the smallest ARL_1 values, its ANOS_1 performances are uniformly worse than those of the WLC and VSS WLC charts when η is small (i.e., $1 < \eta \leq 1.4$ for all values of δ). The discerning reader might have also noticed that, although the VSS WLC chart has a smaller ARL_1 value compared to the WLC chart, its ANOS_1 value is slightly larger than those of the WLC chart in all cases. This can be explained by the fact that, when a process shift occurs, the alert (larger) sample size of a VSS control chart will be used more frequently than the relax (smaller) sample size, thus leading to a larger average sample number in an out-of-control case. When the mean shift size is very large (e.g., $\delta > 2.0$), the performance ranking based on the ANOS_1 metric is $\text{OSPRT} >$

Table 4.29 (ARL₁, ANOS₁) values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (2.0, 2.0)$, for shift sizes $\delta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ and $\eta \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$.

δ	η	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
		(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)
0.2	1.2	(28.15, 140.74)	(13.32, 66.58)	(12.17, 68.75)	(8.34, 85.61)	(5.55, 61.95)
	1.4	(10.62, 53.11)	(5.76, 28.82)	(5.42, 30.52)	(2.75, 31.00)	(2.63, 28.28)
	1.6	(5.37, 26.87)	(3.65, 18.27)	(3.48, 19.37)	(1.86, 18.18)	(1.90, 17.49)
	1.8	(3.37, 16.85)	(2.70, 13.52)	(2.59, 14.27)	(1.55, 12.89)	(1.59, 12.44)
	2.0	(2.44, 12.21)	(2.18, 10.89)	(2.10, 11.42)	(1.40, 10.04)	(1.43, 9.58)
0.4	1.2	(15.44, 77.21)	(9.02, 45.10)	(8.33, 47.00)	(5.56, 61.45)	(2.97, 35.45)
	1.4	(7.64, 38.19)	(4.88, 24.42)	(4.61, 25.85)	(2.44, 26.97)	(2.06, 20.88)
	1.6	(4.44, 22.21)	(3.34, 16.70)	(3.18, 17.67)	(1.78, 16.87)	(1.69, 14.45)
	1.8	(2.99, 14.97)	(2.56, 12.80)	(2.45, 13.48)	(1.52, 12.30)	(1.49, 10.91)
	2.0	(2.26, 11.30)	(2.10, 10.51)	(2.02, 11.01)	(1.39, 9.72)	(1.37, 8.71)
0.6	1.2	(8.49, 42.44)	(5.94, 29.70)	(5.56, 31.22)	(3.49, 40.61)	(2.00, 22.69)
	1.4	(5.32, 26.60)	(3.93, 19.65)	(3.73, 20.79)	(2.10, 22.18)	(1.70, 15.78)
	1.6	(3.57, 17.86)	(2.93, 14.67)	(2.81, 15.50)	(1.67, 15.08)	(1.52, 11.92)
	1.8	(2.61, 13.06)	(2.36, 11.78)	(2.26, 12.39)	(1.47, 11.43)	(1.40, 9.49)
	2.0	(2.07, 10.33)	(1.99, 9.94)	(1.92, 10.40)	(1.36, 9.24)	(1.32, 7.84)
0.8	1.2	(4.96, 24.79)	(4.10, 20.51)	(3.89, 21.68)	(2.37, 27.10)	(1.56, 15.83)
	1.4	(3.73, 18.66)	(3.13, 15.66)	(2.99, 16.55)	(1.81, 17.81)	(1.47, 12.25)
	1.6	(2.85, 14.25)	(2.53, 12.66)	(2.43, 13.33)	(1.56, 13.16)	(1.39, 9.89)
	1.8	(2.26, 11.30)	(2.13, 10.66)	(2.06, 11.19)	(1.42, 10.43)	(1.32, 8.23)
	2.0	(1.88, 9.39)	(1.86, 9.28)	(1.80, 9.69)	(1.33, 8.64)	(1.27, 7.02)
1.0	1.2	(3.15, 15.75)	(3.01, 15.04)	(2.88, 15.93)	(1.80, 18.98)	(1.33, 11.74)
	1.4	(2.71, 13.54)	(2.53, 12.64)	(2.43, 13.33)	(1.59, 14.28)	(1.31, 9.76)
	1.6	(2.29, 11.47)	(2.18, 10.88)	(2.10, 11.44)	(1.45, 11.35)	(1.29, 8.28)
	1.8	(1.95, 9.77)	(1.92, 9.58)	(1.86, 10.03)	(1.36, 9.39)	(1.25, 7.15)
	2.0	(1.70, 8.52)	(1.72, 8.59)	(1.67, 8.96)	(1.30, 8.00)	(1.22, 6.27)
1.5	1.2	(1.49, 7.45)	(1.72, 8.61)	(1.69, 9.08)	(1.26, 9.65)	(1.09, 6.61)
	1.4	(1.51, 7.54)	(1.64, 8.19)	(1.61, 8.60)	(1.27, 8.72)	(1.12, 6.06)
	1.6	(1.48, 7.41)	(1.56, 7.79)	(1.53, 8.14)	(1.26, 7.84)	(1.13, 5.57)
	1.8	(1.43, 7.14)	(1.48, 7.42)	(1.46, 7.73)	(1.24, 7.07)	(1.13, 5.13)
	2.0	(1.36, 6.81)	(1.41, 7.07)	(1.39, 7.35)	(1.22, 6.41)	(1.13, 4.75)
2.0	1.2	(1.09, 5.47)	(1.23, 6.16)	(1.23, 6.44)	(1.10, 6.09)	(1.02, 4.34)
	1.4	(1.13, 5.65)	(1.24, 6.18)	(1.23, 6.44)	(1.12, 5.90)	(1.04, 4.17)
	1.6	(1.15, 5.77)	(1.23, 6.15)	(1.23, 6.40)	(1.14, 5.66)	(1.05, 4.00)
	1.8	(1.16, 5.82)	(1.22, 6.10)	(1.22, 6.33)	(1.15, 5.39)	(1.07, 3.84)
	2.0	(1.16, 5.80)	(1.21, 6.03)	(1.20, 6.25)	(1.15, 5.10)	(1.07, 3.68)

ABS-SPRT > Shewhart $\bar{X}$ - S > WLC > VSS WLC, which is a rather surprising result as we often believe that Shewhart control charts are inferior to memory-type control charts. Overall, the OSPRT chart stands out as the winner of the comparative study, and we highly recommended its use in all SQC applications.

Table 4.30 (ARL₁, ANOS₁) values of the optimal Shewhart $\bar{X}$ - S , WLC, VSS WLC, ABS-SPRT, and OSPRT charts designed using $(\delta_{\max}, \eta_{\max}) = (3.0, 3.0)$, for shift sizes $\delta \in \{0.4, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and $\eta \in \{1.4, 1.7, 2.0, 2.5, 3.0\}$.

δ	η	Shewhart $\bar{X}$ - S	WLC	VSS WLC	ABS-SPRT	OSPRT
		(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)	(ARL ₁ , ANOS ₁)
0.4	1.4	(7.60, 37.98)	(4.78, 23.90)	(4.58, 25.68)	(2.62, 27.30)	(2.04, 20.76)
	1.7	(3.55, 17.76)	(2.78, 13.88)	(2.73, 15.09)	(1.55, 13.25)	(1.50, 11.83)
	2.0	(2.24, 11.20)	(2.01, 10.05)	(2.00, 10.86)	(1.29, 8.74)	(1.30, 8.13)
	2.5	(1.50, 7.50)	(1.47, 7.36)	(1.48, 7.83)	(1.15, 5.69)	(1.16, 5.33)
	3.0	(1.25, 6.24)	(1.25, 6.23)	(1.25, 6.55)	(1.10, 4.32)	(1.10, 4.01)
0.7	1.4	(4.45, 22.27)	(3.43, 17.15)	(3.32, 18.43)	(1.95, 19.36)	(1.53, 13.58)
	1.7	(2.75, 13.75)	(2.37, 11.86)	(2.34, 12.82)	(1.43, 11.32)	(1.34, 9.27)
	2.0	(1.96, 9.80)	(1.85, 9.24)	(1.84, 9.94)	(1.25, 7.99)	(1.24, 6.96)
	2.5	(1.42, 7.11)	(1.42, 7.11)	(1.43, 7.55)	(1.14, 5.45)	(1.14, 4.90)
	3.0	(1.22, 6.08)	(1.23, 6.14)	(1.24, 6.44)	(1.09, 4.21)	(1.09, 3.82)
1.0	1.4	(2.72, 13.61)	(2.47, 12.37)	(2.42, 13.26)	(1.51, 13.36)	(1.27, 9.42)
	1.7	(2.11, 10.55)	(1.97, 9.87)	(1.95, 10.59)	(1.31, 9.28)	(1.22, 7.28)
	2.0	(1.70, 8.49)	(1.66, 8.30)	(1.66, 8.87)	(1.21, 7.09)	(1.18, 5.89)
	2.5	(1.34, 6.71)	(1.36, 6.79)	(1.36, 7.18)	(1.13, 5.12)	(1.12, 4.45)
	3.0	(1.18, 5.92)	(1.20, 6.01)	(1.21, 6.29)	(1.09, 4.06)	(1.08, 3.59)
1.5	1.4	(1.51, 7.57)	(1.61, 8.04)	(1.60, 8.56)	(1.18, 7.77)	(1.09, 5.75)
	1.7	(1.46, 7.30)	(1.49, 7.43)	(1.48, 7.88)	(1.16, 6.56)	(1.10, 5.04)
	2.0	(1.36, 6.80)	(1.38, 6.90)	(1.38, 7.29)	(1.14, 5.61)	(1.10, 4.47)
	2.5	(1.22, 6.09)	(1.24, 6.21)	(1.25, 6.52)	(1.10, 4.48)	(1.08, 3.73)
	3.0	(1.13, 5.66)	(1.15, 5.76)	(1.16, 6.00)	(1.07, 3.73)	(1.07, 3.20)
2.0	1.4	(1.13, 5.66)	(1.22, 6.09)	(1.23, 6.41)	(1.07, 5.12)	(1.03, 3.91)
	1.7	(1.16, 5.81)	(1.21, 6.04)	(1.22, 6.34)	(1.08, 4.79)	(1.04, 3.68)
	2.0	(1.16, 5.80)	(1.19, 5.94)	(1.20, 6.21)	(1.08, 4.42)	(1.05, 3.46)
	2.5	(1.12, 5.62)	(1.14, 5.72)	(1.15, 5.96)	(1.07, 3.84)	(1.05, 3.12)
	3.0	(1.09, 5.43)	(1.10, 5.51)	(1.11, 5.72)	(1.06, 3.37)	(1.05, 2.82)
2.5	1.4	(1.02, 5.12)	(1.06, 5.29)	(1.07, 5.48)	(1.02, 3.73)	(1.01, 2.88)
	1.7	(1.05, 5.23)	(1.07, 5.36)	(1.08, 5.55)	(1.04, 3.67)	(1.02, 2.82)
	2.0	(1.06, 5.30)	(1.08, 5.39)	(1.09, 5.58)	(1.05, 3.55)	(1.02, 2.76)
	2.5	(1.06, 5.31)	(1.08, 5.38)	(1.09, 5.56)	(1.05, 3.28)	(1.03, 2.63)
	3.0	(1.05, 5.25)	(1.06, 5.31)	(1.07, 5.48)	(1.05, 3.01)	(1.03, 2.49)
3.0	1.4	(1.00, 5.01)	(1.01, 5.05)	(1.02, 5.15)	(1.01, 2.93)	(1.00, 2.25)
	1.7	(1.01, 5.05)	(1.02, 5.09)	(1.03, 5.22)	(1.02, 2.94)	(1.01, 2.26)
	2.0	(1.02, 5.09)	(1.03, 5.13)	(1.03, 5.26)	(1.03, 2.92)	(1.01, 2.27)
	2.5	(1.03, 5.13)	(1.03, 5.17)	(1.04, 5.31)	(1.03, 2.82)	(1.02, 2.25)
	3.0	(1.03, 5.14)	(1.03, 5.17)	(1.04, 5.30)	(1.04, 2.68)	(1.02, 2.20)

4.6 Real Industrial Examples

In this section, two industrial examples are provided to illustrate the implementation of the OSPRT chart. Section 4.6.1 presents an application for monitoring tensile strength measurements obtained from a wire pull test using the AEQL-optimal OSPRT chart proposed in Section 4.3.2. In Section 4.6.2, an application for monitoring shear force measurements from a ball bond shear test is demonstrated using the EANOS-optimal OSPRT chart proposed in Section 4.3.4.

4.6.1 Wire Pull Test

The wire bonding process is a standard process used to establish electrical connections between integrated circuits and lead frames in the fabrication of semiconductor packages. Before proceeding to subsequent back-end processes, it is often necessary to perform a wire pull test on a set of samples to ensure that the bond quality adhere to specific standards. The wire pull test is conducted by applying a pulling force on the wire. The force required to break the wire off the lead frame is then measured and monitored via a quality control software. While there is a minimum specification limit for the pull force in practice, it is essential to ensure that the pull force measurements do not go over or below certain control limits. This is because pull values that are too high or too low can indicate a variety of suboptimal conditions, such as incorrect positioning of the test machine, incorrect wire loop geometry, etc (Harman, 2010). Besides, poor setup of apparatus may also surface through an increase in the dispersion of force measurements. Hence, it is crucial to monitor the pull force readings with the OSPRT chart for both upper-sided and lower-sided mean shifts, as well as for an upper-sided standard deviation shift.

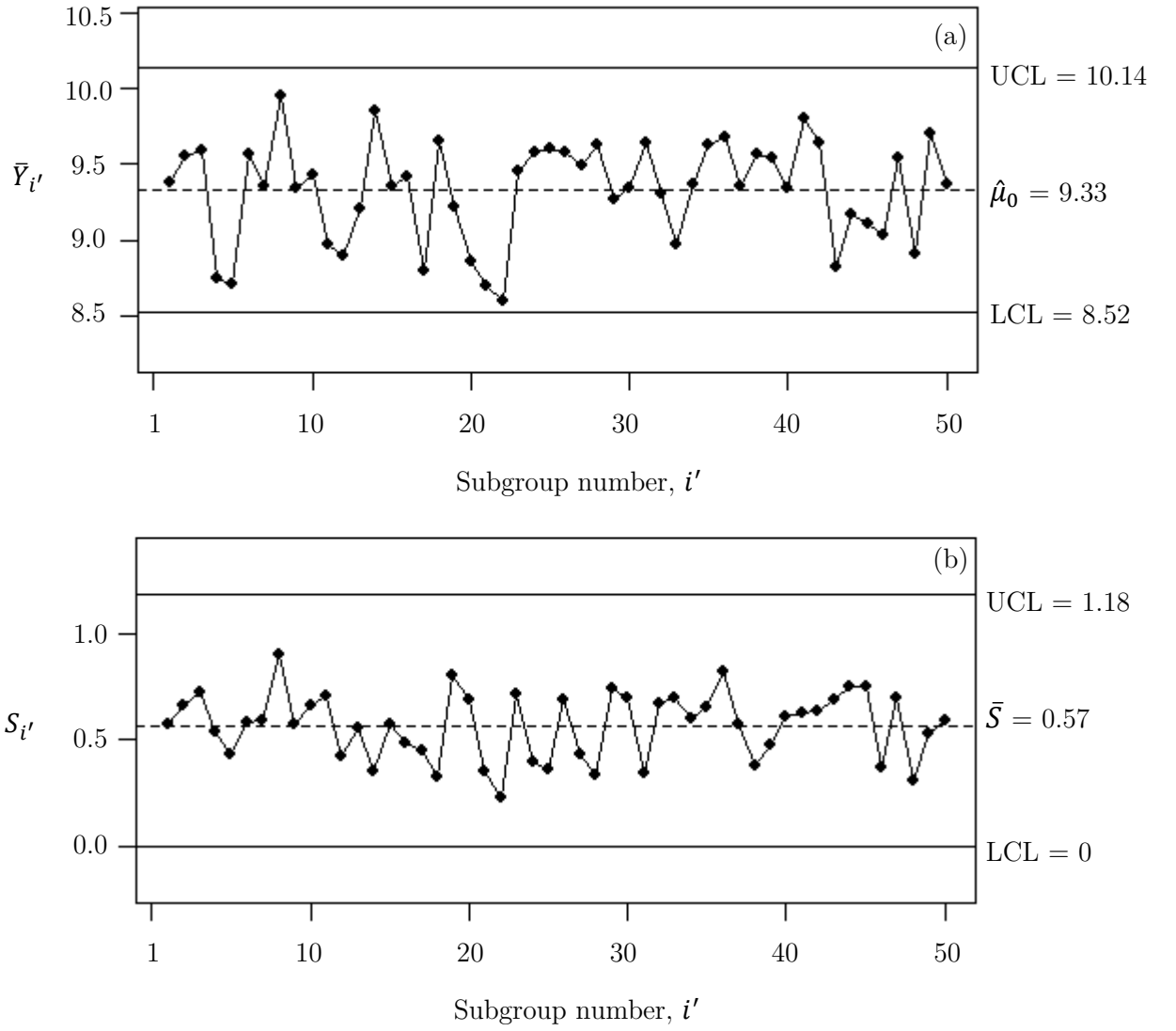


Figure 4.3 The Phase-I Shewhart (a) $\bar{X}$ chart and (b) S chart for monitoring 50 Phase-I samples, each consisting of five pull force measurements.

The dataset used in this section is obtained from a wire pull test provided by an optoelectronic product manufacturing company. The measurements are expressed in units of gram-force (gf). To obtain reliable estimates for the mean and standard deviation of the process, 50 Phase-I samples, each containing five observations, have been collected. The stability of the Phase-I samples $(Y_{1,1}, \dots, Y_{1,5}), \dots, (Y_{50,1}, \dots, Y_{50,5})$ are assessed using a Phase-I Shewhart $\bar{X}$ - S chart with three-sigma control limits (Montgomery, 2019). Figure 4.3(a) displays the Phase-I $\bar{X}$ chart with UCL and LCL equal to $\hat{\mu}_0 \pm 3\hat{\sigma}_0/\sqrt{5} = (8.52, 10.14)$ gf, respectively.

The estimated process parameters are calculated as $\hat{\mu}_0 = \sum_{i'=1}^{50} \bar{Y}_{i'}/50 = 9.33$ gf and $\hat{\sigma}_0 = \bar{S}/c_4 = 0.60$ gf. Here, $\bar{Y}_{i'} = \sum_{j'=1}^5 Y_{i',j'}/5$ is the sample average of the i' th subgroup, $\bar{S} = \sum_{i'=1}^{50} S_{i'}/50$ is the mean of the sample standard deviations, $S_{i'} = \sqrt{\sum_{j'=1}^5 (Y_{i',j'} - \bar{Y}_{i'})^2 / 4}$ is the sample standard deviation of the i' th subgroup, and $c_4 = 0.9400$ is an adjustment factor for the standard deviation estimator. Figure 4.3(b) displays the Phase-I S chart with LCL = 0 gf and UCL = $(c_4 + 3\sqrt{1 - c_4^2}) \hat{\sigma}_0 = 1.18$ gf. From Figure 4.3, all the Phase-I statistics are found to fall within the control limits, indicating that the Phase-I measurements are in-control. Hence, the mean and standard deviation of the pull force measurements are estimated as $\hat{\mu}_0 = 9.33$ gf and $\hat{\sigma}_0 = 0.60$ gf, respectively.

Suppose that the optoelectronic product manufacturing company is currently inspecting five measurements every six hours. Letting $d = 1$ correspond to a duration of six hours, the inspection rate adopted by the company would be computed as $\text{IR} = 5/1 = 5$. Using the design specifications $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 5$, $\bar{n}_{\text{max}} = 15$, $d_{\text{min}} = 0.2$, $\delta_{\text{max}} = 3.00$, and $\eta_{\text{max}} = 3.00$, the optimal charting parameters of the OSPRT chart for an upper-sided mean can be obtained from Table 4.4 as $(\text{ASN}_0, k, \gamma, d, g, h) = (1.419, 0.394, 2.227, 0.284, 0.067, 14.363)$. Assuming that a lower-sided mean shift is equally likely to happen as an upper-sided mean shift, the charting parameters of the OSPRT chart for a lower-sided mean are taken to be identical to those of the upper-sided counterpart. Note that the optimal sampling interval $d = 0.284$ is equivalent to 1 hour and 42 minutes. For the ease of administration, the sampling interval is rounded to 1 hour and 40 minutes in this example.

The decision rule of the OSPRT chart for an upper- and lower-sided mean is described as follows: At any sampling point i ($i = 1, 2, \dots$), for $j = 1, 2, \dots$,

- sampling is terminated and the process is declared as in-control, if

- both $U_{i,j}^{++} \in (-\infty, g]$ and $U_{i,j}^{-+} \in (-\infty, g]$, or
- $U_{i,j}^{++} \in (-\infty, g]$, given that $U_{i,t}^{-+} \in (-\infty, g]$ at some $t < j$, or
- $U_{i,j}^{-+} \in (-\infty, g]$, given that $U_{i,t}^{++} \in (-\infty, g]$ at some $t < j$.
- sampling is terminated and the process is declared as out-of-control, if
 - $U_{i,j}^{++} \in [h, +\infty)$, regardless of the value of $U_{i,j}^{-+}$, or
 - $U_{i,j}^{-+} \in [h, +\infty)$, regardless of the value of $U_{i,j}^{++}$,
- sampling continues as long as none of the conditions above are fulfilled.

Table 4.31 shows the Phase-II control statistics of the AEQL-optimal OSPRT chart for detecting an upper-sided and lower-sided shift in the mean and/or an upper-sided shift in the variance. The control statistics $U_{i,j}^{++}$ and $U_{i,j}^{-+}$ are computed using Equations (4.9a) and (4.9b), respectively. A single plot depicting both the control statistics $U_{i,j}^{++}$ and $U_{i,j}^{-+}$ is given in Figure 4.4, where the horizontal axis shows the exact sampling time of each group of measurements. To distinguish between the upper-sided and lower-sided statistics, a solid line and a solid dot are

Table 4.31 Phase-II summary statistics for monitoring pull force readings using the AEQL-optimal OSPRT chart.

Sampling time	Sample index, i	Sample number, j	$U_{i,j}^{++}$	$U_{i,j}^{-+}$	Conclusion
09:00	1	1	-0.217	-1.830	In-control
10:40	2	1	-1.437	0.584	*
		2	*	-1.265	In-control
12:20	3	1	2.176	-0.510	*
		2	0.468	*	*
		3	-1.001	*	In-control
14:00	4	1	2.603	-0.239	*
		2	0.970	*	*
		3	-1.227	*	In-control
15:40	5	1	3.439	0.308	*
		2	14.112	5.942	*
		3	16.936	5.845	Out-of-control

NOTE: The boldfaced values in **blue** indicate that the statistic falls in the acceptance region, whereas the boldfaced value in **red** indicates that the statistic falls in the rejection region.

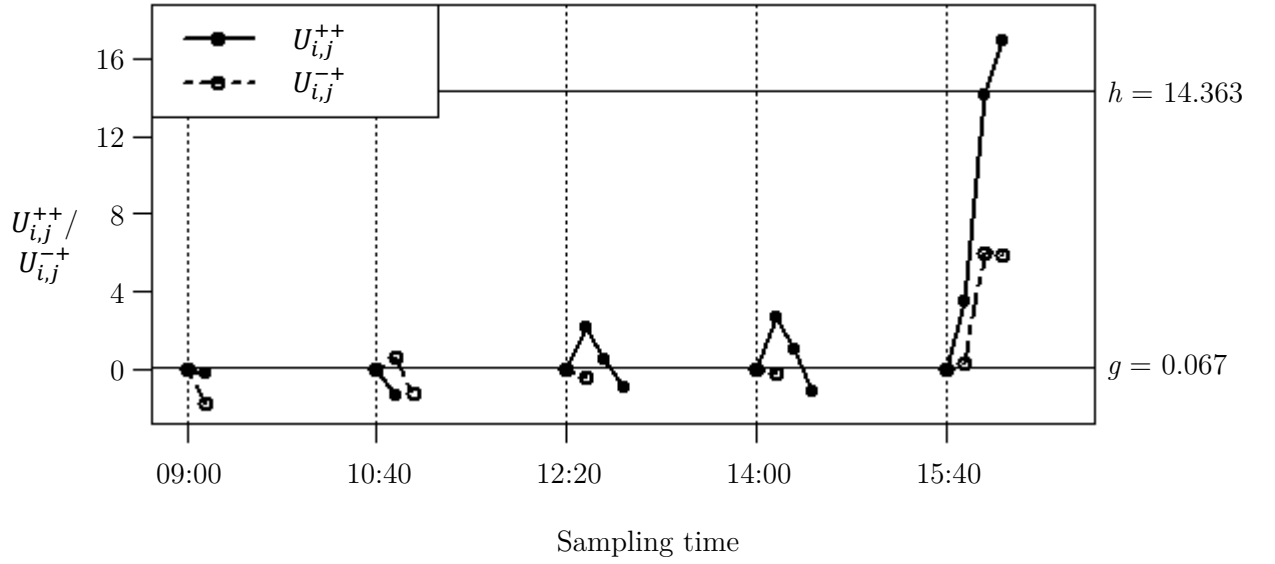


Figure 4.4 The Phase-II OSPRT chart for monitoring pull force measurements of the wire bonding process.

used to represent the trace of $U_{i,j}^{++}$, whereas a dotted line and a hollow dot are used to represent the trace of $U_{i,j}^{-+}$. Note that in this example, the OSPRT chart is assumed to have operated for an extended period of time, thus only a certain part of the data is extracted for the purpose of illustration.

The first sampling time of the extracted data is 09:00. At 09:00, the OSPRT chart quickly concludes that the process is in-control by sampling only one measurement. At the subsequent three sampling times (i.e., 10:40, 12:20, 14:00), both the upper-sided and lower-sided control statistics of the OSPRT chart indicate that there is no obvious issue with the process. At time 15:40, the OSPRT chart emits an out-of-control signal after sampling three observations, indicating that there is an increase in the mean and/or increase in the variance of the pull force measurements. After the signal is produced, the pre-designated out-of-control action plan is immediately executed to investigate the issue.

4.6.2 Ball Bond Shear Test

Apart from the wire pull test, the ball bond shear test is also widely employed in the semiconductor industry for assessing the mechanical strength between wire bonds and bond pads. During the ball shear test, a horizontal force is exerted across the bond ball by a shearing machine, and the minimum force required to shove the ball off the pad is measured. The shear force readings may be affected by a variety of factors, such as an incorrect bonding geometry, changes in the intermetallic compound coverage, changes in the bonding parameters, etc (Lin & Lin, 2020). Hence, it is crucial to monitor the shear forces of a ball bond to ensure that the connection between the bond ball and the bond pad is adequate.

The dataset used in this section is obtained from a ball bond shear test provided by an optoelectronic product manufacturing company. The measurements are expressed in units of grams (g). Prior to implementing the Phase-II OSPRT chart, 50 Phase-I samples, each containing five shear force measurements, have been collected to estimate the process parameters (i.e., the mean and standard deviation). The Shewhart $\bar{X}$ - S chart with three-sigma limits is used to examine the status of the Phase-I samples $(Y_{1,1}, \dots, Y_{1,5}), \dots, (Y_{50,1}, \dots, Y_{50,5})$. The sample mean and standard deviation of the i' th subgroup are first calculated using $\bar{Y}_{i'} = \sum_{j'=1}^5 Y_{i',j'}/5$ and $S_{i'} = \sqrt{\sum_{j'=1}^5 (Y_{i',j'} - \bar{Y}_{i'})^2 / 4}$, respectively. The estimates of the process mean and standard deviation are then calculated as $\hat{\mu}_0 = \sum_{i'=1}^{50} \bar{Y}_{i'} / 50$ and $\hat{\sigma}_0 = \bar{S} / c_4$, respectively, where $\bar{S}$ and c_4 are defined in a similar way as in Section 4.6.1. Using the Phase-I measurements acquired, the (LCL, UCL) of the Phase-I $\bar{X}$ chart are computed as (81.21, 96.59) g, and the graphical display is presented in Figure 4.5(a). On the other hand, the (LCL, UCL) of the Phase-I S chart are calculated as (0, 11.26) g, and the graphical display is presented in Figure 4.5(b). From the figure, it is clear that all the Phase-I observations are in-control. Hence,

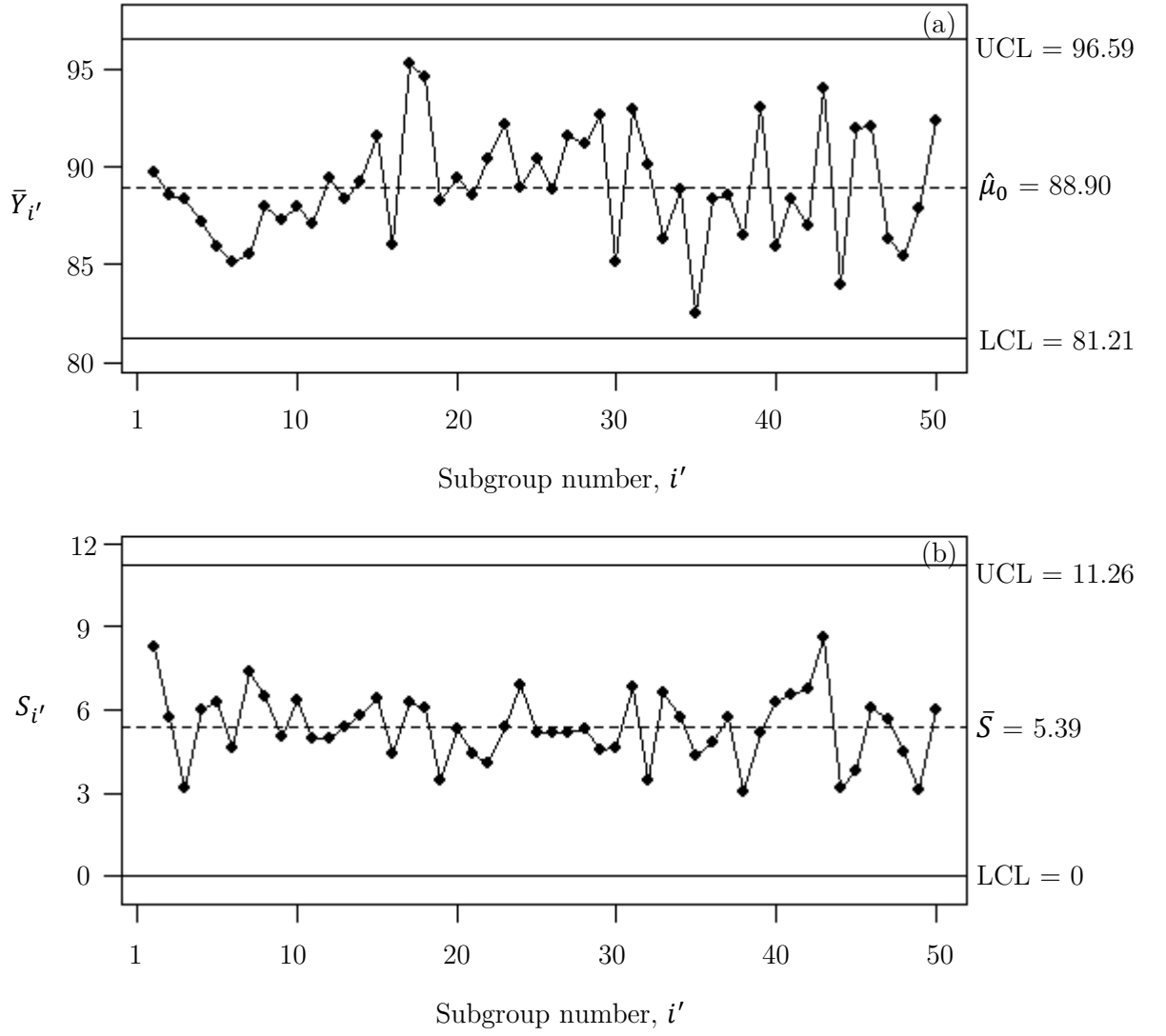


Figure 4.5 The Phase-I Shewhart (a) $\bar{X}$ chart and (b) S chart for monitoring 50 Phase-I samples, each consisting of five shear force measurements.

the parameters of the shear force distribution can be estimated as $\hat{\mu}_0 = 88.90$ g and $\hat{\sigma}_0 = 5.73$ g.

Suppose that the company is currently testing five ball bonds at each sampling point. Using the design specifications $\tau_{\text{ANOS}} = 1500$, $\text{ASN}_0 = 5$, $\delta_{\text{max}} = 3.00$, and $\eta_{\text{max}} = 3.00$, the optimal charting parameters of the one-sided OSPRT chart can be obtained from Table 4.10 as $(k, \gamma, g, h) = (0.470, 1.850, -2.482, 17.123)$. Table 4.32 shows the Phase-II control statistics of the EANOS-optimal OSPRT chart for detecting an upper-sided and lower-sided shift in the mean and/or an

Table 4.32 Phase-II summary statistics for monitoring ball shear force readings using the EANOS-optimal OSPRT chart.

Sample index, i	Sample number, j	$U_{i,j}^{++}$	$U_{i,j}^{+-}$	Conclusion
1	1	-0.423	2.706	*
	2	-1.888	3.292	*
	3	-3.664	2.911	*
	4	*	1.064	*
	5	*	-0.355	*
	6	*	-1.370	*
	7	*	-3.105	In-control
2	1	-1.734	-0.212	*
	2	-1.059	-1.640	*
	3	-2.853	-2.994	In-control
3	1	-1.682	-1.569	*
	2	-3.081	-0.823	*
	3	*	-1.995	*
	4	*	-3.786	In-control
4	1	-1.845	-1.090	*
	2	-3.674	-1.760	*
	3	*	-0.980	*
	4	*	-2.079	*
	5	*	-0.445	*
	6	*	-1.544	*
	7	*	-3.379	In-control
5	1	-1.717	-0.149	*
	2	-3.562	-1.248	*
	3	*	-2.697	In-control
6	1	1.530	-1.043	*
	2	-0.318	-1.922	*
	3	-2.167	-2.937	*
	4	-4.017	*	In-control
7	1	-1.845	-1.099	*
	2	-3.578	-1.307	*
	3	*	-3.142	In-control
8	1	-1.399	0.746	*
	2	-3.240	-0.027	*
	3	*	-1.762	*
	4	*	1.720	*
	5	*	-0.065	*
	6	*	-1.428	*
	7	*	-3.260	In-control
9	1	1.006	5.066	*
	2	0.587	7.780	*
	3	2.464	14.170	*
	4	3.649	19.515	Out-of-control

NOTE: The boldfaced values in **blue** indicate that the statistic falls in the acceptance region, whereas the boldfaced value in **red** indicates that the statistic falls in the rejection region.

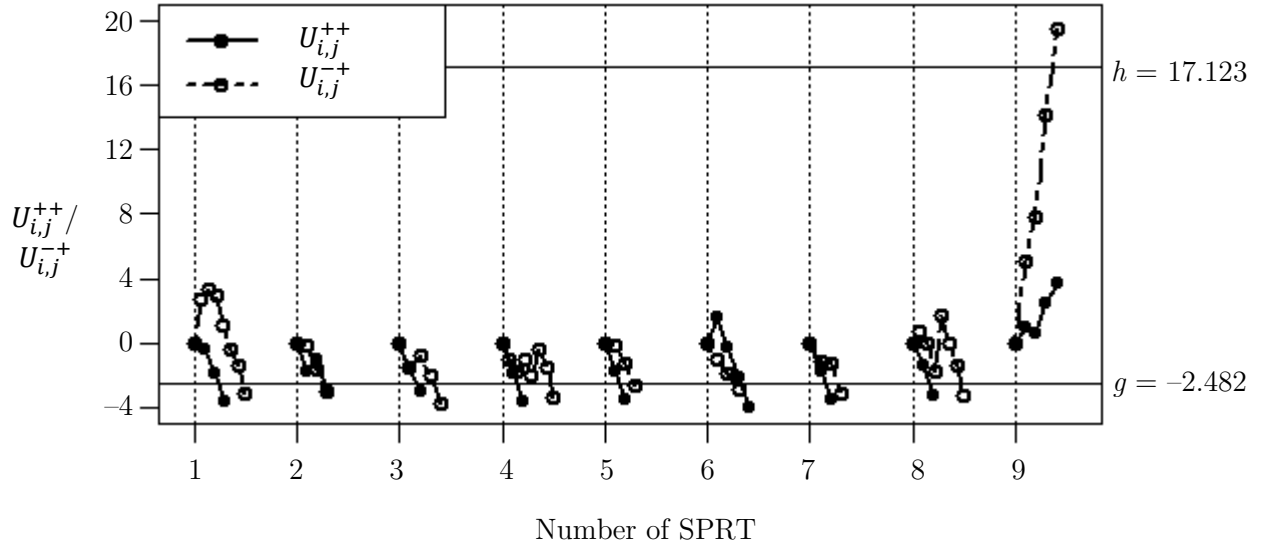


Figure 4.6 The Phase-II OSPRT chart for monitoring ball shear force measurements of the wire bonding process.

upper-sided shift in the variance. The implementation of the OSPRT chart is depicted in Figure 4.6. From Figure 4.6, the OSPRT chart does not indicate any issue for the first eight samples. On the 9th sample, the lower control statistic $U_{i,j}^{-+}$ rises beyond the upper control limit h after four measurements are inspected. The sampling procedure is immediately terminated, and the OSPRT chart produces an out-of-control alarm, indicating a probable downward shift in the process mean and/or increase in the process variance. The total number of observations to signal is recorded as 42. It should be noted that the number of observations inspected averages to $42/9 = 4.667$, which is smaller than the default number of five observations currently practised by the company.

4.7 Conclusions

This chapter introduces a new SPRT control chart for joint monitoring of the mean and dispersion, known as the omnibus SPRT (OSPRT) chart. The full theoretical framework, which includes the formulation of a two-parameter simple

hypothesis test, derivation of the statistic from first principles, and a description of the working mechanism of the OSPRT chart, is outlined in this chapter. The important run length properties of the OSPRT chart, i.e., the ASN, ARL, SDRL, ATS, SDTS, ANOS, etc., are also derived via the Markov chain approach in this chapter.

To develop enhanced designs for the OSPRT chart, we propose four optimisation models based on the minimisation of the ATS, AEQL, ANOS, and EANOS metrics, respectively. We show that, given a deterministic shift point (δ, η) , the OSPRT chart with prescribed reference parameters (see Equations (4.7b) and (4.7c)) indeed achieves the smallest out-of-control ATS and ANOS at the specified shift point. This implies that the ATS- and ANOS-optimal designs for the OSPRT chart involve no independent searches in the parameter space of k and γ , thus reducing the computational complexity of the optimisation procedure. On the other hand, although the AEQL- and EANOS-optimal designs cannot be simplified via any statistical theory, it is possible to restrict the search domain of k and γ according to Equations (4.7b) and (4.7c) for a particular shift domain $[\delta_{\min}, \delta_{\max}] \times [\eta_{\min}, \eta_{\max}]$. By conducting a comparative analysis, we show that the OSPRT chart outperforms four alternative control charts, i.e., the Shewhart $\bar{X}$ -S, WLC, VSS WLC, and ABS-SPRT charts, in cases where only the mean is shifting ($\delta > 0$ and $\eta = 1$) and where both the mean and variance are shifting ($\delta > 0$ and $\eta > 1$). In the aforementioned cases, the OSPRT chart has a uniformly shorter average detection time and a smaller average number of observations inspected to signal compared to its rivals, indicating the OSPRT chart is the most effective control chart (among others) in terms of fault detection as well as savings in sampling cost. To demonstrate the ease of implementation of the OSPRT chart in SQC applications, we have illustrated two applications involving a wire pull test and a ball bond shear test. All the charting parameters tabulated in this chapter are ready to use and can be reproduced via the computer programs published on GitHub.

5 The SPRT Charts under Non-normal Conditions

5.1 Introduction

In this chapter, we investigate the effects of non-normality on the performances of the SPRT chart with estimated process parameters (proposed in Chapter 3) and the OSPRT chart (proposed in Chapter 4). It should be noted that the aforementioned SPRT charts are constructed on the basis that the process data follow the normal distribution. This assumption is motivated by two reasons. First, the majority of the quality variables can be reasonably modelled by the normal distribution. Qiu (2018) explained that, since product quality characteristics depend on a variety of factors (such as machinery, raw materials, and human labour), their distributions are often approximately normal due to the central limit theorem. Naderkhani (2022) further stated that, by appropriate pre-processing techniques, it is possible to transform measurements from a skewed distribution to an approximate Gaussian distribution. Second, it is convenient to derive a control chart for the normal distribution (as a starting point) due to the “nice” properties of the normal distribution. For example, referring to the control statistic of the OSPRT chart in Section 4.2.1, it is relieving to see that the log-likelihood ratio simplifies to an elegant expression when the normal distribution is assumed. The control statistic, on the other hand, may end up having a rather convoluted expression if the gamma density were to be used.

In practice, a simple charting procedure is almost always preferred over a complex charting procedure. While the assumption of normality does bring about some benefits, it is of great interest to study how the performances of the control charts will change when the distributions of data deviate from normality (or symmetry) in various degrees. Qiu (2018) provided some examples of quality variables that have a right skewed distribution, such as economic indices and

lifetimes of products. In several other manufacturing applications, such as those for monitoring the breaking strength of glass fibres, monitoring the weight of bias tyres, etc., measurements are often found to follow a positively skewed distribution, see for example, Bai and Choi (1995), Hossain *et al.* (2022), Lee *et al.* (2022), and Zaka *et al.* (2023). Hence, this section aims to evaluate the performances of the SPRT chart with estimated process parameters and OSPRT chart under the influence of various degrees of skewness.

5.2 A Review on Some Non-normal Distributions

Before the investigation, we provide a brief review on three different statistical distributions, i.e., the gamma, lognormal, and Weibull distributions. These distributions are selected for two main reasons. First, they possess high flexibility in achieving a multitude of shapes and skewness. Second, they are widely used to model a range of quality characteristics in industrial applications, see for example, Huang *et al.* (2016), Qiu (2018), and Lee *et al.* (2022). In the following subsections, three statistical properties of the aforementioned distributions are presented, i.e., the mean (μ_0^*), standard deviation (σ_0^*), and Pearson's moment coefficient of skewness (ρ).

5.2.1 The Gamma Distribution

Let X_γ be the gamma random variable with shape parameter $\alpha' > 0$ and scale parameter $\beta' > 0$. The pdf of X_γ is given as

$$f_{X_\gamma}(x_\gamma) = \frac{1}{(\beta')^{\alpha'} \Gamma(\alpha')} x_\gamma^{\alpha'-1} \exp\left\{-\frac{x_\gamma}{\beta'}\right\}, \quad (5.1)$$

where $\Gamma(\cdot)$ is the gamma function. The formulae for the mean (μ_0^*), standard

deviation (σ_0^*), and coefficient of skewness (ρ) of the gamma distribution are (Zwillinger & Kokoska, 1999)

$$\mu_0^* = \alpha' \beta', \quad (5.2)$$

$$\sigma_0^* = \sqrt{\alpha'} \beta', \quad (5.3)$$

and

$$\rho = \frac{2}{\sqrt{\alpha'}}, \quad (5.4)$$

respectively.

It is important to note that the coefficient of skewness of the gamma distribution (i.e., Equation (5.4)) depends only on the shape parameter α' . As the main focus of the investigation revolves around the skewness of the data, the scale parameter is set as $\beta' = 1$ throughout this chapter for the ease of computation.

5.2.2 The Lognormal Distribution

Let X_{LN} be the lognormal random variable whose location parameter is μ_{LN} and scale parameter is $\sigma_{\text{LN}} > 0$. The pdf of X_{LN} is given as

$$f_{X_{\text{LN}}}(x_{\text{LN}}) = \frac{1}{x_{\text{LN}} \sigma_{\text{LN}} \sqrt{2\pi}} \exp \left\{ -\frac{(\ln x_{\text{LN}} - \mu_{\text{LN}})^2}{2\sigma_{\text{LN}}^2} \right\}. \quad (5.5)$$

The mean, standard deviation, and coefficient of skewness of the lognormal distribution can be calculated using the formulae (Zwillinger & Kokoska, 1999)

$$\mu_0^* = \exp \left(\mu_{\text{LN}} + \frac{\sigma_{\text{LN}}^2}{2} \right), \quad (5.6)$$

$$\sigma_0^* = \sqrt{[\exp(\sigma_{\text{LN}}^2) - 1] \exp(2\mu_{\text{LN}} + \sigma_{\text{LN}}^2)}, \quad (5.7)$$

and

$$\rho = [\exp(\sigma_{\text{LN}}^2) + 2] \sqrt{\exp(\sigma_{\text{LN}}^2) - 1}, \quad (5.8)$$

respectively.

Since the skewness of the lognormal distribution does not depend on its location parameter, it is convenient to set $\mu_{LN} = 0$.

5.2.3 The Weibull Distribution

Let X_W be the Weibull random variable whose shape and scale parameters are $k' > 0$ and $\lambda' > 0$. The pdf of X_W is given as

$$f_{X_W}(x_W) = \frac{k'}{\lambda'} \left(\frac{x_W}{\lambda'}\right)^{k'-1} \exp\left\{-\left(\frac{x_W}{\lambda'}\right)^{k'}\right\}. \quad (5.9)$$

The mean, standard deviation, and coefficient of skewness of the lognormal distribution can be calculated using the formulae (Rinne, 2008)

$$\mu_0^* = \lambda' \Gamma\left(1 + \frac{1}{k'}\right), \quad (5.10)$$

$$\sigma_0^* = \lambda' \sqrt{\Gamma\left(1 + \frac{2}{k'}\right) - \left[\Gamma\left(1 + \frac{1}{k'}\right)\right]^2}, \quad (5.11)$$

and

$$\rho = \frac{\Gamma\left(1 + \frac{3}{k'}\right) - 3\Gamma\left(1 + \frac{1}{k'}\right)\Gamma\left(1 + \frac{2}{k'}\right) + 2\left[\Gamma\left(1 + \frac{1}{k'}\right)\right]^3}{\left\{\Gamma\left(1 + \frac{2}{k'}\right) - \left[\Gamma\left(1 + \frac{1}{k'}\right)\right]^2\right\}^{1.5}}, \quad (5.12)$$

respectively.

It is once again noted that the skewness of the Weibull distribution does not depend on the scale parameter λ' . Hence, λ' is taken as the value 1 for the ease of computation.

Remark 4.1. For all the three distributions above, the mean μ_0^* and standard deviation σ_0^* are related in some way, i.e., they can be written as (implicit) functions

of each other. This is unlike the normal distribution, where the mean and standard deviation are statistically independent.

5.3 The Performances of the SPRT Chart with Estimated Process Parameters under Non-normal Conditions

This section studies the impact of various levels of skewness on the performances of the SPRT chart with estimated process parameters designed under the integrated optimal-GICP framework. Recall from Chapter 3 that the GICP design ensures that the conditional false alarm rate is kept at an acceptably low level, whereas the optimal design aims at reducing the trade-off between the in-control and out-of-control performances due to the inflated control limits. To demonstrate how the performances of the GICP-adjusted SPRT chart would be affected by skewed datasets, we consider the AAEQL-optimal SPRT chart proposed in Section 3.6.2 in this study. It should be noted that the inferences (and hence conclusions) made in this section generalise to other types of optimisation designs (e.g., AATS-optimisation, AANOS-optimisation, and EAANOS-optimisation). Therefore, they are omitted from this section to avoid presenting duplicated results.

In this demonstration, we select the design specifications for the AAEQL-optimal SPRT chart as $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 3$, $\delta_{\min} = 0.1$, $\delta_{\max} = 3.0$, $d_{\min} = 0.2$, $\bar{n}_{\max} = 15$, $m \in \{200, 400, 600, 1000\}$, $p_e = 0.05$, and $\varepsilon = 0.2$. The optimal charting parameters of the AAEQL-optimal SPRT chart designed under the GICP framework can be computed using the optimisation model proposed in Section 3.6.2 and the computer program available on GitHub (labelled with A.4.1). To understand how the performances of the SPRT chart with estimated process parameters would change relative to the known-parameter case, the results for AEQL-optimal SPRT chart with known process parameters (indicated by $m =$

$+\infty$) are also presented for the purpose of comparison. The optimal charting parameters of AEQL-optimal SPRT chart with known process parameters can be computed using the optimisation algorithm outlined in Section 3.4.1 and the computer program available on GitHub (labelled with A.3).

Table 5.1 tabulates the optimal charting parameters of the SPRT chart with estimated process parameters designed for $m \in \{200, 400, 600, 1000, +\infty\}$, assuming that the data follow the normal distribution. Note once again that $m = +\infty$ corresponds to the case where the process parameters are known. As a numeric example, when $m = 600$, the charting parameters of the AAEQL-optimal SPRT chart with GICP-adjusted control limits are $(\text{AASN}_0, \gamma, d, g, h) = (1.624, 0.344, 0.541, 0.618, 7.991)$.

The performances of the SPRT chart with estimated process parameters under skewed data distributions are analysed from two aspects, i.e., when the process is in-control and when the process is out-of-control. In the in-control case, it is of interest to understand how the conditional and unconditional performances of the GICP-adjusted SPRT chart would change for different degrees of skewness of the dataset. Table 5.2 tabulates the exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$, as well as the $(\text{AATS}_0, \text{ASDTS}_0)$ of the AAEQL-optimal SPRT chart, for skewness $\rho \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$. The table is separated into four regions. The first region shows the baseline performance assuming the normal distribution,

Table 5.1 Optimal charting parameters $(\text{AASN}_0, \gamma, d, g, h)$ of the AAEQL-optimal SPRT chart with GICP-adjusted limits for $m \in \{200, 400, 600, 1000, +\infty\}$ when the underlying data follow the normal distribution.

m	Charting parameters				
	AASN_0	γ	d	g	h
200	1.630	0.386	0.543	0.540	8.559
400	1.597	0.332	0.532	0.686	8.634
600	1.624	0.344	0.541	0.618	7.991
1000	1.632	0.363	0.544	0.553	7.297

$+\infty$	1.587	0.380	0.529	0.541	6.327
-----------	-------	-------	-------	-------	-------

Table 5.2 The exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$, AATS_0 , and ASDTS_0 values of the AAEQL-optimal SPRT chart with GICP-adjusted limits, for $\rho \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the gamma, lognormal, and Weibull distributions.

Distribution	Parameter	ρ	$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = +\infty$
			$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$	$\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$
			($\text{AATS}_0, \text{ASDTS}_0$)	($\text{AATS}_0, \text{ASDTS}_0$)	($\text{AATS}_0, \text{ASDTS}_0$)	($\text{AATS}_0, \text{ASDTS}_0$)	($\text{ATS}_0, \text{SDTS}_0$)
Normal	-	0.0	92.77%	91.34%	90.24%	87.69%	-
			(5856.56, > 20000)	(1784.91, 3501.03)	(1091.10, 1638.63)	(724.57, 901.03)	(370.40, 370.13)
Gamma	α'	0.0	92.61%	91.27%	90.07%	87.46%	-
			(5761.50, > 20000)	(1762.43, 3484.92)	(1084.11, 1639.53)	(717.95, 898.54)	(368.36, 368.09)
		0.5	83.05%	77.64%	69.72%	53.16%	-
			(2471.22, 9237.85)	(974.13, 1769.69)	(622.26, 900.15)	(424.31, 515.70)	(229.43, 229.16)
		1.0	69.25%	59.50%	44.35%	20.24%	-
			(1337.99, 4306.31)	(624.67, 1079.48)	(412.39, 573.39)	(285.70, 339.79)	(160.62, 160.36)
		1.5	54.91%	42.33%	24.78%	6.04%	-
			(845.24, 2665.75)	(438.75, 741.40)	(300.90, 411.06)	(211.18, 248.03)	(123.37, 123.11)
		2.0	42.85%	29.42%	13.54%	1.86%	-
			(596.76, 1748.63)	(335.50, 564.88)	(234.77, 318.22)	(168.29, 196.73)	(101.77, 101.51)
		2.5	33.37%	20.86%	7.81%	0.66%	-
			(461.93, 1427.17)	(272.85, 463.47)	(194.62, 263.38)	(142.16, 166.39)	(88.50, 88.23)
		3.0	26.90%	15.30%	4.92%	0.32%	-
			(388.00, 2560.26)	(236.03, 398.79)	(170.75, 231.03)	(126.41, 147.35)	(80.40, 80.13)
Lognormal	σ_{LN}	0.0	92.60%	91.35%	89.99%	87.51%	-
			(5784.73, > 20000)	(1762.91, 3514.18)	(1084.30, 1635.58)	(714.56, 892.05)	(368.35, 368.09)
		0.5	82.66%	77.62%	69.57%	52.68%	-
			(2446.71, 9503.49)	(974.78, 1777.25)	(621.96, 895.81)	(424.17, 511.50)	(228.73, 228.47)
		1.0	68.64%	58.98%	43.60%	19.48%	-
			(1266.82, 4251.24)	(609.89, 1062.75)	(407.41, 565.86)	(282.68, 335.84)	(160.34, 160.08)
		1.5	53.43%	41.40%	24.04%	5.83%	-
			(776.54, 3766.32)	(430.07, 746.43)	(297.50, 403.51)	(211.35, 247.87)	(125.51, 125.24)
		2.0	41.04%	28.91%	13.76%	2.23%	-
			(569.86, 5827.56)	(337.73, 688.56)	(239.33, 332.67)	(174.48, 205.54)	(107.39, 107.13)
		2.5	32.15%	21.24%	9.14%	1.34%	-
			(466.17, 5893.05)	(290.26, 676.18)	(206.92, 317.66)	(154.38, 186.75)	(97.71, 97.45)
		3.0	26.56%	16.94%	7.08%	1.20%	-
			(446.88, > 10000)	(263.34, 1518.45)	(191.88, 456.80)	(144.42, 204.31)	(92.44, 92.18)
Weibull	k'	0.0	93.05%	91.88%	90.70%	88.76%	-
			(6539.93, > 25000)	(1859.32, 3741.83)	(1131.91, 1728.81)	(748.35, 942.33)	(381.74, 381.47)
		0.5	83.30%	78.23%	70.57%	54.71%	-
			(2695.90, 9535.27)	(1008.28, 1864.63)	(644.02, 931.29)	(436.94, 530.52)	(232.78, 232.51)
		1.0	69.90%	59.89%	44.77%	20.93%	-
			(1411.86, 4456.09)	(630.25, 1100.92)	(415.55, 579.67)	(288.65, 344.64)	(160.98, 160.71)
		1.5	55.26%	42.53%	24.88%	6.08%	-
			(873.10, 2744.46)	(441.80, 752.08)	(301.20, 411.58)	(211.48, 249.77)	(122.96, 122.69)
		2.0	42.88%	29.33%	13.45%	1.81%	-
			(597.04, 1820.98)	(337.02, 565.39)	(235.05, 318.21)	(168.33, 196.67)	(101.77, 101.51)
		2.5	33.16%	20.75%	7.89%	0.71%	-
			(457.03, 2305.04)	(277.63, 525.44)	(196.02, 265.84)	(143.99, 167.57)	(89.74, 89.47)
		3.0	26.59%	15.44%	5.38%	0.41%	-
			(378.92, 2261.98)	(238.05, 415.72)	(172.42, 236.30)	(128.95, 151.13)	(83.07, 82.80)

whereas the remaining three regions show results for the gamma, lognormal, and Weibull distributions, respectively. The dominant parameter of each non-normal distribution (i.e., α' , σ_{LN} , and k') is calculated based on the degree of skewness ρ via some numerical root-finding techniques. For instance, when the degree of skewness is set as 1.0 for the Weibull distribution, the corresponding value of k' is determined by solving Equation (5.12) with $\rho = 1.0$. A standard mathematical software should calculate the value as $k' = 1.56391$.

Note that when data follow the normal distribution, all the $(\text{AATS}_0, \text{ASDTS}_0)$ values can be computed using Equations (3.47) and (3.53) from Chapter 3, with the charting parameters tabulated in Table 5.1. However, when data follow the gamma, lognormal, or Weibull distribution, the distributions of the sample mean and sample standard deviation may not be tractable. Besides, the distributions may not be statistically independent when distributions are non-normal (see Remark 4.1). One example is the gamma distribution, whose mean can be expressed as an explicit function of its standard deviation. This implies that Equations (3.47), (3.53), and (3.54) cannot be applied in the computation of the AATS_0 and ASDTS_0 values for non-normal distributions. Due to the intractability of the sampling distributions, we resort to the Monte Carlo simulation method to approximate the performance metrics of the SPRT chart with estimated process parameters. The computer program is available on GitHub (labelled with C.1). As a numeric example, when $m = 400$, $\rho = 1.0$, and the Weibull distribution is assumed, the exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$ is 59.89%, and the $(\text{AATS}_0, \text{ASDTS}_0)$ values are equal to (630.25, 1100.92).

Two main observations are found in the results from Table 5.2.

- (i) When the skewness equals zero ($\rho = 0$), the exceedance probabilities $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$ are found to resemble those of the normal distribution for

all Phase-I sample sizes m . This observation is common for all the three non-normal distributions. It is also noticed that the exceedance probabilities for the normal distribution are adequately high (not exactly equal to 95% due to the tolerance of $\varepsilon = 0.2$), which is expected since the GICP design guarantees a high conditional in-control performance. Analysing the unconditional metrics of the three non-normal distributions when $\rho = 0$, one notices that the Weibull distribution yields $(\text{AATS}_0, \text{ASDTS}_0)$ values that are slightly larger than the normal, gamma, and lognormal distributions. As an example, when $m = 1000$ is used, the $(\text{AATS}_0, \text{ASDTS}_0)$ values of the Weibull distribution $(= (748.35, 942.33))$ are arguably larger than those of the normal $(= (724.57, 901.03))$, gamma $(= (717.95, 898.54))$, and lognormal $(= (714.56, 892.05))$ distributions.

- (ii) When the skewness is positive ($\rho > 0$), the conditional in-control performances are found to deteriorate rapidly for all Phase-I sample sizes m . As an example, when $m = 600$ and the gamma distribution is assumed, increasing the skewness from $\rho = 0.0$ to $\rho = 0.5$ leads to a fall in the exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$ from 90.07% to 69.72%. When the skewness grows larger, the conditional in-control performances worsen even further, with exceedance probabilities dropping to values as low as 0.32%. This indicates that skewed distributions likely cause the GICP-adjusted SPRT chart to perform terribly in the in-control situation. Besides, it is noted that the conditional in-control performances worsen faster for a larger m compared to those of a smaller m , presumably due to a smaller practitioner-to-practitioner variability for a larger Phase-I sample size. While a fall in the ASDTS_0 values (due to an increased skewness) may suggest that the performances have become more consistent, it is worth noting that the in-control time to signal values have been consistent around smaller AATS_0 values, which is undesirable in any process monitoring application.

For all the four distributions, it is found that the $(\text{AATS}_0, \text{ASDTS}_0)$ values converge to their corresponding $(\text{ATS}_0, \text{SDTS}_0)$ values in the known-parameter case as m increases. This is consistent with the observations made in Sections 3.4, 3.5, and 3.6 from Chapter 3. It should also be noted that the exceedance probability $\Pr(\text{CATS}_0 \geq \tau_{\text{ATS}})$ is not meaningful when $m = +\infty$, since the ATS_0 is no longer a random variable.

We now consider the out-of-control performance of the SPRT chart with estimated process parameters under different levels of skewness. Tables 5.3, 5.4, and 5.5 tabulate the out-of-control metrics, i.e., $(\text{AATS}_1, \text{ASDTS}_1)$ values, of the AAEQL-optimal SPRT chart with GICP-adjusted limits, when the underlying data follow the gamma, lognormal, and Weibull distributions, respectively. Similarly, due to the intractable sampling properties of the non-normal distributions, there are no theoretical formulae available for computing the AATS_1 and ASDTS_1 values of the SPRT chart with estimated process parameters. All the results shown in Tables 5.3 to 5.5 are therefore calculated by Monte Carlo simulation programmes. To facilitate comparisons between the chart's performance under each non-normal distribution and the normal distribution, we have provided results for the normal distribution on the top panel of each table. As a numeric example, when $m = 200$, $\rho = 0.5$, and the lognormal distribution is assumed, the dominant parameter is calculated as $\sigma_{\text{LN}} = 0.16405$, and the $(\text{AATS}_1, \text{ASDTS}_1)$ values of the AAEQL-optimal SPRT chart evaluated at $\delta = 1.5$ are equal to $(0.54, 0.49)$ (see Table 5.4).

From Tables 5.3 to 5.5, the following observations can be made:

- (i) When the skewness equals zero ($\rho = 0$), the $(\text{AATS}_1, \text{ASDTS}_1)$ values of the AAEQL-optimal SPRT chart between the three non-normal distributions and the normal distribution are very close to each other. This agrees with our intuition, since the gamma, lognormal, and Weibull distributions with zero

Table 5.3 The AATS₁ and ASDTS₁ values of the AAEQL-optimal SPRT chart with GICP-adjusted limits evaluated at $\delta \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, when $\rho \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and the underlying data follow the gamma distribution.

Parameter			$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = +\infty$
	ρ	δ	(AATS ₁ , ASDTS ₁)	(AATS ₁ , ASDTS ₁)	(AATS ₁ , ASDTS ₁)	(AATS ₁ , ASDTS ₁)	(ATS ₁ , SDTS ₁)
Normal distribution							
-	0.0	0.5	(6.66, 9.59)	(5.02, 5.54)	(4.79, 5.09)	(4.72, 4.89)	(4.61, 4.61)
		1.0	(1.03, 1.03)	(1.06, 1.05)	(1.02, 1.00)	(0.99, 0.97)	(0.98, 0.95)
		1.5	(0.53, 0.48)	(0.55, 0.50)	(0.53, 0.49)	(0.52, 0.47)	(0.51, 0.46)
		2.0	(0.37, 0.29)	(0.38, 0.31)	(0.37, 0.30)	(0.37, 0.29)	(0.36, 0.28)
		2.5	(0.31, 0.21)	(0.31, 0.22)	(0.31, 0.21)	(0.31, 0.21)	(0.30, 0.20)
		3.0	(0.28, 0.17)	(0.28, 0.18)	(0.28, 0.18)	(0.28, 0.17)	(0.27, 0.17)
Gamma distribution							
160000	0.0	0.5	(6.64, 9.46)	(5.01, 5.50)	(4.82, 5.14)	(4.72, 4.90)	(4.62, 4.61)
		1.0	(1.03, 1.03)	(1.06, 1.05)	(1.03, 1.00)	(0.99, 0.97)	(0.98, 0.95)
		1.5	(0.53, 0.48)	(0.55, 0.51)	(0.54, 0.49)	(0.52, 0.47)	(0.51, 0.46)
		2.0	(0.37, 0.29)	(0.38, 0.31)	(0.37, 0.30)	(0.37, 0.29)	(0.36, 0.28)
		2.5	(0.31, 0.21)	(0.31, 0.22)	(0.31, 0.21)	(0.30, 0.21)	(0.30, 0.20)
		3.0	(0.28, 0.17)	(0.28, 0.18)	(0.28, 0.18)	(0.28, 0.17)	(0.27, 0.17)
16.00000	0.5	0.5	(7.61, 12.25)	(5.43, 6.18)	(5.20, 5.63)	(5.11, 5.35)	(4.99, 4.98)
		1.0	(1.09, 1.11)	(1.13, 1.13)	(1.09, 1.07)	(1.05, 1.03)	(1.03, 1.01)
		1.5	(0.54, 0.50)	(0.56, 0.52)	(0.55, 0.51)	(0.53, 0.48)	(0.52, 0.47)
		2.0	(0.36, 0.28)	(0.37, 0.30)	(0.37, 0.29)	(0.36, 0.28)	(0.35, 0.27)
		2.5	(0.29, 0.19)	(0.30, 0.20)	(0.30, 0.20)	(0.29, 0.19)	(0.29, 0.19)
		3.0	(0.29, 0.18)	(0.27, 0.16)	(0.28, 0.16)	(0.27, 0.16)	(0.27, 0.16)
4.00000	1.0	0.5	(8.87, 18.93)	(5.86, 6.97)	(5.63, 6.23)	(5.55, 5.89)	(5.39, 5.38)
		1.0	(1.16, 1.19)	(1.20, 1.20)	(1.15, 1.14)	(1.10, 1.09)	(1.09, 1.07)
		1.5	(0.55, 0.51)	(0.58, 0.55)	(0.56, 0.52)	(0.54, 0.49)	(0.53, 0.48)
		2.0	(0.35, 0.27)	(0.36, 0.29)	(0.35, 0.28)	(0.34, 0.26)	(0.33, 0.25)
		2.5	(0.28, 0.17)	(0.28, 0.18)	(0.28, 0.17)	(0.28, 0.17)	(0.27, 0.16)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
1.77778	1.5	0.5	(10.36, 25.34)	(6.38, 7.97)	(6.11, 6.97)	(6.01, 6.47)	(5.81, 5.81)
		1.0	(1.23, 1.29)	(1.27, 1.28)	(1.22, 1.22)	(1.16, 1.15)	(1.14, 1.12)
		1.5	(0.56, 0.53)	(0.60, 0.57)	(0.58, 0.54)	(0.55, 0.50)	(0.53, 0.49)
		2.0	(0.32, 0.24)	(0.34, 0.27)	(0.33, 0.24)	(0.31, 0.22)	(0.31, 0.22)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
1.00000	2.0	0.5	(12.33, 38.99)	(6.93, 9.24)	(6.60, 7.76)	(6.50, 7.14)	(6.27, 6.27)
		1.0	(1.30, 1.39)	(1.34, 1.37)	(1.28, 1.29)	(1.22, 1.22)	(1.20, 1.18)
		1.5	(0.57, 0.55)	(0.63, 0.61)	(0.59, 0.56)	(0.55, 0.51)	(0.54, 0.50)
		2.0	(0.29, 0.19)	(0.30, 0.21)	(0.28, 0.18)	(0.27, 0.16)	(0.26, 0.15)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
0.64000	2.5	0.5	(14.79, 59.06)	(7.56, 10.83)	(7.15, 8.80)	(7.02, 7.87)	(6.77, 6.76)
		1.0	(1.38, 1.52)	(1.42, 1.46)	(1.35, 1.37)	(1.29, 1.29)	(1.26, 1.24)
		1.5	(0.56, 0.58)	(0.65, 0.65)	(0.60, 0.58)	(0.55, 0.52)	(0.54, 0.50)
		2.0	(0.28, 0.17)	(0.27, 0.17)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
0.44444	3.0	0.5	(17.70, 101.46)	(8.27, 13.06)	(7.74, 10.05)	(7.63, 8.80)	(7.14, 7.13)
		1.0	(1.46, 1.66)	(1.53, 1.58)	(1.44, 1.47)	(1.36, 1.37)	(1.33, 1.32)
		1.5	(0.54, 0.62)	(0.65, 0.69)	(0.58, 0.59)	(0.50, 0.49)	(0.51, 0.46)
		2.0	(0.28, 0.17)	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)

Table 5.4 The AATS₁ and ASDTS₁ values of the AAEQL-optimal SPRT chart with GICP-adjusted limits evaluated at $\delta \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, when $\rho \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and the underlying data follow the lognormal distribution.

Parameter			$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = +\infty$
	ρ	δ	(AATS ₁ , ASDTS ₁)	(AATS ₁ , ASDTS ₁)	(AATS ₁ , ASDTS ₁)	(AATS ₁ , ASDTS ₁)	(ATS ₁ , SDTS ₁)
Normal distribution							
-	0.0	0.5	(6.66, 9.59)	(5.02, 5.54)	(4.79, 5.09)	(4.72, 4.89)	(4.61, 4.61)
		1.0	(1.03, 1.03)	(1.06, 1.05)	(1.02, 1.00)	(0.99, 0.97)	(0.98, 0.95)
		1.5	(0.53, 0.48)	(0.55, 0.50)	(0.53, 0.49)	(0.52, 0.47)	(0.51, 0.46)
		2.0	(0.37, 0.29)	(0.38, 0.31)	(0.37, 0.30)	(0.37, 0.29)	(0.36, 0.28)
		2.5	(0.31, 0.21)	(0.31, 0.22)	(0.31, 0.21)	(0.31, 0.21)	(0.30, 0.20)
		3.0	(0.28, 0.17)	(0.28, 0.18)	(0.28, 0.18)	(0.28, 0.17)	(0.27, 0.17)
Lognormal distribution							
0.00167	0.0	0.5	(6.68, 9.60)	(5.03, 5.55)	(4.81, 5.11)	(4.73, 4.91)	(4.62, 4.61)
		1.0	(1.03, 1.03)	(1.06, 1.05)	(1.03, 1.01)	(0.99, 0.97)	(0.98, 0.95)
		1.5	(0.53, 0.48)	(0.55, 0.50)	(0.54, 0.49)	(0.52, 0.47)	(0.51, 0.46)
		2.0	(0.37, 0.29)	(0.37, 0.31)	(0.37, 0.30)	(0.37, 0.29)	(0.36, 0.28)
		2.5	(0.31, 0.21)	(0.31, 0.22)	(0.31, 0.21)	(0.31, 0.21)	(0.30, 0.20)
		3.0	(0.28, 0.17)	(0.28, 0.18)	(0.28, 0.18)	(0.28, 0.17)	(0.27, 0.17)
0.16405	0.5	0.5	(7.63, 13.31)	(5.43, 6.19)	(5.22, 5.66)	(5.12, 5.36)	(4.99, 4.99)
		1.0	(1.09, 1.10)	(1.13, 1.13)	(1.08, 1.07)	(1.04, 1.02)	(1.03, 1.01)
		1.5	(0.54, 0.49)	(0.56, 0.52)	(0.55, 0.50)	(0.53, 0.48)	(0.52, 0.47)
		2.0	(0.36, 0.28)	(0.37, 0.30)	(0.37, 0.29)	(0.36, 0.28)	(0.35, 0.27)
		2.5	(0.29, 0.19)	(0.30, 0.20)	(0.30, 0.20)	(0.29, 0.19)	(0.29, 0.19)
		3.0	(0.28, 0.16)	(0.27, 0.16)	(0.28, 0.16)	(0.28, 0.16)	(0.27, 0.16)
0.31426	1.0	0.5	(8.95, 19.94)	(5.92, 7.07)	(5.67, 6.31)	(5.56, 5.90)	(5.40, 5.40)
		1.0	(1.15, 1.18)	(1.19, 1.19)	(1.14, 1.13)	(1.09, 1.08)	(1.08, 1.06)
		1.5	(0.54, 0.50)	(0.57, 0.54)	(0.55, 0.51)	(0.53, 0.48)	(0.52, 0.47)
		2.0	(0.35, 0.27)	(0.36, 0.29)	(0.35, 0.28)	(0.34, 0.26)	(0.33, 0.25)
		2.5	(0.28, 0.18)	(0.28, 0.18)	(0.28, 0.18)	(0.28, 0.17)	(0.27, 0.17)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
0.44349	1.5	0.5	(10.91, 40.03)	(6.44, 8.15)	(6.14, 7.05)	(6.02, 6.54)	(5.81, 5.81)
		1.0	(1.20, 1.27)	(1.24, 1.25)	(1.19, 1.19)	(1.13, 1.13)	(1.12, 1.09)
		1.5	(0.54, 0.51)	(0.58, 0.55)	(0.56, 0.51)	(0.53, 0.48)	(0.51, 0.47)
		2.0	(0.33, 0.25)	(0.34, 0.27)	(0.34, 0.25)	(0.32, 0.24)	(0.32, 0.23)
		2.5	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
0.55138	2.0	0.5	(14.83, 296.93)	(7.06, 11.55)	(6.61, 8.17)	(6.49, 7.90)	(6.20, 6.20)
		1.0	(1.25, 3.20)	(1.29, 1.32)	(1.23, 1.24)	(1.17, 1.16)	(1.15, 1.13)
		1.5	(0.53, 0.52)	(0.58, 0.56)	(0.55, 0.51)	(0.52, 0.47)	(0.51, 0.46)
		2.0	(0.31, 0.23)	(0.33, 0.25)	(0.32, 0.23)	(0.30, 0.21)	(0.30, 0.20)
		2.5	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
0.64088	2.5	0.5	(30.62, 5188.91)	(8.29, 275.83)	(7.21, 12.35)	(6.92, 11.44)	(6.56, 6.56)
		1.0	(1.82, 625.89)	(1.33, 1.40)	(1.26, 1.29)	(1.19, 1.19)	(1.17, 1.15)
		1.5	(0.53, 0.53)	(0.58, 0.57)	(0.55, 0.52)	(0.51, 0.47)	(0.50, 0.45)
		2.0	(0.30, 0.21)	(0.31, 0.23)	(0.30, 0.20)	(0.29, 0.18)	(0.28, 0.18)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
0.71557	3.0	0.5	(58.69, 3056.37)	(12.52, 743.25)	(7.84, 24.01)	(7.44, 13.58)	(6.89, 6.89)
		1.0	(7.90, 1973.22)	(1.42, 137.45)	(1.29, 1.34)	(1.21, 1.23)	(1.19, 1.17)
		1.5	(0.52, 5.08)	(0.58, 0.59)	(0.54, 0.52)	(0.50, 0.46)	(0.49, 0.44)
		2.0	(0.29, 0.23)	(0.30, 0.21)	(0.29, 0.19)	(0.28, 0.17)	(0.27, 0.16)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)

Table 5.5 The AATS₁ and ASDTS₁ values of the AAEQL-optimal SPRT chart with GICP-adjusted limits evaluated at $\delta \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, when $\rho \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ and the underlying data follow the Weibull distribution.

Parameter			$m = 200$	$m = 400$	$m = 600$	$m = 1000$	$m = +\infty$
	ρ	δ	(AATS ₁ , ASDTS ₁)	(AATS ₁ , ASDTS ₁)	(AATS ₁ , ASDTS ₁)	(AATS ₁ , ASDTS ₁)	(ATS ₁ , SDTS ₁)
Normal distribution							
-	0.0	0.5	(6.66, 9.59)	(5.02, 5.54)	(4.79, 5.09)	(4.72, 4.89)	(4.61, 4.61)
		1.0	(1.03, 1.03)	(1.06, 1.05)	(1.02, 1.00)	(0.99, 0.97)	(0.98, 0.95)
		1.5	(0.53, 0.48)	(0.55, 0.50)	(0.53, 0.49)	(0.52, 0.47)	(0.51, 0.46)
		2.0	(0.37, 0.29)	(0.38, 0.31)	(0.37, 0.30)	(0.37, 0.29)	(0.36, 0.28)
		2.5	(0.31, 0.21)	(0.31, 0.22)	(0.31, 0.21)	(0.31, 0.21)	(0.30, 0.20)
		3.0	(0.28, 0.17)	(0.28, 0.18)	(0.28, 0.18)	(0.28, 0.17)	(0.27, 0.17)
Weibull distribution							
3.60235	0.0	0.5	(6.54, 9.09)	(4.95, 5.45)	(4.75, 5.04)	(4.67, 4.84)	(4.55, 4.55)
		1.0	(1.03, 1.03)	(1.06, 1.05)	(1.02, 1.00)	(0.99, 0.96)	(0.98, 0.95)
		1.5	(0.53, 0.49)	(0.55, 0.51)	(0.54, 0.49)	(0.52, 0.48)	(0.51, 0.47)
		2.0	(0.37, 0.30)	(0.38, 0.31)	(0.38, 0.31)	(0.37, 0.30)	(0.36, 0.29)
		2.5	(0.31, 0.21)	(0.31, 0.22)	(0.31, 0.22)	(0.31, 0.21)	(0.30, 0.21)
		3.0	(0.28, 0.17)	(0.28, 0.17)	(0.28, 0.17)	(0.28, 0.17)	(0.27, 0.17)
2.21560	0.5	0.5	(7.47, 12.12)	(5.35, 6.07)	(5.14, 5.56)	(5.06, 5.30)	(4.93, 4.92)
		1.0	(1.10, 1.11)	(1.12, 1.12)	(1.09, 1.08)	(1.05, 1.03)	(1.04, 1.01)
		1.5	(0.55, 0.51)	(0.58, 0.54)	(0.56, 0.52)	(0.54, 0.49)	(0.53, 0.48)
		2.0	(0.37, 0.29)	(0.38, 0.31)	(0.37, 0.30)	(0.36, 0.29)	(0.35, 0.28)
		2.5	(0.29, 0.19)	(0.29, 0.20)	(0.29, 0.19)	(0.29, 0.19)	(0.28, 0.18)
		3.0	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
1.56391	1.0	0.5	(8.70, 16.78)	(5.80, 6.83)	(5.58, 6.13)	(5.51, 5.84)	(5.35, 5.35)
		1.0	(1.17, 1.20)	(1.20, 1.20)	(1.16, 1.15)	(1.11, 1.10)	(1.10, 1.08)
		1.5	(0.56, 0.52)	(0.60, 0.56)	(0.58, 0.53)	(0.55, 0.50)	(0.54, 0.49)
		2.0	(0.35, 0.27)	(0.37, 0.30)	(0.36, 0.28)	(0.34, 0.26)	(0.34, 0.26)
		2.5	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
1.21112	1.5	0.5	(10.19, 23.56)	(6.34, 7.89)	(6.07, 6.89)	(5.99, 6.41)	(5.81, 5.80)
		1.0	(1.24, 1.29)	(1.27, 1.28)	(1.22, 1.22)	(1.17, 1.16)	(1.15, 1.13)
		1.5	(0.57, 0.54)	(0.61, 0.58)	(0.59, 0.55)	(0.55, 0.51)	(0.54, 0.50)
		2.0	(0.32, 0.23)	(0.34, 0.26)	(0.32, 0.24)	(0.31, 0.21)	(0.30, 0.20)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
1.00000	2.0	0.5	(12.33, 41.18)	(6.94, 9.42)	(6.60, 7.80)	(6.49, 7.13)	(6.27, 6.27)
		1.0	(1.30, 1.40)	(1.34, 1.37)	(1.28, 1.29)	(1.22, 1.21)	(1.20, 1.18)
		1.5	(0.57, 0.55)	(0.63, 0.61)	(0.59, 0.56)	(0.55, 0.51)	(0.54, 0.50)
		2.0	(0.29, 0.19)	(0.30, 0.21)	(0.28, 0.18)	(0.27, 0.16)	(0.26, 0.15)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
0.86317	2.5	0.5	(15.22, 76.83)	(7.60, 11.25)	(7.15, 8.82)	(7.04, 7.94)	(6.73, 6.73)
		1.0	(1.35, 1.50)	(1.40, 1.44)	(1.33, 1.35)	(1.27, 1.27)	(1.24, 1.22)
		1.5	(0.55, 0.57)	(0.64, 0.63)	(0.59, 0.57)	(0.54, 0.51)	(0.53, 0.48)
		2.0	(0.28, 0.18)	(0.28, 0.18)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
0.76862	3.0	0.5	(19.56, 228.99)	(8.35, 15.32)	(7.70, 10.38)	(7.55, 8.73)	(7.17, 7.17)
		1.0	(1.42, 2.44)	(1.47, 1.54)	(1.38, 1.42)	(1.31, 1.32)	(1.28, 1.26)
		1.5	(0.53, 0.59)	(0.63, 0.65)	(0.57, 0.56)	(0.51, 0.49)	(0.51, 0.46)
		2.0	(0.28, 0.18)	(0.27, 0.17)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		2.5	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)
		3.0	(0.27, 0.16)	(0.27, 0.15)	(0.27, 0.16)	(0.27, 0.16)	(0.26, 0.15)

skewness have an approximately bell shape, which resembles the shape of a normal distribution. Nonetheless, the $(\text{AATS}_1, \text{ASDTS}_1)$ values of the Weibull distribution are found to be slightly lower than those of the normal distribution (and the other two non-normal distributions), with a more pronounced difference observed for a small shift size (i.e., $\delta = 0.5$). For example, when $m = 1000$, the $(\text{AATS}_1, \text{ASDTS}_1)$ values evaluated at $\delta = 0.5$ under the normal $(= (4.72, 4.89))$, gamma $(= (4.72, 4.90))$ (see Table 5.3), and lognormal $(= (4.73, 4.91))$ (see Table 5.4) distributions are all very close to each other, whereas the Weibull distribution produces slightly smaller $(\text{AATS}_1, \text{ASDTS}_1)$ values at $(4.67, 4.84)$ (see Table 5.5) compared to the rest.

- (ii) When the skewness increases from zero ($\rho > 0$), the $(\text{AATS}_1, \text{ASDTS}_1)$ values are found to increase relative to the case when $\rho = 0$, especially when shift sizes are small and moderate ($\delta \leq 1.0$). The worst deterioration has been observed for the smallest value of m (i.e., $m = 200$), with AATS_1 values reaching over three times as large as the AATS_1 value observed in the case of the normal distribution, when δ is small. It is found that, for a combination of a small m and δ alongside a large ρ , the out-of-control metrics are most impacted when data follow the lognormal distribution. In particular, one observes that, when $\rho = 3.0$, $m = 200$, and $\delta = 0.5$, the $(\text{AATS}_1, \text{ASDTS}_1)$ values under the lognormal distribution $(= (58.69, 3056.37))$ (see Table 5.4) are substantially higher than those of the gamma $(= (17.70, 101.46))$ (see Table 5.3) and Weibull $(= (19.56, 228.99))$ (see Table 5.5) distributions. However, as the Phase-I sample size increases, the instability in the performances of the SPRT chart quickly diminishes, and their values tend towards the values in the case where process parameters are known ($m = +\infty$). It is also found that, for moderately large shift sizes ($\delta \geq 1.5$), the performance metrics of the AAEQL-optimal SPRT chart are almost unaffected by extreme skewness in the data.

To summarise, both the conditional and unconditional performances of the SPRT chart with GICP-adjusted limits are adversely impacted when there exists positive skewness in the data. Based on the results, it seems that the conditional in-control performance of the GICP-adjusted SPRT chart with a small Phase-I sample size (e.g., $m = 200$) is much less affected by an increased skewness compared to that with a large Phase-I sample size (e.g., $m = 1000$). The observation is rather counter-intuitive, but not too surprising, since we expect that the performances become more consistent as m increases. If one wishes to redesign the SPRT chart, then a different adjustment needs to be applied to each m due to the difference in the extent to which the in-control performances are affected. Referring to the out-of-control metric, there seems to be no big issue with the overall sensitivity of the SPRT chart under skewed data distributions, except when m and δ are both small and when the data have an extremely skewed distribution. It is advised to keep the skewness of data smaller than 1.0, or to at least perform some transformation (e.g., Box-Cox transformation) on the data before applying the proposed SPRT chart with GICP-adjusted control limits. Practitioners may also impose a stricter EPC (e.g., $p_e = 0.01$ and $\varepsilon = 0.0$) on the GICP design to reduce the degree by which the conditional in-control performances deteriorate. However, one should bear in mind that doing so can lead to a worsened out-of-control sensitivity, due to the trade-off between the in-control and out-of-control performances of a control chart.

5.4 The Performances of the OSPRT Chart under Non-normal Conditions

This section studies the effect of various degrees of skewness on the performances of the OSPRT chart. It is important to note that the statistic for the OSPRT chart has been derived using the pdf of the normal distribution. When the true distribution of data deviates from normality, it is anticipated that the

performances of the OSPRT chart would be adversely affected, depending on how large the skewness is. In this subsection, only the AEQL-optimal OSPRT chart is considered, and the conclusions can be generalised to all other types of optimal designs for the OSPRT chart.

In this demonstration, the design specifications for the AEQL-optimal OSPRT chart are selected as $\tau_{\text{ATS}} = 370.4$, $\text{IR} = 3$, $d_{\min} = 0.2$, $\bar{n}_{\max} = 15$, $(\delta_{\min}, \delta_{\max}) = (0.05, 3.00)$, and $(\eta_{\min}, \eta_{\max}) = (1.05, 3.00)$. The optimal charting parameters of the AEQL-optimal OSPRT chart can be quoted directly from Table 4.3 as $(k, \gamma, d, g, h) = (0.389, 2.288, 0.469, -0.061, 12.596)$. We first consider the case where skewness equals zero. Table 5.6 provides a comparison between the performances of the OSPRT chart under the usual normal distribution, and those under the gamma, lognormal, and Weibull distributions with zero skewness (i.e., $\rho = 0$). The metrics evaluated, i.e., the ATS and SDTS values, are computed using the Monte Carlo simulation programmes available on GitHub (labelled with C.2). As a numeric example, when the gamma distribution with zero skewness is assumed, the (ATS, SDTS) values evaluated at $(\delta, \eta) = (0.5, 1.5)$ are (2.23, 2.22).

From Table 5.6, the (ATS, SDTS) values of the gamma and lognormal distributions with $\rho = 0$ are found to resemble those of the normal distribution closely. This observation is consistent with our intuition, as the gamma and lognormal distribution with zero skewness should have shapes similar to the normal distribution. However, the in-control performances of the OSPRT chart under the symmetric Weibull distribution are found to be very distinctive from the rest. In particular, when $(\delta, \eta) = (0.0, 1.0)$, the $(\text{ATS}_0, \text{SDTS}_0)$ values of the OSPRT chart under the Weibull distribution $(= (760.20, 759.96))$ is more than twice as large as those of the normal $(= (370.26, 370.03))$, gamma $(= (363.76, 363.52))$, and lognormal $(= (363.74, 363.50))$ distributions (see Table 5.6). Referring to the out-of-control performances, there is no substantial difference observed in the $(\text{ATS}_1, \text{SDTS}_1)$ performances among the four distributions, although it may be argued that

Table 5.6 The (ATS, SDTS) values of the AEQL-optimal OSPRT chart evaluated for $\rho = 0.0$, $\delta \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the normal, gamma, lognormal, and Weibull distributions.

Distribution Parameter		Normal -	Gamma $\alpha' = 160000$	Lognormal $\sigma_{LN} = 0.00167$	Weibull $k' = 3.60235$
δ	η	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)
0.0	1.0	(370.26, 370.03)	(363.76, 363.52)	(363.74, 363.50)	(760.20, 759.96)
	1.5	(4.59, 4.59)	(4.59, 4.59)	(4.59, 4.59)	(4.42, 4.42)
	2.0	(1.29, 1.27)	(1.29, 1.27)	(1.29, 1.27)	(1.19, 1.18)
	2.5	(0.79, 0.76)	(0.79, 0.76)	(0.79, 0.76)	(0.74, 0.72)
	3.0	(0.61, 0.58)	(0.61, 0.58)	(0.61, 0.58)	(0.58, 0.55)
0.5	1.0	(21.24, 21.24)	(21.15, 21.14)	(21.15, 21.14)	(22.90, 22.90)
	1.5	(2.23, 2.22)	(2.23, 2.22)	(2.23, 2.22)	(2.12, 2.11)
	2.0	(1.05, 1.03)	(1.05, 1.03)	(1.05, 1.03)	(0.99, 0.97)
	2.5	(0.72, 0.70)	(0.73, 0.70)	(0.73, 0.70)	(0.69, 0.66)
	3.0	(0.58, 0.55)	(0.58, 0.55)	(0.58, 0.55)	(0.56, 0.52)
1.0	1.0	(1.86, 1.85)	(1.86, 1.85)	(1.86, 1.85)	(1.82, 1.81)
	1.5	(1.10, 1.08)	(1.10, 1.08)	(1.10, 1.08)	(1.08, 1.06)
	2.0	(0.80, 0.78)	(0.80, 0.78)	(0.80, 0.78)	(0.77, 0.75)
	2.5	(0.64, 0.61)	(0.64, 0.61)	(0.64, 0.61)	(0.62, 0.59)
	3.0	(0.54, 0.51)	(0.54, 0.51)	(0.54, 0.51)	(0.52, 0.49)
1.5	1.0	(0.61, 0.58)	(0.61, 0.58)	(0.61, 0.58)	(0.61, 0.58)
	1.5	(0.65, 0.62)	(0.65, 0.62)	(0.65, 0.62)	(0.65, 0.62)
	2.0	(0.61, 0.58)	(0.61, 0.58)	(0.61, 0.58)	(0.61, 0.57)
	2.5	(0.55, 0.52)	(0.55, 0.52)	(0.55, 0.52)	(0.54, 0.51)
	3.0	(0.50, 0.46)	(0.50, 0.46)	(0.50, 0.46)	(0.49, 0.45)
2.0	1.0	(0.36, 0.31)	(0.36, 0.31)	(0.36, 0.31)	(0.37, 0.31)
	1.5	(0.45, 0.40)	(0.45, 0.41)	(0.45, 0.41)	(0.46, 0.42)
	2.0	(0.48, 0.44)	(0.48, 0.44)	(0.48, 0.44)	(0.49, 0.45)
	2.5	(0.47, 0.43)	(0.48, 0.43)	(0.48, 0.43)	(0.47, 0.43)
	3.0	(0.45, 0.41)	(0.45, 0.41)	(0.45, 0.41)	(0.45, 0.40)
2.5	1.0	(0.28, 0.20)	(0.28, 0.20)	(0.28, 0.20)	(0.28, 0.21)
	1.5	(0.35, 0.29)	(0.35, 0.29)	(0.35, 0.29)	(0.35, 0.30)
	2.0	(0.39, 0.34)	(0.39, 0.34)	(0.39, 0.34)	(0.40, 0.35)
	2.5	(0.41, 0.36)	(0.41, 0.37)	(0.41, 0.37)	(0.42, 0.37)
	3.0	(0.41, 0.36)	(0.41, 0.36)	(0.41, 0.36)	(0.41, 0.36)
3.0	1.0	(0.25, 0.16)	(0.25, 0.16)	(0.25, 0.16)	(0.25, 0.16)
	1.5	(0.29, 0.22)	(0.29, 0.22)	(0.29, 0.22)	(0.30, 0.23)
	2.0	(0.34, 0.28)	(0.34, 0.28)	(0.34, 0.28)	(0.34, 0.29)
	2.5	(0.36, 0.31)	(0.36, 0.31)	(0.36, 0.31)	(0.37, 0.32)
	3.0	(0.37, 0.32)	(0.37, 0.32)	(0.37, 0.32)	(0.38, 0.33)

the (ATS₁, SDTS₁) values of the OSPRT chart under the symmetric Weibull distribution are slightly lower than those of the other three distributions.

Now, we consider the case where skewness is non-zero (or positive). Tables 5.7, 5.8, and 5.9 tabulate the (ATS, SDTS) values of the AEQL-optimal SPRT chart under the gamma, lognormal, and Weibull distributions, respectively, in the case where the skewness of the data is positive ($\rho > 0$). The degrees of skewness

Table 5.7 The (ATS, SDTS) values of the AEQL-optimal OSPRT chart evaluated for $\rho \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, $\delta \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the gamma distribution.

α'		16.00000	4.00000	1.77778	1.00000	0.64000	0.44444
ρ		0.5	1.0	1.5	2.0	2.5	3.0
δ	η	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)
0.0	1.0	(94.35, 94.12)	(47.45, 47.22)	(32.50, 32.27)	(26.21, 25.98)	(23.17, 22.94)	(21.65, 21.41)
	1.5	(5.18, 5.18)	(6.00, 5.99)	(6.79, 6.79)	(7.20, 7.19)	(7.58, 7.58)	(8.04, 8.04)
	2.0	(1.45, 1.44)	(1.81, 1.80)	(2.50, 2.49)	(3.47, 3.46)	(4.28, 4.27)	(4.83, 4.83)
	2.5	(0.84, 0.82)	(0.93, 0.91)	(1.14, 1.12)	(1.62, 1.61)	(2.46, 2.45)	(3.46, 3.45)
	3.0	(0.63, 0.60)	(0.66, 0.63)	(0.72, 0.69)	(0.86, 0.84)	(1.23, 1.22)	(1.96, 1.95)
0.5	1.0	(15.01, 15.01)	(12.50, 12.50)	(11.46, 11.46)	(11.11, 11.11)	(11.16, 11.16)	(11.44, 11.44)
	1.5	(2.74, 2.73)	(3.34, 3.33)	(3.83, 3.83)	(4.29, 4.29)	(4.78, 4.78)	(5.30, 5.30)
	2.0	(1.27, 1.25)	(1.66, 1.64)	(2.23, 2.22)	(2.73, 2.73)	(3.17, 3.17)	(3.62, 3.62)
	2.5	(0.82, 0.80)	(0.99, 0.97)	(1.33, 1.32)	(1.93, 1.92)	(2.45, 2.44)	(2.86, 2.86)
	3.0	(0.63, 0.60)	(0.70, 0.68)	(0.85, 0.82)	(1.20, 1.18)	(1.92, 1.91)	(2.40, 2.40)
1.0	1.0	(2.18, 2.17)	(2.53, 2.53)	(2.93, 2.92)	(3.36, 3.35)	(3.84, 3.84)	(4.37, 4.36)
	1.5	(1.35, 1.34)	(1.64, 1.63)	(1.94, 1.93)	(2.27, 2.26)	(2.63, 2.62)	(3.02, 3.02)
	2.0	(0.97, 0.96)	(1.25, 1.23)	(1.56, 1.54)	(1.84, 1.83)	(2.15, 2.14)	(2.49, 2.48)
	2.5	(0.74, 0.72)	(0.93, 0.91)	(1.25, 1.24)	(1.59, 1.58)	(1.88, 1.87)	(2.19, 2.18)
	3.0	(0.61, 0.58)	(0.71, 0.68)	(0.92, 0.90)	(1.34, 1.33)	(1.69, 1.68)	(1.98, 1.97)
1.5	1.0	(0.67, 0.64)	(0.73, 0.70)	(0.79, 0.77)	(0.86, 0.84)	(0.91, 0.89)	(0.96, 0.94)
	1.5	(0.75, 0.72)	(0.85, 0.83)	(0.97, 0.95)	(1.09, 1.07)	(1.22, 1.21)	(1.36, 1.34)
	2.0	(0.72, 0.69)	(0.87, 0.85)	(1.01, 0.99)	(1.16, 1.15)	(1.33, 1.31)	(1.51, 1.49)
	2.5	(0.64, 0.61)	(0.79, 0.77)	(1.00, 0.98)	(1.16, 1.15)	(1.35, 1.33)	(1.54, 1.53)
	3.0	(0.56, 0.53)	(0.68, 0.65)	(0.89, 0.87)	(1.14, 1.12)	(1.33, 1.31)	(1.53, 1.52)
2.0	1.0	(0.36, 0.31)	(0.36, 0.30)	(0.34, 0.28)	(0.29, 0.22)	(0.23, 0.14)	(0.23, 0.14)
	1.5	(0.48, 0.44)	(0.51, 0.47)	(0.53, 0.49)	(0.53, 0.50)	(0.52, 0.48)	(0.45, 0.41)
	2.0	(0.54, 0.50)	(0.60, 0.57)	(0.66, 0.63)	(0.71, 0.68)	(0.75, 0.72)	(0.77, 0.75)
	2.5	(0.54, 0.51)	(0.64, 0.61)	(0.73, 0.71)	(0.82, 0.79)	(0.90, 0.88)	(0.98, 0.96)
	3.0	(0.51, 0.48)	(0.61, 0.58)	(0.77, 0.74)	(0.88, 0.86)	(0.99, 0.97)	(1.10, 1.08)
2.5	1.0	(0.27, 0.19)	(0.25, 0.16)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)
	1.5	(0.35, 0.29)	(0.34, 0.28)	(0.32, 0.26)	(0.27, 0.19)	(0.23, 0.14)	(0.23, 0.14)
	2.0	(0.42, 0.37)	(0.44, 0.39)	(0.44, 0.40)	(0.43, 0.38)	(0.38, 0.33)	(0.23, 0.14)
	2.5	(0.45, 0.41)	(0.51, 0.47)	(0.54, 0.50)	(0.56, 0.53)	(0.56, 0.53)	(0.53, 0.49)
	3.0	(0.46, 0.42)	(0.53, 0.50)	(0.61, 0.58)	(0.66, 0.63)	(0.70, 0.67)	(0.72, 0.69)
3.0	1.0	(0.24, 0.15)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)
	1.5	(0.28, 0.21)	(0.27, 0.19)	(0.24, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)
	2.0	(0.34, 0.28)	(0.34, 0.28)	(0.31, 0.25)	(0.26, 0.18)	(0.23, 0.14)	(0.23, 0.14)
	2.5	(0.39, 0.34)	(0.40, 0.35)	(0.40, 0.35)	(0.38, 0.33)	(0.31, 0.24)	(0.23, 0.14)
	3.0	(0.41, 0.36)	(0.45, 0.41)	(0.48, 0.44)	(0.48, 0.44)	(0.46, 0.42)	(0.37, 0.32)

considered are $\rho = \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$. As a numeric example, from Table 5.7, when the underlying distribution is gamma and $\rho = 1.5$, the $(ATS_0, SDTS_0)$ values are equal to $(32.50, 32.27)$.

From the tables, the in-control performances (when $(\delta, \eta) = (0, 1)$) of the OSPRT chart are found to deteriorate considerably as the level of skewness increases. In particular, when ρ increases from 0.0 to 0.5, the ATS_0 values are found

Table 5.8 The (ATS, SDTS) values of the AEQL-optimal OSPRT chart evaluated for $\rho \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, $\delta \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the lognormal distribution.

σ_{LN}		0.16405	0.31426	0.44349	0.55138	0.64088	0.71557
ρ		0.5	1.0	1.5	2.0	2.5	3.0
δ	η	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)
0.0	1.0	(90.72, 90.48)	(46.90, 46.66)	(34.46, 34.22)	(29.73, 29.50)	(27.74, 27.50)	(26.91, 26.68)
	1.5	(5.21, 5.21)	(6.10, 6.10)	(6.97, 6.97)	(7.75, 7.75)	(8.40, 8.40)	(8.93, 8.93)
	2.0	(1.47, 1.46)	(1.84, 1.83)	(2.42, 2.41)	(3.12, 3.11)	(3.82, 3.82)	(4.48, 4.48)
	2.5	(0.85, 0.83)	(0.96, 0.94)	(1.15, 1.13)	(1.43, 1.42)	(1.82, 1.81)	(2.28, 2.27)
	3.0	(0.64, 0.61)	(0.68, 0.66)	(0.75, 0.73)	(0.86, 0.83)	(1.00, 0.98)	(1.20, 1.18)
0.5	1.0	(15.07, 15.07)	(12.95, 12.95)	(12.37, 12.36)	(12.42, 12.42)	(12.76, 12.76)	(13.22, 13.22)
	1.5	(2.75, 2.75)	(3.37, 3.36)	(3.94, 3.93)	(4.44, 4.43)	(4.90, 4.90)	(5.34, 5.34)
	2.0	(1.27, 1.26)	(1.64, 1.63)	(2.10, 2.09)	(2.57, 2.57)	(2.99, 2.98)	(3.33, 3.33)
	2.5	(0.83, 0.80)	(0.99, 0.97)	(1.25, 1.23)	(1.59, 1.57)	(1.96, 1.95)	(2.33, 2.32)
	3.0	(0.63, 0.60)	(0.71, 0.69)	(0.84, 0.81)	(1.01, 0.99)	(1.25, 1.24)	(1.54, 1.52)
1.0	1.0	(2.19, 2.18)	(2.55, 2.54)	(2.92, 2.91)	(3.29, 3.29)	(3.66, 3.65)	(4.01, 4.00)
	1.5	(1.35, 1.34)	(1.63, 1.62)	(1.90, 1.89)	(2.15, 2.14)	(2.40, 2.39)	(2.63, 2.62)
	2.0	(0.97, 0.95)	(1.21, 1.20)	(1.47, 1.46)	(1.70, 1.69)	(1.90, 1.89)	(2.09, 2.08)
	2.5	(0.75, 0.72)	(0.91, 0.89)	(1.13, 1.11)	(1.38, 1.36)	(1.60, 1.59)	(1.78, 1.77)
	3.0	(0.61, 0.58)	(0.71, 0.68)	(0.85, 0.83)	(1.05, 1.03)	(1.27, 1.26)	(1.49, 1.48)
1.5	1.0	(0.66, 0.64)	(0.71, 0.69)	(0.75, 0.73)	(0.78, 0.75)	(0.79, 0.77)	(0.80, 0.77)
	1.5	(0.74, 0.72)	(0.83, 0.81)	(0.91, 0.89)	(0.98, 0.96)	(1.04, 1.02)	(1.08, 1.06)
	2.0	(0.72, 0.69)	(0.84, 0.82)	(0.95, 0.93)	(1.04, 1.02)	(1.12, 1.10)	(1.18, 1.17)
	2.5	(0.64, 0.61)	(0.76, 0.74)	(0.91, 0.88)	(1.03, 1.01)	(1.12, 1.11)	(1.20, 1.19)
	3.0	(0.56, 0.53)	(0.66, 0.63)	(0.79, 0.77)	(0.94, 0.92)	(1.08, 1.06)	(1.18, 1.17)
2.0	1.0	(0.36, 0.31)	(0.35, 0.30)	(0.34, 0.28)	(0.32, 0.25)	(0.29, 0.22)	(0.27, 0.19)
	1.5	(0.48, 0.44)	(0.49, 0.46)	(0.50, 0.46)	(0.50, 0.46)	(0.48, 0.45)	(0.47, 0.43)
	2.0	(0.54, 0.50)	(0.59, 0.55)	(0.62, 0.59)	(0.64, 0.61)	(0.65, 0.62)	(0.65, 0.62)
	2.5	(0.54, 0.50)	(0.61, 0.58)	(0.68, 0.65)	(0.73, 0.70)	(0.76, 0.73)	(0.78, 0.75)
	3.0	(0.51, 0.47)	(0.59, 0.56)	(0.68, 0.65)	(0.76, 0.74)	(0.82, 0.79)	(0.85, 0.83)
2.5	1.0	(0.27, 0.19)	(0.26, 0.17)	(0.24, 0.15)	(0.24, 0.14)	(0.23, 0.14)	(0.23, 0.14)
	1.5	(0.35, 0.29)	(0.34, 0.28)	(0.32, 0.26)	(0.30, 0.23)	(0.28, 0.21)	(0.26, 0.18)
	2.0	(0.42, 0.37)	(0.43, 0.38)	(0.42, 0.38)	(0.41, 0.37)	(0.40, 0.35)	(0.38, 0.32)
	2.5	(0.45, 0.41)	(0.49, 0.45)	(0.51, 0.47)	(0.51, 0.48)	(0.51, 0.47)	(0.50, 0.46)
	3.0	(0.46, 0.41)	(0.51, 0.47)	(0.56, 0.52)	(0.59, 0.55)	(0.60, 0.57)	(0.60, 0.57)
3.0	1.0	(0.24, 0.15)	(0.24, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)
	1.5	(0.28, 0.21)	(0.27, 0.19)	(0.25, 0.17)	(0.24, 0.14)	(0.23, 0.14)	(0.23, 0.14)
	2.0	(0.34, 0.28)	(0.33, 0.27)	(0.32, 0.25)	(0.30, 0.23)	(0.28, 0.20)	(0.26, 0.17)
	2.5	(0.38, 0.33)	(0.39, 0.34)	(0.39, 0.34)	(0.38, 0.32)	(0.36, 0.30)	(0.33, 0.27)
	3.0	(0.41, 0.36)	(0.44, 0.39)	(0.45, 0.41)	(0.45, 0.41)	(0.44, 0.40)	(0.42, 0.38)

to decrease from 370.4 to values as low as 90 (see Tables 5.7 to 5.9). Similar observations can be made with respect to the $SDTS_0$ performances of the OSPRT chart, which indicates that majority of the in-control time of signal values now cluster around a substantially smaller ATS_0 value. Note that the drastic fall in the $(ATS_0, SDTS_0)$ values marks a big red flag, since it implies that the false alarm rates have increased by almost four times compared to the recommended rate of $1/370.4 = 0.0027$. When the distribution becomes more skewed in shape, the $(ATS_0,$

Table 5.9 The (ATS, SDTS) values of the AEQL-optimal OSPRT chart evaluated for $\rho \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, $\delta \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and $\eta \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, when the underlying data follow the Weibull distribution.

k'		2.21560	1.56391	1.21112	1.00000	0.86317	0.76862
ρ		0.5	1.0	1.5	2.0	2.5	3.0
δ	η	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)	(ATS, SDTS)
0.0	1.0	(123.30, 123.07)	(49.37, 49.14)	(32.10, 31.86)	(26.21, 25.98)	(23.89, 23.65)	(23.00, 22.77)
	1.5	(5.04, 5.04)	(5.88, 5.88)	(6.71, 6.70)	(7.20, 7.19)	(7.74, 7.74)	(8.32, 8.32)
	2.0	(1.34, 1.33)	(1.77, 1.76)	(2.53, 2.52)	(3.47, 3.46)	(4.29, 4.29)	(4.84, 4.83)
	2.5	(0.78, 0.76)	(0.89, 0.87)	(1.14, 1.12)	(1.62, 1.61)	(2.30, 2.29)	(3.24, 3.24)
	3.0	(0.59, 0.56)	(0.63, 0.60)	(0.70, 0.68)	(0.86, 0.84)	(1.16, 1.14)	(1.65, 1.64)
0.5	1.0	(14.91, 14.91)	(12.05, 12.05)	(11.17, 11.17)	(11.11, 11.11)	(11.44, 11.44)	(11.96, 11.95)
	1.5	(2.67, 2.67)	(3.28, 3.27)	(3.78, 3.78)	(4.29, 4.29)	(4.82, 4.81)	(5.34, 5.33)
	2.0	(1.22, 1.20)	(1.68, 1.67)	(2.29, 2.28)	(2.73, 2.73)	(3.15, 3.15)	(3.56, 3.55)
	2.5	(0.78, 0.75)	(0.98, 0.96)	(1.38, 1.36)	(1.93, 1.92)	(2.41, 2.40)	(2.75, 2.75)
	3.0	(0.60, 0.57)	(0.68, 0.65)	(0.85, 0.83)	(1.20, 1.18)	(1.69, 1.68)	(2.27, 2.26)
1.0	1.0	(2.14, 2.13)	(2.51, 2.50)	(2.92, 2.91)	(3.36, 3.35)	(3.81, 3.81)	(4.28, 4.27)
	1.5	(1.36, 1.34)	(1.65, 1.64)	(1.95, 1.94)	(2.27, 2.26)	(2.59, 2.58)	(2.91, 2.90)
	2.0	(0.98, 0.96)	(1.29, 1.28)	(1.57, 1.56)	(1.84, 1.83)	(2.11, 2.10)	(2.37, 2.36)
	2.5	(0.73, 0.70)	(0.96, 0.94)	(1.31, 1.30)	(1.59, 1.58)	(1.83, 1.82)	(2.07, 2.06)
	3.0	(0.59, 0.55)	(0.71, 0.68)	(0.96, 0.94)	(1.34, 1.33)	(1.64, 1.63)	(1.86, 1.85)
1.5	1.0	(0.68, 0.66)	(0.75, 0.73)	(0.81, 0.79)	(0.86, 0.84)	(0.88, 0.86)	(0.90, 0.88)
	1.5	(0.77, 0.74)	(0.88, 0.86)	(0.99, 0.97)	(1.09, 1.07)	(1.18, 1.17)	(1.26, 1.24)
	2.0	(0.74, 0.72)	(0.90, 0.88)	(1.03, 1.01)	(1.16, 1.15)	(1.28, 1.27)	(1.40, 1.38)
	2.5	(0.65, 0.62)	(0.84, 0.82)	(1.02, 1.00)	(1.16, 1.15)	(1.30, 1.29)	(1.43, 1.42)
	3.0	(0.56, 0.52)	(0.70, 0.67)	(0.94, 0.92)	(1.14, 1.12)	(1.28, 1.27)	(1.42, 1.40)
2.0	1.0	(0.37, 0.32)	(0.36, 0.31)	(0.34, 0.28)	(0.29, 0.22)	(0.23, 0.14)	(0.23, 0.14)
	1.5	(0.49, 0.46)	(0.52, 0.49)	(0.54, 0.50)	(0.53, 0.50)	(0.51, 0.47)	(0.47, 0.43)
	2.0	(0.56, 0.53)	(0.62, 0.59)	(0.67, 0.65)	(0.71, 0.68)	(0.73, 0.70)	(0.73, 0.70)
	2.5	(0.56, 0.53)	(0.68, 0.65)	(0.75, 0.73)	(0.82, 0.79)	(0.87, 0.85)	(0.90, 0.88)
	3.0	(0.52, 0.48)	(0.65, 0.62)	(0.79, 0.77)	(0.88, 0.86)	(0.95, 0.93)	(1.01, 0.99)
2.5	1.0	(0.27, 0.19)	(0.24, 0.15)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)
	1.5	(0.36, 0.30)	(0.35, 0.29)	(0.32, 0.26)	(0.27, 0.19)	(0.23, 0.14)	(0.23, 0.14)
	2.0	(0.44, 0.39)	(0.45, 0.41)	(0.45, 0.41)	(0.43, 0.38)	(0.39, 0.34)	(0.33, 0.26)
	2.5	(0.48, 0.44)	(0.53, 0.49)	(0.55, 0.52)	(0.56, 0.53)	(0.55, 0.52)	(0.52, 0.49)
	3.0	(0.47, 0.43)	(0.57, 0.54)	(0.62, 0.59)	(0.66, 0.63)	(0.67, 0.65)	(0.67, 0.65)
3.0	1.0	(0.24, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)
	1.5	(0.29, 0.21)	(0.26, 0.18)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)	(0.23, 0.14)
	2.0	(0.35, 0.29)	(0.34, 0.28)	(0.31, 0.25)	(0.26, 0.18)	(0.23, 0.14)	(0.23, 0.14)
	2.5	(0.41, 0.36)	(0.42, 0.37)	(0.41, 0.36)	(0.38, 0.33)	(0.33, 0.27)	(0.24, 0.14)
	3.0	(0.43, 0.38)	(0.47, 0.43)	(0.49, 0.45)	(0.48, 0.44)	(0.46, 0.42)	(0.41, 0.37)

SDTS₀) values are found to decrease further, reaching values as low as 21 for all the three non-normal distributions (which are 10 times smaller than the recommended (ATS₀, SDTS₀) values). One possible explanation is that, since the control statistic of the OSPRT chart involves a “square” function, it is more likely that the control statistic would blow up when measurements are taken from a positively skewed distribution, which then causes the false alarm rate to increase by a substantial degree. However, when the Weibull distribution is assumed, it

seems that the in-control performance of the OSPRT chart is affected by a smaller degree compared to the gamma and lognormal distributions when data are mildly skewed (i.e., $\rho = 0.5$).

Referring to the out-of-control metrics (when $(\delta, \eta) \neq (0, 1)$), one finds that the $(\text{ATS}_1, \text{SDTS}_1)$ values increase generally as the degree of skewness increases in all the three non-normal distributions. For instance, when the lognormal distribution is assumed, the ATS_1 value evaluated at $(\delta, \eta) = (0.5, 2.0)$ is found to increase steadily from 1.27 to 3.33 as the skewness increases from $\rho = 0.5$ to $\rho = 3.0$ (see Table 5.8). Similar inferences can be made with regard to the SDTS_1 performance of the OSPRT chart. However, there are some exceptions noted for cases where δ is large and the case with $(\delta, \eta) = (0.5, 1.0)$. For instance, when the lognormal distribution is assumed (see Table 5.8), the ATS_1 value at $(\delta, \eta) = (2.5, 1.5)$ is found to decrease persistently from 0.35 to 0.26 as the degree of skewness increases from 0.5 to 3.0. To conclude, it is rather concerning that both the in-control and out-of-control performances of the OSPRT chart have deteriorated as a result of an increased skewness. While the situation is unfortunate, it is not beyond our expectation since the OSPRT chart has been formulated using the density of the normal distribution.

As a summary, both the in-control and out-of-control performances of the OSPRT chart are negatively affected by the presence of skewness in data. Although the Weibull distribution seems to pose a relatively smaller effect on the in-control performance of the OSPRT chart compared to the gamma and lognormal distributions, one notices that the ATS_0 value plummets rapidly as the skewness of data increases beyond 1.0. As the OSPRT chart designed under the normal model is sensitive to non-normality, practitioners are strongly advised to perform some transformation (e.g., Box-Cox transformation) on skewed datasets before implementing the proposed OSPRT scheme. Alternatively, practitioners may set a higher threshold τ_{ATS} (e.g., set $\tau_{\text{ATS}} = 3000$) during the design of the OSPRT chart

so that the ATS_0 value drops to an acceptably low level when skewness is present. However, it should be noted that setting a higher τ_{ATS} can lead to reduced out-of-control sensitivity of the OSPRT chart due to the inevitable trade-off between the in-control and out-of-control performances. If the underlying distribution is known upfront (either through historical knowledge or by inferential methods), it is recommended to derive a new OSPRT scheme specific to the distribution of the process. For example, if there is historical evidence that the underlying process follows the gamma distribution, the OSPRT scheme can be reconstructed by using the density function of the gamma distribution. However, due to the substantial theoretical effort associated with such a proposal, subsequent developments will be left for future research.

6 Conclusions

6.1 Introduction

In this thesis, we have identified the relevant areas of improvement and development for the SPRT control chart in the current SQC landscape. A balanced mix of research proposals, including one from each of vertical developments and expansive developments, have been established in this thesis. As the chosen research topics align with modern SQC practices and demand, this thesis serves as a valuable reference for industrial practitioners who would seek to improve the performance of their processes, or to employ a more sophisticated quality control tool. We believe that the proposals made in this thesis would create opportunities for more research extensions and endeavours, which in turn contribute to the goal of achieving sustainable industrialisation, i.e., Sustainable Development Goal 9 according to the United Nations.

We conclude this thesis with a summary and a list of future research ideas in the following subsections. An exhaustive list of research findings and contributions made in this thesis is given in Section 6.2. The potential areas for extension and future research directions are highlighted in Section 6.3.

6.2 Summary of Research Findings and Contributions

Chapters 1 and 2 provide an overview of the current research landscape in the field of SQC. A picture depicting the fundamental separation of research works into two categories, i.e., vertical developments and expansive developments, have been illustrated in Chapter 1 by citing appropriate references from the literature. A literature survey on the historical development of the SPRT chart, the evolution of control charts with estimated process parameters, as well as a review on the

strategies for developing joint monitoring control schemes, have been provided in Chapter 2.

Chapter 3 discusses the impact of Phase-I parameter estimation on the performances of the SPRT chart, as well as developing a robust design for the SPRT chart with estimated process parameters based on the state-of-the-art GICP approach. A list of findings and research contributions is given as follows:

- (i) The full theoretical framework for the SPRT chart with estimated process parameters is developed in Section 3.3. This includes the derivation of the run length properties through the Markov chain approach. Note that these properties are not only useful in assessing the performance of the SPRT chart with estimated process parameters, but they are also important for accelerating subsequent computational and optimisation efforts of the SPRT chart. As Monte Carlo simulations are known to be more computationally demanding than calculations through formulae, an extensive derivation of the theoretical formulae is necessary to encourage reproducibility of the research work and to speed up the optimisation procedure.
- (ii) A thorough investigation on the limitations of the traditional design and unconditional metric matching design is conducted in Sections 3.4 and 3.5, respectively. It is found that the SPRT chart designed under the traditional approach has a 50% chance of producing undesirable false alarm rates, even for a very large Phase-I sample size. Under the unconditional metric matching approach, the conditional in-control performance is found to deteriorate further, revealing exceedance probabilities that are as low as 8% when the number of Phase-I samples is small. These observations motivate the transition from these outdated designs to the more recent GICP design, which is endorsed by many leading researchers working on control charts with estimated process parameters.

- (iii) A novel approach developed by coupling the GICP framework and the concept of optimisation, known as the integrated optimal-GICP design, is proposed in Section 3.6. On the one hand, the GICP framework ensures that the conditional in-control performance can be achieved at a very high probability (90% or 95%). On the other hand, the optimisation design attempts to search the set of charting parameters that achieves the smallest quality loss. Through combining the two designs, we find out that the out-of-control performances of the SPRT chart with estimated process parameters is nearly the same as those of the SPRT chart with known process parameters, for mean shift sizes as small as $\delta = 0.6$. Various parameters used to control the rigidity of the GICP constraint, i.e., p_e and ε , can also be adjusted to suit practitioners' preferences and needs. Besides, we have developed a more efficient stochastic approximation algorithm for the GICP design by modifying the original algorithm proposed by Capizzi and Masarotto (2020). The proposed algorithm is found to converge in a fewer number of iterations compared to the bootstrap technique proposed by Saleh *et al.* (2015b) and the empirical distribution technique proposed by Diko *et al.* (2019b).
- (iv) Four optimisation designs based on the assumptions of deterministic and stochastic shift sizes are outlined in Sections 3.6.1 to 3.6.4, respectively, for the SPRT chart with estimated process parameters. In the case where practitioners have a specialised understanding of a certain process, the AATS-optimal and AANOS-optimal SPRT charts are recommended as they are optimised to detect a particular shift size of the process. In the case where practitioners have limited knowledge about the process (which happens to be the case most of the time), the AAQL-optimal and EAANOS-optimal SPRT chart are recommended due to their overall effectiveness towards a broad range of shift sizes. On the other hand, if practitioners seek to optimise only the detection speed of the SPRT chart, the time-based designs, i.e., the

AATS-optimal and AAQL-optimal designs, are recommended in general. If practitioners wish to optimise both the detection speed and the out-of-control sampling efficiency of the SPRT chart, the AANOS-optimal and EAANOS-optimal designs are recommended.

- (v) The superiority of the SPRT chart over the CUSUM chart is verified in the case where process parameters are estimated in Section 3.7. For both the AAQL-optimal and EAANOS-optimal designs, the GICP-adjusted SPRT chart is found to be more sensitive than the GICP-adjusted CUSUM chart for all shift sizes δ , even though the SPRT chart has a much more variable $CATS_0$ (or $CANOS_0$) distribution compared to the CUSUM chart. If practitioners do not mind employing a VSS control chart in quality control applications, then the SPRT chart is highly recommended over the CUSUM chart.
- (vi) Two real industrial applications are demonstrated in Section 3.8 to showcase the ease of use of the optimal SPRT chart with GICP-adjusted control limits. Practitioners are recommended to use a minimum of $m = 200$ Phase-I individual measurements. Suitable charting parameters for the SPRT chart can be chosen from the tables provided in Sections 3.6.1 to 3.6.4 based on practitioners' customised needs, the available resources, and the in-control policy practised by the factory.

Chapter 4 develops a new omnibus SPRT control chart for joint monitoring of the mean and standard deviation of a normal distribution. A list of findings and research contributions is given as follows:

- (i) The full theoretical framework of the OSPRT chart is developed in Section 4.2. This includes the formulation of a two-parameter hypothesis testing framework, the derivation of the statistic of the OSPRT from first principles, a comprehensive description of the working mechanism of the OSPRT chart,

and the derivation of its run length properties via the Markov chain approach. These theoretical foundations are essential to facilitate subsequent research works and enhancement designs, which could be time-consuming if the proposed scheme is not well-documented. To speed up the computation process, a more efficient algorithm based on the cdf of the standard normal distribution has been developed and given in Appendix A.

- (ii) Four optimisation designs based on the assumptions of deterministic and stochastic shift sizes are outlined in Sections 4.3.1 to 4.3.4, respectively. Similar to Chapter 3, the ATS-optimal and ANOS-optimal designs are recommended only when practitioners have grasped abundant evidence about a particular shift size of a certain process. In majority of the cases, the AEQL-optimal and EANOS-optimal designs are recommended due to their relevance to the assumption that process shift sizes are unknown in practice.
- (iii) The OSPRT chart possesses strong optimality properties as shown in Section 4.3.1. In particular, we prove that the OSPRT chart with prescribed reference parameters minimises the ARL_1 , ATS_1 , and $ANOS_1$ values for any deterministic shift sizes (δ, η) . By virtue of the optimality property, the ATS-optimal design for the OSPRT chart can be completed by performing a one-dimensional parameter search (see Section 4.3.1), whereas the ANOS-optimal design can be completed without involving any independent parameter searches (see Section 4.3.3). The simplification in the optimisation procedure leads to a lower computational time required to complete the design, which in turn fosters a more efficient design strategy for practitioners.
- (iv) The OSPRT chart is shown to outperform the Shewhart $\bar{X}$ - S , WLC, VSS WLC, and ABS-SPRT charts in Section 4.5. In cases where only the process mean is shifting ($\delta > 0$ and $\eta = 1$) or where both the mean and variance are shifting ($\delta > 0$ and $\eta > 1$), the OSPRT chart is found to be more effective

than the four competing control charts for all magnitudes of shift sizes. In cases where only the process variance is shifting ($\delta = 0$ and $\eta > 1$), the OSPRT chart has a slightly lower sensitivity compared to the ABS-SPRT chart for small values of η , but has a comparable performance to the ABS-SPRT chart when η is moderate or large. In terms of the overall effectiveness, the OSPRT chart stands out as the best performer as it produces the lowest AEQL and EANOS values among the five control charts.

- (v) The optimal OSPRT chart is shown to have a higher sampling efficiency compared to the standard FSS control charts and the ABS-SPRT chart. In the ANOS-optimal and EANOS-optimal designs, the OSPRT chart is found to produce the smallest ARL_1 and $ANOS_1$ values compared to the Shewhart $\bar{X}$ - S , WLC, VSS WLC, and ABS-SPRT charts, for a given fixed ASN_0 . It is hence evident that the OSPRT chart is not only highly effective in detecting process shifts, but also offers the best savings in terms of the number of observations required during inspection. As destructive testing is practised in many industries, the OSPRT chart serves as an excellent alternative to many existing control schemes due to its high inspection efficiency.
- (vi) Two real industrial applications are demonstrated in Section 4.6 to showcase the ease of use of the OSPRT scheme. Practitioners are encouraged to select the optimal charting parameters from the tables tabulated in Sections 4.3.1 to 4.3.4 based on practical considerations and the employment capacity of the factory.

Chapter 5 investigates the extent to which the performances of the SPRT chart with estimated process parameters and the OSPRT chart would change under the influence of non-normal distributions. The gamma, lognormal, and Weibull distributions with various degrees of skewness have been considered in the study. The following research findings have been deduced:

- (i) The conditional in-control performance of the GICP-adjusted SPRT chart with estimated process parameters is found to deteriorate rapidly as skewness increases, with exceedance probabilities dropping to percentages as low as 0.32%. However, the out-of-control performances do not seem to deteriorate excessively, except in cases where both the Phase-I sample size and mean shift size are small. When a skewed distribution is expected, practitioners are advised to perform some transformation on the data before applying the proposed GICP-adjusted SPRT chart. Alternatively, practitioners may also adopt a stricter GICP design, bearing in mind that doing so may lead to a reduction in the sensitivity of the SPRT chart.
- (ii) Both the in-control and out-of-control performances of the OSPRT chart are negatively affected by the presence of skewness in datasets. In particular, the false alarm rates of the OSPRT chart may increase by up to ten times when data come from an extremely skewed distribution. The out-of-control detection sensitivity of the OSPRT chart is also found to decline generally for most of the joint shift sizes as the degree of skewness increases. When a skewed distribution is expected, practitioners are advised to perform some transformation on the data before applying the OSPRT chart. Alternatively, practitioners may redesign the OSPRT chart with a larger τ_{ATS} (or τ_{ANOS}), bearing in mind that doing so may reduce the out-of-control sensitivity of the OSPRT chart even further.

6.3 Future Research Directions

While the new developments proposed in this thesis are rather substantive, there is still plenty of room for potential improvement in the research inventions. We provide the following list of suggestions for future research as follows:

- (i) Development of the SPRT chart for specific non-normal distributions. As the proposed SPRT charts are found to perform poorly under non-normal conditions (see Chapter 5 for evaluations under the gamma, lognormal, and Weibull distributions), researchers may consider deriving distribution-specific control statistics for the SPRT chart with estimated process parameters and the OSPRT chart. One example is the SPRT chart developed by Godase *et al.* (2022) for monitoring the scale parameter of the Maxwell distribution. Researchers may derive an SPRT chart for monitoring the parameters of the gamma, lognormal, or Weibull distribution by resorting to the basic principles. Note, however, that the expression for the control statistic may not be elegant, especially if the pdf of the assumed distribution is rather complex.
- (ii) Development of a distribution-free SPRT chart. Instead of developing a distribution-specific statistic, researchers may consider proposing a non-parametric statistic for the SPRT chart with estimated process parameters and the OSPRT chart. One example is the SPRT sign chart proposed by Mahadik and Godase (2023b). Future research may explore the SPRT sign chart with estimated process parameters, or develop the OSPRT sign chart by using the sequential sign test for both the location and scale of an arbitrary distribution. Note that this proposal has the advantage of avoiding a specific parametric distribution, and hence it should perform generally well on all types of distribution. However, one should expect a loss in the statistical power (non-parametric tests can be less powerful than parametric tests) when data follow an approximate normal distribution.
- (iii) A parameter learning strategy for the SPRT chart with estimated process parameters. As the amount of Phase-I data is limited in practice, researchers often face the issue with highly volatile $CATS_0$ (or $CANOS_0$) performances during the application of Phase-II control charts, due to the uncertainty in

the parameter estimates. One solution is to update the parameter estimates by incorporating new Phase-II samples obtained from the process, so that the parameter estimates become more and more accurate. While the idea sounds appealing, there exist some concerns regarding the delay of Phase-II control charts in detecting early process shifts, which could lead to contaminations to the parameter estimates. Therefore, in future research, a cautious parameter learning strategy could be devised to ensure that the process parameters are learnt in a “prudent” way.

- (iv) Enhancement features to improve the detection speed of the SPRT chart. It is possible to introduce a variable sampling interval (VSI) feature to both the SPRT chart with estimated process parameters and the OSPRT chart proposed in this thesis as a future extension. As the SPRT chart is a VSS control chart by the way it is constructed, introducing an additional VSI feature could boost the detection speed of the SPRT chart further, making it comparable to some existing variable-sample-size-variable-sampling-interval control charts in the literature.
- (v) A truncated SPRT control chart. Practitioners are often held back by the fact that the SPRT inspects an uncertain number of observations before reaching a decision, even though there have been well-established theorems proving that the SPRT terminates in a finite number of observations. To make the SPRT chart more operational-friendly, future research may consider devising a truncated version of the SPRT chart with estimated process parameters and a truncated OSPRT chart. This involves setting a ceiling for the number of observations inspected, and a pre-defined decision criterion when the maximum number of observations is reached.
- (vi) Development of economic designs for the SPRT chart. While this thesis focuses on optimising the detection speed and sampling efficiency of the SPRT

chart, it is of great interest to consider alternative design methodologies based on the minimisation of the total costs involved in an inspection cycle (Raim *et al.*, 1988; Zhang *et al.*, 2014b). Future research may consider embarking on an economic design for the SPRT chart with estimated process parameters (or the OSPRT chart). The economic design typically involves multiple cost components associated with monitoring the process, as well as constraints that are relevant to the costing structure of a system. Note that such models are usually huge and are more computational demanding than the usual statistical designs for control charts.

REFERENCES

- Adeoti, O. A., & Malela-Majika, J. C. (2020). Double exponentially weighted moving average control chart with supplementary runs-rules. *Quality Technology & Quantitative Management*, 17(2), 149-172.
- Albers, W., & Kallenberg, W. C. (2004). Estimation in Shewhart control charts: effects and corrections. *Metrika*, 59(4), 207-234.
- Amin, R. W., & Ethridge, R. A. (1998). A note on individual and moving range control charts. *Journal of Quality Technology*, 30(1), 70-74.
- Annadi, H. P., Keats, J. B., Runger, G. C., & Montgomery, D. C. (1995). An adaptive sample size CUSUM control chart. *International Journal of Production Research*, 33(6), 1605-1616.
- Atalay, M., Caner Testik, M., Duran, S., & Weiß, C. H. (2020). Guidelines for automating Phase I of control charts by considering effects on Phase-II performance of individuals control chart. *Quality Engineering*, 32(2), 223-243.
- Aytaçoğlu, B., Driscoll, A. R., & Woodall, W. H. (2022). Controlling the conditional false alarm rate for the MEWMA control chart. *Journal of Quality Technology*, 54(5), 487-502.
- Bai, D. S., & Choi, I. S. (1995). $\bar{X}$ and R control charts for skewed populations. *Journal of Quality Technology*, 27(2), 120-131.
- Braun, W. J. (1999). Run length distributions for estimated attributes charts. *Metrika*, 50(2), 121-129.
- Brook, D. & Evans, D. A. (1972). An approach to the probability distribution of CUSUM run length. *Biometrika*, 59(3), 539-549.

- Calzada, M. E., & Scariano, S. M. (2013). The synthetic t and synthetic EWMA t charts. *Quality Technology & Quantitative Management*, 10(1), 37–56.
- Capizzi, G., & Masarotto, G. (2010). Combined Shewhart–EWMA control charts with estimated parameters. *Journal of Statistical Computation and Simulation*, 80(7), 793-807.
- Capizzi, G., & Masarotto, G. (2020). Guaranteed in-control control chart performance with cautious parameter learning. *Journal of Quality Technology*, 52(4), 385-403.
- Castagliola, P., Celano, G., & Chen, G. (2009). The exact run length distribution and design of the S^2 chart when the in-control variance is estimated. *International Journal of Reliability, Quality and Safety Engineering*, 16(1), 23-38.
- Celano, G., & Castagliola, P. (2016). Design of a Phase II control chart for monitoring the ratio of two normal variables. *Quality and Reliability Engineering International*, 32(1), 291-308.
- Chakraborti, S. (2006). Parameter estimation and design considerations in prospective applications of the $\bar{X}$ chart. *Journal of Applied Statistics*, 33(4), 439-459.
- Chakraborti, S., Human, S. W., & Graham, M. A. (2008). Phase I statistical process control charts: An overview and some results. *Quality Engineering*, 21(1), 52-62.
- Chan, L. K., Hapuarachchi, K. P., & Macpherson, B. D. (1988). Robustness of mean $E(X)$ and R charts. *IEEE Transactions on Reliability*, 37(1), 117-123.
- Chen, G. (1997). The mean and standard deviation of the run length distribution of $\bar{X}$ charts when control limits are estimated. *Statistica Sinica*, 7(3), 789-798.

- Chen, G. (1998). The run length distributions of the R , S and S^2 control charts when is estimated. *Canadian Journal of Statistics*, 26(2), 311-322.
- Chen, G., & Cheng, S. W. (1998). Max chart: combining $\bar{X}$ -bar chart and S chart. *Statistica Sinica*, 8(1), 263-271.
- Chen, G., Cheng, S. W., & Xie, H. (2001). Monitoring process mean and variability with one EWMA chart. *Journal of Quality Technology*, 33(2), 223-233.
- Cheng, S. W., & Thaga, K. (2006). Single variables control charts: an overview. *Quality and Reliability Engineering International*, 22(7), 811-820.
- Cheng, S. W., & Thaga, K. (2010). The Max-CUSUM chart. In H. J. Lenz, P. T. Wilrich, & W. Schmid (Eds.), *Frontiers in Statistical Quality Control 9* (pp. 85-98). Physica-Verlag HD.
- Chong, Z. L., Tan, K. L., Khoo, M. B. C., Teoh, W. L., & Castagliola, P. (2022). Optimal designs of the exponentially weighted moving average (EWMA) median chart for known and estimated parameters based on median run length. *Communications in Statistics-Simulation and Computation*, 51(7), 3660–3684.
- Chong, Z. L., Yeong, W. C., Khaw, K. W., Chew, X. Y., & Teoh, W. L. (2024). Optimal design of the side sensitive group runs double sampling (SSGRDS) $\bar{X}$ scheme with estimated process parameters. *Transactions of the Institute of Measurement and Control*, Article 01423312241249824.
- Chou, Y. M., Mason, R. L., & Young, J. C. (2006). The SPRT control chart for standard deviation based on individual observations. *Quality Technology & Quantitative Management*, 3(3), 335-345.
- Costa, A. F. (1994). $\bar{X}$ charts with variable sample size. *Journal of Quality Technology*, 26(3), 155-163.

- Costa, A. F. (1999). Joint $\bar{X}$ and R charts with variable sample sizes and sampling intervals. *Journal of Quality Technology*, 31(4), 387-397.
- Crowder, S. V. (1987). Computation of ARL for combined individual measurement and moving range charts. *Journal of Quality Technology*, 19(2), 98-102.
- Del Castillo, E. (1996). Run length distributions and economic design of $\bar{X}$ charts with unknown process variance. *Metrika*, 43(12), 189-201.
- Deore, R. E., Mahadik, S. B., & Godase, D. G. (2023). The two-sided SPRT sign charts. *Quality and Reliability Engineering International*, In Press.
- Diko, M. D., Chakraborti, S., & Does, R. J. M. M. (2019a). Guaranteed in-control performance of the EWMA chart for monitoring the mean. *Quality and Reliability Engineering International*, 35(4), 1144-1160.
- Diko, M. D., Chakraborti, S., & Does, R. J. M. M. (2019b). An alternative design of the two-sided CUSUM chart for monitoring the mean when parameters are estimated. *Computers & Industrial Engineering*, 137(11), 106042.
- Domangue, R., & Patch, S. C. (1991). Some omnibus exponentially weighted moving average statistical process monitoring schemes. *Technometrics*, 33(3), 299-313.
- Engmann, G. M., & Han, D. (2022). The optimized CUSUM and EWMA multi-charts for jointly detecting a range of mean and variance change. *Journal of Applied Statistics*, 49(6), 1540-1558.
- Epprecht, E. K., Loureiro, L. D., & Chakraborti, S. (2015). Effect of the amount of Phase I data on the Phase II performance of S^2 and S control charts. *Journal of Quality Technology*, 47(2), 139-155.

- Faraz, A., Heuchenne, C., & Saniga, E. (2018). An exact method for designing Shewhart and S^2 control charts to guarantee in-control performance. *International Journal of Production Research*, 56(7), 2570-2584.
- Faraz, A., Woodall, W. H., & Heuchenne, C. (2015). Guaranteed conditional performance of the S^2 control chart with estimated parameters. *International Journal of Production Research*, 53(14), 4405-4413.
- Gan, F. F. (1995). Joint monitoring of process mean and variance using exponentially weighted moving average control charts. *Technometrics*, 37(4), 446-453.
- Gan, F. F., Ting, K. W., & Chang, T. C. (2004). Interval charting schemes for joint monitoring of process mean and variance. *Quality and Reliability Engineering International*, 20(4), 291-303.
- Gandy, A., & Kvaløy, J. T. (2013). Guaranteed conditional performance of control charts via bootstrap methods. *Scandinavian Journal of Statistics*, 40(4), 647-668.
- Geyer, P. L., Steiner, S. H., & Wesolowsky, G. O. (1996). Optimal SPRT and CUSUM procedures with compressed-limit gauges. *IIE transactions*, 28(6), 489-496.
- Godase, D. G., & Mahadik, S. B. (2019). The SPRT control chart for process dispersion. *Quality and Reliability Engineering International*, 35(6), 1878-1889.
- Godase, D. G., & Mahadik, S. B. (2022). The SPRT sign chart for process dispersion. *Stochastics and Quality Control*, 37(1), 101-106.
- Godase, D. G., Mahadik, S. B., & Rakitzis, A. C. (2022). The SPRT control charts for the Maxwell distribution. *Quality and Reliability Engineering International*, 38(4), 1713-1728.

- Goedhart, R., da Silva, M. M., Schoonhoven, M., Epprecht, E. K., Chakraborti, S., Does, R. J. M. M., & Veiga, Á. (2017a). Shewhart control charts for dispersion adjusted for parameter estimation. *IIE Transactions*, 49(8), 838-848.
- Goedhart, R., Schoonhoven, M., & Does, R. J. M. M. (2016). Correction factors for Shewhart and control charts to achieve desired unconditional ARL. *International Journal of Production Research*, 54(24), 7464-7479.
- Goedhart, R., Schoonhoven, M., & Does, R. J. M. M. (2017b). Guaranteed in-control performance for the Shewhart $\bar{X}$ and X control charts. *Journal of Quality Technology*, 49(2), 155-171.
- Goel, A. L., & Wu, S. M. (1971). Determination of ARL and a contour nomogram for CUSUM charts to control normal mean. *Technometrics*, 13(2), 221-230.
- Hany, M., & Mahmoud, M. A. (2016). An evaluation of the Crosier's CUSUM control chart with estimated parameters. *Quality and Reliability Engineering International*, 32(5), 1825-1835.
- Haridy, S., Maged, A., Alherimi, N., Shamsuzzaman, M., & Al-Ali, S. (2024). Optimization design of control charts: A systematic review. *Quality and Reliability Engineering International*, 40(4), 2122-2157.
- Haridy, S., Wu, Z., Lee, K. M., & Bhuiyan, N. (2013). Optimal average sample number of the SPRT chart for monitoring fraction nonconforming. *European Journal of Operational Research*, 229(2), 411-421.
- Haridy, S., Wu, Z., Lee, K. M., & Rahim, M. A. (2014). An attribute chart for monitoring the process mean and variance. *International Journal of Production Research*, 52(11), 3366-3380.
- Harman, G. (2010). *Wire Bonding in Microelectronics*. McGraw-Hill.

- Hawkins, D. M., & Deng, Q. (2009). Combined charts for mean and variance information. *Journal of Quality Technology*, 41(4), 415-425.
- Hawkins, D. M., and Olwell, D. H. (1998). *Cumulative Sum Charts and Charting for Quality Improvement*. Springer Science + Business Media.
- Hillier, F. S. (1969). $\bar{X}$ - and R-chart control limits based on a small number of subgroups. *Journal of Quality Technology*, 1(1), 17-26.
- Hossain, M. P., Omar, M. H., Riaz, M., & Arafat, S. Y. (2022). On designing a new control chart for Rayleigh distributed processes with an application to monitor glass fiber strength. *Communications in Statistics-Simulation and Computation*, 51(6), 3168-3184.
- Hu, X., & Castagliola, P. (2017). Guaranteed conditional design of the median chart with estimated parameters. *Quality and Reliability Engineering International*, 33(8), 1873-1884.
- Huang, W. H., Wang, H., & Yeh, A. B. (2016). Control charts for the lognormal mean. *Quality and Reliability Engineering International*, 32(4), 1407-1416.
- Jardim, F. S., Chakraborti, S., & Epprecht, E. K. (2019). $\bar{X}$ Chart with estimated parameters: The conditional ARL distribution and new insights. *Production and Operations Management*, 28(6), 1545-1557.
- Jardim, F. S., Chakraborti, S., & Epprecht, E. K. (2020). Two perspectives for designing a Phase II control chart with estimated parameters: The case of the Shewhart $\bar{X}$ Chart. *Journal of Quality Technology*, 52(2), 198-217.
- Jensen, W. A., Jones-Farmer, L. A., Champ, C. W., & Woodall, W. H. (2006). Effects of parameter estimation on control chart properties: A literature review. *Journal of Quality Technology*, 38(4), 349-364.

- Jeske, D. R. (2016). Determining the reference sample size needed to control the accuracy of the conditional in-control ARL of a normal-theory CUSUM. *Quality and Reliability Engineering International*, 32(7), 2499-2504.
- Jones, L. A. (2002). The statistical design of EWMA control charts with estimated parameters. *Journal of Quality Technology*, 34(3), 277-288.
- Jones, L. A., Champ, C. W., & Rigdon, S. E. (2001). The performance of exponentially weighted moving average charts with estimated parameters. *Technometrics*, 43(2), 156-167.
- Jones, L. A., Champ, C. W., & Rigdon, S. E. (2004). The run length distribution of the CUSUM with estimated parameters. *Journal of Quality Technology*, 36(1), 95-108.
- Jones, M. A., & Steiner, S. H. (2012). Assessing the effect of estimation error on risk-adjusted CUSUM chart performance. *International Journal for Quality in Health Care*, 24(2), 176-181.
- Keefe, M. J., Woodall, W. H., & Jones-Farmer, L. A. (2015). The conditional in-control performance of self-starting control charts. *Quality Engineering*, 27(4), 488-499.
- Kemp, K. W. (1958). Formulae for calculating the operating characteristic and the average sample number of some sequential tests. *Journal of the Royal Statistical Society: Series B (Methodological)*, 20(2), 379-386.
- Khaw, K. W., Khoo, M. B. C., Yeong, W. C., & Wu, Z. (2017). Monitoring the coefficient of variation using a variable sample size and sampling interval control chart. *Communications in Statistics-Simulation and Computation*, 46(7), 5772-5794.

- Khoo, M. B. C., Saha, S., Lee, H. C., & Castagliola, P. (2019). Variable sampling interval exponentially weighted moving average median chart with estimated process parameters. *Quality and Reliability Engineering International*, 35(8), 2732-2748.
- Khoo, M. B. C., Teoh, W. L., Castagliola, P., & Lee, M. H. (2013). Optimal designs of the double sampling $\bar{X}$ chart with estimated parameters. *International Journal of Production Economics*, 144(1), 345-357.
- Khusna, H., Mashuri, M., Ahsan, M., Suhartono, S., & Prastyo, D. D. (2020). Bootstrap-based maximum multivariate CUSUM control chart. *Quality Technology & Quantitative Management*, 17(1), 52-74.
- Khusna, H., Mashuri, M., Suhartono, Prastyo, D. D., Lee, M. H., & Ahsan, M. (2019). Residual-based maximum MCUSUM control chart for joint monitoring the mean and variability of multivariate autocorrelated processes. *Production & Manufacturing Research*, 7(1), 364-394.
- Kushner, H. J., and G. G. Yin. (2003). *Stochastic Approximation and Recursive Algorithms and Applications*. Springer.
- Kumar, N., & Chakraborti, S. (2016). Phase II Shewhart-type control charts for monitoring times between events and effects of parameter estimation. *Quality and Reliability Engineering International*, 32(1), 315-328.
- Lee, P. H., Torng, C. C., Lin, C. H., & Chou, C. Y. (2022). Control chart pattern recognition using spectral clustering technique and support vector machine under gamma distribution. *Computers & Industrial Engineering*, 171(9), 108437.
- Li, Y., Pu, X., & Tsung, F. (2009). Adaptive charting schemes based on double sequential probability ratio tests. *Quality and Reliability Engineering International*, 25(1), 21-39.

- Lin, C. T., & Lin, C. J. (2020). Multi-response optimization of wire bonding process for evaluating alternative wire material. *Microelectronics Journal*, 106(12), 104925.
- Mahadik, S. B., & Godase, D. G. (2023a). A two-sided SPRT control chart for process mean. *Journal of Statistical Computation and Simulation*, In Press.
- Mahadik, S. B., & Godase, D. G. (2023b). The SPRT sign chart for process location. *Communications in Statistics-Theory and Methods*, 52(7), 2276-2290.
- Mahadik, S. B., Godase, D. G., & Teoh, W. L. (2021). A two-sided SPRT control chart for process dispersion. *Journal of Statistical Computation and Simulation*, 91(17), 3603-3614.
- Mahmoud, M. A., Henderson, G. R., Epprecht, E. K., & Woodall, W. H. (2010). Estimating the standard deviation in quality-control applications. *Journal of Quality Technology*, 42(4), 348-357.
- Malela-Majika, J. C., Shongwe, S. C., Aslam, M., Chong, Z. L., & Rapoo, E. M. (2021). Distribution-free double-sampling precedence monitoring scheme to detect unknown shifts in the location parameter. *Quality and Reliability Engineering International*, 37(8), 3580–3599.
- Maravelakis, P. E., Panaretos, J., & Psarakis, S. (2002). Effect of estimation of the process parameters on the control limits of the univariate control charts for process dispersion. *Communications in Statistics-Simulation and Computation*, 31(3), 443-461.
- McCracken, A. K., & Chakraborti, S. (2013). Control charts for joint monitoring of mean and variance: An overview. *Quality Technology & Quantitative Management*, 10(1), 17-36.

- Mim, F. N., Khoo, M. B. C., Saha, S., & Castagliola, P. (2022). Revised triple sampling $\bar{X}$ control charts for the mean with known and estimated process parameters. *International Journal of Production Research*, 60(16), 4911-4935.
- Montgomery, D. C. (2019). *Introduction to Statistical Quality Control*. (8th ed.). John Wiley & Sons.
- Moustakides, G. V. (1986). Optimal stopping times for detecting changes in distributions. *The Annals of Statistics*, 14(4), 1379-1387.
- Naderkhani, F. (2022). Time to signal distribution of multivariate bayesian control chart with dual sampling scheme. *International Journal of Production Research*, 60(20), 6124-6147.
- Ou, Y., Chen, N., & Khoo, M. B. C. (2015a). An efficient multivariate control charting mechanism based on SPRT. *International Journal of Production Research*, 53(7), 1937-1949.
- Ou, Y., Wen, D., Wu, Z., & Khoo, M. B. C. (2012a). A comparison study on effectiveness and robustness of control charts for monitoring process mean and variance. *Quality and Reliability Engineering International*, 28(1), 3-17.
- Ou, Y., Wu, Z., Chen, S., & Lee, K. M. (2010). An improved SPRT control chart for monitoring process mean. *The International Journal of Advanced Manufacturing Technology*, 51(4), 1045-1054.
- Ou, Y., Wu, Z., & Goh, T. N. (2011a). A new SPRT chart for monitoring process mean and variance. *International Journal of Production Economics*, 132(2), 303-314.
- Ou, Y., Wu, Z., Khoo, M. B. C., & Chen, N. (2015b). A rational sequential probability ratio test control chart for monitoring process shifts in mean and variance. *Journal of Statistical Computation and Simulation*, 85(9), 1765-1781.

- Ou, Y., Wu, Z., Lee, K. M., & Chen, S. (2012b). An optimal design algorithm of the SPRT chart for minimizing weighted ATS. *International Journal of Production Economics*, 139(2), 564-574.
- Ou, Y., Wu, Z., & Tsung, F. (2012c). A comparison study of effectiveness and robustness of control charts for monitoring process mean. *International Journal of Production Economics*, 135(1), 479-490.
- Ou, Y., Wu, Z., Yu, F. J., & Shamsuzzaman, M. (2011b). An SPRT control chart with variable sampling intervals. *The International Journal of Advanced Manufacturing Technology*, 56(9-12), 1149-1158.
- Ozsan, G., Testik, M. C., & Weiß, C. H. (2010). Properties of the exponential EWMA chart with parameter estimation. *Quality and Reliability Engineering International*, 26(6), 555-569.
- Page, E. S. (1954). Continuous inspection schemes. *Biometrika*, 41(1/2), 100-115.
- Parvin, R., Khoo, M. B., Saha, S., & Teoh, W. L. (2023). Proposed variable sampling interval maximum EWMA and distance EWMA charts with unknown process parameters. *Stat*, 12(1), e605.
- Perry, M. B., & Wang, Z. (2022). A distribution-free joint monitoring scheme for location and scale using individual observations. *Journal of Quality Technology*, 54(2), 144-161.
- Qiu, P. (2018). Some perspectives on nonparametric statistical process control. *Journal of Quality Technology*, 50(1), 49-65.
- Quesenberry, C. P. (1993). The effect of sample size on estimated limits for $\bar{X}$ and X control charts. *Journal of Quality Technology*, 25(4), 237-247.

- Quinino, R. D. C., Cruz, F. R., & Ho, L. L. (2020). Attribute inspection control charts for the joint monitoring of mean and variance. *Computers & Industrial Engineering*, 139(1), 106131.
- Rahim, M. A. (1989). Determination of optimal design parameters of joint $\bar{X}$ and R charts. *Journal of Quality Technology*, 21(1), 65-70.
- Raim, M. A., Lashkari, R. S., & Banerjee, P. K. (1988). Joint economic design of mean and variance control charts. *Engineering Optimization*, 14(1), 65-78.
- Rasay, H., & Alinezhad, E. (2022). Developing an adaptable sequential probability ratio test applicable for lifetime analysis of different continuous distributions. *Quality Technology & Quantitative Management*, 19(4), 511-530.
- Reynolds Jr, M. R., Amin, R. W., & Arnold, J. C. (1990). CUSUM charts with variable sampling intervals. *Technometrics*, 32(4), 371-384.
- Reynolds Jr, M. R., Amin, R. W., Arnold, J. C., & Nachlas, J. A. (1988). $\bar{X}$ charts with variable sampling intervals. *Technometrics*, 30(2), 181-192.
- Reynolds Jr, M. R., Lou, J., Lee, J., & Wang, S. (2013). The design of GLR control charts for monitoring the process mean and variance. *Journal of Quality Technology*, 45(1), 34-60.
- Reynolds Jr, M. R., & Stoumbos, Z. G. (1998). The SPRT chart for monitoring a proportion. *IIE Transactions*, 30(6), 545-561.
- Reynolds Jr, M. R., & Stoumbos, Z. G. (2004a). Control charts and the efficient allocation of sampling resources. *Technometrics*, 46(2), 200-214.
- Reynolds Jr, M. R., & Stoumbos, Z. G. (2004b). Should observations be grouped for effective process monitoring? *Journal of Quality Technology*, 36(4), 343-366.

- Reynolds Jr, M. R., & Stoumbos, Z. G. (2005). Should exponentially weighted moving average and cumulative sum charts be used with Shewhart limits? *Technometrics*, 47(4), 409-424.
- Rigdon, S. E., Cruthis, E. N., & Champ, C. W. (1994). Design strategies for individuals and moving range control charts. *Journal of Quality Technology*, 26(4), 274-287.
- Rinne, H. (2008). *The Weibull Distribution: A Handbook*. CRC Press.
- Roberts, S. W. (1959). Control chart tests based on geometric moving averages. *Technometrics*, 1(3), 239-250.
- Roes, K. C., Does, R. J. M. M., & Schurink, Y. (1993). Shewhart-type control charts for individual observations. *Journal of Quality Technology*, 25(3), 188-198.
- Saccucci, M. S., Amin, R. W., & Lucas, J. M. (1992). Exponentially weighted moving average control schemes with variable sampling intervals. *Communications in Statistics-simulation and Computation*, 21(3), 627-657.
- Saha, S., Khoo, M. B. C., Lee, M. H., & Castagliola, P. (2018). A side-sensitive modified group runs double sampling (SSMGRDS) control chart for detecting mean shifts. *Communications in Statistics-Simulation and Computation*, 47(5), 1353-1369.
- Saha, S., Khoo, M. B. C., Lee, M. H., & Haq, A. (2019). A variable sample size and sampling interval control chart for monitoring the process mean using auxiliary information. *Quality Technology & Quantitative Management*, 16(4), 389-406.
- Saleh, N. A., Mahmoud, M. A., & Abdel-Salam, A. S. G. (2013). The performance of the adaptive exponentially weighted moving average control chart with

- estimated parameters. *Quality and Reliability Engineering International*, 29(4), 595-606.
- Saleh, N. A., Mahmoud, M. A., Jones-Farmer, L. A., Zwetsloot, I. & Woodall, W. H. (2015a). Another look at the EWMA control chart with estimated parameters. *Journal of Quality Technology*, 47(4), 363-382.
- Saleh, N. A., Mahmoud, M. A., Keefe, M. J., & Woodall, W. H. (2015b). The difficulty in designing Shewhart $\bar{X}$ and X control charts with estimated parameters. *Journal of Quality Technology*, 47(2), 127-138.
- Saleh, N. A., Zwetsloot, I. M., Mahmoud, M. A., & Woodall, W. H. (2016). CUSUM charts with controlled conditional performance under estimated parameters. *Quality Engineering*, 28(4), 402-415.
- Saniga, E. M. (1977). Joint economically optimal design of $\bar{X}$ and R control charts. *Management Science*, 24(4), 420-431.
- Saniga, E. M. (1991). Joint statistical design of $\bar{X}$ and R control charts. *Journal of Quality Technology*, 23(2), 156-162.
- Sarmiento, M. G., Jardim, F. S., Chakraborti, S., & Epprecht, E. K. (2022). Design of variance control charts with estimated parameters: A head to head comparison between two perspectives. *Journal of Quality Technology*, 54(3), 249-268.
- Schoonhoven, M., Nazir, H. Z., Riaz, M., & Does, R. J. M. M. (2011a). Robust Location Estimators for the $\bar{X}$ Control Chart. *Journal of Quality technology*, 43(4), 363-379.
- Schoonhoven, M., Riaz, M., & Does, R. J. M. M. (2011b). Design and analysis of control charts for standard deviation with estimated parameters. *Journal of Quality Technology*, 43(4), 307-333.

- Shewhart, W. A. (1931). *Economic Control of Quality of Manufactured Product*.
D. Van Nostrand Company.
- Siegmund, D. (1975). Error probabilities and average sample number of the sequential probability ratio test. *Journal of the Royal Statistical Society: Series B (Methodological)*, 37(3), 394-401.
- Siegmund, D. (1979). Corrected diffusion approximations in certain random walk problems. *Advances in Applied Probability*, 11(4), 701-719.
- Solak, S., & Öner, M. (2020). Sequential decision fusion for abnormality detection via diffusive molecular communications. *IEEE Communications Letters*, 25(3), 825-829.
- Song, Z., Liu, Y., Li, Z., & Zhang, J. (2018). A weighted likelihood ratio test-based chart for monitoring process mean and variability. *Journal of Statistical Computation and Simulation*, 88(7), 1415-1436.
- Sparks, R. S. (2000). CUSUM charts for signalling varying location shifts. *Journal of Quality Technology*, 32(2), 157-171.
- Stoumbos, Z. G. (1993). *Control charts applying a sequential test at fixed sampling intervals with optional sampling at fixed times* [Doctoral dissertation, Virginia Polytechnic Institute].
- Stoumbos, Z. G., & Reynolds Jr, M. R. (1996). Control charts applying a general sequential test at each sampling point. *Sequential Analysis*, 15(2-3), 159-183.
- Stoumbos, Z. G., & Reynolds Jr, M. R. (1997a). Control charts applying a sequential test at fixed sampling intervals. *Journal of Quality Technology*, 29(1), 21-40.

- Stoumbos, Z. G., & Reynolds Jr, M. R. (1997b). Corrected diffusion theory approximations in evaluating properties of SPRT charts for monitoring a process mean. *Nonlinear Analysis: Theory, Methods & Applications*, 30(7), 3987-3996.
- Stoumbos, Z. G., & Reynolds Jr, M. R. (2001). The SPRT control chart for the process mean with samples starting at fixed times. *Nonlinear Analysis: Real World Applications*, 2(1), 1-34.
- Stoumbos, Z. G., Reynolds Jr, M. R., & Woodall, W. H. (2003). Control chart schemes for monitoring the mean and variance of processes subject to sustained shifts and drifts. In C. R. Rao & R. Khattree (Eds.), *Handbook of Statistics: Statistics in Industry* (pp. 553-571). Elsevier Science.
- Taguchi, G. (1980). *Introduction to Off-Line Quality Control*. Central Japan Quality Control Association.
- Tartakovsky, A., Nikiforov, I., & Basseville, M. (2015). *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. CRC Press.
- Teoh, W. L., Ong, L., Khoo, M. B. C., Castagliola, P., & Chong, Z. (2019). The variable sampling interval EWMA $\bar{X}$ chart with estimated process parameters. *Journal of Testing and Evaluation*, 49(2), 1237-1265.
- Testik, M. C. (2007). Conditional and marginal performance of the Poisson CUSUM control chart with parameter estimation. *International Journal of Production Research*, 45(23), 5621-5638.
- Testik, M. C., McCullough, B. D., & Borrar, C. M. (2006). The effect of estimated parameters on Poisson EWMA control charts. *Quality Technology & Quantitative Management*, 3(4), 513-527.
- Wald, A. (1945). Sequential tests of statistical hypotheses. *Annals of Mathematical Statistics*, 16(2), 117-186.

- Wald, A. (1947). *Sequential Analysis*. John Wiley & Sons.
- Wald, A., & Wolfowitz, J. (1948). Optimum character of the sequential probability ratio test. *The Annals of Mathematical Statistics*, 19(3), 326-339.
- Wu, Z., Jiao, J., Yang, M., Liu, Y., & Wang, Z. (2009). An enhanced adaptive CUSUM control chart. *IIE transactions*, 41(7), 642-653.
- Wu, Z., & Tian, Y. (2005). Weighted-loss-function CUSUM chart for monitoring mean and variance of a production process. *International Journal of Production Research*, 43(14), 3027-3044.
- Wu, Z., & Tian, Y. (2006). Weighted-loss-function control charts. *The International Journal of Advanced Manufacturing Technology*, 31(11), 107-115.
- Wu, Z., & Wang, Q. (1997). Optimization of joint $\bar{X}$ and S control charts with asymmetric control limits. *Process Control and Quality*, 3(10), 269-281.
- Wu, Z., Xie, M., & Tian, Y. (2002). Optimization design of the $\bar{X}$ & S charts for monitoring process capability. *Journal of Manufacturing Systems*, 21(2), 83-92.
- Wu, Z., Yang, M., Khoo, M. B. C., & Yu, F. J. (2010). Optimization designs and performance comparison of two CUSUM schemes for monitoring process shifts in mean and variance. *European Journal of Operational Research*, 205(1), 136-150.
- Xie, H. (1999). *Contributions to qualimetry* [Doctoral dissertation, The University of Manitoba].
- Yang, Z., Xie, M., Kuralmani, V., & Tsui, K. L. (2002). On the performance of geometric charts with estimated control limits. *Journal of Quality Technology*, 34(4), 448-458.

- Zaka, A., Jabeen, R., Ahmad, M., Aljohani, H. M., & Helmi, M. M. (2024). Control theory for skewed distribution under operation side of the telecommunication industry and hard-bake process in the semiconductor manufacturing process. *Measurement and Control*, 57(6), 703-723.
- Zhang, J., Zou, C., & Wang, Z. (2010). A control chart based on likelihood ratio test for monitoring process mean and variability. *Quality and Reliability Engineering International*, 26(1), 63-73.
- Zhang, M., Megahed, F. M., & Woodall, W. H. (2014a). Exponential CUSUM charts with estimated control limits. *Quality and Reliability Engineering International*, 30(2), 275-286.
- Zhang, P., Su, Q., Li, C., & Wang, T. (2014b). An economically designed sequential probability ratio test control chart for short-run production. *Computers & Industrial Engineering*, 78(12), 74-83.
- Zhang, S., & Wu, Z. (2007). A CUSUM scheme with variable sample sizes for monitoring process shifts. *The International Journal of Advanced Manufacturing Technology*, 33(7), 977-987.
- Zwillinger, D., & Kokoska, S. (1999). *CRC Standard Probability and Statistics Tables and Formulae*. CRC Press.

APPENDIX A

Derivation of the Formulae for $r_{a,b}$, c_a , P_0 , and $q_{a,(\xi+1)}$ of the OSPRT Chart using the cdf of the Standard Normal Distribution

The simplified expression for $r_{a,b}$ can be written as (see Equation (4.13)):

$$\begin{aligned} r_{a,b} &= \Pr \left[\frac{\Delta(b-a-0.5)+\gamma}{\eta^2} \leq \left(\frac{Z_{i,j}+k}{\eta} \right)^2 \leq \frac{\Delta(b-a+0.5)+\gamma}{\eta^2} \right] \\ &= \Pr \left[\Delta(b-a-0.5) + \gamma \leq (Z_{i,j} + k)^2 \leq \Delta(b-a+0.5) + \gamma \right] \end{aligned} \quad (\text{A.1})$$

To solve the inequality above for $Z_{i,j}$, it is necessary to consider the following three cases:

Case 1: When $\Delta(b-a+0.5) + \gamma \leq 0$, i.e., when both the lower and upper bounds of the inequality are negative or zero:

It is clear that the inequality in Equation (A.1) has no solution, since $(Z_{i,j} + k)^2 \geq 0$ for all real numbers of $Z_{i,j}$ and k . Hence, $r_{a,b} = 0$ in this case.

Case 2: When $\Delta(b-a-0.5) + \gamma \leq 0 < \Delta(b-a+0.5) + \gamma$, i.e., when the lower bound is negative or zero and the upper bound is positive:

The expression in Equation (A.1) is equivalent to

$$\begin{aligned} r_{a,b} &= \Pr \left[(Z_{i,j} + k)^2 \leq \Delta(b-a+0.5) + \gamma \right] \\ &= \Pr \left[-\sqrt{\Delta(b-a+0.5) + \gamma} \leq Z_{i,j} + k \leq \sqrt{\Delta(b-a+0.5) + \gamma} \right] \\ &= \Pr \left[-\sqrt{\Delta(b-a+0.5) + \gamma} - k \leq Z_{i,j} \leq \sqrt{\Delta(b-a+0.5) + \gamma} - k \right]. \end{aligned}$$

Since $Z_{i,j}$ follows the normal distribution $\mathcal{N}(\delta, \eta^2)$, the expression becomes

$$r_{a,b} = \Phi \left[\frac{\sqrt{\Delta(b-a+0.5)+\gamma}-k-\delta}{\eta} \right] - \Phi \left[\frac{-\sqrt{\Delta(b-a+0.5)+\gamma}-k-\delta}{\eta} \right]. \quad (\text{A.2})$$

Case 3: When $\Delta(b - a - 0.5) + \gamma > 0$, i.e., when both the lower and upper bounds are positive:

The expression in Equation (A.1) is equivalent to

$$\begin{aligned}
r_{a,b} &= \Pr\left[-\sqrt{\Delta(b - a + 0.5) + \gamma} \leq Z_{i,j} + k \leq -\sqrt{\Delta(b - a - 0.5) + \gamma}\right] \\
&\quad + \Pr\left[\sqrt{\Delta(b - a - 0.5) + \gamma} \leq Z_{i,j} + k \leq \sqrt{\Delta(b - a + 0.5) + \gamma}\right] \\
&= \Pr\left[-\sqrt{\Delta(b - a + 0.5) + \gamma} - k \leq Z_{i,j} \leq -\sqrt{\Delta(b - a - 0.5) + \gamma} - k\right] \\
&\quad + \Pr\left[\sqrt{\Delta(b - a - 0.5) + \gamma} - k \leq Z_{i,j} \leq \sqrt{\Delta(b - a + 0.5) + \gamma} - k\right] \\
&= \Phi\left[\frac{-\sqrt{\Delta(b-a-0.5)+\gamma-k-\delta}}{\eta}\right] - \Phi\left[\frac{-\sqrt{\Delta(b-a+0.5)+\gamma-k-\delta}}{\eta}\right] + \Phi\left[\frac{\sqrt{\Delta(b-a+0.5)+\gamma-k-\delta}}{\eta}\right] \\
&\quad - \Phi\left[\frac{\sqrt{\Delta(b-a-0.5)+\gamma-k-\delta}}{\eta}\right]. \tag{A.3}
\end{aligned}$$

Similarly, to express c_a in terms of the cdf of the standard normal distribution, first write c_a as (see Equation (4.15)):

$$c_a = \Pr\left[g + \Delta(a - 1) + \gamma \leq (Z_{i,1} + k)^2 \leq g + \Delta a + \gamma\right]. \tag{A.4}$$

To solve the inequality above for $Z_{i,1}$, it is necessary to consider the following three cases:

Case 1: When $g + \Delta a + \gamma \leq 0$, i.e., when both the lower and upper bounds of the inequality are negative or zero:

It is clear that the inequality in Equation (A.4) has no solution, since $(Z_{i,1} + k)^2 \geq 0$ for all real numbers of $Z_{i,1}$ and k . Hence, $c_a = 0$ in this case.

Case 2: When $g + \Delta(a - 1) + \gamma \leq 0 < g + \Delta a + \gamma$, i.e., when the lower bound is negative or zero and the upper bound is positive:

Using the method similar to the derivation of $r_{a,b}$, the expression in Equation (A.4) is equivalent to

$$c_a = \Phi \left[\frac{\sqrt{g+\Delta a+\gamma-k-\delta}}{\eta} \right] - \Phi \left[\frac{-\sqrt{g+\Delta a+\gamma-k-\delta}}{\eta} \right]. \quad (\text{A.5})$$

Case 3: When $g + \Delta(a - 1) + \gamma > 0$, i.e., when both the lower and upper bounds are positive:

Using the method similar to the derivation of $r_{a,b}$, the expression in Equation (A.4) is equivalent to

$$c_a = \Phi \left[\frac{-\sqrt{g+\Delta(a-1)+\gamma-k-\delta}}{\eta} \right] - \Phi \left[\frac{-\sqrt{g+\Delta a+\gamma-k-\delta}}{\eta} \right] + \Phi \left[\frac{\sqrt{g+\Delta a+\gamma-k-\delta}}{\eta} \right] - \Phi \left[\frac{\sqrt{g+\Delta(a-1)+\gamma-k-\delta}}{\eta} \right]. \quad (\text{A.6})$$

To express P_0 in terms of the cdf of the standard normal distribution, first write P_0 as (see Equation (4.18)):

$$P_0 = \Pr \left[(Z_{i,1} + k)^2 \leq g + \gamma \right]. \quad (\text{A.7})$$

To solve the inequality above for $Z_{i,1}$, it is necessary to consider the following two cases:

Case 1: When $g + \gamma \leq 0$, i.e., when the upper bound is negative or zero:

It is clear that the inequality in Equation (A.7) has no solution, since $(Z_{i,1} + k)^2 \geq 0$ for all real numbers of $Z_{i,1}$ and k . Hence, $P_0 = 0$ in this case.

Case 2: When $g + \gamma > 0$, i.e., when the upper bound is positive:

Using the method similar to the derivation of $r_{a,b}$, the expression in Equation (A.7) is equivalent to

$$P_0 = \Phi \left[\frac{\sqrt{g+\gamma}-k-\delta}{\eta} \right] - \Phi \left[\frac{-\sqrt{g+\gamma}-k-\delta}{\eta} \right]. \quad (\text{A.8})$$

To express $q_{a,(\xi+1)}$ in terms of the cdf of the standard normal distribution, first write $q_{a,(\xi+1)}$ as (see Equation (4.19)):

$$q_{a,(\xi+1)} = \Pr \left[(Z_{i,j} + k)^2 \leq \Delta(0.5 - a) + \gamma \right]. \quad (\text{A.9})$$

To solve the inequality above for $Z_{i,j}$, it is necessary to consider the following two cases:

Case 1: When $\Delta(0.5 - a) + \gamma \leq 0$, i.e., when the upper bound is negative or zero:

It is clear that the inequality in Equation (A.9) has no solution, since $(Z_{i,j} + k)^2 \geq 0$ for all real numbers of $Z_{i,j}$ and k . Hence, $q_{a,(\xi+1)} = 0$ in this case.

Case 2: When $\Delta(0.5 - a) + \gamma > 0$, i.e., when the upper bound is positive:

Using the method similar to the derivation of $r_{a,b}$, the expression in Equation (A.9) is equivalent to

$$q_{a,(\xi+1)} = \Phi \left[\frac{\sqrt{\Delta(0.5-a)+\gamma}-k-\delta}{\eta} \right] - \Phi \left[\frac{-\sqrt{\Delta(0.5-a)+\gamma}-k-\delta}{\eta} \right]. \quad (\text{A.10})$$